Epic-FHIR — API category: Clinical General (sections 10 – 10.3.9)

Source: Epic on FHIR (fhir.epic.com) documentation, mirrored 2026-11. Text-only edition for AI ingestion (NotebookLM); pages follow the source order (its table of contents when one exists).

10 API category: Clinical General

File: site/apis/10-Clinical-General/index.html

R4 (66)

API | Method | URL template | Type
AdverseEvent.Read (R4) | GET | /api/FHIR/R4/AdverseEvent/{ID} | Industry-Standard
AdverseEvent.Search (R4) | GET | /api/FHIR/R4/AdverseEvent?seriousness={seriousness}&study={study}&subject={subject} | Industry-Standard
AllergyIntolerance.Create (Patient Chart) (R4) | POST | api/FHIR/R4/AllergyIntolerance | Industry-Standard
AllergyIntolerance.Read (Patient Chart) (R4) | GET | api/FHIR/R4/AllergyIntolerance/{ID} | USCDI
AllergyIntolerance.Search (Patient Chart) (R4) | GET | /api/FHIR/R4/AllergyIntolerance?clinical-status={clinical-status}&patient={patient} | USCDI
Condition.Create (Problems) (R4) | POST | api/FHIR/R4/Condition | Industry-Standard
Condition.Read (Care Plan Problem) (R4) | GET | /api/FHIR/R4/Condition/{ID} | USCDI
Condition.Read (Dental Finding) (R4) | GET | /api/FHIR/R4/Condition/{ID} | Industry-Standard
Condition.Read (Encounter Diagnosis) (R4) | GET | api/FHIR/R4/Condition/{ID} | USCDI
Condition.Read (Genomics) (R4) | | api/FHIR/R4/Condition/{ID} | Industry-Standard
Condition.Read (Health Concerns) (R4) | GET | api/FHIR/R4/Condition/{ID} | USCDI
Condition.Read (Infection) (R4) | GET | /api/FHIR/R4/Condition/{ID} | Industry-Standard
Condition.Read (Medical History) (R4) | GET | /api/FHIR/R4/Condition/{ID} | Industry-Standard
Condition.Read (Problems) (R4) | GET | api/FHIR/R4/Condition/{ID} | USCDI
Condition.Read (Reason for Visit) (R4) | GET | /api/FHIR/R4/Condition/{ID} | Industry-Standard
Condition.Search (Care Plan Problem) (R4) | GET | /api/FHIR/R4/Condition?_id={_id} | USCDI
Condition.Search (Dental Finding) (R4) | GET | /api/FHIR/R4/Condition?category={category}&clinical-status={clinical-status}&patient={patient} | Industry-Standard
Condition.Search (Encounter Diagnosis) (R4) | GET | api/FHIR/R4/Condition?patient={patient}&subject={subject}&category={category}&encounter={encounter}&_id={_id} | USCDI
Condition.Search (Genomics) (R4) | GET | api/FHIR/R4/Condition?category={category}&patient={patient} | Industry-Standard
Condition.Search (Health Concerns) (R4) | GET | api/FHIR/R4/Condition?patient={patient}&subject={subject}&category={category}&clinical-status={clinical-status}&_id={_id} | USCDI
Condition.Search (Infection) (R4) | GET;POST | /api/FHIR/R4/Condition?abatement-date={abatement-date}&category={category}&clinical-status={clinical-status}&code={code}&encounter={encounter}&onset-date={onset-date}&patient={patient}&recorded-date={recorded-date}&subject={subject} | Industry-Standard
Condition.Search (Medical History) (R4) | GET | /api/FHIR/R4/Condition?category={category}&patient={patient}&subject={subject} | Industry-Standard
Condition.Search (Problems) (R4) | GET | api/FHIR/R4/Condition?subject={subject}&patient={patient}&clinical-status={clinical-status}&category={category} | USCDI
Condition.Search (Reason for Visit) (R4) | GET | /api/FHIR/R4/Condition?category={category}&encounter={encounter}&patient={patient}&subject={subject} | Industry-Standard
FamilyMemberHistory.Read (R4) | GET | /api/FHIR/R4/FamilyMemberHistory/{ID} | Industry-Standard
FamilyMemberHistory.Search (R4) | GET;POST | /api/FHIR/R4/FamilyMemberHistory?patient={patient} | Industry-Standard
Flag.Read (Health Concerns) (R4) | GET | /api/FHIR/R4/Flag/{ID} | Industry-Standard
Flag.Read (Infection) (R4) | GET | /api/FHIR/R4/Flag/{ID} | Industry-Standard
Flag.Search (Health Concerns) (R4) | GET | /api/FHIR/R4/Flag?category={category}&patient={patient}&status={status}&subject={subject} | Industry-Standard
Flag.Search (Infection) (R4) | GET | /api/FHIR/R4/Flag?category={category}&patient={patient}&status={status}&subject={subject} | Industry-Standard
List.Read (Allergies) (R4) | GET | /api/FHIR/R4/List/{ID} | Industry-Standard
List.Read (Family History) (R4) | GET | /api/FHIR/R4/List/{ID} | Industry-Standard
List.Read (Hospital Problems) (R4) | GET | /api/FHIR/R4/List/{ID} | Industry-Standard
List.Read (Problems) (R4) | GET | /api/FHIR/R4/List/{ID} | Industry-Standard
List.Search (Allergies) (R4) | GET;POST | /api/FHIR/R4/List?code={code}&patient={patient}&subject={subject} | Industry-Standard
List.Search (Family History) (R4) | GET | /api/FHIR/R4/List?code={code}&patient={patient} | Industry-Standard
List.Search (Hospital Problems) (R4) | GET;POST | /api/FHIR/R4/List?code={code}&encounter={encounter}&patient={patient}&subject={subject} | Industry-Standard
List.Search (Problems) (R4) | GET;POST | /api/FHIR/R4/List?code={code}&patient={patient}&subject={subject} | Industry-Standard
Observation.Read (Care Experience Preference) (R4) | GET | /api/FHIR/R4/Observation/{ID} | USCDI
Observation.Read (Genomics) (R4) | GET | /api/FHIR/R4/Observation/{ID} | Industry-Standard
Observation.Read (Phenotype) (R4) | GET | /api/FHIR/R4/Observation/{ID} | Industry-Standard
Observation.Read (SDOH Assessments) (R4) | GET | /api/FHIR/R4/Observation/{ID} | USCDI
Observation.Read (Social History) (R4) | GET | api/FHIR/R4/Observation/{ID} | USCDI
Observation.Search (Care Experience Preference) (R4) | GET | /api/FHIR/R4/Observation?based-on={based-on}&category={category}&code={code}&date={date}&focus={focus}&issued={issued}&patient={patient}&subject={subject}&_count={_count}&_id={_id}&_include={_include}&_lastupdated={_lastupdated}&_revinclude={_revinclude} | USCDI
Observation.Search (Genomics) (R4) | GET | /api/FHIR/R4/Observation?patient={patient}&category=genomics&code={code}&based-on={based-on} | Industry-Standard
Observation.Search (Phenotype) (R4) | GET | /api/FHIR/R4/Observation?based-on={based-on}&category={category}&code={code}&date={date}&focus={focus}&issued={issued}&patient={patient}&subject={subject}&_count={_count}&_id={_id}&_include={_include}&_revinclude={_revinclude} | Industry-Standard

Observation.Search (SDOH Assessments) (R4) | GET | /api/FHIR/R4/Observation?category={category}&code={code}&date={date}&patient={patient}&subject={subject} | USCDI
Observation.Search (Social History) (R4) | GET | api/FHIR/R4/Observation?patient={patient}&subject={subject}&category={category}&code={code}&issued={issued} | USCDI
Procedure.Read (Nursing Intervention) (R4) | GET | /api/FHIR/R4/Procedure/{ID} | Industry-Standard
Procedure.Read (Orders) (R4) | GET | api/FHIR/R4/Procedure/{ID} | USCDI
Procedure.Read (Patient-Reported Surgical History) (R4) | GET | /api/FHIR/R4/Procedure/{ID} | Industry-Standard
Procedure.Read (Restricting Intervention) (R4) | GET | /api/FHIR/R4/Procedure/{ID} | Industry-Standard
Procedure.Read (SDOH Intervention) (R4) | GET | /api/FHIR/R4/Procedure/{ID} | USCDI
Procedure.Read (Surgeries) (R4) | GET | /api/FHIR/R4/Procedure/{ID} | USCDI
Procedure.Search (Nursing Intervention) (R4) | GET | /api/FHIR/R4/Procedure?category={category}&date={date}&patient={patient}&subject={subject}&_id={_id} | Industry-Standard
Procedure.Search (Orders) (R4) | GET | api/FHIR/R4/Procedure?date={date}&patient={patient}&subject={subject}&category={category} | USCDI
Procedure.Search (Patient-Reported Surgical History) (R4) | GET | /api/FHIR/R4/Procedure?category=387713003&patient={patient}&subject={subject} | Industry-Standard
Procedure.Search (Restricting Intervention) (R4) | GET | /api/FHIR/R4/Procedure?category={category}&patient={patient}&subject={subject} | Industry-Standard
Procedure.Search (SDOH Intervention) (R4) | GET | /api/FHIR/R4/Procedure?category={category}&code={code}&date={date}&patient={patient}&subject={subject} | USCDI
Procedure.Search (Surgeries) (R4) | GET | /api/FHIR/R4/Procedure?date={date}&patient={patient}&subject={subject}&category=387713003 | USCDI
ResearchStudy.Read (R4) | GET | api/FHIR/R4/ResearchStudy/{ID} | Industry-Standard
ResearchStudy.Search (R4) | GET | api/FHIR/R4/ResearchStudy?identifier={identifier} | Industry-Standard
ResearchSubject.Read (R4) | GET | /api/FHIR/R4/ResearchSubject/{ID} | Industry-Standard
ResearchSubject.Search (R4) | GET;POST | /api/FHIR/R4/ResearchSubject?patient={patient}&status={status}&study={study} | Industry-Standard
Substance.Read (R4) | GET | /api/FHIR/R4/Substance/{ID} | Industry-Standard
Substance.Search (R4) | GET | /api/FHIR/R4/Substance?_id={_id} | Industry-Standard

STU3 (21)

API | Method | URL template | Type
AllergyIntolerance.Create (STU3) | POST | api/FHIR/STU3/AllergyIntolerance | Industry-Standard
AllergyIntolerance.Read (STU3) | GET | api/FHIR/STU3/AllergyIntolerance/{ID} | USCDI
AllergyIntolerance.Search (STU3) | GET | /api/FHIR/STU3/AllergyIntolerance?patient={patient} | USCDI
Condition.Create (Problems) (STU3) | POST | api/FHIR/STU3/Condition | Industry-Standard
Condition.Read (Encounter Diagnosis, Problems) (STU3) | GET | api/FHIR/STU3/Condition/{ID} | USCDI
Condition.Search (Encounter Diagnosis, Problems) (STU3) | GET | /api/FHIR/STU3/Condition?subject={subject}&patient={patient}&clinical-status={clinical-status}&category={category}&encounter={encounter} | USCDI
Flag.Read (Health Concern) (STU3) | GET | /api/FHIR/STU3/Flag/{ID} | Industry-Standard
Flag.Read (Infection) (STU3) | GET | /api/FHIR/STU3/Flag/{ID} | Industry-Standard
Flag.Search (Health Concern) (STU3) | GET | /api/FHIR/STU3/Flag?category={category}&patient={patient}&status={status}&subject={subject}&_id={_id} | Industry-Standard
Flag.Search (Infection) (STU3) | GET | /api/FHIR/STU3/Flag?category={category}&patient={patient}&status={status}&subject={subject}&_id={_id} | Industry-Standard
Observation.$lastn (Social History) (STU3) | GET | /api/FHIR/STU3/Observation/$lastn?category={category}&code={code}&max={max}&patient={patient}&subject={subject} | Industry-Standard
Observation.Read (Social History) (STU3) | GET | api/FHIR/STU3/Observation/{ID} | USCDI
Observation.Search (Social History) (STU3) | GET | api/FHIR/STU3/Observation?patient={patient}&category={category}&subject={subject}&issued={issued}&code={code} | USCDI
Procedure.Read (Nursing Intervention) (STU3) | GET | /api/FHIR/STU3/Procedure/{ID} | Industry-Standard
Procedure.Read (Orders, Surgeries) (STU3) | GET | api/FHIR/STU3/Procedure/{ID} | USCDI
Procedure.Read (Restricting Intervention) (STU3) | GET | /api/FHIR/STU3/Procedure/{ID} | Industry-Standard
Procedure.Read (Surgical History) (STU3) | GET | /api/FHIR/STU3/Procedure/{ID} | Industry-Standard
Procedure.Search (Nursing Intervention) (STU3) | GET | /api/FHIR/STU3/Procedure?category={category}&date={date}&patient={patient}&subject={subject}&_id={_id} | Industry-Standard
Procedure.Search (Orders, Surgeries) (STU3) | GET | api/FHIR/STU3/Procedure?patient={patient}&subject={subject}&date={date}&category={category} | USCDI
Procedure.Search (Restricting Intervention) (STU3) | GET | /api/FHIR/STU3/Procedure?category={category}&patient={patient}&subject={subject} | Industry-Standard
Procedure.Search (Surgical History) (STU3) | GET | /api/FHIR/STU3/Procedure?category=387713003&patient={patient}&subject={subject} | Industry-Standard

DSTU2 (9)

API | Method | URL template | Type
AllergyIntolerance.Read (DSTU2) | GET | api/FHIR/DSTU2/AllergyIntolerance/{ID} | USCDI
AllergyIntolerance.Search (DSTU2) | GET | /api/FHIR/DSTU2/AllergyIntolerance?patient={patient}&subject={subject} | USCDI
Condition.Read (Problems) (DSTU2) | GET | api/FHIR/DSTU2/Condition/{ID} | USCDI
Condition.Search (Problems) (DSTU2) | GET | api/FHIR/DSTU2/Condition?patient={patient}&_id={_id}&clinicalStatus={clinicalStatus}&category={category}&subject={subject} | USCDI
FamilyMemberHistory.Search (DSTU2) | GET | api/FHIR/DSTU2/FamilyMemberHistory?patient={patient}&subject={subject} | Industry-Standard
Observation.Read (Social History) (DSTU2) | GET | api/FHIR/DSTU2/Observation/{ID} | USCDI

Observation.Search (Social History) (DSTU2) | GET | api/FHIR/DSTU2/Observation?code={code}&patient={patient}&subject={subject}&category={category}&issued={issued} | USCDI
Procedure.Read (Orders) (DSTU2) | GET | api/FHIR/DSTU2/Procedure/{ID} | USCDI
Procedure.Search (Orders) (DSTU2) | GET | api/FHIR/DSTU2/Procedure?patient={patient}&subject={subject}&date={date} | USCDI

10.1 Clinical General: R4

File: site/apis/10-Clinical-General/1-R4/index.html

* AdverseEvent.Read (R4)
* AdverseEvent.Search (R4)
* AllergyIntolerance.Create (Patient Chart) (R4)
* AllergyIntolerance.Read (Patient Chart) (R4)
* AllergyIntolerance.Search (Patient Chart) (R4)
* Condition.Create (Problems) (R4)
* Condition.Read (Care Plan Problem) (R4)
* Condition.Read (Dental Finding) (R4)
* Condition.Read (Encounter Diagnosis) (R4)
* Condition.Read (Genomics) (R4)
* Condition.Read (Health Concerns) (R4)
* Condition.Read (Infection) (R4)
* Condition.Read (Medical History) (R4)
* Condition.Read (Problems) (R4)
* Condition.Read (Reason for Visit) (R4)
* Condition.Search (Care Plan Problem) (R4)
* Condition.Search (Dental Finding) (R4)
* Condition.Search (Encounter Diagnosis) (R4)
* Condition.Search (Genomics) (R4)
* Condition.Search (Health Concerns) (R4)
* Condition.Search (Infection) (R4)
* Condition.Search (Medical History) (R4)
* Condition.Search (Problems) (R4)
* Condition.Search (Reason for Visit) (R4)
* FamilyMemberHistory.Read (R4)
* FamilyMemberHistory.Search (R4)
* Flag.Read (Health Concerns) (R4)
* Flag.Read (Infection) (R4)
* Flag.Search (Health Concerns) (R4)
* Flag.Search (Infection) (R4)
* List.Read (Allergies) (R4)
* List.Read (Family History) (R4)
* List.Read (Hospital Problems) (R4)
* List.Read (Problems) (R4)
* List.Search (Allergies) (R4)
* List.Search (Family History) (R4)
* List.Search (Hospital Problems) (R4)
* List.Search (Problems) (R4)
* Observation.Read (Care Experience Preference) (R4)
* Observation.Read (Genomics) (R4)
* Observation.Read (Phenotype) (R4)
* Observation.Read (SDOH Assessments) (R4)
* Observation.Read (Social History) (R4)
* Observation.Search (Care Experience Preference) (R4)
* Observation.Search (Genomics) (R4)
* Observation.Search (Phenotype) (R4)

* Observation.Search (SDOH Assessments) (R4)
* Observation.Search (Social History) (R4)
* Procedure.Read (Nursing Intervention) (R4)
* Procedure.Read (Orders) (R4)
* Procedure.Read (Patient-Reported Surgical History) (R4)
* Procedure.Read (Restricting Intervention) (R4)
* Procedure.Read (SDOH Intervention) (R4)
* Procedure.Read (Surgeries) (R4)
* Procedure.Search (Nursing Intervention) (R4)
* Procedure.Search (Orders) (R4)
* Procedure.Search (Patient-Reported Surgical History) (R4)
* Procedure.Search (Restricting Intervention) (R4)
* Procedure.Search (SDOH Intervention) (R4)
* Procedure.Search (Surgeries) (R4)
* ResearchStudy.Read (R4)
* ResearchStudy.Search (R4)
* ResearchSubject.Read (R4)
* ResearchSubject.Search (R4)
* Substance.Read (R4)
* Substance.Search (R4)

10.1.1 AdverseEvent.Read (R4)

File: site/apis/10-Clinical-General/1-R4/981-AdverseEvent-Read-R4.html
Epic API ID | 981
FHIR version | R4
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | February 2020
HTTP method | GET
URL template | /api/FHIR/R4/AdverseEvent/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=981

Description

The AdverseEvent resource returns data about an event that caused unintended physical injury resulting from or contributed to by medical care, a research study, or other healthcare setting factors. These events might require additional monitoring, treatment, or hospitalization, or might result in the death of a patient.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | String | False | | | The Adverse Event FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
AdverseEvent | AdverseEvent | False | | | A single instance of a FHIR AdverseEvent resource.
AdverseEvent.actuality | String | False | | | The nature of the adverse event, either real or potential.
AdverseEvent.extension (also: adverse-event-comments) | Extension | True | Yes | | This element is returned starting in the May 2022 version of Epic. Free-text comments associated with an adverse event.
AdverseEvent.extension (also: adverse-event-expected) | Extension | True | Yes | | Whether this adverse event is expected.
AdverseEvent.extension (also: adverse-event-resolved-date) | Extension | True | Yes | | The date the adverse event was documented as ended.
AdverseEvent.extension (also: adverse-event-severity-history) | Extension | False | Yes | | A record of the historic severities for a particular adverse event.
AdverseEvent.extension.extension (also: adverse-event-severity-history-line) | Extension | False | Yes | | A specific instance in the severity history for a particular adverse event.
AdverseEvent.extension.extension.extension (also: adverse-event-severity) | Extension | False | Yes | | The grade of an adverse event at a point in time.

AdverseEvent.extension.extension.extension (also: adverse-event-severity-start-date) | Extension | False | Yes | | The date the adverse event was documented as started.
AdverseEvent.category | CodeableConcept | False | Yes | | CodeableConcept value.
AdverseEvent.date | String | False | | | When the event occurred.
AdverseEvent.encounter | Reference | False | | | A FHIR reference.
AdverseEvent.event | CodeableConcept | False | | | Type of the event itself in relation to the subject.
AdverseEvent.id | String | False | | | The Adverse Event FHIR ID.
AdverseEvent.identifier | Identifier | False | | | Business identifier for the event.
AdverseEvent.location | Reference | False | | | A FHIR reference.
AdverseEvent.outcome | CodeableConcept | False | | | The outcome of the adverse event. Currently providing only "resolved" and "ongoing."
AdverseEvent.recordedDate | String | False | | | When the event was recorded from the audit trail. This is when the event was first recorded.
AdverseEvent.recorder | Reference | False | | | The recorder of the adverse event. This is the user who first recorded the adverse event.
AdverseEvent.seriousness | CodeableConcept | False | | | The seriousness of the event. This is populated with "Serious" or "Non-Serious."
AdverseEvent.severity | CodeableConcept | False | | | The severity of the adverse event itself, in direct relation to the subject.
AdverseEvent.study | Reference | False | Yes | | The research study related to this adverse event.
AdverseEvent.subject | Reference | False | | | Subject impacted by the event, such as the patient.
AdverseEvent.suspectEntity | SuspectEntity | False | Yes | | The suspected agent causing the adverse event.

Sample request

```
https://hostname/instance/api/FHIR/R4/AdverseEvent/eBrj0mrZZ9-WmgLrAXW.ZQmF3xBGWbDn1vkbtSszAQnY3
```

Sample response

```
<AdverseEvent xmlns="http://hl7.org/fhir">
  <id value="eBrj0mrZZ9-WmgLrAXW.ZQmF3xBGWbDn1vkbtSszAQnY3"/>
  <extension url="http://open.epic.com/FHIR/StructureDefinition/adverse-event-expected">
    <valueCodeableConcept>
      <coding>
        <system value="urn:oid:1.2.840.114350.1.72.1.7.7.10.768076.7056"/>
        <code value="0"/>
        <display value="No"/>
      </coding>
      <text value="0"/>
    </valueCodeableConcept>
  </extension>
  <extension url="http://open.epic.com/FHIR/StructureDefinition/adverse-event-severity-history">
    <extension url="http://open.epic.com/FHIR/StructureDefinition/adverse-event-severity-history-line">
      <extension url="http://open.epic.com/FHIR/StructureDefinition/adverse-event-severity">
        <valueCodeableConcept>
          <coding>
            <system value="https://www.hl7.org/fhir/valueset-adverse-event-severity.html"/>
            <code value="mild"/>
            <display value="Mild"/>
          </coding>
          <coding>
            <system value="urn:oid:1.2.840.114350.1.13.861.1.7.10.768076.7010"/>
            <code value="1"/>
            <display value="Mild"/>
          </coding>
          <text value="Mild"/>
        </valueCodeableConcept>
      </extension>
      <extension url="http://open.epic.com/FHIR/StructureDefinition/adverse-event-severity-start-date">
        <valueDate value="2019-11-01"/>
      </extension>
    </extension>
    <extension url="http://open.epic.com/FHIR/StructureDefinition/adverse-event-severity-history-line">
      <extension url="http://open.epic.com/FHIR/StructureDefinition/adverse-event-severity">
        <valueCodeableConcept>
          <coding>
            <system value="https://www.hl7.org/fhir/valueset-adverse-event-severity.html"/>
            <code value="severe"/>
            <display value="Severe"/>
          </coding>
          <coding>
            <system value="urn:oid:1.2.840.114350.1.13.861.1.7.10.768076.7010"/>
            <code value="3"/>
            <display value="Severe"/>
          </coding>
          <text value="Severe"/>
        </valueCodeableConcept>
      </extension>
      <extension url="http://open.epic.com/FHIR/StructureDefinition/adverse-event-severity-start-date">
        <valueDate value="2019-11-08"/>
      </extension>
    </extension>
  </extension>
  <identifier>
    <use value="usual"/>
    <system value="urn:oid:1.2.840.114350.1.13.861.1.7.2.768076"/>
    <value value="161246"/>
  </identifier>
  <actuality value="actual"/>
  <event>
```

```
    <coding>
      <system value="https://www.meddra.org/"/>
      <code value="10016256"/>
      <display value="Fatigue"/>
    </coding>
    <coding>
      <system value="urn:oid:1.2.840.114350.1.13.861.1.7.2.696871"/>
      <code value="888001016"/>
      <display value="Fatigue"/>
    </coding>
    <text value="Fatigue"/>
  </event>
  <subject>
    <reference value="Patient/eea8Msv-9WjX-ffbBcv-4rw3"/>
    <display value="Research, Ellie"/>
  </subject>
  <date value="2019-11-01"/>
  <recordedDate value="2019-11-08T15:05:40Z"/>
  <seriousness>
    <coding>
      <system value="https://www.hl7.org/fhir/valueset-adverse-event-seriousness.html"/>
      <code value="Serious"/>
      <display value="Serious"/>
    </coding>
    <text value="Serious"/>
  </seriousness>
  <severity>
    <coding>
      <system value="https://www.hl7.org/fhir/valueset-adverse-event-severity.html"/>
      <code value="severe"/>
      <display value="Severe"/>
    </coding>
    <coding>
      <system value="urn:oid:1.2.840.114350.1.13.861.1.7.10.768076.7010"/>
      <code value="3"/>
      <display value="Severe"/>
    </coding>
    <text value="Severe"/>
  </severity>
  <outcome>
    <coding>
      <system value="https://www.hl7.org/fhir/valueset-adverse-event-outcome.html"/>
      <code value="ongoing"/>
      <display value="Ongoing"/>
    </coding>
    <text value="Ongoing"/>
  </outcome>
  <recorder>
    <reference value="Practitioner/e7gJ3G0n9-Dhv8jsHuiI9pw3"/>
    <display value="Steven Van Manen"/>
  </recorder>
  <suspectEntity>
    <instance>
      <reference value="ResearchStudy/eAO.XWJpIicOp3xl5CLneUQ3"/>
      <display value="Breast Cancer Stage II-BRAC0142"/>
    </instance>
    <causality>
      <assessment>
        <coding>
          <system value="https://www.hl7.org/fhir/valueset-adverse-event-causality-assess.html"/>
          <code value="Probably-Likely"/>
          <display value="Probably/Likely"/>
        </coding>
        <coding>
          <system value="urn:oid:1.2.840.114350.1.13.861.1.7.10.768076.7021"/>
          <code value="4"/>
          <display value="Probable"/>
        </coding>
        <text value="Probably/Likely"/>
      </assessment>
    </causality>
  </suspectEntity>
  <study>
    <reference value="ResearchStudy/eAO.XWJpIicOp3xl5CLneUQ3"/>
    <display value="Breast Cancer Stage II-BRAC0142"/>
  </study>
</AdverseEvent>
```

10.1.2 AdverseEvent.Search (R4)

File: site/apis/10-Clinical-General/1-R4/982-AdverseEvent-Search-R4.html
Epic API ID | 982
FHIR version | R4
Categories | Clinical General
API type | Industry-Standard
USCDI | No

Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | February 2020
HTTP method | GET
URL template | /api/FHIR/R4/AdverseEvent?seriousness={seriousness}&study={study}&subject={subject}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=982

Description

The AdverseEvent resource returns data about an event that caused unintended physical injury resulting from or contributed to by medical care, a research study, or other healthcare setting factors. These events might require additional monitoring, treatment, or hospitalization, or might result in the death of a patient.

A user or staff member accessing this FHIR resource must have security to vew adverse events for research studies.

This resource requires additional setup if the organization has custom category list values for adverse event grades or research study attribution.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
research-subject | String | Available starting in Epic version August 2024 and later. This parameter must be specified unless both study and subject are specified. | | | A ResearchSubject FHIR ID
seriousness | String | True | | | Seriousness of the event. Possible values are: Serious Non-Serious
study | String | In Epic version May 2024 and earlier, both study and subject are required. In Epic version August 2024 and later, either study+subject, or research-subject is required. | | | A ResearchStudy FHIR ID.
subject | String | In Epic version May 2024 and earlier, both study and subject are required. In Epic version August 2024 and later, either study+subject, or research-subject is required. | | | A Patient FHIR ID.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
actuality | String | True | | | The nature of the adverse event, either real or potential.
category | String | True | | | Unsupported
date | String | True | | | When the event occurred.
event | String | True | | | Type of the event itself in relation to the subject.
location | String | True | | | Unsupported
recorder | String | True | | | The recorder of the adverse event. This is the user who first recorded the adverse event.
resultingcondition | String | True | | | Unsupported
severity | String | True | | | The severity of the adverse event itself, in direct relation to the subject.
substance | String | True | | | Refers to the specific entity that caused the adverse event.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | BackboneElement | False | Yes | | An array of entries included in this bundle.
Bundle.entry.resource | ResourceContainer | False | | | A container for a single FHIR resource.
Bundle.entry.resource.AdverseEvent | AdverseEvent | False | | | A single instance of a FHIR AdverseEvent resource.
Bundle.entry.resource.AdverseEvent.actuality | String | False | | | The nature of the adverse event, either real or potential.
Bundle.entry.resource.AdverseEvent.extension (also: adverse-event-comments) | Extension | True | Yes | | This element is returned starting in the May 2022 version of Epic. Free-text comments associated with an adverse event.
Bundle.entry.resource.AdverseEvent.extension (also: adverse-event-expected) | Extension | True | Yes | | Whether this adverse event is expected.
Bundle.entry.resource.AdverseEvent.extension (also: adverse-event-resolved-date) | Extension | False | Yes | | The date the adverse event was documented as ended.
Bundle.entry.resource.AdverseEvent.extension (also: adverse-event-severity-history) | Extension | False | Yes | | A record of the historic severities for a particular adverse event.
Bundle.entry.resource.AdverseEvent.extension.extension (also: adverse-event-severity-history-line) | Extension | False | Yes | | A specific instance in the severity history for a particular adverse event.
Bundle.entry.resource.AdverseEvent.extension.extension.extension (also: adverse-event-severity) | Extension | False | Yes | | The grade of an adverse event at a point in time.
Bundle.entry.resource.AdverseEvent.extension.extension.extension (also: adverse-event-severity-start-date) | Extension | False | Yes | | The date the adverse event was documented as started.
Bundle.entry.resource.AdverseEvent.date | String | True | | | When the event occurred.
Bundle.entry.resource.AdverseEvent.event | CodeableConcept | False | | | Type of the event itself in relation to the subject.
Bundle.entry.resource.AdverseEvent.identifier | Identifier | True | | | Business identifier for the event.
Bundle.entry.resource.AdverseEvent.outcome | CodeableConcept | True | | | The outcome of the adverse event. Currently providing only "resolved" and "ongoing."
Bundle.entry.resource.AdverseEvent.recordedDate | String | True | | | When the event was recorded from the audit trail. This is when the event was first recorded.
Bundle.entry.resource.AdverseEvent.recorder | Reference | True | | | The recorder of the adverse event. This is the user who first recorded the adverse event.

Bundle.entry.resource.AdverseEvent.seriousness | CodeableConcept | True | | | The seriousness of the event. This is populated with "Serious" or "Non-Serious."
Bundle.entry.resource.AdverseEvent.severity | CodeableConcept | True | | | The severity of the adverse event itself, in direct relation to the subject.
Bundle.entry.resource.AdverseEvent.study | Reference | False | Yes | | The research study related to this adverse event.
Bundle.entry.resource.AdverseEvent.subject | Reference | False | | | Subject impacted by the event, such as the patient.
Bundle.entry.resource.AdverseEvent.suspectEntity | SuspectEntity | True | Yes | | The suspected agent causing the adverse event.
Bundle.entry.resource.AdverseEvent.suspectEntity.causality | Causality | False | Yes | | Information about the possible cause of the event.
Bundle.entry.resource.AdverseEvent.suspectEntity.causality.assessment | CodeableConcept | False | | | Assessment of whether the entity caused the event.
Bundle.entry.resource.AdverseEvent.suspectEntity.instance | Reference | False | | | Refers to the specific entity that caused the adverse event.

Sample request

```
https://hostname/instance/api/FHIR/R4/AdverseEvent?subject=eea8Msv-9WjX-
    ffbBcv-4rw3&study=eAO.XWJpIicOp3xl5CLneUQ3&seriousness=serious
```

Sample response


```
<Bundle xmlns="http://hl7.org/fhir">
  <type value="searchset"/>
  <total value="1"/>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/R4/AdverseEvent?subject=eea8Msv-9WjX-
    ffbBcv-4rw3&amp;study=eAO.XWJpIicOp3xl5CLneUQ3&amp;seriousness=serious"/>
  </link>
  <entry>
    <link>
      <relation value="self"/>
      <url value="https://hostname/instance/api/FHIR/R4/AdverseEvent/eBrj0mrZZ9-WmgLrAXW.ZQmF3xBGWbDn1vkbtSszAQnY3"/>
    </link>
    <fullUrl value="https://hostname/instance/api/FHIR/R4/AdverseEvent/eBrj0mrZZ9-WmgLrAXW.ZQmF3xBGWbDn1vkbtSszAQnY3"/>
    <resource>
      <AdverseEvent>
        <id value="eBrj0mrZZ9-WmgLrAXW.ZQmF3xBGWbDn1vkbtSszAQnY3"/>
        <extension url="http://open.epic.com/FHIR/StructureDefinition/adverse-event-expected">
          <valueCodeableConcept>
            <coding>
              <system value="urn:oid:1.2.840.114350.1.72.1.7.7.10.768076.7056"/>
              <code value="0"/>
              <display value="No"/>
            </coding>
            <text value="0"/>
          </valueCodeableConcept>
        </extension>
        <extension url="http://open.epic.com/FHIR/StructureDefinition/adverse-event-severity-history">
          <extension url="http://open.epic.com/FHIR/StructureDefinition/adverse-event-severity-history-line">
            <extension url="http://open.epic.com/FHIR/StructureDefinition/adverse-event-severity">
              <valueCodeableConcept>
                <coding>
                  <system value="https://www.hl7.org/fhir/valueset-adverse-event-severity.html"/>
                  <code value="mild"/>
                  <display value="Mild"/>
                </coding>
                <coding>
                  <system value="urn:oid:1.2.840.114350.1.13.861.1.7.10.768076.7010"/>
                  <code value="1"/>
                  <display value="Mild"/>
                </coding>
                <text value="Mild"/>
              </valueCodeableConcept>
            </extension>
            <extension url="http://open.epic.com/FHIR/StructureDefinition/adverse-event-severity-start-date">
              <valueDate value="2019-11-01"/>
            </extension>
          </extension>
          <extension url="http://open.epic.com/FHIR/StructureDefinition/adverse-event-severity-history-line">
            <extension url="http://open.epic.com/FHIR/StructureDefinition/adverse-event-severity">
              <valueCodeableConcept>
                <coding>
                  <system value="https://www.hl7.org/fhir/valueset-adverse-event-severity.html"/>
                  <code value="severe"/>
                  <display value="Severe"/>
                </coding>
                <coding>
                  <system value="urn:oid:1.2.840.114350.1.13.861.1.7.10.768076.7010"/>
                  <code value="3"/>
                  <display value="Severe"/>
                </coding>
                <text value="Severe"/>
              </valueCodeableConcept>
            </extension>
            <extension url="http://open.epic.com/FHIR/StructureDefinition/adverse-event-severity-start-date">
              <valueDate value="2019-11-08"/>
            </extension>
          </extension>
        </extension>
```

```
      <identifier>
        <use value="usual"/>
        <system value="urn:oid:1.2.840.114350.1.13.861.1.7.2.768076"/>
        <value value="161246"/>
      </identifier>
      <actuality value="actual"/>
      <event>
        <coding>
          <system value="https://www.meddra.org/"/>
          <code value="10016256"/>
          <display value="Fatigue"/>
        </coding>
        <coding>
          <system value="urn:oid:1.2.840.114350.1.13.861.1.7.2.696871"/>
          <code value="888001016"/>
          <display value="Fatigue"/>
        </coding>
        <text value="Fatigue"/>
      </event>
      <subject>
        <reference value="Patient/eea8Msv-9WjX-ffbBcv-4rw3"/>
        <display value="Research, Ellie"/>
      </subject>
      <date value="2019-11-01"/>
      <recordedDate value="2019-11-08T15:05:40Z"/>
      <seriousness>
        <coding>
          <system value="https://www.hl7.org/fhir/valueset-adverse-event-seriousness.html"/>
          <code value="Serious"/>
          <display value="Serious"/>
        </coding>
        <text value="Serious"/>
      </seriousness>
      <severity>
        <coding>
          <system value="https://www.hl7.org/fhir/valueset-adverse-event-severity.html"/>
          <code value="severe"/>
          <display value="Severe"/>
        </coding>
        <coding>
          <system value="urn:oid:1.2.840.114350.1.13.861.1.7.10.768076.7010"/>
          <code value="3"/>
          <display value="Severe"/>
        </coding>
        <text value="Severe"/>
      </severity>
      <outcome>
        <coding>
          <system value="https://www.hl7.org/fhir/valueset-adverse-event-outcome.html"/>
          <code value="ongoing"/>
          <display value="Ongoing"/>
        </coding>
        <text value="Ongoing"/>
      </outcome>
      <recorder>
        <reference value="Practitioner/e7gJ3G0n9-Dhv8jsHuiI9pw3"/>
        <display value="Steven Van Manen"/>
      </recorder>
      <suspectEntity>
        <instance>
          <reference value="ResearchStudy/eAO.XWJpIicOp3xl5CLneUQ3"/>
          <display value="Breast Cancer Stage II-BRAC0142"/>
        </instance>
        <causality>
          <assessment>
            <coding>
              <system value="https://www.hl7.org/fhir/valueset-adverse-event-causality-assess.html"/>
              <code value="Probably-Likely"/>
              <display value="Probably/Likely"/>
            </coding>
            <coding>
              <system value="urn:oid:1.2.840.114350.1.13.861.1.7.10.768076.7021"/>
              <code value="4"/>
              <display value="Probable"/>
            </coding>
            <text value="Probably/Likely"/>
          </assessment>
        </causality>
      </suspectEntity>
      <study>
        <reference value="ResearchStudy/eAO.XWJpIicOp3xl5CLneUQ3"/>
        <display value="Breast Cancer Stage II-BRAC0142"/>
      </study>
    </AdverseEvent>
  </resource>
  <search>
    <mode value="match"/>
  </search>
</entry>
```

```
</Bundle>
```

10.1.3 AllergyIntolerance.Create (Patient Chart) (R4)

File: site/apis/10-Clinical-General/1-R4/945-AllergyIntolerance-Create-Patient-Chart-R4.html
Epic API ID | 945
FHIR version | R4
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | November 2019
HTTP method | POST
URL template | api/FHIR/R4/AllergyIntolerance
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=945

Description

The AllergyIntolerance resource defines clinical information about a patient's allergic response to a substance. It defines the substance that elicited the response, as well as when the reaction occurred, the severity, and the type of reaction noted.

The AllergyIntolerance resource accommodates creating allergies by adding them to a temporary holding tank. Information in the holding tank is not yet in the patient's chart until reconciled by a clinician. Once a clinician reconciles the allergy, the allergy becomes a part of the patient's chart and is searchable.

The HL7v2 Incoming Patient Administration interface can also file allergy information, both to the holding tank and directly to the patient chart (without clinical review).

The response of a Create depends on the Prefer header:

Prefer: return=minimal

returns the full URL to the resource in the Location header.

Prefer: return=representation

returns both the full URL to the resource in the Location header, as well as the full FHIR resource in the response body.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
AllergyIntolerance (R4) | AllergyIntolerance | | | | A single AllergyIntollerance resource.
AllergyIntolerance (R4).clinicalStatus | CodeableConcept | | | 0..1 | The current status of the allergy.
AllergyIntolerance (R4).clinicalStatus.coding | Coding | | Yes | 0..* | Coding value.
AllergyIntolerance (R4).clinicalStatus.coding.code | String | | | 0..1 | Supported values for creation include: * active * resolved The provided clinicalStatus must also match the endDate. For example, an active allergy must not have an endDate in the past, and a resolved allergy must have an endDate in the past.
AllergyIntolerance (R4).clinicalStatus.coding.system | String | | | 0..1 | Expected to be http://hl7.org/fhir/ValueSet/allergyintolerance-clinical
AllergyIntolerance (R4).clinicalStatus.coding.version | String | | | 0..1 | Expected to be 4.0.0
AllergyIntolerance (R4).code | CodeableConcept | | | 0..* | The allergen name and codes.
AllergyIntolerance (R4).code.coding | Coding | | Yes | 0..* | Coding value.
AllergyIntolerance (R4).code.coding.code | String | | | 0..1 | The unique code for the substance.
AllergyIntolerance (R4).code.coding.display (also: display) | String | | | 0..1 | The name of the substance.
AllergyIntolerance (R4).code.coding.system | String | | | 0..1 | The coding system for the allergen code.
AllergyIntolerance (R4).criticality | code | | | 0..1 | Refers to the risk of clinical harm of a reaction to the substance. Supported values are: unable-to-assess high low
AllergyIntolerance (R4).note | Annotation | | Yes | 0..* | Any comments associated with the allergy.
AllergyIntolerance (R4).onsetDateTime | dateTime | | | 0..1 | The date the allergy was identified.
AllergyIntolerance (R4).onsetPeriod | Period | | | 0..1 | The date range when the allergy was identified.
AllergyIntolerance (R4).onsetPeriod.end | String | | | 0..1 | The date the allergy was resolved.
AllergyIntolerance (R4).onsetPeriod.start | String | | | 0..1 | The date the allergy was identified.
AllergyIntolerance (R4).patient | Reference | | | 1..1 | Reference to a patient resource for whom the allergy is relevant.
AllergyIntolerance (R4).reaction | BackboneElement | | Yes | 0..* | List of known reactions the patient had to the substance.
AllergyIntolerance (R4).reaction.description | string | | | 0..1 | The name of the reaction.
AllergyIntolerance (R4).reaction.manifestation (also: manifestation) | CodeableConcept | | Yes | 1..1 | A single reaction manifestation - clinical symptoms/signs associated with the event.
AllergyIntolerance (R4).reaction.manifestation.coding | Coding | | Yes | 0..* | Coding value.
AllergyIntolerance (R4).reaction.manifestation.coding.code | String | | | 0..1 | Coded value for reaction.
AllergyIntolerance (R4).reaction.manifestation.coding.display | String | | | 0..1 | Reaction display name.
AllergyIntolerance (R4).reaction.manifestation.coding.system | String | | | 0..1 | Coding system (e.g. http://hl7.org/fhir/ValueSet/clinical-findings).
AllergyIntolerance (R4).reaction.severity | code | | | 0..1 | Severity of the event. Supported values: mild moderate severe
AllergyIntolerance (R4).type | code | | | 0..1 | Underlying mechanism, if known. Supported values: allergy intolerance

AllergyIntolerance (R4).verificationStatus | CodeableConcept | | | 0..1 | How certain it is that the patient has an allergy to the identified substance. Only a value of "unconfirmed" is supported, other values are ignored. Allergies must be reconciled into the chart in Epic to be considered confirmed.
AllergyIntolerance (R4).verificationStatus.coding | Coding | | Yes | 0..* | Coding value.
AllergyIntolerance (R4).verificationStatus.coding.code | String | | | 0..1 | The coded verification status value.
AllergyIntolerance (R4).verificationStatus.coding.system | String | | | 0..1 | Expected to be http://terminology.hl7.org/CodeSystem/allergyintolerance-verification
AllergyIntolerance (R4).verificationStatus.coding.version | String | | | 0..1 | Expected to be 4.0.0

Response elements

Name | Type | Optional | Array | Cardinality | Description
AllergyIntolerance (R4) | AllergyIntolerance
AllergyIntolerance (R4).category | code | | Yes | 0..* | The category of the allergy. The following default values are supported, but additional values may be configured by each Epic organization: food medication environment biologic
AllergyIntolerance (R4).clinicalStatus | CodeableConcept | | | 1..1 | The current status of the allergy. Possible values include: active resolved inactive If no known allergies are on file, the value is "active".
AllergyIntolerance (R4).code | CodeableConcept | | | 0..* | The allergen name and codes (RxNorm, UNII, NDF-RT, or SNOMED).
AllergyIntolerance (R4).criticality | code | | | 0..1 | Refers to the risk of clinical harm of a reaction to the substance. Possible values: unable-to-assess high low
AllergyIntolerance (R4).id | id | | | 0..1 | The AllergyIntolerance FHIR ID.
AllergyIntolerance (R4).note | Annotation | | Yes | 0..* | Any comments associated with the allergy.
AllergyIntolerance (R4).onsetDateTime | dateTime | | | 0..1 | If an allergy has a noted date, that date is specified here. This element is returned in the following locales dependent on Epic version: August 2023 and later: returned in Dutch locales. November 2024 and later: returned in all locales.
AllergyIntolerance (R4).onsetPeriod | Period | | | 0..1 | If an allergy has a noted date, that date or date range for non-specific dates is specified here. This element is returned in the following locales dependent on Epic version: May 2023 and earlier: returned in all locales. August 2023 - August 2024: returned in non-Dutch locales only. November 2024 and later: no longer returned in any locale.
AllergyIntolerance (R4).patient | Reference | | | 1..1 | Reference to a Patient resource for whom the allergy is relevant.
AllergyIntolerance (R4).reaction | BackboneElement | | Yes | 0..* | List of known reactions the patient had to the substance.
AllergyIntolerance (R4).reaction.description | string | | | 0..1 | The name of the reaction.
AllergyIntolerance (R4).reaction.manifestation | CodeableConcept | | Yes | 0..* | A single reaction the patient had to the substance.
AllergyIntolerance (R4).reaction.severity | code | | | 0..1 | The noted reaction severity to the substance. This value is not returned for allergies documented directly in the Epic, and is only returned for allergies received from sources external to the Epic instance.
AllergyIntolerance (R4).type | code | | | 0..1 | Identification of the underlying physiological mechanism for the reaction risk. Possible values: allergy intolerance
AllergyIntolerance (R4).verificationStatus | CodeableConcept | | | 1..1 | If the allergy is on the patient's chart in Epic, or if no known allergies has been confirmed, this element specifies "confirmed". For newly added allergies still needing reconciliation, unconfirmed lack of allergies, or if no allergies are on file, this element specifies "unconfirmed".

Sample request

```
POST https://hostname/instance/api/FHIR/R4/AllergyIntolerance
{
  "resourceType": "AllergyIntolerance",
  "clinicalStatus": {
    "coding": [
      {
        "system": "http://terminology.hl7.org/CodeSystem/allergyintolerance-clinical",
        "version": "4.0.0",
        "code": "active",
        "display": "Active"
      }
    ]
  },
  "verificationStatus": {
    "coding": [
      {
        "system": "http://terminology.hl7.org/CodeSystem/allergyintolerance-verification",
        "version": "4.0.0",
        "code": "unconfirmed",
        "display": "Unconfirmed"
      }
    ]
  },
  "category": [
    "medication"
  ],
  "code": {
    "coding": [
      {
        "system": "2.16.840.1.113883.6.88",
        "code": "18235"
      }
    ],
    "text": "PENICILLIN"
  },
  "patient": {
    "reference": "Patient/eivR2DL-WAG8Nxx3xBYUu7A3"
  },
  "onsetPeriod": {
    "start": "2022-04-12"
  },
  "recordedDate": "2022-04-12T18:32:10Z",
```

```
  "reaction": [
    {
      "manifestation": [
        {
          "text": "Hives"
        }
      ],
      "description": "Hives",
      "severity": "severe"
    }
  ]
}
```

Sample response

```
Status code: Created (201)
Content-Type: application/json; charset=utf-8
Location: AllergyIntolerance/e5tJ.hqE.w72xbNuoy6jwDLihCnKB5gygaRZ.jFwJvb.baCsbmdfR2bCLwA9DL95n3
```

Change log

Epic Version | Description | Type

November 2022 | Code systems for the following code sets now return a URL in place of an OID: ActCode, ICD-10-CM, ICD-10-CA, ICD-10-NL, SNOMED System URL OID ActCode STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode 2.16.840.1.113883.5.4 ICD-10-CM http://hl7.org/fhir/sid/icd-10-cm 2.16.840.1.113883.6.90 ICD-10-CA http://hl7.org/fhir/sid/icd-10-ca 2.16.840.1.113883.6.94 ICD-10-NL http://hl7.org/fhir/sid/icd-10-nl 2.16.840.1.113883.6.3.2 SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 | Breaking change

System | URL | OID

ActCode | STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode | 2.16.840.1.113883.5.4

ICD-10-CM | http://hl7.org/fhir/sid/icd-10-cm | 2.16.840.1.113883.6.90

ICD-10-CA | http://hl7.org/fhir/sid/icd-10-ca | 2.16.840.1.113883.6.94

ICD-10-NL | http://hl7.org/fhir/sid/icd-10-nl | 2.16.840.1.113883.6.3.2

SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96

November 2022 | Epic has updated the behavior of certain modifier elements in FHIR write APIs to ensure the data in the incoming request body matches the data filed to the database and to comply with the HL7 FHIR specification. Previously, the modifier element was ignored in certain situations, and an undocumented default value was saved in its place. Now, if the modifier element is included and is set to an unaccepted value, a fatal error logs, preventing the request from filing to the database. The FHIR request returns an OperationOutcome with the invalid element and what value is accepted for that element. Element Accepted Incoming Value Default Value clinicalStatus active active verificationStatus unconfirmed unconfirmed | Substantive change

Element | Accepted Incoming Value | Default Value

clinicalStatus | active | active

verificationStatus | unconfirmed | unconfirmed

10.1.4 AllergyIntolerance.Read (Patient Chart) (R4)

File: site/apis/10-Clinical-General/1-R4/946-AllergyIntolerance-Read-Patient-Chart-R4.html

Epic API ID | 946

FHIR version | R4

Categories | Clinical General

API type | USCDI

A la carte | Industry-Standard Level 1 API

USCDI | Yes

Available in inpatient and other contexts | Yes

ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base

ONC certification versions | November 2019

HTTP method | GET

URL template | api/FHIR/R4/AllergyIntolerance/{ID}

Supports OAuth 2.0 | True

Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients

Performs scope validation | False

Source | https://fhir.epic.com/Specifications?api=946

Description

The AllergyIntolerance resource defines clinical information about a patient's allergic response to a substance. It defines the substance that elicited the response, as well as when the reaction occurred, the severity, and the type of reaction noted.

The HL7v2 Outgoing Patient Administration Interface is an alternative to this resource, recommended in cases where real-time updates of allergy information for a broad group of patients are necessary.

Request parameters

Name | Type | Optional | Array | Cardinality | Description

ID | token | Required | | | The AllergyIntolerance FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description

AllergyIntolerance (R4) | AllergyIntolerance

AllergyIntolerance (R4).category | code | | Yes | 0..* | The category of the allergy. The following default values are supported, but additional values may be configured by each Epic organization: food medication environment biologic

AllergyIntolerance (R4).clinicalStatus | CodeableConcept | | | 1..1 | The current status of the allergy. Possible values include: active resolved inactive If no known allergies are on file, the value is "active".

AllergyIntolerance (R4).code | CodeableConcept | | | 0..* | The allergen name and codes (RxNorm, UNII, NDF-RT, or SNOMED).

AllergyIntolerance (R4).criticality | code | | | 0..1 | Refers to the risk of clinical harm of a reaction to the substance. Possible values: unable-to-assess high low

AllergyIntolerance (R4).id | id | | | 0..1 | The AllergyIntolerance FHIR ID.

AllergyIntolerance (R4).note | Annotation | | Yes | 0..* | Any comments associated with the allergy.

AllergyIntolerance (R4).onsetDateTime | dateTime | | | 0..1 | If an allergy has a noted date, that date is specified here. This element is returned in the following locales dependent on Epic version: August 2023 and later: returned in Dutch locales. November 2024 and later: returned in all locales.

AllergyIntolerance (R4).onsetPeriod | Period | | | 0..1 | If an allergy has a noted date, that date or date range for non-specific dates is specified here. This element is returned in the following locales dependent on Epic version: May 2023 and earlier: returned in all locales. August 2023 - August 2024: returned in non-Dutch locales only. November 2024 and later: no longer returned in any locale.

AllergyIntolerance (R4).patient | Reference | | | 1..1 | Reference to a Patient resource for whom the allergy is relevant.

AllergyIntolerance (R4).reaction | BackboneElement | | Yes | 0..* | List of known reactions the patient had to the substance.

AllergyIntolerance (R4).reaction.description | string | | | 0..1 | The name of the reaction.

AllergyIntolerance (R4).reaction.manifestation | CodeableConcept | | Yes | 0..* | A single reaction the patient had to the substance.

AllergyIntolerance (R4).reaction.severity | code | | | 0..1 | The noted reaction severity to the substance. This value is not returned for allergies documented directly in the Epic, and is only returned for allergies received from sources external to the Epic instance.

AllergyIntolerance (R4).recordedDate | dateTime | | | 0..1 | The date the allergy was recorded in Epic. Called "assertedDate" in STU3.

AllergyIntolerance (R4).recorder | Reference | | | 0..1 | Reference to the practitioner who updated the allergy.

AllergyIntolerance (R4).type | code | | | 0..1 | Identification of the underlying physiological mechanism for the reaction risk. Possible values: allergy intolerance

AllergyIntolerance (R4).verificationStatus | CodeableConcept | | | 1..1 | If the allergy is on the patient's chart in Epic, or if no known allergies has been confirmed, this element specifies "confirmed". For newly added allergies still needing reconciliation, unconfirmed lack of allergies, or if no allergies are on file, this element specifies "unconfirmed".

Sample request

```
GET https://hostname/instance/api/FHIR/R4/AllergyIntolerance/eeJxm9Vi8-QmUQuWDhBMklw3
```

Sample response


```
{
  "resourceType": "AllergyIntolerance",
  "id": "eDDkI1cAyDYgpTJheyIFMmg3",
  "clinicalStatus": {
    "coding": [
      {
        "system": "http://terminology.hl7.org/CodeSystem/allergyintolerance-clinical",
        "version": "4.0.0",
        "code": "active",
        "display": "Active"
      }
    ],
    "text": "Active"
  },
  "verificationStatus": {
    "coding": [
      {
        "system": "http://terminology.hl7.org/CodeSystem/allergyintolerance-verification",
        "version": "4.0.0",
        "code": "confirmed",
        "display": "Confirmed"
      }
    ],
    "text": "Confirmed"
  },
  "category": [
    "medication"
  ],
  "criticality": "high",
  "code": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "764146007",
        "display": "PENICILLINS"
      },
      {
        "system": "urn:oid:2.16.840.1.113883.3.26.1.5",
        "code": "N0000005863"
      }
    ],
    "text": "PENICILLINS"
  },
  "patient": {
    "reference": "Patient/e63wRTbPfr1p8UW81d8Seiw3",
    "display": "Mychart, Teddy"
  },
  "onsetDateTime": "2006-07-07",
```

```
  "recordedDate": "2006-07-07T05:00:00Z",
  "recorder": {
    "reference": "Practitioner/eodo7Bj9dk-LtJSthMwARpw3",
    "type": "Practitioner",
    "display": "Fhirdoc Testdoctor"
  },
  "reaction": [
    {
      "manifestation": [
        {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "126485001",
              "display": "Urticaria (disorder)"
            }
          ],
          "text": "Hives"
        }
      ],
      "description": "Hives"
    },
    {
      "manifestation": [
        {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "41829006",
              "display": "Dietary finding (finding)"
            }
          ],
          "text": "Itching"
        }
      ],
      "description": "Itching"
    }
  ]
}
```


Change log

Epic Version	Description	Type
November 2024	AllergyIntolerance now returns onsetDateTime. The onsetPeriod element is no longer used.	Breaking change
August 2023	AllergyIntolerance now returns onsetDateTime in Dutch locales and returns onsetPeriod as a range for all other locales.	Breaking change
November 2022	AllergyIntolerance no longer returns reaction-level severities for an allergen documented in Epic. Instead, allergen-level severity is now returned as the criticality element.	Breaking change

10.1.5 AllergyIntolerance.Search (Patient Chart) (R4)

File: site/apis/10-Clinical-General/1-R4/947-AllergyIntolerance-Search-Patient-Chart-R4.html	
Epic API ID	947
FHIR version	R4
Categories	Clinical General
API type	USCDI
A la carte	Industry-Standard Level 1 API
USCDI	Yes
Available in inpatient and other contexts	Yes
ONC certification products	EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions	November 2019
HTTP method	GET
URL template	/api/FHIR/R4/AllergyIntolerance?clinical-status={clinical-status}&patient={patient}
Supports OAuth 2.0	True
Supported user types	Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation	False
Source	https://fhir.epic.com/Specifications?api=947

Description

The AllergyIntolerance resource defines clinical information about a patient's allergic response to a substance. It defines the substance that elicited the response, as well as when the reaction occurred, the severity, and the type of reaction noted. The AllergyIntolerance resource can accommodate search by ID and by patient, allowing it to return a list of allergies. If a patient has no active allergies, an AllergyIntolerance resource is returned indicating whether the patient's allergies have never been reviewed (are not on file), or if they have been reviewed and it has been determined that they have no known allergies.

Unconfirmed AllergyIntolerance resources that have yet to be reconciled into a patient's chart by a clinician are only returned when searching for them by their FHIR IDs in _id. Once a clinician reconciles an allergy, a new AllergyIntolerance resource associated with the reconciled allergy is returned in AllergyIntolerance.Search results, and the previously-unconfirmed AllergyIntolerance in the "holding tank" no longer appears in the results.

The HL7v2 Outgoing Patient Administration Interface is an alternative to this resource, recommended in cases where real-time updates of allergy information for a broad group of patients are necessary.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
_count | number | Optional | Yes | | Number of results per page.
_id | token | Optional | Yes | | FHIR resource IDs for the desired resources. Can be used in combination with _count, _include, and _revInclude. All others are ignored.
_include | string | Optional | Yes | | Used to indicate which referenced resources to include in the search bundle.
_revinclude | string | Optional | Yes | | Used to include Provenance or AuditEvent resources that refer to search results, when available.
clinical-status | token | Optional | Yes | | The current status of the allergy. Only "active" is accepted. If not specified, both active and inactive/resolved allergies are returned.
patient | reference | Required | Yes | | Reference to a patient resource.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
category | token | Optional | Yes | | The category of the allergy. The following default values are supported, but additional values may be configured by each Epic organization: food medication environment biologic
code | token | Optional | Yes | | Code that identifies the allergy or intolerance.
criticality | token | Optional | Yes | | Refers to the risk of clinical harm of a reaction to the substance. Supported values are: unable-to-assess high low
date | date | Optional | Yes | | The date the allergy was recorded in Epic.
identifier | token | Optional | Yes | | The AllergyIntolerance FHIR ID.
manifestation | token | Optional | Yes | | A single reaction the patient had to the substance.
onset | date | Optional | Yes | | If an allergy has a noted date, that date can be specified here.
recorder | reference | Optional | Yes | | Reference to the practitioner who updated the allergy.
severity | token | Optional | Yes | | The degree of the patient's reaction. Supported values are: mild moderate severe
type | token | Optional | Yes | | Identification of the underlying physiological mechanism for the reaction risk. Supported values: allergy intolerance
verification-status | token | Optional | Yes | | If the allergy is on the patient's chart, this element specifies "confirmed". If it is a newly created allergy and and still needs to be reconciled, it is "unconfirmed". For no allergies on file or unconfirmed lack of allergies, this value is "unconfirmed". For confirmed no known allergies or no active allergies, this value is "confirmed".

Response elements

Name | Type | Optional | Array | Cardinality | Description
AllergyIntolerance (R4) | AllergyIntolerance
AllergyIntolerance (R4).category | code | | Yes | 0..* | The category of the allergy. The following default values are supported, but additional values may be configured by each Epic organization: food medication environment biologic
AllergyIntolerance (R4).clinicalStatus | CodeableConcept | | | 1..1 | The current status of the allergy. Possible values include: active resolved inactive If no known allergies are on file, the value is "active".
AllergyIntolerance (R4).code | CodeableConcept | | | 0..* | The allergen name and codes (RxNorm, UNII, NDF-RT, or SNOMED).
AllergyIntolerance (R4).criticality | code | | | 0..1 | Refers to the risk of clinical harm of a reaction to the substance. Possible values: unable-to-assess high low
AllergyIntolerance (R4).id | id | | | 0..1 | The AllergyIntolerance FHIR ID.
AllergyIntolerance (R4).note | Annotation | | Yes | 0..* | Any comments associated with the allergy.
AllergyIntolerance (R4).onsetDateTime | dateTime | | | 0..1 | If an allergy has a noted date, that date is specified here. This element is returned in the following locales dependent on Epic version: August 2023 and later: returned in Dutch locales. November 2024 and later: returned in all locales.
AllergyIntolerance (R4).onsetPeriod | Period | | | 0..1 | If an allergy has a noted date, that date or date range for non-specific dates is specified here. This element is returned in the following locales dependent on Epic version: May 2023 and earlier: returned in all locales. August 2023 - August 2024: returned in non-Dutch locales only. November 2024 and later: no longer returned in any locale.
AllergyIntolerance (R4).patient | Reference | | | 1..1 | Reference to a Patient resource for whom the allergy is relevant.
AllergyIntolerance (R4).reaction | BackboneElement | | Yes | 0..* | List of known reactions the patient had to the substance.
AllergyIntolerance (R4).reaction.description | string | | | 0..1 | The name of the reaction.
AllergyIntolerance (R4).reaction.manifestation | CodeableConcept | | Yes | 0..* | A single reaction the patient had to the substance.
AllergyIntolerance (R4).reaction.severity | code | | | 0..1 | The noted reaction severity to the substance. This value is not returned for allergies documented directly in the Epic, and is only returned for allergies received from sources external to the Epic instance.
AllergyIntolerance (R4).recordedDate | dateTime | | | 0..1 | The date the allergy was recorded in Epic. Called "assertedDate" in STU3.
AllergyIntolerance (R4).recorder | Reference | | | 0..1 | Reference to the practitioner who updated the allergy.
AllergyIntolerance (R4).type | code | | | 0..1 | Identification of the underlying physiological mechanism for the reaction risk. Possible values: allergy intolerance
AllergyIntolerance (R4).verificationStatus | CodeableConcept | | | 1..1 | If the allergy is on the patient's chart in Epic, or if no known allergies has been confirmed, this element specifies "confirmed". For newly added allergies still needing reconciliation, unconfirmed lack of allergies, or if no allergies are on file, this element specifies "unconfirmed".

Sample request

```
GET https://hostname/instance/api/FHIR/R4/AllergyIntolerance?patient=edO5RwPdFu.l85iSJJ8z5-w3
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 2,
  "link": [
```

```
  {
    "relation": "self",
    "url": "https://hostname/instance/api/FHIR/R4/AllergyIntolerance?clinical-
  status=active&patient=edO5RwPdFu.l85iSJJ8z5-w3"
  }
],
"entry": [
  {
    "link": [
      {
        "relation": "self",
        "url": "https://hostname/instance/api/FHIR/R4/AllergyIntolerance/ezfHOflYGLQUJHPkTABenag3"
      }
    ],
    "fullUrl": "https://hostname/instance/api/FHIR/R4/AllergyIntolerance/ezfHOflYGLQUJHPkTABenag3",
    "resource": {
      "resourceType": "AllergyIntolerance",
      "id": "ezfHOflYGLQUJHPkTABenag3",
      "clinicalStatus": {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/allergyintolerance-clinical",
            "version": "4.0.0",
            "code": "active",
            "display": "Active"
          }
        ],
        "text": "Active"
      },
      "verificationStatus": {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/allergyintolerance-verification",
            "version": "4.0.0",
            "code": "confirmed",
            "display": "Confirmed"
          }
        ],
        "text": "Confirmed"
      },
      "type": "allergy",
      "category": [
        "medication"
      ],
      "criticality": "high",
      "code": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "764146007",
            "display": "PENICILLINS"
          },
          {
            "system": "urn:oid:2.16.840.1.113883.3.26.1.5",
            "code": "N0000005863"
          },
          {
            "system": "urn:oid:2.16.840.1.113883.3.26.1.5",
            "code": "N0000005873"
          },
          {
            "system": "urn:oid:2.16.840.1.113883.3.26.1.5",
            "code": "N0000005840"
          }
        ],
        "text": "PENICILLINS"
      },
      "patient": {
        "reference": "Patient/edO5RwPdFu.l85iSJJ8z5-w3",
        "display": "Ambulatory, Abraham"
      },
      "onsetDateTime": "2024-03-25",
      "recordedDate": "2024-03-25T14:42:08Z",
      "recorder": {
        "reference": "Practitioner/eodo7Bj9dk-LtJSthMwARpw3",
        "type": "Practitioner",
        "display": "Fhirdoc Testdoctor"
      },
      "reaction": [
        {
          "manifestation": [
            {
              "coding": [
                {
                  "system": "http://snomed.info/sct",
                  "code": "39579001",
                  "display": "Anaphylaxis (disorder)"
                }
              ],
```

```
            "text": "Anaphylaxis"
          }
        ],
        "description": "Anaphylaxis"
      }
    ]
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/AllergyIntolerance/eIKvPnE-1VjHmuDXY2uOveA3"
    }
  ],
  "fullUrl": "https://hostname/instance/api/FHIR/R4/AllergyIntolerance/eIKvPnE-1VjHmuDXY2uOveA3",
  "resource": {
    "resourceType": "AllergyIntolerance",
    "id": "eIKvPnE-1VjHmuDXY2uOveA3",
    "clinicalStatus": {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/allergyintolerance-clinical",
          "version": "4.0.0",
          "code": "active",
          "display": "Active"
        }
      ],
      "text": "Active"
    },
    "verificationStatus": {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/allergyintolerance-verification",
          "version": "4.0.0",
          "code": "confirmed",
          "display": "Confirmed"
        }
      ],
      "text": "Confirmed"
    },
    "type": "allergy",
    "category": [
      "food"
    ],
    "criticality": "high",
    "code": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "3718001",
          "display": "MILK (COW)"
        }
      ],
      "text": "MILK (COW)"
    },
    "patient": {
      "reference": "Patient/edO5RwPdFu.l85iSJJ8z5-w3",
      "display": "Ambulatory, Abraham"
    },
    "onsetDateTime": "2024-03-25",
    "recordedDate": "2024-03-25T14:55:39Z",
    "reaction": [
      {
        "manifestation": [
          {
            "coding": [
              {
                "system": "http://snomed.info/sct",
                "code": "62315008",
                "display": "Diarrhea (finding)"
              }
            ],
            "text": "Diarrhea"
          }
        ],
        "description": "Diarrhea"
      },
      {
        "manifestation": [
          {
            "coding": [
              {
                "system": "http://snomed.info/sct",
                "code": "422587007",
                "display": "Nausea (finding)"
```

```
                }
              ],
              "text": "GI intolerance"
            }
          ],
          "description": "GI intolerance"
        }
      ]
    },
    "search": {
      "mode": "match"
    }
  }
 ]
}
```

Change log

Epic Version | Description | Type
November 2024 | AllergyIntolerance now returns onsetDateTime. The onsetPeriod element is no longer used. | Breaking change
August 2023 | AllergyIntolerance now returns onsetDateTime in Dutch locales and returns onsetPeriod as a range for all other locales. | Breaking change
November 2022 | AllergyIntolerance no longer returns reaction-level severities for an allergen documented in Epic. Instead, allergen-level severity is now returned as the criticality element. | Breaking change

10.1.6 Condition.Create (Problems) (R4)

File: site/apis/10-Clinical-General/1-R4/949-Condition-Create-Problems-R4.html
Epic API ID | 949
FHIR version | R4
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Keywords | Problems;Problem List
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | November 2019
HTTP method | POST
URL template | api/FHIR/R4/Condition
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=949

Description

The Condition.Create (Problems) API can be used to submit new problem list items to the problem list. Problem list items submitted by this API are not saved directly to the patient chart. Instead, they are saved to a temporary holding tank. Providers review the information in the holding tank, make any necessary adjustments, then reconcile the problem list item into the patient's chart.

If a problem list item submitted by this API has not been reconciled when a provider views the patient's problem list, the provider will see a banner indicating that there are unreconciled problem list items in the holding tank.

When a new problem list item is submitted via this API, the system will perform duplicate checks to ensure duplicate problem list items are not created.

The response of a Create depends on the Prefer header:

Prefer: return=minimal

returns the full URL to the resource in the Location header.

Prefer: return=representation

returns both the full URL to the resource in the Location header, as well as the full FHIR resource in the response body.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
Condition (R4) | Condition | | | | An instance of the R4 Condition resource.
Condition (R4).abatementDateTime | dateTime | Optional | | 0..1 | The date or estimated date that the problem resolved or went into remission. Available starting the May 2023 version of Epic.
Condition (R4).abatementPeriod | Period | Optional | | 0..1 | The date or estimated date that the problem resolved or went into remission. Available starting the May 2023 version of Epic. If abatementDateTime is populated, it is used instead.
Condition (R4).abatementPeriod.end | String | | | 0..1 | The date or estimated date that the condition resolved or went into remission. Available starting in the May 2023 version of Epic.
Condition (R4).clinicalStatus | CodeableConcept | Conditional | | 0..1 | The clinical status of the condition. Required if the problem is not active based on the date in abatementDateTime or abatementPeriod.end.
Condition (R4).clinicalStatus.coding | Coding | | Yes | 1..1 | The code and system for the clinical status of the condition.

Condition (R4).clinicalStatus.coding.code (also: code) | String | Optional | | 1..1 | Only values of "active" or "resolved" are supported. We do not support creating inactive problems. The status must be "resolved" if the abatementDateTime or abatementPeriod.end date is in the past. Otherwise, the status must be "active".

Condition (R4).clinicalStatus.coding.system | String | | | 1..1 | The system value must be http://terminology.hl7.org/CodeSystem/condition-clinical

Condition (R4).code | CodeableConcept | Required | | 1..* | The code for the problem list item. The code and code system are used to match to a discrete diagnosis record. ICD-10 and SNOMED codes are supported and preferred. Other code systems can be supported if mappings for those codes are configured. If no discrete record is found automatically, the provider will be prompted to manually select the correct diagnosis during reconciliation. If a discrete record is found automatically, the provider will still have the opportunity to adjust the selected diagnosis during reconciliation.

Condition (R4).code.coding | Coding | Required | Yes | 1..* | Coding value.

Condition (R4).code.coding.code | String | Conditional | | 1..1 | The code for the condition.

Condition (R4).code.coding.display | String | Conditional | | 0..1 | The name of the problem. Either code.coding.display or code.text must be populated.

Condition (R4).code.text | String | Conditional | | 0..1 | The name of the problem. Either code.coding.display or code.text must be populated.

Condition (R4).note | Annotation | Optional | Yes | 0..1 | Free text comments.

Condition (R4).note.text | String | Optional | | 0..1 | The comment text. A length of 450 characters is supported.

Condition (R4).onsetDateTime | dateTime | Conditional | | | The date or estimated date when the problem began. If onsetPeriod and onsetDateTime are both populated, onsetDateTime is used.

Condition (R4).onsetPeriod | Period | Required | | | The date or estimated date when the problem began. If onsetPeriod and onsetDateTime are both populated, onsetDateTime is used. Starting in the May 2023 version of Epic, if abatementDateTime or abatementPeriod is set, their values are used instead of the value in onsetPeriod to determine when the condition ended.

Condition (R4).onsetPeriod.end | String | Optional | | | When the problem ended. This field is only supported for backwards compatibility and should not be used. Instead, use abatementDateTime or abatementPeriod to communicate the problem end date.

Condition (R4).onsetPeriod.start | String | Conditional | | | The date or estimated date when the problem began.

Condition (R4).subject | Reference | Required | | 1..1 | The reference to the Patient resource for the patient who has the condition.

Condition (R4).verificationStatus | CodeableConcept | | | 0..1 | The verification status that supports the clinical status of the condition. Only a value of "provisional" is supported. Problems must be reconciled into the chart before they can be verified.

Condition (R4).verificationStatus.coding | Coding | | Yes | 1..1 | The code and system for the verification status of the condition.

Condition (R4).verificationStatus.coding.code (also: code) | String | Optional | | 1..1 | Only a value of "provisional" is supported.

Condition (R4).verificationStatus.coding.system | String | | | 1..1 | The system value must be http://terminology.hl7.org/CodeSystem/condition-ver-status

Response elements

Name | Type | Optional | Array | Cardinality | Description

Condition (R4) | Condition | | | | An instance of the R4 Condition resource.

Condition (R4).abatementDateTime | dateTime | | | 0..1 | The date the condition was resolved. Starting in the February 2024 version of Epic, abatementDateTime is used to communicate the resolution date. In November 2023 and earlier versions of Epic, abatementPeriod is used to communicate the resolution date.

Condition (R4).abatementPeriod | Period | | | 0..1 | The date the condition is resolved. In the November 2023 and earlier versions of Epic, abatementPeriod is used to communicate the resolution date. Starting in the February 2024 version of Epic, abatementDateTime is used to communicate the resolution date.

Condition (R4).category | CodeableConcept | | Yes | 0..* | Always includes category code "problem-list-item". If a problem list item is marked with a class of "Health Concern", indicating that the problem is of special concern, a category code of "health-concern" is also included. This can also include additional category codes, such as "sdoh" for problems related to social determinates of health. This category depends on configuration of diagnosis groupers defined by the Epic organization.

Condition (R4).clinicalStatus | CodeableConcept | | | 1..1 | The current clinical status of the problem list item. The following Condition Clinical Status Codes are supported: active - Active problem list items resolved - Resolved problem list items inactive - Deleted problem list items

Condition (R4).code | CodeableConcept | | | 1..1 | The code that identifies the specific type of problem. Examples of code sytems that are often used to code problem list items include ICD-10, SNOMED, ICPC Basic Health Care Classification, and Intelligent Medical Objects ProblemIT and others. If there are codes available to send, they're included in this element. If there are no codes to send, this element includes the uncoded, free text problem name. Starting in the February 2024 version of Epic or with special updates to November 2023, August 2023, or May 2023 version of Epic, each code also sends its display name and the provider-entered problem name is sent if available. Common code system values: ICD-10: http://hl7.org/fhir/sid/icd-10-cm SNOMED: http://snomed.info/sct

Condition (R4).extension (assertedDateTime) (also: extension (assertedDateTime)) | Extension | | Yes | 0..* | The date the condition was diagnosed. URL: http://hl7.org/fhir/StructureDefinition/condition-assertedDate

Condition (R4).extension (smartdata) (also: extension (smartdata)) | Extension | | Yes | | When documenting a problem, various questions, also known as SmartData Elements, can be conditionally presented to the clinician. The smartdata extension on Condition (Problems) refernces the clinician's answers, which are available in the Observation (SmartData Elements) API.

Condition (R4).id | id | | | 1..1 | The condition FHIR ID.

Condition (R4).meta | Meta | | | | Metadata about the resource.

Condition (R4).meta.lastUpdated | String | | | 0..1 | The instant the condition was last updated.

Condition (R4).note | Annotation | | Yes | 1..* | Free text comments for the note. Can be an Overview note or, starting in the February 2025 version of Epic, an Assessment & Plan note.

Condition (R4).note.authorReference | Reference | | | | Who wrote the note. This element is returned starting in the following Epic versions, dependent on note type: May 2025: Problem overview notes February 2025: Assessment & Plan notes

Condition (R4).note.extension (annotationType) (also: extension (annotationType)) | Extension | | Yes | | This extension is returned starting in the following Epic versions, dependent on note type: May 2025: Problem overview notes February 2025: Assessment & Plan notes The following valueCodeableConcept codings are supported: Assessment & Plan notes: LOINC code 51847-2 (Evaluation + Plan note) Problem overview notes: LOINC code 68608-9 (Summary note) URL: http://hl7.org/fhir/StructureDefinition/annotationType

Condition (R4).note.extension (data-conversion-warning) (also: extension (data-conversion-warning)) | Extension | | Yes | | This extension is returned starting in the following Epic versions, dependent on note type: May 2025: Problem overview notes February 2025: Assessment & Plan notes One or more of the following warnings can be returned depending on RTF to plan text conversion: Reason for warning Warning text Formatting was lost due to unicode characters or tables being present in the original note. Formatting of this note is different from the original. The original note was RTF. Formatting of this note might be different from the original. There were one or more images in the original note. Images from the original note were not included. There were one or more media attachments in the original note. Attached media from the original note were not included. URL:

http://open.epic.com/FHIR/StructureDefinition/extension/data-conversion-warning
Reason for warning | Warning text
Formatting was lost due to unicode characters or tables being present in the original note. | Formatting of this note is different from the original.
The original note was RTF. | Formatting of this note might be different from the original.
There were one or more images in the original note. | Images from the original note were not included.
There were one or more media attachments in the original note. | Attached media from the original note were not included.
Condition (R4).note.text | String | | | The content of the note. In the February 2025 version of Epic and earlier versions, this note text can include prepended data conversion warnings similar to those documented in the “extension (data-conversion-warning)” response element.
Condition (R4).note.time | String | | | The time that the note was signed. This element is returned starting in the following Epic versions, dependent on note type: May 2025: Problem overview notes February 2025: Assessment & Plan notes
Condition (R4).onsetDateTime | dateTime | | 0..1 | Starting in the February 2024 version of Epic, if a condition has a noted date, it appears here.
Condition (R4).onsetPeriod | Period | | | 0..1 | In the November 2023 and earlier versions of Epic, if a condition has a noted date, it appears here.
Condition (R4).recordedDate | dateTime | | | 1..1 | The date the condition was recorded.
Condition (R4).recorder | Reference | | | 0..1 | Reference to the practitioner who updated the problem list.
Condition (R4).severity | CodeableConcept | | | 0..1 | The default priority of the problem. Note that Epic support two types of priorities for problems: A default priority that is specified when the problem is initially created. This value is available in the Condition.severity property. A user-specific priority that users can adjust based on the relevance of the problem to their specialty. For example, a dermatologist might rate eczema as a high priority while a cardiologist would rate it as low. User-level priorities allow each clinician to see the list organized in a way that is most useful to them. This value is displayed to users Epic's problem list activity. This value is not supported in the FHIR API.
Condition (R4).stage | BackboneElement | | Yes | 0..* | For problems that represent cancer, staging information can be documented as part of the problem. Cancer staging usually involves the identification of the location, size, and the extent of metastasis of the tumor.
Condition (R4).stage.assessment | Reference | | Yes | 1..* | SmartData element associated with cancer staging for this condition.
Condition (R4).subject | Reference | | | 1..1 | The patient associated with the condition.
Condition (R4).verificationStatus | CodeableConcept | | | 1..1 | If this is on the patient's local chart, the value is "confirmed". If it is outside data, it is "provisional". If it is deleted, it is "entered-in-error".

Sample request

```
POST https://hostname/instance/api/FHIR/R4/Condition
{
  "resourceType": "Condition",
  "clinicalStatus": {
    "coding": [
      {
        "system": "http://terminology.hl7.org/CodeSystem/condition-clinical",
        "version": "4.0.0",
        "code": "active",
        "display": "Active"
      }
    ],
    "text": "Active"
  },
  "verificationStatus": {
    "coding": [
      {
        "system": "http://terminology.hl7.org/CodeSystem/condition-ver-status",
        "version": "4.0.0",
        "code": "provisional",
        "display": "Provisional"
      }
    ],
    "text": "Provisional"
  },
  "category": [
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/condition-category",
          "code": "problem-list-item",
          "display": "Problem List Item"
        }
      ],
      "text": "Problem List Item"
    }
  ],
  "code": {
    "coding": [
      {
        "system": "http://hl7.org/fhir/sid/icd-9-cm",
        "code": "487.1",
        "display": "Influenza due to seasonal influenza virus"
      }
    ],
    "text": "Influenza due to seasonal influenza virus"
  },
  "subject": {
    "reference": "Patient/efmOH6ghpsJDh0NDQXIpasg3"
  },
  "onsetDateTime": "2025-08-11",
  "note": [
    {
      "text": "Patient presented with persistent cough and fever."
    }
  ]
```

```
}
```

Sample response

```
Status code: Created (201)
Content-Type: application/json; charset=utf-8
Location: Condition/LOJfXTZpcKKZjcKEEYPRf0pXwDu05QWm0IvjmWpxSb1Um0vs94DjjPBYambS5xhQ
```

Change log

Epic Version | Description | Type
November 2022 | Code systems for the following code sets now return a URL in place of an OID: ActCode, ICD-10-CM, ICD-10-CA, ICD-10-NL, SNOMED System URL OID ActCode STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode 2.16.840.1.113883.5.4 ICD-10-CM http://hl7.org/fhir/sid/icd-10-cm 2.16.840.1.113883.6.90 ICD-10-CA http://hl7.org/fhir/sid/icd-10-ca 2.16.840.1.113883.6.94 ICD-10-NL http://hl7.org/fhir/sid/icd-10-nl 2.16.840.1.113883.6.3.2 SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 | Breaking change
System | URL | OID
ActCode | STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode | 2.16.840.1.113883.5.4
ICD-10-CM | http://hl7.org/fhir/sid/icd-10-cm | 2.16.840.1.113883.6.90
ICD-10-CA | http://hl7.org/fhir/sid/icd-10-ca | 2.16.840.1.113883.6.94
ICD-10-NL | http://hl7.org/fhir/sid/icd-10-nl | 2.16.840.1.113883.6.3.2
SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96
November 2022 | Epic has updated the behavior of certain modifier elements in FHIR write APIs to ensure the data in the incoming request body matches the data filed to the database and to comply with the HL7 FHIR specification. Previously, the modifier element was ignored in certain situations, and an undocumented default value was saved in its place. Now, if the modifier element is included and is set to an unaccepted value, a fatal error logs, preventing the request from filing to the database. The FHIR request returns an OperationOutcome with the invalid element and what value is accepted for that element. Element Accepted Incoming Value Default Value clinicalStatus active active verificationStatus provisional provisional | Substantive change
Element | Accepted Incoming Value | Default Value
clinicalStatus | active | active
verificationStatus | provisional | provisional

10.1.7 Condition.Read (Care Plan Problem) (R4)

File: site/apis/10-Clinical-General/1-R4/10047-Condition-Read-Care-Plan-Problem-R4.html
Epic API ID | 10047
FHIR version | R4
Categories | Clinical General
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | February 2021
HTTP method | GET
URL template | /api/FHIR/R4/Condition/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10047

Description

This web service retrieves a care plan problem from a patient's care plan.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | False | | | The condition FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Condition | Condition | False | | | An instance of the R4 Condition resource.
Condition.abatementDateTime | dateTime | True | | | The date the condition was resolved.
Condition.category | CodeableConcept | False | Yes | | Always "care-plan-problem".
Condition.clinicalStatus | CodeableConcept | False | | | The clinical status, which can be active, resolved, or inactive.
Condition.code | CodeableConcept | False | | | The display name for the condition.
Condition.id | id | False | | | The condition FHIR ID.
Condition.note | Annotation | True | Yes | | Free text comments. Omitted when this resource is used in the patient context.
Condition.onsetDateTime | dateTime | True | | | The date the condition began.
Condition.recordedDate | dateTime | False | | | The date the condition was documented.
Condition.recorder | Reference | False | | | The practitioner who noted the condition.
Condition.severity | CodeableConcept | True | | | The priority of the problem.
Condition.subject | Reference | False | | | The patient associated with the condition.

Sample request

```
https://hostname/instance/api/FHIR/R4/Condition/eG0hRQR7MUQBcS.AfHqMYXQ3
```

Sample response

```
<Condition xmlns="http://hl7.org/fhir">
  <id value="eG0hRQR7MUQBcS.AfHqMYXQ3"/>
  <clinicalStatus>
    <coding>
      <system value="http://terminology.hl7.org/CodeSystem/condition-clinical"/>
      <version value="4.0.0"/>
      <code value="active"/>
      <display value="Active"/>
    </coding>
    <text value="Active"/>
  </clinicalStatus>
  <category>
    <coding>
      <system value="http://open.epic.com/FHIR/StructureDefinition/condition-category"/>
      <code value="care-plan-problem"/>
      <display value="Care Plan Problem"/>
    </coding>
    <text value="Care Plan Problem"/>
  </category>
  <severity>
    <coding>
      <system value="urn:oid:1.2.840.114350.1.13.861.1.7.4.768066.85"/>
      <display value="Low"/>
    </coding>
    <text value="Low"/>
  </severity>
  <code>
    <text value="MyC3 Problem"/>
  </code>
  <subject>
    <reference value="Patient/eS.3bWuX-xYAfV7T5h12w5Q3"/>
    <display value="Test, Patient"/>
  </subject>
  <onsetDateTime value="2020-10-26"/>
  <recordedDate value="2020-10-07T16:12:37Z"/>
  <recorder>
    <reference value="Practitioner/etgK7TEYmpXBluVvFVNOlvA3"/>
    <display value="Test Provider, PHARMD"/>
  </recorder>
</Condition>
```

10.1.8 Condition.Read (Dental Finding) (R4)

File: site/apis/10-Clinical-General/1-R4/10066-Condition-Read-Dental-Finding-R4.html
Epic API ID | 10066
FHIR version | R4
Categories | Clinical General, Clinical Specialty
API type | Industry-Standard
API group | Dental Care Plan Group
USCDI | No
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | May 2021
HTTP method | GET
URL template | /api/FHIR/R4/Condition/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10066

Description

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | False | | | FHIR identifier for the target Condition resource.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Condition | Condition | False | | | An instance of the R4 Condition resource.
Condition.bodySite | CodeableConcept | True | Yes | | Locations of the finding including tooth, surfaces, and sites for furcation and suppuration findings.
Condition.category | CodeableConcept | True | Yes | | Always “dental-finding” for this resource.
Condition.clinicalStatus | CodeableConcept | True | | | The clinical status of a finding. Active or resolved.

Condition.code | CodeableConcept | True | | | The diagnosis code linked to a finding with coding system name.
Condition.encounter | Reference | True | | | The encounter this finding was documented in.
Condition.note | Annotation | True | Yes | | Note or annotation details.
Condition.note.text | String | True | | | The comment associated with this finding.
Condition.recordedDate | dateTime | True | | | The date the finding was first documented.
Condition.recorder | Reference | True | | | The provider who documented the finding.
Condition.subject | Reference | False | | | The patient or subject of this finding.

Sample request

```
https://hostname/instance/api/FHIR/R4/Condition/e.CYlLh27G5TiC0kZdum1Af0iUHvGK3k3zIubbaLVgpk3
```

Sample response

```
<Condition>
  <id value="e.CYlLh27G5TiC0kZdum1Af0iUHvGK3k3zIubbaLVgpk3"/>
  <clinicalStatus>
    <coding>
      <system value="http://terminology.hl7.org/CodeSystem/condition-clinical"/>
      <version value="4.0.0"/>
      <code value="active"/>
      <display value="Active"/>
    </coding>
  </clinicalStatus>
  <category>
    <text value="dental-finding"/>
  </category>
  <bodySite>
    <coding>
      <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.3.866976.94010"/>
      <code value="SFm"/>
    </coding>
    <text value="Teeth (Universal): SFm"/>
  </bodySite>
  <bodySite>
    <coding>
      <system value="urn:oid:1.2.840.114350.1.72.1.7.7.10.866976.94030"/>
      <code value="4"/>
      <display value="Incisal"/>
    </coding>
    <text value="Surfaces: Incisal"/>
  </bodySite>
  <subject>
    <reference value="Patient/e6DDE5.4tu-jjWXY8CtuVKw3"/>
    <type value="Patient"/>
    <display value="Nadal, Rafael"/>
  </subject>
  <encounter>
    <reference value="Encounter/eT7iIeZGfhZP8WXwImOgZBr3R17PGztDaeBtmBq9psAI3"/>
    <type value="Encounter"/>
  </encounter>
  <recordedDate value="2020-12-21T19:48:23Z"/>
  <recorder>
    <reference value="Practitioner/eNzt.GkOzCBShLeRDkLPGfw3"/>
    <type value="Practitioner"/>
    <display value="Gordon Lofgren, DDS"/>
  </recorder>
  <note>
    <text value="Caries class"/>
  </note>
</Condition>
```

10.1.9 Condition.Read (Encounter Diagnosis) (R4)

File: site/apis/10-Clinical-General/1-R4/950-Condition-Read-Encounter-Diagnosis-R4.html
Epic API ID | 950
FHIR version | R4
Categories | Clinical General
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | November 2019
HTTP method | GET
URL template | api/FHIR/R4/Condition/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=950

Description

This web service retrieves a single Condition from a patient's chart that represents an encounter diagnosis. An encounter diagnosis can either be documented as an admission diagnosis, visit diagnosis, or discharge diagnosis.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | Required | | | FHIR identifier for the target Condition resource.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Condition (R4) | Condition | | | | An instance of the R4 Condition resource.
Condition (R4).abatementDateTime | dateTime | | | 0..1 | Starting in the February 2024 version of Epic, if the Condition has a resolved date, it appears here. Only applicable to visit diagnoses with an attached problem.
Condition (R4).category | CodeableConcept | | Yes | 1..* | Always includes the category code of encounter-diagnosis. Also includes one of the following category codes: visit-diagnosis, admission-diagnosis, or discharge-diagnosis.
Condition (R4).clinicalStatus | CodeableConcept | | | 1..1 | The clinical status, which can be active, resolved, or inactive. Starting in the February 2024 version of Epic, if the encounter diagnosis is linked to a problem, it uses the clinical status of the linked problem. Only visit diagnoses can be linked to a problem.
Condition (R4).code | CodeableConcept | | | 1..1 | External codings for the Condition. If a list of codes is sent, they're listed in this element. Otherwise, this element uses the problem name from the name element. Starting in the February 2024 version of Epic or with special updates to November 2023, August 2023, or May 2023 version of Epic, each code also sends its display name and the provider-entered problem name is sent if available.
Condition (R4).encounter | Reference | | | 1..1 | Reference to an Encounter resource that the Condition is attached to.
Condition (R4).evidence | BackboneElement | | Yes | 0..1 | Supporting evidence for the Condition. This can return a linked Condition resource.
Condition (R4).evidence.detail | Reference | | Yes | 1..1 | Detailed information for the evidence.
Condition (R4).evidence.detail.display | String | | | 1..1 | The display name of the linked Condition. Returns the patient-friendly name for the problem when called in a patient context.
Condition (R4).evidence.detail.reference | String | | | 1..1 | A reference to the linked Condition.
Condition (R4).extension (assertedDateTime) (also: extension (assertedDateTime)) | Extension | | Yes | 0..1 | The date the Condition was diagnosed. Only applies to visit diagnoses with an attached problem.
Condition (R4).id | id | | | 1..1 | The Condition FHIR ID.
Condition (R4).note | Annotation | | Yes | 0..* | Free text comment.
Condition (R4).onsetDateTime | dateTime | | | 0..1 | Starting in the February 2024 version of Epic, if a Condition has a noted date, it appears here. Only applies to visit diagnoses with an attached problem.
Condition (R4).recordedDate | dateTime | | | 0..1 | The date the Condition was recorded. This property only applies to visit diagnoses. For a visit diagnosis with an attached problem, this is the entry date of the problem. For a visit diagnosis without an attached problem, this is the entry date of the visit diagnosis.
Condition (R4).recorder | Reference | | | 0..1 | Reference to the practitioner who updated the encounter diagnosis.
Condition (R4).subject | Reference | | | 1..1 | The patient associated with the Condition.
Condition (R4).verificationStatus | CodeableConcept | | | 0..1 | Admission and discharge diagnoses have a verificationStatus of "confirmed". Other visit diagnoses can be mapped by the organization to standard FHIR values found here.

Sample request

```
GET https://hostname/instance/api/FHIR/R4/Condition/ej5R7X.4RdaFNR5vWdsTpVTtccmkRkWJcKixc5XVY4cQ3
```

Sample response


```
{
  "resourceType": "Condition",
  "id": "eykG2p947BoIJRKQrvI9unC-Ov7r3qgrqibe9l1g9QyT-gowm6CIhUVWYQtPQ5m7gDPrE9mW9aMn8k-m8C3TUzw3",
  "extension": [
    {
      "valueDateTime": "2023-06-28",
      "url": "http://hl7.org/fhir/StructureDefinition/condition-assertedDate"
    }
  ],
  "clinicalStatus": {
    "coding": [
      {
        "system": "http://terminology.hl7.org/CodeSystem/condition-clinical",
        "version": "4.0.0",
        "code": "active",
        "display": "Active"
      }
    ],
    "text": "Active"
  },
  "verificationStatus": {
    "coding": [
      {
        "system": "http://terminology.hl7.org/CodeSystem/condition-ver-status",
        "version": "4.0.0",
        "code": "confirmed",
        "display": "Confirmed"
      }
    ],
    "text": "Confirmed"
  },
```

```
  "category": [
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/condition-category",
          "code": "encounter-diagnosis",
          "display": "Encounter Diagnosis"
        }
      ],
      "text": "Encounter Diagnosis"
    },
    {
      "coding": [
        {
          "system": "http://open.epic.com/FHIR/StructureDefinition/condition-category",
          "code": "visit-diagnosis",
          "display": "Visit Diagnosis"
        }
      ],
      "text": "Visit Diagnosis"
    }
  ],
  "code": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "400097005",
        "display": "Ingrowing Nail (Disorder)"
      },
      {
        "system": "http://hl7.org/fhir/sid/icd-9-cm",
        "code": "703.0",
        "display": "Ingrowing Nail"
      },
      {
        "system": "http://hl7.org/fhir/sid/icd-10-cm",
        "code": "L60.0",
        "display": "Ingrowing Nail"
      }
    ],
    "text": "Ingrowing Nail"
  },
  "subject": {
    "reference": "Patient/euI3cZMuLFNu6kO9mg4ScEg3",
    "display": "Test, Patient"
  },
  "encounter": {
    "reference": "Encounter/edrYhTQC4e9jviqejxOouP40Y6gzmMFnZFGpX5rWqx2A3",
    "identifier": {
      "use": "usual",
      "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.3.698084.8",
      "value": "10003083454"
    },
    "display": "Office Visit"
  },
  "onsetDateTime": "2023-06-28",
  "recordedDate": "2023-06-28",
  "evidence": [
    {
      "detail": [
        {
          "reference": "Condition/emYq8NzSEaKrCZtPWXa-GIioNncX-zeoLZG4utC3iD5w3",
          "display": "Ingrowing Nail"
        }
      ]
    }
  ]
}
```


Change log

Epic Version | Description | Type
February 2024 | The USCDIv3 specification for Condition will move the Condition's start date and time from onsetPeriod to onsetDatetime and end date and time from abatementPeriod to abatementDatetime fields. This could impact the evaluation of a Condition if a Condition consumer is not prepared for the transition. | Breaking change

10.1.10 Condition.Read (Genomics) (R4)

File: site/apis/10-Clinical-General/1-R4/1074-Condition-Read-Genomics-R4.html
Epic API ID | 1074
FHIR version | R4
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes

ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | November 2020
URL template | api/FHIR/R4/Condition/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=1074

Description

This web service retrieves a genomic indicator from a patient's chart. Genomic indicators represent the implications of genetic lab results for a given patient. For example, a genomic indicator, created based on the presence of several variants in the patient's genetic test results, might indicate a sensitivity to the drug warfarin. The system might then show decision support information to clinicians prescribing warfarin, triggered by the genomic indicator in the patient's chart, so that they are aware of additional considerations for specific medication. This resource can be queried by a patient ID.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | Required | | | FHIR identifier for the target Condition resource.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Condition (R4) | Condition | | | | An instance of the R4 Condition resource.
Condition (R4).category | CodeableConcept | | Yes | 1..* | Always "genomics" for patient genomic indicators.
Condition (R4).code | CodeableConcept | | | 0..* | LOINC and SNOMED codes representative of the genomic indicator.
Condition (R4).evidence | BackboneElement | | Yes | 0..* | The Observation (Labs) or DiagnosticReport (Results) resource that indicated the presence of this Condition.
Condition (R4).evidence.detail | Reference | | Yes | 0..* | Reference to an Observation (Labs) or DiagnosticReport (Results) resource that indicated the presence of this Condition.
Condition (R4).id | id | | | 1..1 | The indicator FHIR ID.
Condition (R4).recordedDate | dateTime | | | 1..1 | The date the indicator was recorded.
Condition (R4).subject | Reference | | | 1..1 | The patient associated with the indicator.

Sample request

```
GET https://hostname/instance/api/FHIR/R4/Condition/eV82zCR467Wt0UhpfS8mwXg3
```

Sample response


```
{
  "resourceType": "Condition",
  "id": "eV82zCR467Wt0UhpfS8mwXg3",
  "category": [
    {
      "coding": [
        {
          "system": "https://open.epic.com/FHIR/StructureDefinition/condition-category",
          "code": "genomics",
          "display": "Genomic Indicators"
        }
      ],
      "text": "Genomic Indicators"
    }
  ],
  "code": {
    "text": "Lynch Syndrome"
  },
  "subject": {
    "reference": "Patient/eOrFj.KdMJPmguszyMsRIjQ3",
    "display": "Bellsandwhistles, Sheila"
  },
  "recordedDate": "2024-03-19T17:12:10Z",
  "evidence": [
    {
      "detail": [
        {
          "reference": "DiagnosticReport/e3MlX5a0JkepBJvp4Lhf0W9Upm5kqXcOYE9ajagL.vGM3",
          "display": "TEMPUS XT"
        }
      ]
    }
  ]
}
```


10.1.11 Condition.Read (Health Concerns) (R4)

File: site/apis/10-Clinical-General/1-R4/984-Condition-Read-Health-Concerns-R4.html
Epic API ID | 984

FHIR version | R4
Categories | Clinical General
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Keywords | Health Concerns
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | February 2020
HTTP method | GET
URL template | api/FHIR/R4/Condition/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=984

Description

This API provides access to two data elements:

* The subset of problems on the problem list that have been marked as health concerns.

* The health status of the patient.

Health Concerns

Problems on the problem list may be marked with a class of "Health Concern," indicating that the problem is of special concern.

Marking a problem on the problem list as a health concern may not be a common workflow at many health systems. Prior to using this API, work with clinical teams to verify that the ability to mark problems as health concerns is enabled, and that end users are trained to flag problems as concerns.

Health Status

Documenting a general, specialty-independent health status for a patient may not be a common workflow at many health systems. Prior to using this API, it is essential to work closely with clinical teams to verify that that the ability to capture health status is enabled, and that end users are trained to evaluate and document the health status.

Because the health status is a single assessment, many Condition elements do not apply, thus Condition resources that represent health status include a smaller subset of elements than health concerns. For health status, severity, onsetPeriod, abatementPeriod, and note elements are not populated.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | Required | | | FHIR identifier for the target Condition resource.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Condition (R4) | Condition | | | | An instance of the R4 Condition resource.
Condition (R4).abatementDateTime | dateTime | | | 0..1 | Starting in the February 2024 version of Epic, this is the date the health concern was resolved. For versions prior to February 2024, this field is always empty, and the abatement date can be found in abatementPeriod instead. Health status conditions will not have an abatement date.
Condition (R4).abatementPeriod | Period | | | 0..1 | In the November 2023 version of Epic and earlier, this is the date the health concern was resolved. For versions after November 2023, this field is always empty, and the abatement date can be found in abatementDateTime instead. Health status conditions will not have an abatement date.
Condition (R4).category | CodeableConcept | | Yes | 1..* | Health Concerns Always includes category code "problem-list-item". If a problem list item is marked with a class of "Health Concern", indicating that the problem is of special concern, a category code of "health-concern" will also be included. This can also include additional category codes, such as 'sdoh' for problems related to social determinates of health. This category depends on configuration of diagnosis groupers defined by the Epic organization. Health Status Always includes category code "health-concern".
Condition (R4).clinicalStatus | CodeableConcept | | | 1..1 | The current status of the health concern or health status. Health Concerns The following Condition Clinical Status Codes are supported: FHIR Status Epic Status active active conditions resolved resolved conditions inactive deleted conditions Health Status Health status Conditions will always have a status of 'active'.
FHIR Status | Epic Status
active | active conditions
resolved | resolved conditions
inactive | deleted conditions
Condition (R4).code | CodeableConcept | | | 1..1 | The code that identifies the specific type of problem. Health Concerns Examples of code sytems that are often used to code problem list items include ICD-10, SNOMED, ICPC Basic Health Care Classification, and Intelligent Medical Objects ProblemIT and others. If there are codes available to send, they're included in this element. If there are no codes to send, this element will include the un-coded, free text problem name. Starting in the February 2024 version of Epic, each code also sends its display name and the provider-entered problem name is sent if available. Health Status Health Status will be coded using SNOMED CT.
Condition (R4).id | id | | | 1..1 | The condition FHIR ID.
Condition (R4).meta | Meta | | | | Metadata about the resource.
Condition (R4).meta.lastUpdated | String | | | 0..1 | The instant the health concern was last updated.
Condition (R4).note | Annotation | | Yes | 0..* | Free text comments for the note. Health Concerns Can be an Overview note or, starting in the February 2025 version of Epic, an Assessment & Plan note. Health Status Notes are not supported for Health Status

Condition (R4).note.authorReference | Reference | | | A reference to the Practitioner resource for the user who wrote the note.

Condition (R4).note.extension (annotationType) (also: extension (annotationType)) | Extension | | Yes | | For Assessment & Plan notes, this is always 51847-2. For Overview notes, this is always 68608-9.

Condition (R4).note.extension (data-conversion-warning) (also: extension (data-conversion-warning)) | Extension | | Yes | | The following warnings may be returned if the note content is converted from RTF to plain text. Reason for warning Warning text Formatting was lost due to unicode characters or tables being present in the original note. Formatting of this note is different from the original The original note was RTF. Formatting of this note might be different from the original There was an image in the original note Images from the original note were not included URL: http://open.epic.com/FHIR/StructureDefinition/extension/data-conversion-warning

Reason for warning | Warning text

Formatting was lost due to unicode characters or tables being present in the original note. | Formatting of this note is different from the original

The original note was RTF. | Formatting of this note might be different from the original

There was an image in the original note | Images from the original note were not included

Condition (R4).note.time | String | | | The time that the note was signed.

Condition (R4).onsetDateTime | dateTime | | | 0..1 | Health Concerns Starting in the February 2024 version of Epic, if a condition has a noted date, it appears here. Prior to the February 2024 version of Epic, the condition noted date is sent in onsetPeriod. Health Status The instant the health status was recorded.

Condition (R4).onsetPeriod | Period | | | 0..1 | In the November 2023 version of Epic and earlier, the date the problem was noted. Starting in the Febuary 2025 version of Epic, the condition noted date is sent in onsetDateTime.

Condition (R4).recordedDate | dateTime | | | 0..1 | The date the condition was recorded.

Condition (R4).recorder | Reference | | | 0..1 | Reference to the practitioner who updated the Health Concern or Health Status.

Condition (R4).severity | CodeableConcept | | | 0..1 | Health Concern The default priority of the problem. Note that Epic support two types of priorities for problems: A default priority that is specified when the problem is initially created. This value is available in the Condition.severity property. A user-specific priority that users can adjust based on the relevance of the problem to their specialty. For example, a dermatologist might rate eczema as a high priority while a cardiologist would rate it as low. User-level priorities allow each clinician to see the list organized in a way that is most useful to them. This value is displayed to users Epic's problem list activity. This value is not supported in the FHIR API. Health Status Health Status conditions will not have a severity.

Condition (R4).stage | BackboneElement | | Yes | | For problems that represent cancer, staging information can be documented as part of the problem. Cancer staging usually involves the identification of the location, size, and the extent of metastasis of the tumor.

Condition (R4).stage.assessment | Reference | | Yes | | SmartData Element associated with cancer staging for this condition.

Condition (R4).subject | Reference | | | 1..1 | The patient associated with the condition.

Condition (R4).verificationStatus | CodeableConcept | | | 1..1 | Health Concerns Always a status of 'confirmed' Health Status Always a status of 'confirmed'.

Sample request

```
GET https://hostname/instance/api/FHIR/R4/Condition/elu6EBsg9gAduSUJjAncRpg3
```

Sample response


```
{
  "resourceType": "Condition",
  "id": "elu6EBsg9gAduSUJjAncRpg3",
  "extension": [
    {
      "valueDateTime": "2025-04-09",
      "url": "http://hl7.org/fhir/StructureDefinition/condition-assertedDate"
    }
  ],
  "clinicalStatus": {
    "coding": [
      {
        "system": "http://terminology.hl7.org/CodeSystem/condition-clinical",
        "version": "4.0.0",
        "code": "active",
        "display": "Active"
      }
    ],
    "text": "Active"
  },
  "verificationStatus": {
    "coding": [
      {
        "system": "http://terminology.hl7.org/CodeSystem/condition-ver-status",
        "version": "4.0.0",
        "code": "confirmed",
        "display": "Confirmed"
      }
    ],
    "text": "Confirmed"
  },
  "category": [
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/condition-category",
          "code": "problem-list-item",
          "display": "Problem List Item"
        }
      ],
      "text": "Problem List Item"
    },
    {
      "coding": [
```

```
        {
          "system": "http://loinc.org",
          "code": "75310-3",
          "display": "Health concerns Document"
        },
        {
          "system": "http://hl7.org/fhir/us/core/CodeSystem/condition-category",
          "code": "health-concern",
          "display": "Health Concern"
        }
      ],
      "text": "Health Concern"
    }
  ],
  "severity": {
    "coding": [
      {
        "system": "urn:oid:2.16.840.1.113883.6.96",
        "code": "6736007",
        "display": "Moderate"
      }
    ],
    "text": "Medium"
  },
  "code": {
    "coding": [
      {
        "system": "http://hl7.org/fhir/sid/icd-10-cm",
        "code": "M54.2",
        "display": "Cervicalgia"
      },
      {
        "system": "http://snomed.info/sct",
        "code": "81680005",
        "display": "Neck Pain"
      },
      {
        "system": "http://hl7.org/fhir/sid/icd-9-cm",
        "code": "723.1",
        "display": "Pain in the neck"
      },
      {
        "system": "urn:oid:2.16.840.1.113883.3.247.1.1",
        "code": "370896",
        "display": "Pain in the neck"
      }
    ],
    "text": "Pain in the neck"
  },
  "subject": {
    "reference": "Patient/eUEKdnYPKuCXlSq8WIYaPTA3",
    "display": "Bugsy, Beau Dakota"
  },
  "onsetDateTime": "2025-03-15",
  "recordedDate": "2025-04-23",
  "note": [
    {
      "extension": [
        {
          "valueString": "Formatting of this note might be different from the original.",
          "url": "http://open.epic.com/FHIR/StructureDefinition/extension/data-conversion-warning"
        },
        {
          "valueCodeableConcept": {
            "coding": [
              {
                "system": "http://loinc.org",
                "code": "68608-9",
                "display": "Summary note"
              }
            ],
            "text": "Problem Overview"
          },
          "url": "http://hl7.org/fhir/StructureDefinition/annotationType"
        }
      ],
      "authorReference": {
        "reference": "Practitioner/ejb8.EbaQ8UyziPoyHy.J3A3",
        "display": "Attending Physician Inpatient, MD"
      },
      "time": "2025-04-23T19:05:36Z",
      "text": "This is a sample problem note\r\n"
    }
  ]
}
```


Change log

Epic Version | Description | Type
May 2025 | Previously data loss warnings in the conversion of notes were prepended to the note content (the note.text element in the FHIR resource). Now this warning is returned in a note.extension element. | Breaking change
February 2024 | The USCDIv3 specification for Condition will move the Condition's start date and time from onsetPeriod to onsetDatetime and end date and time from abatementPeriod to abatementDatetime fields. This could impact the evaluation of a Condition if a Condition consumer is not prepared for the transition. | Breaking change
November 2022 | Code systems for the following code sets now return a URL in place of an OID: ActCode, ICD-10-CM, ICD-10-CA, ICD-10-NL, SNOMED System URL OID ActCode STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode 2.16.840.1.113883.5.4 ICD-10-CM http://hl7.org/fhir/sid/icd-10-cm 2.16.840.1.113883.6.90 ICD-10-CA http://hl7.org/fhir/sid/icd-10-ca 2.16.840.1.113883.6.94 ICD-10-NL http://hl7.org/fhir/sid/icd-10-nl 2.16.840.1.113883.6.3.2 SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 | Breaking change
System | URL | OID
ActCode | STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode | 2.16.840.1.113883.5.4
ICD-10-CM | http://hl7.org/fhir/sid/icd-10-cm | 2.16.840.1.113883.6.90
ICD-10-CA | http://hl7.org/fhir/sid/icd-10-ca | 2.16.840.1.113883.6.94
ICD-10-NL | http://hl7.org/fhir/sid/icd-10-nl | 2.16.840.1.113883.6.3.2
SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96

10.1.12 Condition.Read (Infection) (R4)

File: site/apis/10-Clinical-General/1-R4/10140-Condition-Read-Infection-R4.html
Epic API ID | 10140
FHIR version | R4
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | August 2021
HTTP method | GET
URL template | /api/FHIR/R4/Condition/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10140

Description

Staphylococcus aureus

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | False | | | FHIR identifier for the target Condition resource.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Condition | Condition | False | | | An instance of the R4 Condition resource.
Condition.abatementDateTime | dateTime | True | | | The date and time the infection was resolved.
Condition.category | CodeableConcept | False | Yes | | Always "infection"
Condition.clinicalStatus | CodeableConcept | False | | | Always either “active” or “inactive”.
Condition.code | CodeableConcept | False | | | The display name for the infection and the SNOMED code if one exists.
Condition.encounter | Reference | True | | | The encounter associated with the infection. Only applies to encounter-level infections.
Condition.id | id | False | | | The FHIR ID
Condition.note | Annotation | True | Yes | | Free-text comments.
Condition.onsetDateTime | dateTime | True | | | The date the patient acquired the infection.
Condition.recordedDate | dateTime | False | | | The date and time the infection was added to the patient’s chart.
Condition.subject | Reference | False | | | The patient associated with the infection.

Sample request

```
https://hostname/instance/api/FHIR/R4/Condition/conditionid
```

Sample response

```
<Condition xmlns="http://hl7.org/fhir">
  <id value="ePMnpp6lhpta6YiS9-QsmBA3"/>
  <clinicalStatus>
    <coding>
      <system value="http://terminology.hl7.org/CodeSystem/condition-clinical"/>
      <version value="4.0.0"/>
      <code value="active"/>
      <display value="Active"/>
    </coding>
    <text value="Active"/>
```

```
  </clinicalStatus>
  <category>
    <coding>
      <system value="http://open.epic.com/FHIR/StructureDefinition/condition-category"/>
      <code value="infection"/>
      <display value="Infection"/>
    </coding>
    <text value="Infection"/>
  </category>
  <code>
    <coding>
      <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.4.698084.10375"/>
      <code value="9"/>
      <display value="Measles"/>
    </coding>
    <coding>
      <system value="http://snomed.info/sct"/>
      <code value="14189004"/>
      <display value="Measles"/>
    </coding>
    <text value="Measles"/>
  </code>
  <subject>
    <reference value="Patient/eC4MDFhhJAHRxZCUXt6ozHA3"/>
    <display value="Dunwoody, Terence"/>
  </subject>
  <onsetDateTime xmlns="http://hl7.org/fhir" value="2021-02-12"/>
  <recordedDate value="2021-02-19T20:39:13Z"/>
</Condition>
```

10.1.13 Condition.Read (Medical History) (R4)

File: site/apis/10-Clinical-General/1-R4/10302-Condition-Read-Medical-History-R4.html
Epic API ID | 10302
FHIR version | R4
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Condition/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10302

Description

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | False | | | FHIR identifier for the target Condition resource.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Condition | Condition | False | | | An instance of the R4 Condition resource.
Condition.category | CodeableConcept | True | Yes | | Always “medical-history” for patient medical history.
Condition.code | CodeableConcept | False | | | Array of ICD codes and SNOMED concepts
Condition.note | Annotation | True | Yes | | The comment and display name for positive medical history, or the comment for pertinent negative medical history. Not shown in MyChart context.
Condition.onsetDateTime | dateTime | True | | | The date when the medical problem began for the patient. Only applies to positive medical history.
Condition.onsetPeriod | Period | True | | | The date range when the medical problem began for the patient. Only applies to positive medical history. Applicable for dates saved as fuzzy dates.
Condition.onsetString | String | True | | | The date when the medical problem began for the patient. Only applies to positive medical history. Applicable for dates saved as a string instead of a date format.
Condition.recordedDate | dateTime | True | | | The date when the pertinent negative was noted on the patient’s chart. Only applies to pertinent negative medical history.
Condition.subject | Reference | False | | | The patient ID
Condition.verificationStatus | CodeableConcept | True | | | Always “confirmed” for positive medical history and “refuted” for pertinent negative medical history.

Sample request

```
https://hostname/instance/api/FHIR/R4/Condition/eS2z.azU1DtzQ-pLPAY0rghf9Xa50JGqRrwv9MrBWXa5ph3
```

Sample response

```
<Condition xmlns="http://hl7.org/fhir">
  <id value=" eS2z.azU1DtzQ-pLPAY0rghf9Xa50JGqRrwv9MrBWXa5ph3"/>
```

```
  <verificationStatus>
    <coding>
      <system value="http://terminology.hl7.org/CodeSystem/condition-ver-status"/>
      <version value="4.0.0"/>
      <code value="confirmed"/>
      <display value="Confirmed"/>
    </coding>
    <text value="Confirmed"/>
  </verificationStatus>
  <category>
    <coding>
      <system value="http://open.epic.com/FHIR/StructureDefinition/condition-category"/>
      <code value="medical-history"/>
      <display value="Medical History"/>
    </coding>
    <text value="Medical History"/>
  </category>
  <code>
    <coding>
      <system value="http://snomed.info/sct"/>
      <code value="39423001"/>
      <display value="Stuttering (finding)"/>
    </coding>
    <coding>
      <system value="http://hl7.org/fhir/sid/icd-9-cm"/>
      <code value="315.35"/>
      <display value="Stammer"/>
    </coding>
    <coding>
      <system value="urn:oid:2.16.840.1.113883.6.90"/>
      <code value="F80.81"/>
      <display value="Stammer"/>
    </coding>
    <text value="Stammer"/>
  </code>
  <subject>
    <reference value="Patient/ebb8tmJBI1FH-fdAzdqISZg3"/>
    <display value="Stanley, James"/>
  </subject>
</Condition>
```

Change log

Epic Version | Description | Type
May 2023 (Special Updates to February 2023) | Code systems in the Condition.category element now return a URL in place of an OID. Typical code systems include: System URL OID SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 LOINC http://loinc.org 2.16.840.1.113883.6.1 | Breaking change
System | URL | OID
SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96
LOINC | http://loinc.org | 2.16.840.1.113883.6.1

10.1.14 Condition.Read (Problems) (R4)

File: site/apis/10-Clinical-General/1-R4/951-Condition-Read-Problems-R4.html
Epic API ID | 951
FHIR version | R4
Categories | Clinical General
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Keywords | Problems;Problem List
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | November 2019
HTTP method | GET
URL template | api/FHIR/R4/Condition/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=951

Description

This API retrieves a single problem from the patient's chart. There are two types of problems that this API supports:

* Problems on the patient's problem list in the chart.

* Problems that have been submitted to the holding tank, but have not yet been reconciled into the patient's problem list.

Request parameters

Name | Type | Optional | Array | Cardinality | Description

ID | token | Required | | | The Problem FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description

Condition (R4) | Condition | | | | An instance of the R4 Condition resource.

Condition (R4).abatementDateTime | dateTime | | | 0..1 | The date the condition was resolved. Starting in the February 2024 version of Epic, abatementDateTime is used to communicate the resolution date. In November 2023 and earlier versions of Epic, abatementPeriod is used to communicate the resolution date.

Condition (R4).abatementPeriod | Period | | | 0..1 | The date the condition is resolved. In the November 2023 and earlier versions of Epic, abatementPeriod is used to communicate the resolution date. Starting in the February 2024 version of Epic, abatementDateTime is used to communicate the resolution date.

Condition (R4).category | CodeableConcept | | Yes | 0..* | Always includes category code "problem-list-item". If a problem list item is marked with a class of "Health Concern", indicating that the problem is of special concern, a category code of "health-concern" is also included. This can also include additional category codes, such as "sdoh" for problems related to social determinates of health. This category depends on configuration of diagnosis groupers defined by the Epic organization.

Condition (R4).clinicalStatus | CodeableConcept | | | 1..1 | The current clinical status of the problem list item. The following Condition Clinical Status Codes are supported: active - Active problem list items resolved - Resolved problem list items inactive - Deleted problem list items

Condition (R4).code | CodeableConcept | | | 1..1 | The code that identifies the specific type of problem. Examples of code sytems that are often used to code problem list items include ICD-10, SNOMED, ICPC Basic Health Care Classification, and Intelligent Medical Objects ProblemIT and others. If there are codes available to send, they're included in this element. If there are no codes to send, this element includes the uncoded, free text problem name. Starting in the February 2024 version of Epic or with special updates to November 2023, August 2023, or May 2023 version of Epic, each code also sends its display name and the provider-entered problem name is sent if available. Common code system values: ICD-10: http://hl7.org/fhir/sid/icd-10-cm SNOMED: http://snomed.info/sct

Condition (R4).extension (assertedDateTime) (also: extension (assertedDateTime)) | Extension | | Yes | 0..* | The date the condition was diagnosed. URL: http://hl7.org/fhir/StructureDefinition/condition-assertedDate

Condition (R4).extension (smartdata) (also: extension (smartdata)) | Extension | | Yes | | When documenting a problem, various questions, also known as SmartData Elements, can be conditionally presented to the clinician. The smartdata extension on Condition (Problems) refernces the clinician's answers, which are available in the Observation (SmartData Elements) API.

Condition (R4).id | id | | | 1..1 | The condition FHIR ID.

Condition (R4).meta | Meta | | | | Metadata about the resource.

Condition (R4).meta.lastUpdated | String | | | 0..1 | The instant the condition was last updated.

Condition (R4).note | Annotation | | Yes | 1..* | Free text comments for the note. Can be an Overview note or, starting in the February 2025 version of Epic, an Assessment & Plan note.

Condition (R4).note.authorReference | Reference | | | | Who wrote the note. This element is returned starting in the following Epic versions, dependent on note type: May 2025: Problem overview notes February 2025: Assessment & Plan notes

Condition (R4).note.extension (annotationType) (also: extension (annotationType)) | Extension | | Yes | | This extension is returned starting in the following Epic versions, dependent on note type: May 2025: Problem overview notes February 2025: Assessment & Plan notes The following valueCodeableConcept codings are supported: Assessment & Plan notes: LOINC code 51847-2 (Evaluation + Plan note) Problem overview notes: LOINC code 68608-9 (Summary note) URL: http://hl7.org/fhir/StructureDefinition/annotationType

Condition (R4).note.extension (data-conversion-warning) (also: extension (data-conversion-warning)) | Extension | | Yes | | This extension is returned starting in the following Epic versions, dependent on note type: May 2025: Problem overview notes February 2025: Assessment & Plan notes One or more of the following warnings can be returned depending on RTF to plan text conversion: Reason for warning Warning text Formatting was lost due to unicode characters or tables being present in the original note. Formatting of this note is different from the original. The original note was RTF. Formatting of this note might be different from the original. There were one or more images in the original note. Images from the original note were not included. There were one or more media attachments in the original note. Attached media from the original note were not included. URL: http://open.epic.com/FHIR/StructureDefinition/extension/data-conversion-warning

Reason for warning | Warning text

Formatting was lost due to unicode characters or tables being present in the original note. | Formatting of this note is different from the original.

The original note was RTF. | Formatting of this note might be different from the original.

There were one or more images in the original note. | Images from the original note were not included.

There were one or more media attachments in the original note. | Attached media from the original note were not included.

Condition (R4).note.text | String | | | | The content of the note. In the February 2025 version of Epic and earlier versions, this note text can include prepended data conversion warnings similar to those documented in the “extension (data-conversion-warning)” response element.

Condition (R4).note.time | String | | | | The time that the note was signed. This element is returned starting in the following Epic versions, dependent on note type: May 2025: Problem overview notes February 2025: Assessment & Plan notes

Condition (R4).onsetDateTime | dateTime | | | 0..1 | Starting in the February 2024 version of Epic, if a condition has a noted date, it appears here.

Condition (R4).onsetPeriod | Period | | | 0..1 | In the November 2023 and earlier versions of Epic, if a condition has a noted date, it appears here.

Condition (R4).recordedDate | dateTime | | | 1..1 | The date the condition was recorded.

Condition (R4).recorder | Reference | | | 0..1 | Reference to the practitioner who updated the problem list.

Condition (R4).severity | CodeableConcept | | | 0..1 | The default priority of the problem. Note that Epic support two types of priorities for problems: A default priority that is specified when the problem is initially created. This value is available in the Condition.severity property. A user-specific priority that users can adjust based on the relevance of the problem to their specialty. For example, a dermatologist might rate eczema as a high priority while a cardiologist would rate it as low. User-level priorities allow each clinician to see the list organized in a way that is most useful to them. This value is displayed to users Epic's problem list activity. This value is not supported in the FHIR API.

Condition (R4).stage | BackboneElement | | Yes | 0..* | For problems that represent cancer, staging information can be documented as part of the problem. Cancer staging usually involves the identification of the location, size, and the extent of metastasis of the tumor.

Condition (R4).stage.assessment | Reference | | Yes | 1..* | SmartData element associated with cancer staging for this condition.

Condition (R4).subject | Reference | | | 1..1 | The patient associated with the condition.

Condition (R4).verificationStatus | CodeableConcept | | | 1..1 | If this is on the patient's local chart, the value is "confirmed". If it is outside data, it is "provisional". If it is deleted, it is "entered-in-error".

Sample request

```
GET https://hostname/instance/api/FHIR/R4/Condition/evA9Cs4FuqWRemAD9B321RvjhDZCOh4cL0mszsMdP89s3
```

Sample response


```
{
  "resourceType": "Condition",
  "id": "evA9Cs4FuqWRemAD9B321RvjhDZCOh4cL0mszsMdP89s3",
  "extension": [
    {
      "valueDateTime": "2025-04-15",
      "url": "http://hl7.org/fhir/StructureDefinition/condition-assertedDate"
    }
  ],
  "clinicalStatus": {
    "coding": [
      {
        "system": "http://terminology.hl7.org/CodeSystem/condition-clinical",
        "version": "4.0.0",
        "code": "active",
        "display": "Active"
      }
    ],
    "text": "Active"
  },
  "verificationStatus": {
    "coding": [
      {
        "system": "http://terminology.hl7.org/CodeSystem/condition-ver-status",
        "version": "4.0.0",
        "code": "confirmed",
        "display": "Confirmed"
      }
    ],
    "text": "Confirmed"
  },
  "category": [
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/condition-category",
          "code": "problem-list-item",
          "display": "Problem List Item"
        }
      ],
      "text": "Problem List Item"
    },
    {
      "coding": [
        {
          "system": "http://loinc.org",
          "code": "75310-3",
          "display": "Health concerns Document"
        },
        {
          "system": "http://hl7.org/fhir/us/core/CodeSystem/condition-category",
          "code": "health-concern",
          "display": "Health Concern"
        }
      ],
      "text": "Health Concern"
    },
    {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "64572001"
        }
      ]
    }
  ],
  "code": {
    "coding": [
      {
        "system": "http://hl7.org/fhir/sid/icd-10-cm",
        "code": "W59.22XA",
        "display": "Struck by turtle, initial encounter"
      },
      {
        "system": "http://snomed.info/sct",
        "code": "418589001",
        "display": "Accidental Physical Contact With Animal"
      },
      {
        "system": "http://hl7.org/fhir/sid/icd-9-cm",
        "code": "E906.8",
        "display": "Struck by turtle"
      },
      {
        "system": "urn:oid:2.16.840.1.113883.3.247.1.1",
        "code": "1623504",
        "display": "Struck by turtle"
```

```
      }
    ],
    "text": "Struck by turtle"
  },
  "subject": {
    "reference": "Patient/eLFQugzW.CQz3w48A427N.w3",
    "display": "Fhirson, Fhirboy"
  },
  "onsetDateTime": "2023-11-14",
  "recordedDate": "2023-11-14",
  "note": [
    {
      "extension": [
        {
          "valueString": "Formatting of this note might be different from the original.",
          "url": "http://open.epic.com/FHIR/StructureDefinition/extension/data-conversion-warning"
        },
        {
          "valueCodeableConcept": {
            "coding": [
              {
                "system": "http://loinc.org",
                "code": "68608-9",
                "display": "Summary note"
              }
            ],
            "text": "Problem Overview"
          },
          "url": "http://hl7.org/fhir/StructureDefinition/annotationType"
        }
      ],
      "authorReference": {
        "reference": "Practitioner/eSCE9sLNDEsbNOTQOi595ww3",
        "display": "Doctor Doctorson"
      },
      "time": "2025-08-03T22:26:40Z",
      "text": "This is an overview note regarding this patient's unfortunate incident with the turtle.\r\n"
    },
    {
      "extension": [
        {
          "valueString": "Formatting of this note might be different from the original.",
          "url": "http://open.epic.com/FHIR/StructureDefinition/extension/data-conversion-warning"
        },
        {
          "valueCodeableConcept": {
            "coding": [
              {
                "system": "http://loinc.org",
                "code": "51847-2",
                "display": "Evaluation + Plan note"
              }
            ],
            "text": "Assessment & Plan"
          },
          "url": "http://hl7.org/fhir/StructureDefinition/annotationType"
        }
      ],
      "authorReference": {
        "reference": "Practitioner/eodo7Bj9dk-LtJSthMwARpw3",
        "display": "Doctor Doctorson"
      },
      "time": "2025-04-14T19:43:16Z",
      "text": "A&P note\r\n"
    }
  ]
}
```


Change log

Epic Version | Description | Type

May 2025 | Previously data loss warnings in the conversion of notes were prepended to the note content (the note.text element in the FHIR resource). Now this warning is returned in a note.extension element. | Breaking change

February 2024 | The USCDIv3 specification for Condition will move the Condition's start date and time from onsetPeriod to onsetDatetime and end date and time from abatementPeriod to abatementDatetime fields. This could impact the evaluation of a Condition if a Condition consumer is not prepared for the transition. | Breaking change

May 2023 (Special Updates to February 2023) | Code systems in the Condition.category element now return a URL in place of an OID. Typical code systems include: System URL OID SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 LOINC http://loinc.org 2.16.840.1.113883.6.1 | Breaking change

System | URL | OID

SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96

LOINC | http://loinc.org | 2.16.840.1.113883.6.1

November 2022 | Code systems in the Condition.code element for the following code sets now return a URL in place of an OID: ActCode, ICD-10-CM, ICD-10-CA, ICD-10-NL, SNOMED System URL OID ActCode STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode 2.16.840.1.113883.5.4 ICD-10-CM http://hl7.org/fhir/sid/icd-10-cm 2.16.840.1.113883.6.90 ICD-10-CA http://hl7.org/fhir/sid/icd-10-ca 2.16.840.1.113883.6.94 ICD-10-NL http://hl7.org/fhir/sid/icd-10-nl 2.16.840.1.113883.6.3.2 SNOMED

http://snomed.info/sct 2.16.840.1.113883.6.96 | Breaking change
System | URL | OID
ActCode | STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode | 2.16.840.1.113883.5.4
ICD-10-CM | http://hl7.org/fhir/sid/icd-10-cm | 2.16.840.1.113883.6.90
ICD-10-CA | http://hl7.org/fhir/sid/icd-10-ca | 2.16.840.1.113883.6.94
ICD-10-NL | http://hl7.org/fhir/sid/icd-10-nl | 2.16.840.1.113883.6.3.2
SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96

10.1.15 Condition.Read (Reason for Visit) (R4)

File: site/apis/10-Clinical-General/1-R4/10187-Condition-Read-Reason-for-Visit-R4.html
Epic API ID | 10187
FHIR version | R4
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Condition/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10187

Description

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | False | | | FHIR identifier for the target Condition resource.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Condition | Condition | False | | | An instance of the R4 Condition resource.
Condition.bodySite | CodeableConcept | True | Yes | | The location of the condition.
Condition.code | CodeableConcept | True | | | External codings for the condition. If a list of codes is sent, they're listed in this element.
Condition.encounter | Reference | True | | | The encounter associated with the condition
Condition.id | id | False | | | The condition FHIR ID
Condition.note | Annotation | True | Yes | | Free text comments
Condition.onsetDateTime | dateTime | True | | | The date the condition began.
Condition.recordedDate | dateTime | True | | | The date the condition was recorded.
Condition.recorder | Reference | True | | | The provider associated with recording the condition
Condition.subject | Reference | False | | | The patient associated with the condition

Sample request

```
https://hostname/instance/api/FHIR/R4/Condition/egFptNetytmse51RiokqI2Z3QBHsNx8GHXDYi7EhR.f9fR-tfX6ms3qkivkLc2Bd13
```

Sample response

```
<condition xmlns="http://hl7.org/fhir">
  <id value="egFptNetytmse51RiokqI2Z3QBHsNx8GHXDYi7EhR.f9fR-tfX6ms3qkivkLc2Bd13"/>
  <category>
    <coding>
      <system value="http://open.epic.com/FHIR/StructureDefinition/condition-category"/>
      <code value="reason-for-visit"/>
      <display value="Reason for Visit"/>
    </coding>
    <text value="Reason for Visit"/>
  </category>
  <code>
    <coding>
      <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.2.728286"/>
      <code value="43294"/>
      <display value="Rash"/>
    </coding>
    <text value="Rash"/>
  </code>
  <bodysite>
    <coding>
      <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.2.866976"/>
      <code value="11347"/>
      <display value="Right Leg"/>
    </coding>
    <text value="Right Leg"/>
  </bodysite>
  <subject>
```

```
    <reference value="Patient/e56rHiD8cHF60CyLEUJ-1Ew3"/>
    <display value="Shepherd, Amelia"/>
  </subject>
  <encounter>
    <reference value="Encounter/eeM8KWAwqj5oZbB7ROBexzAvNiJBO0VmJdfkt2kHvJ.s3"/>
  </encounter>
  <recordeddate value="2021-07-28T17:03:39Z"/>
  <recorder>
    <reference value="Practitioner/ehchKMZoNW8B.Ycy.dUrFtg3"/>
    <display value="Eva Ward, MD"/>
  </recorder>
  <note>
    <text value="This is my comment."/>
  </note>
</condition>
```

10.1.16 Condition.Search (Care Plan Problem) (R4)

File: site/apis/10-Clinical-General/1-R4/10048-Condition-Search-Care-Plan-Problem-R4.html
Epic API ID | 10048
FHIR version | R4
Categories | Clinical General
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | February 2021
HTTP method | GET
URL template | /api/FHIR/R4/Condition?_id={_id}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10048

Description

This web service searches for care plan problems in patient care plans.

Currently, this Search interaction supports only searching by _id. For this reason, CarePlan.Read should typically be used instead of this resource.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | BackboneElement | False | Yes | | An array of entries included in this bundle.
Bundle.entry.resource | ResourceContainer | False | | | A container for a single FHIR resource.
Bundle.entry.resource.Condition | Condition | False | | | An instance of the R4 Condition resource.
Bundle.entry.resource.Condition.abatementDateTime | dateTime | True | | | The date the condition was resolved.
Bundle.entry.resource.Condition.category | CodeableConcept | False | Yes | | Always "care-plan-problem".
Bundle.entry.resource.Condition.clinicalStatus | CodeableConcept | False | | | The clinical status, which can be active, resolved, or inactive.
Bundle.entry.resource.Condition.code | CodeableConcept | False | | | The display name for the condition.
Bundle.entry.resource.Condition.id | id | False | | | The condition FHIR ID.
Bundle.entry.resource.Condition.note | Annotation | True | Yes | | Free text comments. Omitted when this resource is used in the patient context.
Bundle.entry.resource.Condition.onsetDateTime | dateTime | True | | | The date the condition began.
Bundle.entry.resource.Condition.recordedDate | dateTime | False | | | The date the condition was documented.
Bundle.entry.resource.Condition.recorder | Reference | False | | | The practitioner who noted the condition.
Bundle.entry.resource.Condition.severity | CodeableConcept | False | | | The priority of the problem.
Bundle.entry.resource.Condition.subject | Reference | False | | | The patient associated with the condition.

Sample request

```
https://hostname/instance/api/FHIR/R4/Condition?_id=eG0hRQR7MUQBcS.AfHqMYXQ3&ehiExport=1
```

Sample response

```
<Bundle xmlns="http://hl7.org/fhir">
  <type value="searchset"/>
  <total value="1"/>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/R4/Condition?_id=eG0hRQR7MUQBcS.AfHqMYXQ3&amp;ehiExport=1"/>
  </link>
  <entry>
    <link>
      <relation value="self"/>
      <url value="https://hostname/instance/api/FHIR/R4/Condition/eG0hRQR7MUQBcS.AfHqMYXQ3"/>
    </link>
```

```
    <fullUrl value="https://hostname/instance/api/FHIR/R4/Condition/eG0hRQR7MUQBcS.AfHqMYXQ3"/>
    <resource>
      <Condition>
        <id value="eG0hRQR7MUQBcS.AfHqMYXQ3"/>
        <clinicalStatus>
          <coding>
            <system value="http://terminology.hl7.org/CodeSystem/condition-clinical"/>
            <version value="4.0.0"/>
            <code value="active"/>
            <display value="Active"/>
          </coding>
          <text value="Active"/>
        </clinicalStatus>
        <category>
          <coding>
            <system value="http://open.epic.com/FHIR/StructureDefinition/condition-category"/>
            <code value="care-plan-problem"/>
            <display value="Care Plan Problem"/>
          </coding>
          <text value="Care Plan Problem"/>
        </category>
        <severity>
          <coding>
            <system value="urn:oid:1.2.840.114350.1.13.861.1.7.4.768066.85"/>
            <display value="Low"/>
          </coding>
          <text value="Low"/>
        </severity>
        <code>
          <text value="MyC3 Problem"/>
        </code>
        <subject>
          <reference value="Patient/eS.3bWuX-xYAfV7T5h12w5Q3"/>
          <display value="Test, Patient"/>
        </subject>
        <onsetDateTime value="2020-10-26"/>
        <recordedDate value="2020-10-07T16:12:37Z"/>
        <recorder>
          <reference value="Practitioner/etgK7TEYmpXBluVvFVNOlvA3"/>
          <display value="Test Provider, PHARMD"/>
        </recorder>
      </Condition>
    </resource>
    <search>
      <mode value="match"/>
    </search>
  </entry>
</Bundle>
```

10.1.17 Condition.Search (Dental Finding) (R4)

File: site/apis/10-Clinical-General/1-R4/10067-Condition-Search-Dental-Finding-R4.html
Epic API ID | 10067
FHIR version | R4
Categories | Clinical General, Clinical Specialty
API type | Industry-Standard
API group | Dental Care Plan Group
USCDI | No
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | May 2021
HTTP method | GET
URL template | /api/FHIR/R4/Condition?category={category}&clinical-status={clinical-status}&patient={patient}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10067

Description

The Dental Finding sub-resource returns a patient's dental findings. Dental findings are individual ailments affecting the patient and can be documented on individual teeth and surfaces. They can be documented in the hard tissue or periodontal views. This sub-resource can return findings by ID or patient.

The search interaction enables the lookup of one or more Condition resources, filtered by the specified parameters.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
category | token | Optional in Epic version August 2021 and later. Required in earlier versions. | | | Points a search to the correct sub-resource. Specify "dental-finding" for Condition. Starting in August 2021, multiple category values are allowed if the requestor is authorized to search multiple Condition categories.

clinical-status | token | True | | | The clinical status of a finding. Addressed or unaddressed.
patient | reference | False | | | Patient to search for data for.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
abatement-string | String | True | | | Unsupported
asserter | String | True | | | Unsupported
body-site | String | True | | | Location of the finding.
evidence | token | True | | | Unsupported
evidence-detail | reference | True | | | Unsupported
identifier | String | True | | | Unsupported
onset-info | string | True | | | Unsupported
severity | token | True | | | Unsupported
stage | token | True | | | Unsupported
verification-status | token | True | | | Unsupported

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | BackboneElement | False | Yes | | An array of entries included in this bundle.
Bundle.entry.resource | ResourceContainer | False | | | A container for a single FHIR resource.
Bundle.entry.resource.Condition | Condition | False | | | An instance of the R4 Condition resource.
Bundle.entry.resource.Condition.bodySite | CodeableConcept | True | Yes | | Locations of the finding including tooth, surfaces, and sites for furcation and suppuration findings.
Bundle.entry.resource.Condition.category | CodeableConcept | True | Yes | | Always "dental-finding" for this sub-resource.
Bundle.entry.resource.Condition.clinicalStatus | CodeableConcept | True | | | The clinical status of a finding. Active or resolved.
Bundle.entry.resource.Condition.code | CodeableConcept | True | | | The diagnosis code linked to a finding with coding system name.
Bundle.entry.resource.Condition.encounter | Reference | True | | | The encounter this finding was documented in.
Bundle.entry.resource.Condition.note | Annotation | True | Yes | | Note or annotation details.
Bundle.entry.resource.Condition.note.text | String | True | | | The comment associated with this finding.
Bundle.entry.resource.Condition.recordedDate | dateTime | True | | | The date the finding was first documented.
Bundle.entry.resource.Condition.recorder | Reference | True | | | The provider who documented the finding.
Bundle.entry.resource.Condition.subject | Reference | False | | | The patient or subject of this finding.

Sample request

```
https://hostname/instance/api/FHIR/R4/Condition?category=dental-finding&subject=e6DDE5.4tu-jjWXY8CtuVKw3&clinical-
    status=active,resolved
```

Sample response

```
<Bundle xmlns="http://hl7.org/fhir">
<type value="searchset" />
<total value="19" />
<link>
<relation value="self" />
<url value="https://hostname/instance/api/FHIR/R4/Condition?category=dental-finding&amp;subject=e6DDE5.4tu-
    jjWXY8CtuVKw3&amp;clinical-status=active,resolved&amp;ehiExport=1" />
</link>
<entry>
<link>
<relation value="self" />
<url value="https://hostname/instance/api/FHIR/R4/Condition/e.CYlLh27G5TiC0kZdum1Af0iUHvGK3k3zIubbaLVgpk3" />
</link>
<fullUrl value="https://hostname/instance/api/FHIR/R4/Condition/e.CYlLh27G5TiC0kZdum1Af0iUHvGK3k3zIubbaLVgpk3" />
<resource>
<Condition>
<id value="e.CYlLh27G5TiC0kZdum1Af0iUHvGK3k3zIubbaLVgpk3" />
<clinicalStatus>
<coding>
<system value="http://terminology.hl7.org/CodeSystem/condition-clinical" />
<version value="4.0.0" />
<code value="active" />
<display value="Active" />
</coding>
</clinicalStatus>
<category>
<text value="dental-finding" />
</category>
<bodySite>
<coding>
<system value="urn:oid:1.2.840.114350.1.13.5325.1.7.3.866976.94010" />
<code value="SFm" />
</coding>
<text value="Teeth (Universal): SFm" />
</bodySite>
<bodySite>
<coding>
<system value="urn:oid:1.2.840.114350.1.72.1.7.7.10.866976.94030" />
```

```
<code value="4" />
<display value="Incisal" />
</coding>
<text value="Surfaces: Incisal" />
</bodySite>
<subject>
<reference value="Patient/e6DDE5.4tu-jjWXY8CtuVKw3" />
<type value="Patient" />
<display value="Nadal, Rafael" />
</subject>
<encounter>
<reference value="Encounter/eT7iIeZGfhZP8WXwImOgZBr3R17PGztDaeBtmBq9psAI3" />
<type value="Encounter" />
</encounter>
<recordedDate value="2020-12-21T19:48:23Z" />
<recorder>
<reference value="Practitioner/eNzt.GkOzCBShLeRDkLPGfw3" />
<type value="Practitioner" />
<display value="Gordon Lofgren, DDS" />
</recorder>
<note>
<text value="Caries class" />
</note>
</Condition>
```


10.1.18 Condition.Search (Encounter Diagnosis) (R4)

File: site/apis/10-Clinical-General/1-R4/952-Condition-Search-Encounter-Diagnosis-R4.html
Epic API ID | 952
FHIR version | R4
Categories | Clinical General
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | November 2019
HTTP method | GET
URL template | api/FHIR/R4/Condition?patient={patient}&subject={subject}&category={category}&encounter={encounter}&_id={_id}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=952

Description

This web service retrieves Conditions from a patient's chart that were documented during their encounters. This includes any data found in the patient's diagnoses across all encounters. An encounter diagnosis can either be documented as an admission diagnosis, visit diagnosis, or discharge diagnosis.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
_count | number | Optional | Yes | | Number of results per page.
_id | token | Optional | Yes | | FHIR resource IDs for the desired resources. Can be used in combination with _count, _include, and _revInclude. All others will be ignored.
_include | string | Optional | Yes | | Used to indicate which referenced resources to include in the search bundle.
_revinclude | string | Optional | Yes | | Used to include Provenance or AuditEvent resources that refer to search results, when available.
category | token | Optional | Yes | | Always "encounter-diagnosis" for this API.
encounter | reference | Conditional | Yes | | Reference to an Encounter resource that the Condition is attached to. Either patient, subject, or encounter must be provided.
patient | reference | Conditional | Yes | | Reference to a Patient resource the Condition is for. Either patient, subject, or encounter must be provided.
subject | reference | Conditional | Yes | | Reference to a Patient resource the Condition is for. Either patient, subject, or encounter must be provided.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
abatement-date | date | Optional | Yes | | Supported starting in the February 2024 version of Epic. A Condition's resolved date. Only applicable to visit diagnoses with an attached problem.
clinical-status | token | Optional | Yes | | The clinical status, which can be active, resolved, or inactive. Starting in the February 2024 version of Epic, if the encounter diagnosis is linked to a problem, it uses the clinical status of the linked problem. Only visit diagnoses can be linked to a problem.
code | token | Optional | Yes | | External codings for the Condition.
onset-date | date | Optional | Yes | | Supported starting in the February 2024 version of Epic. A Condition's noted date. Only applies to visit diagnoses with an attached problem.
recorded-date | date | Optional | Yes | | The date the Condition was recorded. Only applies to visit diagnoses.
severity | token | Optional | Yes | | The priority of the problem.
verification-status | token | Optional | Yes | | Admission and discharge diagnoses have a verificationStatus of "confirmed". Other visit diagnoses can be mapped by the organization to standard FHIR values found here.

Response elements

Name | Type | Optional | Array | Cardinality | Description

Condition (R4) | Condition | | | | An instance of the R4 Condition resource.

Condition (R4).abatementDateTime | dateTime | | | 0..1 | Starting in the February 2024 version of Epic, if the Condition has a resolved date, it appears here. Only applicable to visit diagnoses with an attached problem.

Condition (R4).category | CodeableConcept | | Yes | 1..* | Always includes the category code of encounter-diagnosis. Also includes one of the following category codes: visit-diagnosis, admission-diagnosis, or discharge-diagnosis.

Condition (R4).clinicalStatus | CodeableConcept | | | 1..1 | The clinical status, which can be active, resolved, or inactive. Starting in the February 2024 version of Epic, if the encounter diagnosis is linked to a problem, it uses the clinical status of the linked problem. Only visit diagnoses can be linked to a problem.

Condition (R4).code | CodeableConcept | | | 1..1 | External codings for the Condition. If a list of codes is sent, they're listed in this element. Otherwise, this element uses the problem name from the name element. Starting in the February 2024 version of Epic or with special updates to November 2023, August 2023, or May 2023 version of Epic, each code also sends its display name and the provider-entered problem name is sent if available.

Condition (R4).encounter | Reference | | | 1..1 | Reference to an Encounter resource that the Condition is attached to.

Condition (R4).evidence | BackboneElement | | Yes | 0..1 | Supporting evidence for the Condition. This can return a linked Condition resource.

Condition (R4).evidence.detail | Reference | | Yes | 1..1 | Detailed information for the evidence.

Condition (R4).evidence.detail.display | String | | | 1..1 | The display name of the linked Condition. Returns the patient-friendly name for the problem when called in a patient context.

Condition (R4).evidence.detail.reference | String | | | 1..1 | A reference to the linked Condition.

Condition (R4).extension (assertedDateTime) (also: extension (assertedDateTime)) | Extension | | Yes | 0..1 | The date the Condition was diagnosed. Only applies to visit diagnoses with an attached problem.

Condition (R4).id | id | | | 1..1 | The Condition FHIR ID.

Condition (R4).note | Annotation | | Yes | 0..* | Free text comment.

Condition (R4).onsetDateTime | dateTime | | | 0..1 | Starting in the February 2024 version of Epic, if a Condition has a noted date, it appears here. Only applies to visit diagnoses with an attached problem.

Condition (R4).recordedDate | dateTime | | | 0..1 | The date the Condition was recorded. This property only applies to visit diagnoses. For a visit diagnosis with an attached problem, this is the entry date of the problem. For a visit diagnosis without an attached problem, this is the entry date of the visit diagnosis.

Condition (R4).recorder | Reference | | | 0..1 | Reference to the practitioner who updated the encounter diagnosis.

Condition (R4).subject | Reference | | | 1..1 | The patient associated with the Condition.

Condition (R4).verificationStatus | CodeableConcept | | | 0..1 | Admission and discharge diagnoses have a verificationStatus of "confirmed". Other visit diagnoses can be mapped by the organization to standard FHIR values found here.

Sample request

```
GET https://hostname/instance/api/FHIR/R4/Condition?patient=ezHbeFCm6Ih8kgxuHUAdvJQ3&category=encounter-diagnosis
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 4,
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Condition?category=encounter-
    diagnosis&patient=euI3cZMuLFNu6kO9mg4ScEg3"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url": "https://hostname/instance/api/FHIR/R4/Condition/eykG2p947BoIJRKQrvI9unC-
    Ov7r3qgrqibe9l1g9QyTyIFwlqzWmYi8-4D-WYCYHkG82nZ356KvwZyXzg6K84Q3"
        }
      ],
      "fullUrl": "https://hostname/instance/api/FHIR/R4/Condition/eykG2p947BoIJRKQrvI9unC-
    Ov7r3qgrqibe9l1g9QyTyIFwlqzWmYi8-4D-WYCYHkG82nZ356KvwZyXzg6K84Q3",
      "resource": {
        "resourceType": "Condition",
        "id": "eykG2p947BoIJRKQrvI9unC-Ov7r3qgrqibe9l1g9QyTyIFwlqzWmYi8-4D-WYCYHkG82nZ356KvwZyXzg6K84Q3",
        "category": [
          {
            "coding": [
              {
                "system": "http://terminology.hl7.org/CodeSystem/condition-category",
                "code": "encounter-diagnosis",
                "display": "Encounter Diagnosis"
              }
            ],
            "text": "Encounter Diagnosis"
          },
          {
            "coding": [
              {
                "system": "http://open.epic.com/FHIR/StructureDefinition/condition-category",
                "code": "visit-diagnosis",
```

```
            "display": "Visit Diagnosis"
          }
        ],
        "text": "Visit Diagnosis"
      }
    ],
    "code": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "45030009",
          "display": "Disorder Of Eye Movements (Disorder)"
        },
        {
          "system": "http://hl7.org/fhir/sid/icd-9-cm",
          "code": "378.9",
          "display": "Unspecified Disorder Of Eye Movements"
        },
        {
          "system": "http://hl7.org/fhir/sid/icd-10-cm",
          "code": "H51.9",
          "display": "Unspecified Disorder Of Binocular Movement"
        }
      ],
      "text": "Disorder Of Eye Movements"
    },
    "subject": {
      "reference": "Patient/euI3cZMuLFNu6kO9mg4ScEg3",
      "display": "Test, Patient"
    },
    "encounter": {
      "reference": "Encounter/edrYhTQC4e9jviqejxOouP40Y6gzmMFnZFGpX5rWqx2A3",
      "identifier": {
        "use": "usual",
        "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.3.698084.8",
        "value": "10003083454"
      },
      "display": "Office Visit"
    }
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Condition/eykG2p947BoIJRKQrvI9unC-
Ov7r3qgrqibe9l1g9QyTuL1-ThiNjK4aD.NfjJvwE-WRLDBJRgrYrfKi0b56Sgg3"
    }
  ],
  "fullUrl": "https://hostname/instance/api/FHIR/R4/Condition/eykG2p947BoIJRKQrvI9unC-
Ov7r3qgrqibe9l1g9QyTuL1-ThiNjK4aD.NfjJvwE-WRLDBJRgrYrfKi0b56Sgg3",
  "resource": {
    "resourceType": "Condition",
    "id": "eykG2p947BoIJRKQrvI9unC-Ov7r3qgrqibe9l1g9QyTuL1-ThiNjK4aD.NfjJvwE-WRLDBJRgrYrfKi0b56Sgg3",
    "category": [
      {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/condition-category",
            "code": "encounter-diagnosis",
            "display": "Encounter Diagnosis"
          }
        ],
        "text": "Encounter Diagnosis"
      },
      {
        "coding": [
          {
            "system": "http://open.epic.com/FHIR/StructureDefinition/condition-category",
            "code": "visit-diagnosis",
            "display": "Visit Diagnosis"
          }
        ],
        "text": "Visit Diagnosis"
      }
    ],
    "code": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "102614006",
          "display": "Generalized Abdominal Pain (Finding)"
        },
        {
          "system": "http://hl7.org/fhir/sid/icd-9-cm",
          "code": "789.07",
```

```
          "display": "Abdominal Pain, Generalized"
        },
        {
          "system": "http://hl7.org/fhir/sid/icd-10-cm",
          "code": "R10.84",
          "display": "Generalized Abdominal Pain"
        }
      ],
      "text": "Generalized Abdominal Pain"
    },
    "subject": {
      "reference": "Patient/euI3cZMuLFNu6kO9mg4ScEg3",
      "display": "Test, Patient"
    },
    "encounter": {
      "reference": "Encounter/edrYhTQC4e9jviqejxOouP40Y6gzmMFnZFGpX5rWqx2A3",
      "identifier": {
        "use": "usual",
        "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.3.698084.8",
        "value": "10003083454"
      },
      "display": "Office Visit"
    }
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Condition/f.YSLhz35JcBV-O-liCozZyIaZdi6pMzLrGQRYc5mCyw4"
    }
  ],
  "fullUrl": "https://hostname/instance/api/FHIR/R4/Condition/f.YSLhz35JcBV-O-liCozZyIaZdi6pMzLrGQRYc5mCyw4",
  "resource": {
    "resourceType": "Condition",
    "id": "f.YSLhz35JcBV-O-liCozZyIaZdi6pMzLrGQRYc5mCyw4",
    "verificationStatus": {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/condition-ver-status",
          "version": "4.0.0",
          "code": "confirmed",
          "display": "Confirmed"
        }
      ],
      "text": "Confirmed"
    },
    "category": [
      {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/condition-category",
            "code": "encounter-diagnosis",
            "display": "Encounter Diagnosis"
          }
        ],
        "text": "Encounter Diagnosis"
      },
      {
        "coding": [
          {
            "system": "http://open.epic.com/FHIR/StructureDefinition/condition-category",
            "code": "admission-diagnosis",
            "display": "Admission Diagnosis"
          }
        ],
        "text": "Admission Diagnosis"
      }
    ],
    "code": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "271681002",
          "display": "Stomach ache (finding)"
        },
        {
          "system": "http://hl7.org/fhir/sid/icd-9-cm",
          "code": "536.8",
          "display": "Dyspepsia and other specified disorders of function of stomach"
        },
        {
          "system": "http://hl7.org/fhir/sid/icd-10-cm",
          "code": "R10.9",
          "display": "Unspecified abdominal pain"
        }
```

```
      ],
      "text": "Stomach pain"
    },
    "subject": {
      "reference": "Patient/euI3cZMuLFNu6kO9mg4ScEg3",
      "display": "Test, Patient"
    },
    "encounter": {
      "reference": "Encounter/ewL9gXw2nLic-gSUTPRfL9U-WyuJkq.LAN7LvME2KEkk3",
      "identifier": {
        "use": "usual",
        "system": "urn:oid:1.2.840.114350.1.13.861.1.7.3.698084.8",
        "value": "10011719379"
      },
      "display": "Hospital Encounter"
    }
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Condition/fompkdkOEJtZgyRUM-qkcIT.LwGWez9oqjuYu7qPVcTU4"
    }
  ],
  "fullUrl": "https://hostname/instance/api/FHIR/R4/Condition/fompkdkOEJtZgyRUM-qkcIT.LwGWez9oqjuYu7qPVcTU4",
  "resource": {
    "resourceType": "Condition",
    "id": "fompkdkOEJtZgyRUM-qkcIT.LwGWez9oqjuYu7qPVcTU4",
    "verificationStatus": {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/condition-ver-status",
          "version": "4.0.0",
          "code": "confirmed",
          "display": "Confirmed"
        }
      ],
      "text": "Confirmed"
    },
    "category": [
      {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/condition-category",
            "code": "encounter-diagnosis",
            "display": "Encounter Diagnosis"
          }
        ],
        "text": "Encounter Diagnosis"
      },
      {
        "coding": [
          {
            "system": "http://open.epic.com/FHIR/StructureDefinition/condition-category",
            "code": "discharge-diagnosis",
            "display": "Discharge Diagnosis"
          }
        ],
        "text": "Discharge Diagnosis"
      }
    ],
    "code": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "22298006",
          "display": "Myocardial infarction (disorder)"
        },
        {
          "system": "http://hl7.org/fhir/sid/icd-9-cm",
          "code": "410.90",
          "display": "Acute myocardial infarction, unspecified site, episode of care unspecified"
        },
        {
          "system": "http://hl7.org/fhir/sid/icd-10-cm",
          "code": "I21.9",
          "display": "Acute myocardial infarction, unspecified (HCC)"
        }
      ],
      "text": "Heart attack (HCC)"
    },
    "subject": {
      "reference": "Patient/euI3cZMuLFNu6kO9mg4ScEg3",
      "display": "Test, Patient"
    },
```

```
        "encounter": {
          "reference": "Encounter/eoT.mCCK.MRBHt4eu8i1TR4U4YAfY2ffV2A1EBuPHU.Q3",
          "identifier": {
            "use": "usual",
            "system": "urn:oid:1.2.840.114350.1.13.861.1.7.3.698084.8",
            "value": "10011574519"
          },
          "display": "Hospital Encounter"
        }
      },
      "search": {
        "mode": "match"
      }
    }
  ]
}
```


Change log

Epic Version | Description | Type
February 2024 | The USCDIv3 specification for Condition will move the Condition's start date and time from onsetPeriod to onsetDatetime and end date and time from abatementPeriod to abatementDatetime fields. This could impact the evaluation of a Condition if a Condition consumer is not prepared for the transition. | Breaking change

10.1.19 Condition.Search (Genomics) (R4)

File: site/apis/10-Clinical-General/1-R4/1075-Condition-Search-Genomics-R4.html
Epic API ID | 1075
FHIR version | R4
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | November 2020
HTTP method | GET
URL template | api/FHIR/R4/Condition?category={category}&patient={patient}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=1075

Description

This web service retrieves a genomic indicator from a patient's chart. Genomic indicators represent the implications of genetic lab results for a given patient. For example, a genomic indicator, created based on the presence of several variants in the patient's genetic test results, might indicate a sensitivity to the drug warfarin. The system might then show decision support information to clinicians prescribing warfarin, triggered by the genomic indicator in the patient's chart, so that they are aware of additional considerations for specific medication.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
category | token | Conditional | Yes | 0..* | Always "genomics" for this API.
patient | reference | Conditional | Yes | 0..1 | Reference to a patient resource the indicator is for.
subject | reference | Conditional | Yes | 0..1 | Reference to a patient resource the indicator is for.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
evidence-detail | reference | Optional | Yes | 0..* | Reference to an Observation (Labs) or DiagnosticReport (Results) resource associated with the target condition.
recorded-date | date | Optional | Yes | 0..1 | The date the indicator was recorded.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Condition (R4) | Condition | | | | An instance of the R4 Condition resource.
Condition (R4).category | CodeableConcept | | Yes | 1..* | Always "genomics" for patient genomic indicators.
Condition (R4).code | CodeableConcept | | | 0..* | LOINC and SNOMED codes representative of the genomic indicator.
Condition (R4).evidence | BackboneElement | | Yes | 0..* | The Observation (Labs) or DiagnosticReport (Results) resource that indicated the presence of this Condition.
Condition (R4).evidence.detail | Reference | | Yes | 0..* | Reference to an Observation (Labs) or DiagnosticReport (Results) resource that indicated the presence of this Condition.
Condition (R4).id | id | | | 1..1 | The indicator FHIR ID.
Condition (R4).recordedDate | dateTime | | | 1..1 | The date the indicator was recorded.
Condition (R4).subject | Reference | | | 1..1 | The patient associated with the indicator.

Sample request

```
GET https://hostname/instance/api/FHIR/R4/Condition?patient=eOrFj.KdMJPmguszyMsRIjQ3&category=genomics
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 1,
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Condition?category=genomics&patient=eOrFj.KdMJPmguszyMsRIjQ3"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url": "https://hostname/instance/api/FHIR/R4/Condition/eV82zCR467Wt0UhpfS8mwXg3"
        }
      ],
      "fullUrl": "https://hostname/instance/api/FHIR/R4/Condition/eV82zCR467Wt0UhpfS8mwXg3",
      "resource": {
        "resourceType": "Condition",
        "id": "eV82zCR467Wt0UhpfS8mwXg3",
        "category": [
          {
            "coding": [
              {
                "system": "https://open.epic.com/FHIR/StructureDefinition/condition-category",
                "code": "genomics",
                "display": "Genomic Indicators"
              }
            ],
            "text": "Genomic Indicators"
          }
        ],
        "code": {
          "text": "Lynch Syndrome"
        },
        "subject": {
          "reference": "Patient/eOrFj.KdMJPmguszyMsRIjQ3",
          "display": "Bellsandwhistles, Sheila"
        },
        "recordedDate": "2024-03-19T17:12:10Z",
        "evidence": [
          {
            "detail": [
              {
                "reference": "DiagnosticReport/e3MlX5a0JkepBJvp4Lhf0W9Upm5kqXcOYE9ajagL.vGM3",
                "display": "TEMPUS XT"
              }
            ]
          }
        ]
      },
      "search": {
        "mode": "match"
      }
    }
  ]
}
```


10.1.20 Condition.Search (Health Concerns) (R4)

File: site/apis/10-Clinical-General/1-R4/985-Condition-Search-Health-Concerns-R4.html
Epic API ID | 985
FHIR version | R4
Categories | Clinical General
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Keywords | Health Concerns
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | February 2020
HTTP method | GET
URL template | api/FHIR/R4/Condition?patient={patient}&subject={subject}&category={category}&clinical-status={clinical-status}&_id={_id}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients

Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=985

Description

This API provides access to two data elements:

* The subset of problems on the problem list that have been marked as health concerns.

* The health status of the patient.

Health Concerns

Problems on the problem list may be marked with a class of "Health Concern," indicating that the problem is of special concern.

Marking a problem on the problem list as a health concern may not be a common workflow at many health systems. Prior to using this API, work with clinical teams to verify that the ability to mark problems as health concerns is enabled, and that end users are trained to flag problems as concerns.

Health Status

Documenting a general, specialty-independent health status for a patient may not be a common workflow at many health systems. Prior to using this API, it will be essential to work closely with clinical teams to verify that that the ability to capture health status is enabled, and that end users are trained to evaluate and document the health status.

Because the health status is a single assessment, many Condition elements do not apply, thus Condition resources that represent health status include a smaller subset of elements than health concerns. For health status, severity, onsetPeriod, abatementPeriod, and note elements are not populated.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
_count | number | Optional | Yes | | Number of results per page.
_id | token | Optional | Yes | | FHIR resource IDs for the desired resources. Can be used in combination with _count, _include, and _revInclude. All others will be ignored.
_include | string | Optional | Yes | | Used to indicate which referenced resources to include in the search bundle.
_revinclude | string | Optional | Yes | | Used to include Provenance or AuditEvent resources that refer to search results, when available.
category | token | Optional | Yes | | Always health-concern for the Condition.Search (Health Concern) API. Starting in the November 2024 version of Epic, the problem-list-item category can also be used to return health concerns on the patient's problem list.
clinical-status | token | Optional | Yes | | The current status of the health concern or health status. Health Concerns The following Condition Clinical Status Codes are supported: active - active conditions resolved - resolved conditions inactive - deleted conditions Health Status Note that health status Conditions will always have a status of 'active'.
patient | reference | Conditional | Yes | | Reference to a patient resource the condition is for. Either a patient or subject parameter must be provided. If both are provided, they must match exactly.
subject | reference | Conditional | Yes | | Reference to a patient resource the condition is for. Either a patient or subject parameter must be provided. If both are provided, they must match exactly.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
abatement-date | date | Optional | Yes | | The date the health concern was resolved. Health Concerns Health concerns can have an abatement date that you can use this search parameter to filter on. Health Status The patient's Health Status will not have an abatement date, so searches that filter on abatement-date will exclude Health Status Conditions.
code | token | Optional | Yes | | The code that identifies the specific type of problem. Health Concerns Examples of code sytems that are often used to code problem list items include ICD-10, SNOMED CT, ICPC Basic Health Care Classification, and Intelligent Medical Objects ProblemIT and others. Health Status Health Status will be coded using SNOMED CT.
onset-date | date | Optional | Yes | | The date the problem was noted.
recorded-date | date | Optional | Yes | | The date the condition was recorded.
severity | token | Optional | Yes | | The priority of the problem.
stage | token | Optional | Yes | | For cancer-related health concerns, the stage or grade of the condition.
verification-status | token | Optional | Yes | | The verification status of the health concern or health status. For this API, all Conditions will have a verificationStatus of 'confirmed'.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Condition (R4) | Condition | | | | An instance of the R4 Condition resource.
Condition (R4).abatementDateTime | dateTime | | | 0..1 | Starting in the February 2024 version of Epic, this is the date the health concern was resolved. For versions prior to February 2024, this field is always empty, and the abatement date can be found in abatementPeriod instead. Health status conditions will not have an abatement date.
Condition (R4).abatementPeriod | Period | | | 0..1 | In the November 2023 version of Epic and earlier, this is the date the health concern was resolved. For versions after November 2023, this field is always empty, and the abatement date can be found in abatementDateTime instead. Health status conditions will not have an abatement date.
Condition (R4).category | CodeableConcept | | Yes | 1..* | Health Concerns Always includes category code "problem-list-item". If a problem list item is marked with a class of "Health Concern", indicating that the problem is of special concern, a category code of "health-concern" will also be included. This can also include additional category codes, such as 'sdoh' for problems related to social determinates of health. This category depends on configuration of diagnosis groupers defined by the Epic organization. Health Status Always includes category code "health-concern".

Condition (R4).clinicalStatus | CodeableConcept | | | 1..1 | The current status of the health concern or health status. Health Concerns The following Condition Clinical Status Codes are supported: FHIR Status Epic Status active active conditions resolved resolved conditions inactive deleted conditions Health Status Health status Conditions will always have a status of 'active'.

FHIR Status | Epic Status

active | active conditions

resolved | resolved conditions

inactive | deleted conditions

Condition (R4).code | CodeableConcept | | | 1..1 | The code that identifies the specific type of problem. Health Concerns Examples of code sytems that are often used to code problem list items include ICD-10, SNOMED, ICPC Basic Health Care Classification, and Intelligent Medical Objects ProblemIT and others. If there are codes available to send, they're included in this element. If there are no codes to send, this element will include the un-coded, free text problem name. Starting in the February 2024 version of Epic, each code also sends its display name and the provider-entered problem name is sent if available. Health Status Health Status will be coded using SNOMED CT.

Condition (R4).id | id | | | 1..1 | The condition FHIR ID.

Condition (R4).meta | Meta | | | | Metadata about the resource.

Condition (R4).meta.lastUpdated | String | | | 0..1 | The instant the health concern was last updated.

Condition (R4).note | Annotation | | Yes | 0..* | Free text comments for the note. Health Concerns Can be an Overview note or, starting in the February 2025 version of Epic, an Assessment & Plan note. Health Status Notes are not supported for Health Status

Condition (R4).note.authorReference | Reference | | | | A reference to the Practitioner resource for the user who wrote the note.

Condition (R4).note.extension (annotationType) (also: extension (annotationType)) | Extension | | Yes | | For Assessment & Plan notes, this is always 51847-2. For Overview notes, this is always 68608-9.

Condition (R4).note.extension (data-conversion-warning) (also: extension (data-conversion-warning)) | Extension | | Yes | | The following warnings may be returned if the note content is converted from RTF to plain text. Reason for warning Warning text Formatting was lost due to unicode characters or tables being present in the original note. Formatting of this note is different from the original The original note was RTF. Formatting of this note might be different from the original There was an image in the original note Images from the original note were not included URL: http://open.epic.com/FHIR/StructureDefinition/extension/data-conversion-warning

Reason for warning | Warning text

Formatting was lost due to unicode characters or tables being present in the original note. | Formatting of this note is different from the original

The original note was RTF. | Formatting of this note might be different from the original

There was an image in the original note | Images from the original note were not included

Condition (R4).note.time | String | | | | The time that the note was signed.

Condition (R4).onsetDateTime | dateTime | | | 0..1 | Health Concerns Starting in the February 2024 version of Epic, if a condition has a noted date, it appears here. Prior to the February 2024 version of Epic, the condition noted date is sent in onsetPeriod. Health Status The instant the health status was recorded.

Condition (R4).onsetPeriod | Period | | | 0..1 | In the November 2023 version of Epic and earlier, the date the problem was noted. Starting in the Febuary 2025 version of Epic, the condition noted date is sent in onsetDateTime.

Condition (R4).recordedDate | dateTime | | | 0..1 | The date the condition was recorded.

Condition (R4).recorder | Reference | | | 0..1 | Reference to the practitioner who updated the Health Concern or Health Status.

Condition (R4).severity | CodeableConcept | | | 0..1 | Health Concern The default priority of the problem. Note that Epic support two types of priorities for problems: A default priority that is specified when the problem is initially created. This value is available in the Condition.severity property. A user-specific priority that users can adjust based on the relevance of the problem to their specialty. For example, a dermatologist might rate eczema as a high priority while a cardiologist would rate it as low. User-level priorities allow each clinician to see the list organized in a way that is most useful to them. This value is displayed to users Epic's problem list activity. This value is not supported in the FHIR API. Health Status Health Status conditions will not have a severity.

Condition (R4).stage | BackboneElement | | Yes | | For problems that represent cancer, staging information can be documented as part of the problem. Cancer staging usually involves the identification of the location, size, and the extent of metastasis of the tumor.

Condition (R4).stage.assessment | Reference | | Yes | | SmartData Element associated with cancer staging for this condition.

Condition (R4).subject | Reference | | | 1..1 | The patient associated with the condition.

Condition (R4).verificationStatus | CodeableConcept | | | 1..1 | Health Concerns Always a status of 'confirmed' Health Status Always a status of 'confirmed'.

Sample request

```
GET https://hostname/instance/api/FHIR/R4/Condition?patient=eLFQugzW.CQz3w48A427N.w3&category=health-concern
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 2,
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Condition?category=health-concern&patient=eLFQugzW.CQz3w48A427N.w3"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url": "https://hostname/instance/api/FHIR/R4/Condition/eFxu5PSp73K5YGIXo-XTsBwkIv7ppHTBXdmKE39VDRYo3"
        }
      ],
      "fullUrl": "https://hostname/instance/api/FHIR/R4/Condition/eFxu5PSp73K5YGIXo-XTsBwkIv7ppHTBXdmKE39VDRYo3",
      "resource": {
        "resourceType": "Condition",
        "id": "eFxu5PSp73K5YGIXo-XTsBwkIv7ppHTBXdmKE39VDRYo3",
        "extension": [
          {
```

```
        "valueDateTime": "2025-04-15",
        "url": "http://hl7.org/fhir/StructureDefinition/condition-assertedDate"
      },
      {
        "valueReference": {
          "reference": "Observation/fXGm1sOStdoN2VJpSgEN6LA4",
          "display": "DIAGNOSES/PROBLEMS - ONCOLOGY - CANCER - CANCER DISEASE STATUS"
        },
        "url": "http://open.epic.com/FHIR/StructureDefinition/extension/smartdata"
      },
      {
        "valueReference": {
          "reference": "Observation/fsH0LqfEMBF4U2cVUrlJcrg4",
          "display": "DIAGNOSES/PROBLEMS - ONCOLOGY - CANCER - CANCER DISEASE STATUS - REASON FOR CANCER DISEASE
STATUS"
        },
        "url": "http://open.epic.com/FHIR/StructureDefinition/extension/smartdata"
      },
      {
        "valueReference": {
          "reference": "Observation/fDgxSro-tBmldqmMEnrYqkg4",
          "display": "WORKFLOW - ONCOLOGY - EOM EPISODES START DATES - EOM CANCER DISEASE STATUS"
        },
        "url": "http://open.epic.com/FHIR/StructureDefinition/extension/smartdata"
      }
    ],
    "clinicalStatus": {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/condition-clinical",
          "version": "4.0.0",
          "code": "active",
          "display": "Active"
        }
      ],
      "text": "Active"
    },
    "verificationStatus": {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/condition-ver-status",
          "version": "4.0.0",
          "code": "confirmed",
          "display": "Confirmed"
        }
      ],
      "text": "Confirmed"
    },
    "category": [
      {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/condition-category",
            "code": "problem-list-item",
            "display": "Problem List Item"
          }
        ],
        "text": "Problem List Item"
      },
      {
        "coding": [
          {
            "system": "http://loinc.org",
            "code": "75310-3",
            "display": "Health concerns Document"
          },
          {
            "system": "http://hl7.org/fhir/us/core/CodeSystem/condition-category",
            "code": "health-concern",
            "display": "Health Concern"
          }
        ],
        "text": "Health Concern"
      },
      {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "64572001"
          }
        ]
      }
    ],
    "severity": {
      "coding": [
        {
          "system": "urn:oid:2.16.840.1.113883.6.96",
          "code": "24484000",
          "display": "Severe"
```

```
        }
      ],
      "text": "High"
    },
    "code": {
      "coding": [
        {
          "system": "http://hl7.org/fhir/sid/icd-10-cm",
          "code": "C34.11",
          "display": "Malignant neoplasm of upper lobe, right bronchus or lung (HCC)"
        },
        {
          "system": "http://snomed.info/sct",
          "code": "724060008",
          "display": "Malignant Neoplasm Of Right Upper Lobe Of Lung"
        },
        {
          "system": "http://hl7.org/fhir/sid/icd-9-cm",
          "code": "162.3",
          "display": "Malignant neoplasm of upper lobe of right lung (HCC)"
        },
        {
          "system": "urn:oid:2.16.840.1.113883.3.247.1.1",
          "code": "5726649",
          "display": "Malignant neoplasm of upper lobe of right lung (HCC)"
        }
      ],
      "text": "Malignant neoplasm of upper lobe of right lung (HCC)"
    },
    "subject": {
      "reference": "Patient/eLFQugzW.CQz3w48A427N.w3",
      "display": "zzFhir, Fhirboy"
    },
    "onsetDateTime": "2025-04-15",
    "recordedDate": "2025-04-15"
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Condition/eXg3HgZvtqpIEwXth.XAo8A3"
    }
  ],
  "fullUrl": "https://hostname/instance/api/FHIR/R4/Condition/eXg3HgZvtqpIEwXth.XAo8A3",
  "resource": {
    "resourceType": "Condition",
    "id": "eXg3HgZvtqpIEwXth.XAo8A3",
    "clinicalStatus": {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/condition-clinical",
          "version": "4.0.0",
          "code": "active",
          "display": "Active"
        }
      ],
      "text": "Active"
    },
    "verificationStatus": {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/condition-ver-status",
          "version": "4.0.0",
          "code": "confirmed",
          "display": "Confirmed"
        }
      ],
      "text": "Confirmed"
    },
    "category": [
      {
        "coding": [
          {
            "system": "http://loinc.org",
            "code": "75310-3",
            "display": "Health concerns Document"
          },
          {
            "system": "http://hl7.org/fhir/us/core/CodeSystem/condition-category",
            "code": "health-concern",
            "display": "Health Concern"
          }
        ],
        "text": "Health Concern"
      }
```

```
        ],
        "code": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "161901003",
              "display": "Chronic Sick"
            }
          ],
          "text": "Chronically ill"
        },
        "subject": {
          "reference": "Patient/eLFQugzW.CQz3w48A427N.w3",
          "display": "Patient, Test"
        },
        "onsetDateTime": "2025-04-16T17:00:00Z",
        "recordedDate": "2025-04-16T17:43:55Z",
        "recorder": {
          "reference": "Practitioner/eodo7Bj9dk-LtJSthMwARpw3",
          "type": "Practitioner",
          "display": "Test Provider"
        }
      },
      "search": {
        "mode": "match"
      }
    }
  ]
}
```


Change log

Epic Version | Description | Type

May 2025 | Previously data loss warnings in the conversion of notes were prepended to the note content (the note.text element in the FHIR resource). Now this warning is returned in a note.extension element. | Breaking change

February 2024 | The USCDIv3 specification for Condition will move the Condition's start date and time from onsetPeriod to onsetDatetime and end date and time from abatementPeriod to abatementDatetime fields. This could impact the evaluation of a Condition if a Condition consumer is not prepared for the transition. | Breaking change

November 2022 | Code systems for the following code sets now return a URL in place of an OID: ActCode, ICD-10-CM, ICD-10-CA, ICD-10-NL, SNOMED System URL OID ActCode STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode 2.16.840.1.113883.5.4 ICD-10-CM http://hl7.org/fhir/sid/icd-10-cm 2.16.840.1.113883.6.90 ICD-10-CA http://hl7.org/fhir/sid/icd-10-ca 2.16.840.1.113883.6.94 ICD-10-NL http://hl7.org/fhir/sid/icd-10-nl 2.16.840.1.113883.6.3.2 SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 | Breaking change

System | URL | OID

ActCode | STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode | 2.16.840.1.113883.5.4

ICD-10-CM | http://hl7.org/fhir/sid/icd-10-cm | 2.16.840.1.113883.6.90

ICD-10-CA | http://hl7.org/fhir/sid/icd-10-ca | 2.16.840.1.113883.6.94

ICD-10-NL | http://hl7.org/fhir/sid/icd-10-nl | 2.16.840.1.113883.6.3.2

SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96

10.1.21 Condition.Search (Infection) (R4)

File: site/apis/10-Clinical-General/1-R4/10141-Condition-Search-Infection-R4.html

Epic API ID | 10141

FHIR version | R4

Categories | Clinical General

API type | Industry-Standard

USCDI | No

Available in inpatient and other contexts | Yes

ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base

ONC certification versions | August 2021

HTTP method | GET;POST

URL template | /api/FHIR/R4/Condition?abatement-date={abatement-date}&category={category}&clinical-status={clinical-status}&code={code}&encounter={encounter}&onset-date={onset-date}&patient={patient}&recorded-date={recorded-date}&subject={subject}

Supports OAuth 2.0 | True

Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients

Performs scope validation | False

Source | https://fhir.epic.com/Specifications?api=10141

Description

Staphylococcus aureus

Request parameters

Name | Type | Optional | Array | Cardinality | Description

abatement-date | date | True | | | The date the infection was resolved.

category | token | True | | | Always "infection" for this API. Starting in August 2021, multiple category values are allowed if the requestor is authorized to search multiple Condition categories.

clinical-status | token | True | | | Status of the infection. Infections can have a status of “active” or “inactive.” If no status is specified, all infections are returned.
code | token | True | | | SNOMED code associated with the infection.
encounter | reference | Either patient, subject, or encounter must be specified. | | | Reference to an Encounter resource associated with an infection. If no encounter is specified, all patient- and encounter-level infections are returned.
onset-date | date | True | | | The date the patient acquired the infection.
patient | reference | Either patient, subject, or encounter must be specified. | | | Reference to a patient resource the infection is for.
recorded-date | date | True | | | The date the infection was added to the patient’s chart.
subject | reference | Either patient, subject, or encounter must be specified. | | | Reference to a patient resource the infection is for.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
abatement-string | String | True | | | The date and time the infection was resolved.
asserter | String | True | | | Unsupported
body-site | String | True | | | Unsupported
evidence | token | True | | | Unsupported
evidence-detail | reference | True | | | Unsupported
identifier | String | True | | | Unsupported
onset-info | string | True | | | The date the patient acquired the infection.
severity | token | True | | | Unsupported
stage | token | True | | | Unsupported
verification-status | token | True | | | Unsupported

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | BackboneElement | False | Yes | | An array of entries included in this bundle.
Bundle.entry.resource | ResourceContainer | False | | | A container for a single FHIR resource.
Bundle.entry.resource.Condition | Condition | False | | | An instance of the R4 Condition resource.
Bundle.entry.resource.Condition.abatementDateTime | dateTime | True | | | The date and time the infection was resolved.
Bundle.entry.resource.Condition.category | CodeableConcept | False | Yes | | Always “infection”
Bundle.entry.resource.Condition.clinicalStatus | CodeableConcept | False | | | Always either “active” or “inactive”
Bundle.entry.resource.Condition.code | CodeableConcept | False | | | The display name for the infection and the SNOMED code if one exists.
Bundle.entry.resource.Condition.encounter | Reference | True | | | The encounter associated with the infection. Only applies to encounter-level infections.
Bundle.entry.resource.Condition.id | id | False | | | The FHIR ID
Bundle.entry.resource.Condition.note | Annotation | True | Yes | | Free-text comments
Bundle.entry.resource.Condition.onsetDateTime | dateTime | True | | | The date the patient acquired the infection
Bundle.entry.resource.Condition.recordedDate | dateTime | False | | | The date and time the infection was added to the patient’s chart
Bundle.entry.resource.Condition.subject | Reference | False | | | The patient associated with the infection.

Sample request

```
https://hostname/instance/api/FHIR/R4/Condition?patient=eC4MDFhhJAHRxZCUXt6ozHA3&category=infection
```

Sample response

```
<Bundle xmlns="http://hl7.org/fhir">
  <type value="searchset"/>
  <total value="1"/>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/R4/Condition?category=infection&amp;patient=eC4MDFhhJAHRxZCUXt6ozHA3&
    amp;code=9"/>
  </link>
  <entry>
    <link>
      <relation value="self"/>
      <url value="https://hostname/instance/api/FHIR/R4/Condition/ePMnpp6lhpta6YiS9-QsmBA3"/>
    </link>
    <fullUrl value="https://hostname/instance/api/FHIR/R4/Condition/ePMnpp6lhpta6YiS9-QsmBA3"/>
    <resource>
      <Condition>
        <id value="ePMnpp6lhpta6YiS9-QsmBA3"/>
        <clinicalStatus>
          <coding>
            <system value="http://terminology.hl7.org/CodeSystem/condition-clinical"/>
            <version value="4.0.0"/>
            <code value="active"/>
            <display value="Active"/>
          </coding>
          <text value="Active"/>
        </clinicalStatus>
        <category>
          <coding>
            <system value="http://open.epic.com/FHIR/StructureDefinition/condition-category"/>
            <code value="infection"/>
            <display value="Infection"/>
```

```
          </coding>
          <text value="Infection"/>
        </category>
        <code>
          <coding>
            <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.4.698084.10375"/>
            <code value="9"/>
            <display value="Measles"/>
          </coding>
          <coding>
            <system value="http://snomed.info/sct"/>
            <code value="14189004"/>
            <display value="Measles"/>
          </coding>
          <text value="Measles"/>
        </code>
        <subject>
          <reference value="Patient/eC4MDFhhJAHRxZCUXt6ozHA3"/>
          <display value="Dunwoody, Terence"/>
        </subject>
        <onsetDateTime value="2021-02-12"/>
        <recordedDate value="2021-02-19T20:39:13Z"/>
      </Condition>
    </resource>
    <search>
      <mode value="match"/>
    </search>
  </entry>
</Bundle>
```


10.1.22 Condition.Search (Medical History) (R4)

File: site/apis/10-Clinical-General/1-R4/10314-Condition-Search-Medical-History-R4.html
Epic API ID | 10314
FHIR version | R4
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Condition?category={category}&patient={patient}&subject={subject}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10314

Description

Request parameters

Name | Type | Optional | Array | Cardinality | Description
category | token | True | | | Always “medical-history” for this API. Multiple category values are allowed if the requestor is authorized to search multiple Condition categories.
patient | reference | Either patient or subject must be specified. If both are specified, they must reference the same patient. | | | The Patient FHIR resource ID.
subject | reference | Either patient or subject must be specified. If both are specified, they must reference the same patient. | | | Subject of the condition.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
abatement-date | date | True | | | Unsupported
abatement-string | String | True | | | Unsupported
asserter | String | True | | | Unsupported
body-site | String | True | | | Unsupported
clinical-status | token | True | | | Unsupported
code | token | True | | | ICD code or SNOMED concept
encounter | reference | True | | | Unsupported
evidence | token | True | | | Unsupported
evidence-detail | reference | True | | | Unsupported
identifier | String | True | | | Unsupported
onset-date | date | True | | | The date when the medical problem began for the patient. Only applies to positive medical history.
onset-info | string | True | | | The date when the medical problem began for the patient. Only applies to positive medical history. Applicable for dates saved as a string instead of a date format. Otherwise, use onset-date.
recorded-date | date | True | | | For pertinent negative medical history, the date when the pertinent negative was noted on the patient’s chart.
severity | token | True | | | Unsupported
stage | token | True | | | Unsupported
verification-status | token | True | | | Use “confirmed” for positive medical history and “refuted” for pertinent negative medical history.

Response elements

Name | Type | Optional | Array | Cardinality | Description

Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.

Bundle.entry | BackboneElement | False | Yes | | An array of entries included in this bundle.

Bundle.entry.resource | ResourceContainer | False | | | A container for a single FHIR resource.

Bundle.entry.resource.Condition | Condition | False | | | An instance of the R4 Condition resource.

Bundle.entry.resource.Condition.category | CodeableConcept | True | Yes | | Always “medical-history” for patient medical history.

Bundle.entry.resource.Condition.code | CodeableConcept | False | | | Array of ICD codes and SNOMED concepts.

Bundle.entry.resource.Condition.note | Annotation | True | Yes | | The comment and display name for positive medical history, or the comment for pertinent negative medical history. Not shown in MyChart context.

Bundle.entry.resource.Condition.onsetDateTime | dateTime | True | | | The date when the medical problem began for the patient. Only applies to positive medical history.

Bundle.entry.resource.Condition.onsetPeriod | Period | True | | | The date range when the medical problem began for the patient. Only applies to positive medical history. Applicable for dates saved as fuzzy dates.

Bundle.entry.resource.Condition.onsetString | String | True | | | The date when the medical problem began for the patient. Only applies to positive medical history. Applicable for dates saved as a string instead of a date format.

Bundle.entry.resource.Condition.recordedDate | dateTime | True | | | The date when the pertinent negative was noted on the patient’s chart. Only applies to pertinent negative medical history.

Bundle.entry.resource.Condition.subject | Reference | False | | | The patient ID.

Bundle.entry.resource.Condition.verificationStatus | CodeableConcept | True | | | Always “confirmed” for positive medical history and “refuted” for pertinent negative medical history.

Sample request

```
https://hostname/instance/api/FHIR/R4/Condition?patient=ebb8tmJBI1FH-fdAzdqISZg3&category=medical-history
```

Sample response

```
<Bundle xmlns="http://hl7.org/fhir">
    <type value="searchset" />
    <total value="2" />
    <link>
        <relation value="self" />
        <url value="https://hostname/instance/api/FHIR/R4/Condition?patient=ebb8tmJBI1FH-fdAzdqISZg3&category=medical-
    history" />
    </link>
    <entry>
        <link>
            <relation value="self" />
            <url value="https://hostname/instance/api/FHIR/R4/Condition/esG9.LwMjNQmdDHbomjHegUxZzQ28jhTpnrZtIiXv35A3"
    />
        </link>
        <fullUrl value="https://hostname/instance/api/FHIR/R4/Condition/esG9.LwMjNQmdDHbomjHegUxZzQ28jhTpnrZtIiXv35A3"
    />
        <resource>
            <Condition>
                <id value="esG9.LwMjNQmdDHbomjHegUxZzQ28jhTpnrZtIiXv35A3" />
                <verificationStatus>
                    <coding>
                        <system value="http://terminology.hl7.org/CodeSystem/condition-ver-status" />
                        <version value="4.0.0" />
                        <code value="confirmed" />
                        <display value="Confirmed" />
                    </coding>
                    <text value="Confirmed" />
                </verificationStatus>
                <category>
                    <coding>
                        <system value="http://open.epic.com/FHIR/StructureDefinition/condition-category" />
                        <code value="medical-history" />
                        <display value="Medical History" />
                    </coding>
                    <text value="Medical History" />
                </category>
                <code>
                    <coding>
                        <system value="http://snomed.info/sct" />
                        <code value="195967001" />
                        <display value="Asthma (disorder)" />
                    </coding>
                    <coding>
                        <system value="http://hl7.org/fhir/sid/icd-9-cm" />
                        <code value="493.90" />
                        <display value="Asthma" />
                    </coding>
                    <coding>
                        <system value="urn:oid:2.16.840.1.113883.6.90" />
                        <code value="J45.909" />
                        <display value="Asthma" />
                    </coding>
                    <text value="Asthma" />
                </code>
                <subject>
                    <reference value="Patient/ebb8tmJBI1FH-fdAzdqISZg3" />
```

```
                    <display value="Stanley, James" />
                </subject>
                <onsetPeriod>
                    <start value="1991-01-01" />
                    <end value="1991-12-31" />
                </onsetPeriod>
            </Condition>
        </resource>
        <search>
            <mode value="match" />
        </search>
    </entry>
    <entry>
        <link>
            <relation value="self" />
            <url value="https://hostname/instance/api/FHIR/R4/Condition/esG9.LwMjNQmdDHbomjHegR42AImiOPW2PbQW4A0r2CU3"
    />
        </link>
        <fullUrl value="https://hostname/instance/api/FHIR/R4/Condition/esG9.LwMjNQmdDHbomjHegR42AImiOPW2PbQW4A0r2CU3"
    />
        <resource>
            <Condition>
                <id value="esG9.LwMjNQmdDHbomjHegR42AImiOPW2PbQW4A0r2CU3" />
                <verificationStatus>
                    <coding>
                        <system value="http://terminology.hl7.org/CodeSystem/condition-ver-status" />
                        <version value="4.0.0" />
                        <code value="confirmed" />
                        <display value="Confirmed" />
                    </coding>
                    <text value="Confirmed" />
                </verificationStatus>
                <category>
                    <coding>
                        <system value="http://open.epic.com/FHIR/StructureDefinition/condition-category" />
                        <code value="medical-history" />
                        <display value="Medical History" />
                    </coding>
                    <text value="Medical History" />
                </category>
                <code>
                    <coding>
                        <system value="http://snomed.info/sct" />
                        <code value="39423001" />
                        <display value="Stuttering (finding)" />
                    </coding>
                    <coding>
                        <system value="http://hl7.org/fhir/sid/icd-9-cm" />
                        <code value="315.35" />
                        <display value="Stammer" />
                    </coding>
                    <coding>
                        <system value="urn:oid:2.16.840.1.113883.6.90" />
                        <code value="F80.81" />
                        <display value="Stammer" />
                    </coding>
                    <text value="Stammer" />
                </code>
                <subject>
                    <reference value="Patient/ebb8tmJBI1FH-fdAzdqISZg3" />
                    <display value="Stanley, James" />
                </subject>
            </Condition>
        </resource>
        <search>
            <mode value="match" />
        </search>
    </entry>
</Bundle>
```


Change log

Epic Version | Description | Type

May 2023 (Special Updates to February 2023) | Code systems in the Condition.category element now return a URL in place of an OID. Typical code systems include: System URL OID SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 LOINC http://loinc.org 2.16.840.1.113883.6.1 | Breaking change

System | URL | OID

SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96

LOINC | http://loinc.org | 2.16.840.1.113883.6.1

10.1.23 Condition.Search (Problems) (R4)

File: site/apis/10-Clinical-General/1-R4/953-Condition-Search-Problems-R4.html

Epic API ID | 953

FHIR version | R4

Categories | Clinical General
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Keywords | Problems;Problem List
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | November 2019
HTTP method | GET
URL template | api/FHIR/R4/Condition?subject={subject}&patient={patient}&clinical-status={clinical-status}&category={category}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=953

Description

This API retrieves problems from a patient's chart. This includes any problems list entries across all encounters.

For encounter-specific diagnosis information, see Condition.Search (Encounter Diagnosis).

For problems addressed during a hospital encounter, see List.Search (Hospital Problems).

The problems returned in this API are similar to those returned by the List.Search (Problems) API. However, List.Search (Problems) is intended to represent the clinically-reviewed list of active problems at a specific point of time by a specific clinician. Thus, List.Search (Problems) does not include deleted or resolved problems, and it includes an AuditEvent representing when a clinician last marked the problem list as reviewed. In contrast. Condition.Search (Problems) can be used to get the full history of the patient's problem list.

Unconfirmed Condition resources that have yet to be reconciled into a patient's chart by a clinician are only returned when searching for them by their FHIR IDs in _id. Once a clinician reconciles a problem, a new Condition resource associated with the reconciled problem is returned in Condition.Search results, and the previously-unconfirmed Condition in the "holding tank" no longer appears in the results.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
_count | number | Optional | Yes | | Number of results per page.
_id | token | Optional | Yes | | FHIR resource IDs for the desired resources. Can be used in combination with _count, _include, and _revInclude. All others will be ignored.
_include | string | Optional | Yes | | Used to indicate which referenced resources to include in the search bundle.
_revinclude | string | Optional | Yes | | Used to include Provenance or AuditEvent resources that refer to search results, when available.
category | token | Optional | Yes | | Always problem-list-item for the Condition (Problem List) search API.
clinical-status | token | Optional | Yes | | The current status of the problem list item. The following Condition Clinical Status Codes are supported: active - Active problem list items. resolved - Resolved problem list items. inactive - Deleted problem list items.
patient | reference | Conditional | Yes | | Reference to a patient resource the condition is for. Either a patient or subject parameter must be provided. If both are provided, they must match exactly.
subject | reference | Conditional | Yes | | Reference to a patient resource the condition is for. Either a patient or subject parameter must be provided. If both are provided, they must match exactly.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
abatement-date | date | Optional | Yes | | The date the condition was resolved.
code | token | Optional | Yes | | The code that identifies the specific type of problem. Examples of code sytems that are often used to code problem list items include ICD-10, SNOMED, ICPC Basic Health Care Classification, and Intelligent Medical Objects ProblemIT and others.
onset-date | date | Optional | Yes | | The date a condition was noted.
onset-info | string | Optional | Yes | | The date a condition was noted.
recorded-date | date | Optional | Yes | | The date the condition was recorded.
severity | token | Optional | Yes | | The priority of the problem.
stage | token | Optional | Yes | | For cancer-related problems, the stage or grade of the condition.
verification-status | token | Optional | Yes | | If this is on the patient's local chart, the value is "confirmed". If it is outside data, it is "provisional". If it is deleted, it is "entered-in-error".

Response elements

Name | Type | Optional | Array | Cardinality | Description
Condition (R4) | Condition | | | | An instance of the R4 Condition resource.
Condition (R4).abatementDateTime | dateTime | | | 0..1 | The date the condition was resolved. Starting in the February 2024 version of Epic, abatementDateTime is used to communicate the resolution date. In November 2023 and earlier versions of Epic, abatementPeriod is used to communicate the resolution date.
Condition (R4).abatementPeriod | Period | | | 0..1 | The date the condition is resolved. In the November 2023 and earlier versions of Epic, abatementPeriod is used to communicate the resolution date. Starting in the February 2024 version of Epic, abatementDateTime is used to communicate the resolution date.
Condition (R4).category | CodeableConcept | | Yes | 0..* | Always includes category code "problem-list-item". If a problem list item is marked with a class of "Health Concern", indicating that the problem is of special concern, a category code of "health-concern" is also included. This can also

include additional category codes, such as "sdoh" for problems related to social determinates of health. This category depends on configuration of diagnosis groupers defined by the Epic organization.

Condition (R4).clinicalStatus | CodeableConcept | | | 1..1 | The current clinical status of the problem list item. The following Condition Clinical Status Codes are supported: active - Active problem list items resolved - Resolved problem list items inactive - Deleted problem list items

Condition (R4).code | CodeableConcept | | | 1..1 | The code that identifies the specific type of problem. Examples of code sytems that are often used to code problem list items include ICD-10, SNOMED, ICPC Basic Health Care Classification, and Intelligent Medical Objects ProblemIT and others. If there are codes available to send, they're included in this element. If there are no codes to send, this element includes the uncoded, free text problem name. Starting in the February 2024 version of Epic or with special updates to November 2023, August 2023, or May 2023 version of Epic, each code also sends its display name and the provider-entered problem name is sent if available. Common code system values: ICD-10: http://hl7.org/fhir/sid/icd-10-cm SNOMED: http://snomed.info/sct

Condition (R4).extension (assertedDateTime) (also: extension (assertedDateTime)) | Extension | | Yes | 0..* | The date the condition was diagnosed. URL: http://hl7.org/fhir/StructureDefinition/condition-assertedDate

Condition (R4).extension (smartdata) (also: extension (smartdata)) | Extension | | Yes | | When documenting a problem, various questions, also known as SmartData Elements, can be conditionally presented to the clinician. The smartdata extension on Condition (Problems) refernces the clinician's answers, which are available in the Observation (SmartData Elements) API.

Condition (R4).id | id | | | 1..1 | The condition FHIR ID.

Condition (R4).meta | Meta | | | | Metadata about the resource.

Condition (R4).meta.lastUpdated | String | | | 0..1 | The instant the condition was last updated.

Condition (R4).note | Annotation | | Yes | 1..* | Free text comments for the note. Can be an Overview note or, starting in the February 2025 version of Epic, an Assessment & Plan note.

Condition (R4).note.authorReference | Reference | | | | Who wrote the note. This element is returned starting in the following Epic versions, dependent on note type: May 2025: Problem overview notes February 2025: Assessment & Plan notes

Condition (R4).note.extension (annotationType) (also: extension (annotationType)) | Extension | | Yes | | This extension is returned starting in the following Epic versions, dependent on note type: May 2025: Problem overview notes February 2025: Assessment & Plan notes The following valueCodeableConcept codings are supported: Assessment & Plan notes: LOINC code 51847-2 (Evaluation + Plan note) Problem overview notes: LOINC code 68608-9 (Summary note) URL: http://hl7.org/fhir/StructureDefinition/annotationType

Condition (R4).note.extension (data-conversion-warning) (also: extension (data-conversion-warning)) | Extension | | Yes | | This extension is returned starting in the following Epic versions, dependent on note type: May 2025: Problem overview notes February 2025: Assessment & Plan notes One or more of the following warnings can be returned depending on RTF to plan text conversion: Reason for warning Warning text Formatting was lost due to unicode characters or tables being present in the original note. Formatting of this note is different from the original. The original note was RTF. Formatting of this note might be different from the original. There were one or more images in the original note. Images from the original note were not included. There were one or more media attachments in the original note. Attached media from the original note were not included. URL: http://open.epic.com/FHIR/StructureDefinition/extension/data-conversion-warning

Reason for warning | Warning text

Formatting was lost due to unicode characters or tables being present in the original note. | Formatting of this note is different from the original.

The original note was RTF. | Formatting of this note might be different from the original.

There were one or more images in the original note. | Images from the original note were not included.

There were one or more media attachments in the original note. | Attached media from the original note were not included.

Condition (R4).note.time | String | | | | The time that the note was signed. This element is returned starting in the following Epic versions, dependent on note type: May 2025: Problem overview notes February 2025: Assessment & Plan notes

Condition (R4).onsetDateTime | dateTime | | | 0..1 | Starting in the February 2024 version of Epic, if a condition has a noted date, it appears here.

Condition (R4).onsetPeriod | Period | | | 0..1 | In the November 2023 and earlier versions of Epic, if a condition has a noted date, it appears here.

Condition (R4).recordedDate | dateTime | | | 1..1 | The date the condition was recorded.

Condition (R4).recorder | Reference | | | 0..1 | Reference to the practitioner who updated the problem list.

Condition (R4).severity | CodeableConcept | | | 0..1 | The default priority of the problem. Note that Epic support two types of priorities for problems: A default priority that is specified when the problem is initially created. This value is available in the Condition.severity property. A user-specific priority that users can adjust based on the relevance of the problem to their specialty. For example, a dermatologist might rate eczema as a high priority while a cardiologist would rate it as low. User-level priorities allow each clinician to see the list organized in a way that is most useful to them. This value is displayed to users Epic's problem list activity. This value is not supported in the FHIR API.

Condition (R4).stage | BackboneElement | | Yes | 0..* | For problems that represent cancer, staging information can be documented as part of the problem. Cancer staging usually involves the identification of the location, size, and the extent of metastasis of the tumor.

Condition (R4).stage.assessment | Reference | | Yes | 1..* | SmartData element associated with cancer staging for this condition.

Condition (R4).subject | Reference | | | 1..1 | The patient associated with the condition.

Condition (R4).verificationStatus | CodeableConcept | | | 1..1 | If this is on the patient's local chart, the value is "confirmed". If it is outside data, it is "provisional". If it is deleted, it is "entered-in-error".

Sample request

```
GET https://hostname/instance/api/FHIR/R4/Condition?patient=eLFQugzW.CQz3w48A427N.w3&category=problem-list-item
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 2,
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Condition?category=problem-list-
    item&patient=eLFQugzW.CQz3w48A427N.w3"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url": "https://hostname/instance/api/FHIR/R4/Condition/evA9Cs4FuqWRemAD9B321RvjhDZCOh4cL0mszsMdP89s3"
```

```
  }
],
"fullUrl": "https://hostname/instance/api/FHIR/R4/Condition/evA9Cs4FuqWRemAD9B321RvjhDZCOh4cL0mszsMdP89s3",
"resource": {
  "resourceType": "Condition",
  "id": "evA9Cs4FuqWRemAD9B321RvjhDZCOh4cL0mszsMdP89s3",
  "clinicalStatus": {
    "coding": [
      {
        "system": "http://terminology.hl7.org/CodeSystem/condition-clinical",
        "version": "4.0.0",
        "code": "active",
        "display": "Active"
      }
    ],
    "text": "Active"
  },
  "verificationStatus": {
    "coding": [
      {
        "system": "http://terminology.hl7.org/CodeSystem/condition-ver-status",
        "version": "4.0.0",
        "code": "confirmed",
        "display": "Confirmed"
      }
    ],
    "text": "Confirmed"
  },
  "category": [
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/condition-category",
          "code": "problem-list-item",
          "display": "Problem List Item"
        }
      ],
      "text": "Problem List Item"
    },
    {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "64572001"
        }
      ]
    }
  ],
  "code": {
    "coding": [
      {
        "system": "http://hl7.org/fhir/sid/icd-10-cm",
        "code": "W59.22XA",
        "display": "Struck by turtle, initial encounter"
      },
      {
        "system": "http://snomed.info/sct",
        "code": "418589001",
        "display": "Accidental Physical Contact With Animal"
      },
      {
        "system": "http://hl7.org/fhir/sid/icd-9-cm",
        "code": "E906.8",
        "display": "Struck by turtle"
      },
      {
        "system": "urn:oid:2.16.840.1.113883.3.247.1.1",
        "code": "1623504",
        "display": "Struck by turtle"
      }
    ],
    "text": "Struck by turtle"
  },
  "subject": {
    "reference": "Patient/eLFQugzW.CQz3w48A427N.w3",
    "display": "zzFhir, Fhirwoman"
  },
  "onsetDateTime": "2023-11-14",
  "recordedDate": "2023-11-14",
  "note": [
    {
      "extension": [
        {
          "valueString": "Formatting of this note might be different from the original.",
          "url": "http://open.epic.com/FHIR/StructureDefinition/extension/data-conversion-warning"
        },
        {
          "valueCodeableConcept": {
            "coding": [
```

```
                {
                  "system": "http://loinc.org",
                  "code": "68608-9",
                  "display": "Summary note"
                }
              ],
              "text": "Problem Overview"
            },
            "url": "http://hl7.org/fhir/StructureDefinition/annotationType"
          }
        ],
        "authorReference": {
          "reference": "Practitioner/eodo7Bj9dk-LtJSthMwARpw3",
          "display": "Fhirdoc Doctorman"
        },
        "time": "2025-04-14T19:36:00Z",
        "text": "Example text for an Overview Note.\r\n"
      },
      {
        "extension": [
          {
            "valueString": "Formatting of this note might be different from the original.",
            "url": "http://open.epic.com/FHIR/StructureDefinition/extension/data-conversion-warning"
          },
          {
            "valueCodeableConcept": {
              "coding": [
                {
                  "system": "http://loinc.org",
                  "code": "51847-2",
                  "display": "Evaluation + Plan note"
                }
              ],
              "text": "Assessment & Plan"
            },
            "url": "http://hl7.org/fhir/StructureDefinition/annotationType"
          }
        ],
        "authorReference": {
          "reference": "Practitioner/eodo7Bj9dk-LtJSthMwARpw3",
          "display": "Fhirdoc Doctorman"
        },
        "time": "2025-04-14T19:43:16Z",
        "text": "Example text for an A&P note.\r\n"
      }
    ]
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Condition/ezll22nlZbksIjn.BBwqdhvCVweWcxzxQDpkK3323qA03"
    }
  ],
  "fullUrl": "https://hostname/instance/api/FHIR/R4/Condition/ezll22nlZbksIjn.BBwqdhvCVweWcxzxQDpkK3323qA03",
  "resource": {
    "resourceType": "Condition",
    "id": "ezll22nlZbksIjn.BBwqdhvCVweWcxzxQDpkK3323qA03",
    "clinicalStatus": {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/condition-clinical",
          "version": "4.0.0",
          "code": "resolved",
          "display": "Resolved"
        }
      ],
      "text": "Resolved"
    },
    "verificationStatus": {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/condition-ver-status",
          "version": "4.0.0",
          "code": "confirmed",
          "display": "Confirmed"
        }
      ],
      "text": "Confirmed"
    },
    "category": [
      {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/condition-category",
```

```
              "code": "problem-list-item",
              "display": "Problem List Item"
            }
          ],
          "text": "Problem List Item"
        },
        {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "64572001"
            }
          ]
        }
      ],
      "code": {
        "coding": [
          {
            "system": "http://hl7.org/fhir/sid/icd-10-cm",
            "code": "W61.62XA",
            "display": "Struck by duck, initial encounter"
          },
          {
            "system": "http://snomed.info/sct",
            "code": "418589001",
            "display": "Accidental Physical Contact With Animal"
          },
          {
            "system": "http://hl7.org/fhir/sid/icd-9-cm",
            "code": "E906.8",
            "display": "Struck by duck"
          },
          {
            "system": "urn:oid:2.16.840.1.113883.3.247.1.1",
            "code": "1624982",
            "display": "Struck by duck"
          }
        ],
        "text": "Struck by duck"
      },
      "subject": {
        "reference": "Patient/eLFQugzW.CQz3w48A427N.w3",
        "display": "zzFhir, Fhirwoman"
      },
      "onsetDateTime": "2025-03-13",
      "abatementDateTime": "2025-03-13",
      "recordedDate": "2025-03-13"
    },
    "search": {
      "mode": "match"
    }
  }
 ]
}
```


Change log

Epic Version | Description | Type

May 2025 | Previously data loss warnings in the conversion of notes were prepended to the note content (the note.text element in the FHIR resource). Now this warning is returned in a note.extension element. | Breaking change

February 2024 | The USCDIv3 specification for Condition will move the Condition's start date and time from onsetPeriod to onsetDatetime and end date and time from abatementPeriod to abatementDatetime fields. This could impact the evaluation of a Condition if a Condition consumer is not prepared for the transition. | Breaking change

May 2023 (Special Updates to February 2023) | Code systems in the Condition.category element now return a URL in place of an OID. Typical code systems include: System URL OID SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 LOINC http://loinc.org 2.16.840.1.113883.6.1 | Breaking change

System | URL | OID

SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96

LOINC | http://loinc.org | 2.16.840.1.113883.6.1

November 2022 | Code systems in the Condition.code element for the following code sets now return a URL in place of an OID: ActCode, ICD-10-CM, ICD-10-CA, ICD-10-NL, SNOMED System URL OID ActCode STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode 2.16.840.1.113883.5.4 ICD-10-CM http://hl7.org/fhir/sid/icd-10-cm 2.16.840.1.113883.6.90 ICD-10-CA http://hl7.org/fhir/sid/icd-10-ca 2.16.840.1.113883.6.94 ICD-10-NL http://hl7.org/fhir/sid/icd-10-nl 2.16.840.1.113883.6.3.2 SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 | Breaking change

System | URL | OID

ActCode | STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode | 2.16.840.1.113883.5.4

ICD-10-CM | http://hl7.org/fhir/sid/icd-10-cm | 2.16.840.1.113883.6.90

ICD-10-CA | http://hl7.org/fhir/sid/icd-10-ca | 2.16.840.1.113883.6.94

ICD-10-NL | http://hl7.org/fhir/sid/icd-10-nl | 2.16.840.1.113883.6.3.2

SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96

10.1.24 Condition.Search (Reason for Visit) (R4)

File: site/apis/10-Clinical-General/1-R4/10208-Condition-Search-Reason-for-Visit-R4.html
Epic API ID | 10208
FHIR version | R4
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Condition?category={category}&encounter={encounter}&patient={patient}&subject={subject}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10208

Description

The resource retrieves reasons for visit from a patient's encounters. This includes any reason for visit information that was recorded in the patient's chart by a clinician during any previous encounter.

This resource can be queried by a patient ID or an encounter ID.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
category | token | True | | | Always "reason-for-visit" for this API. Multiple category values are allowed if the requestor is authorized to search multiple Condition categories.
encounter | reference | True | | | Reference to an encounter resource the condition is for.
patient | reference | Either patient or subject must be provided. | | | Reference to a patient resource the condition is for.
subject | reference | Either patient or subject must be provided. | | | Reference to a patient resource the condition is for.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
abatement-date | date | True | | | Unsupported
abatement-string | String | True | | | Unsupported
asserter | String | True | | | Unsupported
body-site | String | True | | | The location of the condition.
clinical-status | token | True | | | Unsupported
code | token | True | | | External codings for the condition.
evidence | token | True | | | Unsupported
evidence-detail | reference | True | | | Unsupported
identifier | String | True | | | Unsupported
onset-date | date | True | | | The date the condition began.
onset-info | string | True | | | Unsupported
recorded-date | date | True | | | The date the condition was recorded.
severity | token | True | | | Unsupported
stage | token | True | | | Unsupported
verification-status | token | True | | | Unsupported

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | BackboneElement | False | Yes | | An array of entries included in this bundle.
Bundle.entry.resource | ResourceContainer | False | | | A container for a single FHIR resource.
Bundle.entry.resource.Condition | Condition | False | | | An instance of the R4 Condition resource.
Bundle.entry.resource.Condition.bodySite | CodeableConcept | True | Yes | | The location of the condition
Bundle.entry.resource.Condition.code | CodeableConcept | True | | | External codings for the condition. If a list of codes is sent, they're listed in this element.
Bundle.entry.resource.Condition.encounter | Reference | True | | | The encounter associated with the condition
Bundle.entry.resource.Condition.id | id | False | | | The condition FHIR ID
Bundle.entry.resource.Condition.note | Annotation | True | Yes | | Free text comments
Bundle.entry.resource.Condition.onsetDateTime | dateTime | True | | | The date the condition began
Bundle.entry.resource.Condition.recordedDate | dateTime | True | | | The date the condition was recorded
Bundle.entry.resource.Condition.recorder | Reference | True | | | The provider associated with recording the condition
Bundle.entry.resource.Condition.subject | Reference | False | | | The patient associated with the condition

Sample request

```
https://hostname/instance/api/FHIR/R4/Condition?patient=ezHbeFCm6Ih8kgxuHUAdvJQ3&category=reason-for-visit
```

Sample response

```
<condition xmlns="http://hl7.org/fhir">
  <id value="egFptNetytmse51RiokqI2Z3QBHsNx8GHXDYi7EhR.f9fR-tfX6ms3qkivkLc2Bd13"/>
```

```
  <category>
    <coding>
      <system value="http://open.epic.com/FHIR/StructureDefinition/condition-category"/>
      <code value="reason-for-visit"/>
      <display value="Reason for Visit"/>
    </coding>
    <text value="Reason for Visit"/>
  </category>
  <code>
    <coding>
      <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.2.728286"/>
      <code value="43294"/>
      <display value="Rash"/>
    </coding>
    <text value="Rash"/>
  </code>
  <bodysite>
    <coding>
      <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.2.866976"/>
      <code value="11347"/>
      <display value="Right Leg"/>
    </coding>
    <text value="Right Leg"/>
  </bodysite>
  <subject>
    <reference value="Patient/e56rHiD8cHF60CyLEUJ-1Ew3"/>
    <display value="Shepherd, Amelia"/>
  </subject>
  <encounter>
    <reference value="Encounter/eeM8KWAwqj5oZbB7ROBexzAvNiJBO0VmJdfkt2kHvJ.s3"/>
  </encounter>
  <recordeddate value="2021-07-28T17:03:39Z"/>
  <recorder>
    <reference value="Practitioner/ehchKMZoNW8B.Ycy.dUrFtg3"/>
    <display value="Eva Ward, MD"/>
  </recorder>
  <note>
    <text value="This is my comment."/>
  </note>
</condition>
```

10.1.25 FamilyMemberHistory.Read (R4)

File: site/apis/10-Clinical-General/1-R4/10158-FamilyMemberHistory-Read-R4.html
Epic API ID | 10158
FHIR version | R4
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | August 2021
HTTP method | GET
URL template | /api/FHIR/R4/FamilyMemberHistory/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10158

Description

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | String | False | | | FHIR ID for this FamilyMemberHistory instance

Response elements

Name | Type | Optional | Array | Cardinality | Description
FamilyMemberHistory | FamilyMemberHistory | False
FamilyMemberHistory.bornDate | String | True | | | Family member’s birth date
FamilyMemberHistory.bornPeriod | Period | True | | | Range of dates between which family member was born
FamilyMemberHistory.condition | Condition | True | Yes | | Condition that the related person had
FamilyMemberHistory.condition.code | CodeableConcept | False | | | Condition that the related person had
FamilyMemberHistory.condition.note | Annotation | True | Yes | | Comment on a single condition
FamilyMemberHistory.condition.onsetAge | Age | True | | | How old the family member was at the time of condition onset
FamilyMemberHistory.condition.onsetRange | Range | True | | | How old the family member was at the time of condition onset and how old they were when it ended.
FamilyMemberHistory.dataAbsentReason | CodeableConcept | True | | | Reason a family member history may not exist, such as "unable-to-obtain".
FamilyMemberHistory.date | String | True | | | When history was recorded or last updated

FamilyMemberHistory.deceasedBoolean | Boolean | True | | | Whether or not the family member is deceased
FamilyMemberHistory.id | String | False | | | FHIR ID for this FamilyMemberHistory instance
FamilyMemberHistory.identifier | Identifier | True | Yes | | Unique ID for the family member history
FamilyMemberHistory.name | String | True | | | The family member described
FamilyMemberHistory.note | Annotation | True | Yes | | General note about related person
FamilyMemberHistory.relationship | CodeableConcept | False | | | Relationship to the patient, or "FAMMEMB" if the relationship is not clear.
FamilyMemberHistory.sex | CodeableConcept | False | | | Family member's biological sex
FamilyMemberHistory.status | String | False | | | Completion status for family member's history. Either "health-unknown" or "completed".

Sample request

```
https://hostname/instance/api/FHIR/R4/FamilyMemberHistory/ePM.Qux3IY3tsLDfCmpxnDhhUJ4WWDI.DRtTk-LVsHD03
```

Sample response

```
<FamilyMemberHistory xmlns="http://hl7.org/fhir">
  <id value="ePM.Qux3IY3tsLDfCmpxnDhhUJ4WWDI.DRtTk-LVsHD03"/>
  <status value="completed"/>
  <patient>
    <reference value="Patient/eV6Ofju2.rCOp6TCunVM9Rg3"/>
    <display value="Smith, Sally"/>
  </patient>
  <date value="2021-07-05"/>
  <name value="Paula"/>
  <relationship>
    <coding>
      <system value="http://terminology.hl7.org/CodeSystem/v3-RoleCode"/>
      <code value="NMTH"/>
      <display value="Natural mother"/>
    </coding>
    <coding>
      <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.10.698084.19350"/>
      <code value="1"/>
      <display value="Mother"/>
    </coding>
  </relationship>
  <sex>
    <coding>
      <system value="http://hl7.org/fhir/administrative-gender"/>
      <code value="female"/>
    </coding>
  </sex>
  <bornDate value="1968-07-05"/>
  <deceasedBoolean value="false"/>
  <note>
    <text value="Estranged"/>
  </note>
  <condition>
    <code>
      <coding>
        <system value="http://snomed.info/sct"/>
        <code value="160266009"/>
        <display value="No family history of"/>
      </coding>
      <coding>
        <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.10.698084.19360"/>
        <code value="1702"/>
        <display value="Healthy"/>
      </coding>
    </code>
    <note>
      <text value="Mother is healthy"/>
    </note>
  </condition>
</FamilyMemberHistory>
```

10.1.26 FamilyMemberHistory.Search (R4)

File: site/apis/10-Clinical-General/1-R4/10159-FamilyMemberHistory-Search-R4.html
Epic API ID | 10159
FHIR version | R4
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | August 2021
HTTP method | GET;POST
URL template | /api/FHIR/R4/FamilyMemberHistory?patient={patient}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False

Source | https://fhir.epic.com/Specifications?api=10159

Description

Request parameters

Name | Type | Optional | Array | Cardinality | Description
patient | String | False | | | FHIR Patient ID If provided, all of the family members associated with the patient are returned, as well as other Family History data items.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
code | String | True | | | Condition that the related person had
date | String | True | | | When history was recorded or last updated
identifier | String | True | | | Unsupported
instantiates-canonical | String | True | | | Unsupported
relationship | String | True | | | Relationship to the patient
sex | String | True | | | Family member’s biological sex
status | String | True | | | Completion status for family member’s history. Either "health-unknown" or "completed".

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | BackboneElement | False | Yes | | An array of entries included in this bundle.
Bundle.entry.resource | ResourceContainer | False | | | A container for a single FHIR resource.
Bundle.entry.resource.FamilyMemberHistory | FamilyMemberHistory | False
Bundle.entry.resource.FamilyMemberHistory.bornDate | String | True | | | Family member’s birth date
Bundle.entry.resource.FamilyMemberHistory.bornPeriod | Period | True | | | Range of dates between which family member was born
Bundle.entry.resource.FamilyMemberHistory.condition | Condition | True | Yes | | Condition that the related person had
Bundle.entry.resource.FamilyMemberHistory.condition.code | CodeableConcept | False | | | Condition that the related person had
Bundle.entry.resource.FamilyMemberHistory.condition.note | Annotation | True | Yes | | Comment on a single condition
Bundle.entry.resource.FamilyMemberHistory.condition.onsetAge | Age | True | | | How old the family member was at the time of condition onset
Bundle.entry.resource.FamilyMemberHistory.condition.onsetRange | Range | True | | | How old the family member was at the time of condition onset and how old they were when it ended.
Bundle.entry.resource.FamilyMemberHistory.dataAbsentReason | CodeableConcept | True | | | Reason a family member history may not exist, such as "unable-to-obtain"
Bundle.entry.resource.FamilyMemberHistory.date | String | True | | | When history was recorded or last updated
Bundle.entry.resource.FamilyMemberHistory.deceasedBoolean | Boolean | False | | | Whether or not the family member is deceased
Bundle.entry.resource.FamilyMemberHistory.id | String | False | | | FHIR ID for this FamilyMemberHistory instance
Bundle.entry.resource.FamilyMemberHistory.identifier | Identifier | True | Yes | | Unique ID for the family member history
Bundle.entry.resource.FamilyMemberHistory.name | String | True | | | The family member described
Bundle.entry.resource.FamilyMemberHistory.note | Annotation | True | Yes | | General note about related person
Bundle.entry.resource.FamilyMemberHistory.relationship | CodeableConcept | False | | | Relationship to the patient. If the relationship is not clear, "FAMMEMB" is sent.
Bundle.entry.resource.FamilyMemberHistory.sex | CodeableConcept | False | | | Family member’s biological sex
Bundle.entry.resource.FamilyMemberHistory.status | String | False | | | Completion status for family member’s history. Either "health-unknown" or "completed".

Sample request

```
https://hostname/instance/api/FHIR/R4/FamilyMemberHistory?patient=eV6Ofju2.rCOp6TCunVM9Rg3
```

Sample response

```
<Bundle xmlns="http://hl7.org/fhir">
  <type value="searchset"/>
  <total value="5"/>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/R4/FamilyMemberHistory?patient=eV6Ofju2.rCOp6TCunVM9Rg3"/>
  </link>
  <entry>
    <link>
      <relation value="self"/>
      <url value="https://hostname/instance/api/FHIR/R4/FamilyMemberHistory/ePM.Qux3IY3tsLDfCmpxnDhhUJ4WWDI.DRtTk-
    LVsHD03"/>
    </link>
    <fullUrl value="https://hostname/instance/api/FHIR/R4/FamilyMemberHistory/ePM.Qux3IY3tsLDfCmpxnDhhUJ4WWDI.DRtTk-
    LVsHD03"/>
    <resource>
      <FamilyMemberHistory>
        <id value="ePM.Qux3IY3tsLDfCmpxnDhhUJ4WWDI.DRtTk-LVsHD03"/>
        <status value="completed"/>
        <patient>
          <reference value="Patient/eV6Ofju2.rCOp6TCunVM9Rg3"/>
          <display value="Smith, Sally"/>
```

```
      </patient>
      <date value="2021-07-05"/>
      <name value="Paula"/>
      <relationship>
        <coding>
          <system value="http://terminology.hl7.org/CodeSystem/v3-RoleCode"/>
          <code value="NMTH"/>
          <display value="Natural mother"/>
        </coding>
        <coding>
          <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.10.698084.19350"/>
          <code value="1"/>
          <display value="Mother"/>
        </coding>
      </relationship>
      <sex>
        <coding>
          <system value="http://hl7.org/fhir/administrative-gender"/>
          <code value="female"/>
        </coding>
      </sex>
      <bornDate value="1968-07-05"/>
      <deceasedBoolean value="false"/>
      <note>
        <text value="Estranged"/>
      </note>
      <condition>
        <code>
          <coding>
            <system value="http://snomed.info/sct"/>
            <code value="160266009"/>
            <display value="No family history of"/>
          </coding>
          <coding>
            <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.10.698084.19360"/>
            <code value="1702"/>
            <display value="Healthy"/>
          </coding>
        </code>
        <note>
          <text value="Mother is healthy"/>
        </note>
      </condition>
    </FamilyMemberHistory>
  </resource>
  <search>
    <mode value="match"/>
  </search>
</entry>
<entry>
  <link>
    <relation value="self"/>
    <url
  value="https://hostname/instance/api/FHIR/R4/FamilyMemberHistory/ePM.Qux3IY3tsLDfCmpxnDiFB6Z7eKudu2cH4dK2ZLDg3"/>
  </link>
  <fullUrl
  value="https://hostname/instance/api/FHIR/R4/FamilyMemberHistory/ePM.Qux3IY3tsLDfCmpxnDiFB6Z7eKudu2cH4dK2ZLDg3"/>
  <resource>
    <FamilyMemberHistory>
      <id value="ePM.Qux3IY3tsLDfCmpxnDiFB6Z7eKudu2cH4dK2ZLDg3"/>
      <status value="completed"/>
      <patient>
        <reference value="Patient/eV6Ofju2.rCOp6TCunVM9Rg3"/>
        <display value="Smith, Sally"/>
      </patient>
      <date value="2021-07-05"/>
      <name value="Joe"/>
      <relationship>
        <coding>
          <system value="http://terminology.hl7.org/CodeSystem/v3-RoleCode"/>
          <code value="NFTH"/>
          <display value="Natural father"/>
        </coding>
        <coding>
          <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.10.698084.19350"/>
          <code value="2"/>
          <display value="Father"/>
        </coding>
      </relationship>
      <sex>
        <coding>
          <system value="http://hl7.org/fhir/administrative-gender"/>
          <code value="male"/>
        </coding>
      </sex>
      <deceasedBoolean value="true"/>
      <note>
        <text value="no genetic component to cancer"/>
      </note>
```

```
      <condition>
        <code>
          <coding>
            <system value="http://snomed.info/sct"/>
            <code value="363418001"/>
            <display value="Malignant tumor of pancreas"/>
          </coding>
          <coding>
            <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.10.698084.19360"/>
            <code value="53313"/>
            <display value="Pancreatic cancer"/>
          </coding>
        </code>
        <onsetAge>
          <value value="45"/>
        </onsetAge>
        <note>
          <text value="stage 4 at dx"/>
        </note>
      </condition>
      <condition>
        <code>
          <coding>
            <system value="http://snomed.info/sct"/>
            <code value="444026000"/>
            <display value="Multiple environmental allergies"/>
          </coding>
          <coding>
            <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.10.698084.19360"/>
            <code value="200"/>
            <display value="Allergies"/>
          </coding>
        </code>
        <onsetAge>
          <value value="5"/>
        </onsetAge>
        <note>
          <text value="pollen"/>
        </note>
      </condition>
    </FamilyMemberHistory>
  </resource>
  <search>
    <mode value="match"/>
  </search>
</entry>
<entry>
  <link>
    <relation value="self"/>
    <url
  value="https://hostname/instance/api/FHIR/R4/FamilyMemberHistory/ePM.Qux3IY3tsLDfCmpxnDhOstn1GTFlXhxiGglTaVdA3"/>
  </link>
  <fullUrl
  value="https://hostname/instance/api/FHIR/R4/FamilyMemberHistory/ePM.Qux3IY3tsLDfCmpxnDhOstn1GTFlXhxiGglTaVdA3"/>
  <resource>
    <FamilyMemberHistory>
      <id value="ePM.Qux3IY3tsLDfCmpxnDhOstn1GTFlXhxiGglTaVdA3"/>
      <status value="completed"/>
      <patient>
        <reference value="Patient/eV6Ofju2.rCOp6TCunVM9Rg3"/>
        <display value="EHI, Sally"/>
      </patient>
      <date value="2021-07-05"/>
      <name value="Ellen"/>
      <relationship>
        <coding>
          <system value="http://terminology.hl7.org/CodeSystem/v3-RoleCode"/>
          <code value="FAMMEMB"/>
          <display value="Family member"/>
        </coding>
        <coding>
          <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.10.698084.19350"/>
          <code value="17"/>
          <display value="Other"/>
        </coding>
      </relationship>
      <sex>
        <coding>
          <system value="http://hl7.org/fhir/administrative-gender"/>
          <code value="female"/>
        </coding>
      </sex>
      <deceasedBoolean value="false"/>
    </FamilyMemberHistory>
  </resource>
  <search>
    <mode value="match"/>
  </search>
</entry>
```

```
  <entry>
    <link>
      <relation value="self"/>
      <url
    value="https://hostname/instance/api/FHIR/R4/FamilyMemberHistory/ePM.Qux3IY3tsLDfCmpxnDreCpIDxiDYvSwSdd2GZMl83"/>
    </link>
    <fullUrl
    value="https://hostname/instance/api/FHIR/R4/FamilyMemberHistory/ePM.Qux3IY3tsLDfCmpxnDreCpIDxiDYvSwSdd2GZMl83"/>
    <resource>
      <FamilyMemberHistory>
        <id value="ePM.Qux3IY3tsLDfCmpxnDreCpIDxiDYvSwSdd2GZMl83"/>
        <status value="completed"/>
        <patient>
          <reference value="Patient/eV6Ofju2.rCOp6TCunVM9Rg3"/>
          <display value="EHI, Sally"/>
        </patient>
        <date value="2021-07-05"/>
        <name value="Lou"/>
        <relationship>
          <coding>
            <system value="http://terminology.hl7.org/CodeSystem/v3-RoleCode"/>
            <code value="FAMMEMB"/>
            <display value="Family member"/>
          </coding>
          <coding>
            <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.10.698084.19350"/>
            <code value="17"/>
            <display value="Other"/>
          </coding>
        </relationship>
        <sex>
          <coding>
            <system value="http://hl7.org/fhir/administrative-gender"/>
            <code value="male"/>
          </coding>
        </sex>
        <deceasedBoolean value="false"/>
      </FamilyMemberHistory>
    </resource>
    <search>
      <mode value="match"/>
    </search>
  </entry>
  <entry>
    <link>
      <relation value="self"/>
      <url
    value="https://hostname/instance/api/FHIR/R4/FamilyMemberHistory/ePM.Qux3IY3tsLDfCmpxnDveFxWGXW2.MzRylZeKEzKw3"/>
    </link>
    <fullUrl
    value="https://hostname/instance/api/FHIR/R4/FamilyMemberHistory/ePM.Qux3IY3tsLDfCmpxnDveFxWGXW2.MzRylZeKEzKw3"/>
    <resource>
      <FamilyMemberHistory>
        <id value="ePM.Qux3IY3tsLDfCmpxnDveFxWGXW2.MzRylZeKEzKw3"/>
        <status value="completed"/>
        <patient>
          <reference value="Patient/eV6Ofju2.rCOp6TCunVM9Rg3"/>
          <display value="EHI, Sally"/>
        </patient>
        <date value="2021-07-05"/>
        <relationship>
          <coding>
            <system value="http://terminology.hl7.org/CodeSystem/v3-RoleCode"/>
            <code value="HSIS"/>
            <display value="Half-sister"/>
          </coding>
          <coding>
            <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.10.698084.19350"/>
            <code value="28"/>
            <display value="Half-sister"/>
          </coding>
        </relationship>
        <sex>
          <coding>
            <system value="http://hl7.org/fhir/administrative-gender"/>
            <code value="female"/>
          </coding>
        </sex>
        <deceasedBoolean value="false"/>
      </FamilyMemberHistory>
    </resource>
    <search>
      <mode value="match"/>
    </search>
  </entry>
</Bundle>
```

10.1.27 Flag.Read (Health Concerns) (R4)

File: site/apis/10-Clinical-General/1-R4/10197-Flag-Read-Health-Concerns-R4.html
Epic API ID | 10197
FHIR version | R4
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Flag/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10197

Description

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | String | False | | | The Flag FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Flag | Flag | False | | | A single instance of a Flag resource.
Flag.category | CodeableConcept | True | Yes | | The Epic category ID for the condition, labeled “health-concern,” and the LOINC code for the condition, labeled “condition.”
Flag.code | CodeableConcept | False | | | The diagnosis code for the condition. Starting in the February 2024 version of Epic or with special updates to November 2023, August 2023, or May 2023 version of Epic, each code also sends its display name and the provider-entered problem name is sent if available.
Flag.id | String | False | | | The Flag FHIR ID.
Flag.period | Period | True | | | The period during which the condition was marked as active.
Flag.period.end | String | True | | | Date when the condition was marked as resolved.
Flag.period.start | String | True | | | Date when the condition was added.
Flag.status | String | False | | | The status of the condition: active inactive entered in error
Flag.subject | Reference | False | | | A reference to the Patient resource.

Sample request

```
https://hostname/instance/api/FHIR/R4/Flag/eyy2ff4sfMS2r1RFpnKxJaP.VLBEBhq5JiMivDW.0klM3
```

Sample response


```
{
  "resourceType": "Flag",
  "id": "enpfRSY57pmn6IxrZD.2eDUnlKr9n60ip6xWB55Ihlws3",
  "extension": [
    {
      "valueReference": {
        "reference": "Condition/emYq8NzSEaKrCZtPWXa-GIioNncX-zeoLZG4utC3iD5w3",
        "display": "Eczema"
      },
      "url": "http://hl7.org/fhir/StructureDefinition/flag-detail"
    }
  ],
  "status": "active",
  "category": [
    {
      "coding": [
        {
          "system": "http://open.epic.com/FHIR/StructureDefinition/flag-category",
          "code": "health-concern",
          "display": "Health Concern Flag"
        },
        {
          "system": "http://loinc.org",
          "code": "75323-6",
          "display": "condition"
        }
      ],
      "text": "Health Concern Flag"
    }
  ],
  "code": {
    "coding": [
      {
        "system": "http://hl7.org/fhir/sid/icd-10-cm",
        "code": "L30.9",
```

```
        "display": "Dermatitis, unspecified"
      },
      {
        "system": "http://snomed.info/sct",
        "code": "43116000",
        "display": "Eczema"
      }
    ],
    "text": "Eczema"
  },
  "subject": {
    "reference": "Patient/euI3cZMuLFNu6kO9mg4ScEg3",
    "display": "Test, Patient"
  },
  "period": {
    "start": "2023-06-28"
  }
}
```


Change log

Epic Version | Description | Type
November 2022 | Code systems for the following code sets now return a URL in place of an OID: ActCode, ICD-10-CM, ICD-10-CA, ICD-10-NL, SNOMED System URL OID ActCode STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode 2.16.840.1.113883.5.4 ICD-10-CM http://hl7.org/fhir/sid/icd-10-cm 2.16.840.1.113883.6.90 ICD-10-CA http://hl7.org/fhir/sid/icd-10-ca 2.16.840.1.113883.6.94 ICD-10-NL http://hl7.org/fhir/sid/icd-10-nl 2.16.840.1.113883.6.3.2 SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 | Breaking change
System | URL | OID
ActCode | STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode | 2.16.840.1.113883.5.4
ICD-10-CM | http://hl7.org/fhir/sid/icd-10-cm | 2.16.840.1.113883.6.90
ICD-10-CA | http://hl7.org/fhir/sid/icd-10-ca | 2.16.840.1.113883.6.94
ICD-10-NL | http://hl7.org/fhir/sid/icd-10-nl | 2.16.840.1.113883.6.3.2
SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96

10.1.28 Flag.Read (Infection) (R4)

File: site/apis/10-Clinical-General/1-R4/10201-Flag-Read-Infection-R4.html
Epic API ID | 10201
FHIR version | R4
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Flag/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10201

Description

This web service retrieves high-level infection information from a patient's chart. An infection might be a laboratory-confirmed infection or colonization, such as influenza or methicillin-resistant Staphylococcus aureus, or a placeholder for a suspected infection, such as respiratory rule-out or suspected COVID-19.

Like the Condition (Infection) web service, this web service evaluates infection (INF) records, but it returns only a brief overview of key information about those records, such as whether they are active and when they were resolved. If you need to return more detailed information, use the Condition (Infection) web service instead.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | String | False | | | The Flag FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Flag | Flag | False | | | A single instance of a Flag resource.
Flag.category | CodeableConcept | True | Yes | | The Epic category ID for the infection, labeled “infection,” and the LOINC code for the infection, labeled “condition.”
Flag.code | CodeableConcept | False | | | The diagnosis code for the infection.
Flag.encounter | Reference | True | | | A reference to the Encounter resource. Used only for encounter-level infections.
Flag.extension (also: flag-detail) | Extension | False | Yes | | A reference to the Condition (Infection) resource.
Flag.id | String | False | | | The Flag FHIR ID.
Flag.period | Period | False | | | The period during which the infection was marked as active.
Flag.period.end | String | True | | | Date when the infection was marked as resolved.

Flag.period.start | String | True | | | Date when the infection was added.
Flag.status | String | False | | | The status of the infection: active or inactive.
Flag.subject | Reference | False | | | A reference to the Patient resource.
OperationOutcome | OperationOutcome | False
OperationOutcome.contained | ResourceContainer | False | Yes | | A container for a single FHIR resource.
OperationOutcome.extension | Extension | False | Yes
OperationOutcome.id | String | False
OperationOutcome.implicitRules | String | False
OperationOutcome.issue | Issue | False | Yes
OperationOutcome.language | String | False
OperationOutcome.meta | Meta | False
OperationOutcome.modifierExtension | Extension | False | Yes
OperationOutcome.text | Narrative | False

Sample request

```
https://hostname/instance/api/FHIR/R4/Flag/eFbv9LL28ikxKaUddrVLr-IVq.wf.k7BR.QGEqq8hfEM3
```

Sample response


```
{
  "resourceType": "Flag",
  "id": "eFbv9LL28ikxKaUddrVLr-IVq.wf.k7BR.QGEqq8hfEM3",
  "extension": [
    {
      "valueReference": {
        "reference": "Condition/erEuPdnY2OKSUOSC-Y9M6DQ3",
        "display": "Influenza"
      },
      "url": "http://hl7.org/fhir/StructureDefinition/flag-detail"
    }
  ],
  "status": "active",
  "category": [
    {
      "coding": [
        {
          "system": "http://open.epic.com/FHIR/StructureDefinition/flag-category",
          "code": "infection",
          "display": "Infection Flag"
        },
        {
          "system": "http://loinc.org",
          "code": "75323-6",
          "display": "condition"
        }
      ],
      "text": "Infection Flag"
    }
  ],
  "code": {
    "coding": [
      {
        "system": "urn:oid:1.2.840.114350.1.13.861.1.7.4.698084.10375",
        "code": "3",
        "display": "Influenza"
      }
    ],
    "text": "Influenza"
  },
  "subject": {
    "reference": "Patient/exnWmPch0AyoUp7ZBplVf.Q3",
    "display": "DXL, Contacts"
  },
  "period": {
    "start": "2021-07-27"
  }
}
```


10.1.29 Flag.Search (Health Concerns) (R4)

File: site/apis/10-Clinical-General/1-R4/10198-Flag-Search-Health-Concerns-R4.html
Epic API ID | 10198
FHIR version | R4
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Flag?category={category}&patient={patient}&status={status}&subject={subject}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients

Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10198

Request parameters

Name | Type | Optional | Array | Cardinality | Description
category | String | True | | | Always "health-concern" for this web service.
patient | String | Either Patient or Subject must be provided. If both are provided, they must match. | | | Reference to a Patient resource the Flag is for.
status | String | True | | | The status of the condition: active, inactive, or entered in error.
subject | String | Either Patient or Subject must be provided. If both are provided, they must match. | | | Reference to a Patient resource the Flag is for.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
author | String | True | | | Unsupported
date | String | True | | | The period during which the condition was marked as active.
identifier | String | True | | | Unsupported

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | BackboneElement | False | Yes | | An array of entries included in this bundle.
Bundle.entry.resource | ResourceContainer | False | | | A container for a single FHIR resource.
Bundle.entry.resource.Flag | Flag | False | | | A single instance of a Flag resource.
Bundle.entry.resource.Flag.category | CodeableConcept | True | Yes | | The Epic category ID for the condition, labeled “health-concern,” and the LOINC code for the condition, labeled “condition.”
Bundle.entry.resource.Flag.code | CodeableConcept | False | | | The diagnosis code for the condition. Starting in the February 2024 version of Epic or with special updates to November 2023, August 2023, or May 2023 version of Epic, each code also sends its display name and the provider-entered problem name is sent if available.
Bundle.entry.resource.Flag.extension (also: flag-detail) | Extension | True | Yes | | A reference to the Condition (Health Concern) resource.
Bundle.entry.resource.Flag.id | String | False | | | The Flag FHIR ID.
Bundle.entry.resource.Flag.period | Period | True | | | The period during which the condition was marked as active.
Bundle.entry.resource.Flag.period.end | String | True | | | Date when the condition was marked as resolved.
Bundle.entry.resource.Flag.period.start | String | True | | | Date when the condition was added.
Bundle.entry.resource.Flag.status | String | False | | | The status of the condition: active, inactive, or entered in error.
Bundle.entry.resource.Flag.subject | Reference | False | | | A reference to the Patient resource.

Sample request

```
https://hostname/instance/api/FHIR/R4/Flag?patient=exnWmPch0AyoUp7ZBplVf.Q3&category=health-concern
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 1,
  "link": [
    {
      "relation": "self",
      "url": " https://hostname/instance/api/FHIR/R4/Flag?category=health-concern&patient=euI3cZMuLFNu6kO9mg4ScEg3"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url": " https://hostname/instance/api/FHIR/R4/Flag/enpfRSY57pmn6IxrZD.2eDUnlKr9n60ip6xWB55Ihlws3"
        }
      ],
      "fullUrl": " https://hostname/instance/api/FHIR/R4/Flag/enpfRSY57pmn6IxrZD.2eDUnlKr9n60ip6xWB55Ihlws3",
      "resource": {
        "resourceType": "Flag",
        "id": "enpfRSY57pmn6IxrZD.2eDUnlKr9n60ip6xWB55Ihlws3",
        "extension": [
          {
            "valueReference": {
              "reference": "Condition/emYq8NzSEaKrCZtPWXa-GIioNncX-zeoLZG4utC3iD5w3",
              "display": "Nauseous"
            },
            "url": "http://hl7.org/fhir/StructureDefinition/flag-detail"
          }
        ],
        "status": "active",
        "category": [
          {
            "coding": [
              {
```

```
              "system": "http://open.epic.com/FHIR/StructureDefinition/flag-category",
              "code": "health-concern",
              "display": "Health Concern Flag"
            },
            {
              "system": "http://loinc.org",
              "code": "75323-6",
              "display": "condition"
            }
          ],
          "text": "Health Concern Flag"
        }
      ],
      "code": {
        "coding": [
          {
            "system": "http://hl7.org/fhir/sid/icd-10-cm",
            "code": "R11.0",
            "display": "Nausea"
          },
          {
            "system": "http://snomed.info/sct",
            "code": "422587007",
            "display": "Nausea"
          }
        ],
        "text": "Nauseous"
      },
      "subject": {
        "reference": "Patient/euI3cZMuLFNu6kO9mg4ScEg3",
        "display": "Test, Patient"
      },
      "period": {
        "start": "2023-06-28"
      }
    },
    "search": {
      "mode": "match"
    }
  }
 ]
}
```


Change log

Epic Version	Description	Type
November 2022	Code systems for the following code sets now return a URL in place of an OID: ActCode, ICD-10-CM, ICD-10-CA, ICD-10-NL, SNOMED System URL OID ActCode STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode 2.16.840.1.113883.5.4 ICD-10-CM http://hl7.org/fhir/sid/icd-10-cm 2.16.840.1.113883.6.90 ICD-10-CA http://hl7.org/fhir/sid/icd-10-ca 2.16.840.1.113883.6.94 ICD-10-NL http://hl7.org/fhir/sid/icd-10-nl 2.16.840.1.113883.6.3.2 SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96	Breaking change

System	URL	OID
ActCode	STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode	2.16.840.1.113883.5.4
ICD-10-CM	http://hl7.org/fhir/sid/icd-10-cm	2.16.840.1.113883.6.90
ICD-10-CA	http://hl7.org/fhir/sid/icd-10-ca	2.16.840.1.113883.6.94
ICD-10-NL	http://hl7.org/fhir/sid/icd-10-nl	2.16.840.1.113883.6.3.2
SNOMED	http://snomed.info/sct	2.16.840.1.113883.6.96

10.1.30 Flag.Search (Infection) (R4)

File: site/apis/10-Clinical-General/1-R4/10202-Flag-Search-Infection-R4.html	
Epic API ID	10202
FHIR version	R4
Categories	Clinical General
API type	Industry-Standard
USCDI	No
Available in inpatient and other contexts	Yes
HTTP method	GET
URL template	/api/FHIR/R4/Flag?category={category}&patient={patient}&status={status}&subject={subject}
Supports OAuth 2.0	True
Supported user types	Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation	False
Source	https://fhir.epic.com/Specifications?api=10202

Description

This web service retrieves high-level infection information from a patient's chart. An infection might be a laboratory-confirmed infection or colonization, such as influenza or methicillin-resistant Staphylococcus aureus, or a placeholder for a suspected infection, such as respiratory rule-out or suspected COVID-19.

Like the Condition (Infection) web service, this web service evaluates infection (INF) records, but it returns only a brief overview of key information about those records, such as whether they are active and when they were resolved. If you need to return more detailed information, use the Condition (Infection) web service instead.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
category | String | True | | | Always "infection" for this web service.
encounter | String | True | | | A reference to the Encounter resource. Used only for encounter-level infections.
patient | String | Either patient or subject must be provided. If both are provided, they must match. | | | Reference to a Patient resource the Flag is for.
status | String | True
subject | String | Either patient or subject must be provided. If both are provided, they must match. | | | Reference to a Patient resource the Flag is for.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
author | String | True | | | Unsupported
date | String | True | | | The period during which the infection was marked as active.
identifier | String | True | | | Unsupported

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | BackboneElement | False | Yes | | An array of entries included in this bundle.
Bundle.entry.resource | ResourceContainer | False | | | A container for a single FHIR resource.
Bundle.entry.resource.Flag | Flag | False | | | A single instance of a Flag resource.
Bundle.entry.resource.Flag.category | CodeableConcept | True | Yes | | The Epic category ID for the infection, labeled “infection,” and the LOINC code for the infection, labeled “condition.”
Bundle.entry.resource.Flag.code | CodeableConcept | False | | | The diagnosis code for the infection.
Bundle.entry.resource.Flag.encounter | Reference | True | | | A reference to the Encounter resource. Used only for encounter-level infections.
Bundle.entry.resource.Flag.extension (also: flag-detail) | Extension | True | Yes | | A reference to the Condition (Infection) resource.
Bundle.entry.resource.Flag.id | String | False | | | The Flag FHIR ID.
Bundle.entry.resource.Flag.period | Period | True | | | The period during which the infection was marked as active.
Bundle.entry.resource.Flag.period.start | String | True | | | Date when the infection was added.
Bundle.entry.resource.Flag.status | String | False | | | The status of the infection: active or inactive.
Bundle.entry.resource.Flag.subject | Reference | False | | | A reference to the Patient resource.

Sample request

```
https://hostname/instance/api/FHIR/R4/Flag?patient=exnWmPch0AyoUp7ZBplVf.Q3&category=infection
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 3,
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Flag?patient=exnWmPch0AyoUp7ZBplVf.Q3&category=infection"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url": "https://hostname/instance/api/FHIR/R4/Flag/e0k81lAJpUaaBWo6PSvirZJ8SPlUWfSwHCDA53Ab9-4A3"
        }
      ],
      "fullUrl": "https://hostname/instance/api/FHIR/R4/Flag/e0k81lAJpUaaBWo6PSvirZJ8SPlUWfSwHCDA53Ab9-4A3",
      "resource": {
        "resourceType": "Flag",
        "id": "e0k81lAJpUaaBWo6PSvirZJ8SPlUWfSwHCDA53Ab9-4A3",
        "extension": [
          {
            "valueReference": {
              "reference": "Condition/e73ENgCh7IMBe3fbv4ds.eQ3",
              "display": "ICON EL 2"
            },
            "url": "http://hl7.org/fhir/StructureDefinition/flag-detail"
          }
        ],
        "status": "entered-in-error",
        "category": [
          {
            "coding": [
              {
                "system": "http://open.epic.com/FHIR/StructureDefinition/flag-category",
```

```
            "code": "infection",
            "display": "Infection Flag"
          },
          {
            "system": "http://loinc.org",
            "code": "75323-6",
            "display": "condition"
          }
        ],
        "text": "Infection Flag"
      }
    ],
    "code": {
      "coding": [
        {
          "system": "urn:oid:1.2.840.114350.1.13.861.1.7.4.698084.10375",
          "code": "66002",
          "display": "ICON EL 2"
        }
      ],
      "text": "ICON EL 2"
    },
    "subject": {
      "reference": "Patient/exnWmPch0AyoUp7ZBplVf.Q3",
      "display": "DXL, Contacts"
    },
    "period": {
      "start": "2021-09-08",
      "end": "2021-09-09T01:12:16Z"
    },
    "encounter": {
      "reference": "Encounter/eVCZHFbcaQME1z8q58dVMPrvvP82VAgtKbe4oVzwUB4A3",
      "identifier": {
        "use": "usual",
        "system": "urn:oid:1.2.840.114350.1.13.861.1.7.3.698084.8",
        "value": "10001585941"
      },
      "display": "History"
    }
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Flag/eG0-zfyMqT1h29FBrsnrhdaNBCP-ydWPSUr44UbL4qfY3"
    }
  ],
  "fullUrl": "https://hostname/instance/api/FHIR/R4/Flag/eG0-zfyMqT1h29FBrsnrhdaNBCP-ydWPSUr44UbL4qfY3",
  "resource": {
    "resourceType": "Flag",
    "id": "eG0-zfyMqT1h29FBrsnrhdaNBCP-ydWPSUr44UbL4qfY3",
    "extension": [
      {
        "valueReference": {
          "reference": "Condition/elWQkOUZuvIYmpvkxzMAUXw3",
          "display": "ICON EL 1"
        },
        "url": "http://hl7.org/fhir/StructureDefinition/flag-detail"
      }
    ],
    "status": "active",
    "category": [
      {
        "coding": [
          {
            "system": "http://open.epic.com/FHIR/StructureDefinition/flag-category",
            "code": "infection",
            "display": "Infection Flag"
          },
          {
            "system": "http://loinc.org",
            "code": "75323-6",
            "display": "condition"
          }
        ],
        "text": "Infection Flag"
      }
    ],
    "code": {
      "coding": [
        {
          "system": "urn:oid:1.2.840.114350.1.13.861.1.7.4.698084.10375",
          "code": "66001",
          "display": "ICON EL 1"
        },
```

```
        {
          "system": "http://snomed.info/sct",
          "code": "73430006",
          "display": "Sleep Apnea"
        }
      ],
      "text": "ICON EL 1"
    },
    "subject": {
      "reference": "Patient/exnWmPch0AyoUp7ZBplVf.Q3",
      "display": "DXL, Contacts"
    },
    "period": {
      "start": "2021-09-08"
    },
    "encounter": {
      "reference": "Encounter/eyAizba4J4yI-R43baHTOg7nY3a3mztFu57ZsgPsfmGI3",
      "identifier": {
        "use": "usual",
        "system": "urn:oid:1.2.840.114350.1.13.861.1.7.3.698084.8",
        "value": "10001459125"
      },
      "display": "Office Visit"
    }
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Flag/eFbv9LL28ikxKaUddrVLr-IVq.wf.k7BR.QGEqq8hfEM3"
    }
  ],
  "fullUrl": "https://hostname/instance/api/FHIR/R4/Flag/eFbv9LL28ikxKaUddrVLr-IVq.wf.k7BR.QGEqq8hfEM3",
  "resource": {
    "resourceType": "Flag",
    "id": "eFbv9LL28ikxKaUddrVLr-IVq.wf.k7BR.QGEqq8hfEM3",
    "extension": [
      {
        "valueReference": {
          "reference": "Condition/erEuPdnY2OKSUOSC-Y9M6DQ3",
          "display": "Influenza"
        },
        "url": "http://hl7.org/fhir/StructureDefinition/flag-detail"
      }
    ],
    "status": "active",
    "category": [
      {
        "coding": [
          {
            "system": "http://open.epic.com/FHIR/StructureDefinition/flag-category",
            "code": "infection",
            "display": "Infection Flag"
          },
          {
            "system": "http://loinc.org",
            "code": "75323-6",
            "display": "condition"
          }
        ],
        "text": "Infection Flag"
      }
    ],
    "code": {
      "coding": [
        {
          "system": "urn:oid:1.2.840.114350.1.13.861.1.7.4.698084.10375",
          "code": "3",
          "display": "Influenza"
        }
      ],
      "text": "Influenza"
    },
    "subject": {
      "reference": "Patient/exnWmPch0AyoUp7ZBplVf.Q3",
      "display": "DXL, Contacts"
    },
    "period": {
      "start": "2021-07-27"
    }
  },
  "search": {
    "mode": "match"
  }
}
```

```
  ]
}
```

10.1.31 List.Read (Allergies) (R4)

File: site/apis/10-Clinical-General/1-R4/10146-List-Read-Allergies-R4.html
Epic API ID | 10146
FHIR version | R4
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | August 2021
HTTP method | GET
URL template | /api/FHIR/R4/List/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10146

Description

The List (Allergies) API is used to represent the current allergies in a patient's Allergy list in their chart.

Details about each allergy can be found with a subsequent Read request to the AllergyIntolerance resource.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | String | False | | | The List FHIR ID

Response elements

Name | Type | Optional | Array | Cardinality | Description
List | List | False | | | An instance of the R4 List FHIR resource.
List.code | CodeableConcept | False | | | The code representing the type of list searched. Always "allergies".
List.emptyReason | CodeableConcept | True | | | If the patient has no allergies documented, the API returns an empty reason of "notstarted".
List.entry | Entry | False | Yes | | A single entry in the list.
List.entry.deleted | Boolean | True | | | Code representing whether the allergy is deleted.
List.entry.item | Reference | False | | | List of FHIR IDs returned by the AllergyIntolerance resource search, excluding external allergies.
List.id | String | False | | | FHIR identifier for this List resource instance.
List.mode | String | False | | | The processing mode that applies to this list. Always "working".
List.status | String | False | | | Status of the medication list. Always "current" because each list represents only the patient's current allergies.
List.subject | Reference | False | | | Reference to the Patient resource with whom the List is associated.
List.title | String | False | | | The title representing this entity. Always "Allergy List".

Sample request

```
https://hostname/instance/api/FHIR/R4/List/ehyYsfQIl0V4MLni61iNl7Y1vM.JjR7Z4CWTKvcUmResxzros6bbLu2cTM2.T7Cpn3
```

Sample response

```
{
  "resourceType": "List",
  "id": "ehyYsfQIl0V4MLni61iNl7Y1vM.JjR7Z4CWTKvcUmResxzros6bbLu2cTM2.T7Cpn3",
  "status": "current",
  "mode": "working",
  "title": "Allergy List",
  "code": {
    "coding": [
      {
        "system": "http://open.epic.com/FHIR/StructureDefinition/list-type",
        "code": "allergies"
      }
    ]
  },
  "subject": {
    "reference": "Patient/ej1cY-oWY4m.DqW8JzxfpnA3",
    "display": "Everett, Interrobang"
  },
  "entry": [
    {
      "item": {
        "reference": "AllergyIntolerance/etjkduCPTph32r0ps6dnsV8IttW7J7C9Jr8gMkuzxk0c3",
        "display": "SACCHARIN"
      }
    },
    {
      "item": {
```

```
        "reference": "AllergyIntolerance/eCOp53AN6NdmycU7JhWYBgjk9m2U1TEbc9V.sMGFkwek3",
        "display": "IBUPROFEN"
      }
    },
    {
      "item": {
        "reference": "AllergyIntolerance/eZSyM9qErwsMxO4SjlHRY2fn9bpopgf7HCzzeWKUutyo3",
        "display": "ACETAMINOPHEN"
      }
    },
    {
      "item": {
        "reference": "AllergyIntolerance/el3HQNl0791lTDNKvlYUciGnUS5K1F3vwB1zrBtj49Uo3",
        "display": "PENICILLINS"
      }
    }
  ]
}
```


10.1.32 List.Read (Family History) (R4)

File: site/apis/10-Clinical-General/1-R4/10150-List-Read-Family-History-R4.html
Epic API ID | 10150
FHIR version | R4
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | August 2021
HTTP method | GET
URL template | /api/FHIR/R4/List/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10150

Description

The List (Family History) API defines the list family members that have been documented in a patient's family history, as well as patient-level information that is documented alongside the family history, including adoptive parents, fertility status, and notes about the family history as a whole, referred to as pedigree annotations. Each family member is returned as a FamilyMemberHistory resource.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | String | False | | | The List FHIR ID

Response elements

Name | Type | Optional | Array | Cardinality | Description
List | List | False | | | An instance of the R4 List FHIR resource.
List.code | CodeableConcept | False | | | Code "pedigree-list" in Epic's "/list-type" definition
List.entry | Entry | False | Yes | | FamilyMemberHistory references for patient's family history.
List.id | String | False | | | FHIR identifier for this List resource instance.
List.mode | String | False | | | Always "working"
List.status | String | False | | | Always "current"
List.title | String | False | | | Always "Pedigree List"

Sample request

```
https://hostname/instance/api/FHIR/R4/List/epz8xX7jOTqs7.On.3GKrfupkrN3CHnG7JlM3DZRNJgQ3
```

Sample response

```
<List xmlns="http://hl7.org/fhir">
  <id value="epz8xX7jOTqs7.On.3GKrfupkrN3CHnG7JlM3DZRNJgQ3"/>
  <status value="current"/>
  <mode value="working"/>
  <title value="Pedigree List"/>
  <code>
    <coding>
      <system value="http://open.epic.com/FHIR/StructureDefinition/list-type"/>
      <code value="pedigree-list"/>
    </coding>
  </code>
  <subject>
    <reference value="Patient/eV6Ofju2.rCOp6TCunVM9Rg3"/>
    <display value="Smith, Sally"/>
```

```
  </subject>
  <entry>
    <item>
      <reference value="FamilyMemberHistory/ePM.Qux3IY3tsLDfCmpxnDhhUJ4WWDI.DRtTk-LVsHD03"/>
      <display value="Paula (Mother)"/>
    </item>
  </entry>
  <entry>
    <item>
      <reference value="FamilyMemberHistory/ePM.Qux3IY3tsLDfCmpxnDiFB6Z7eKudu2cH4dK2ZLDg3"/>
      <display value="Joe (Father)"/>
    </item>
  </entry>
  <entry>
    <item>
      <reference value="FamilyMemberHistory/ePM.Qux3IY3tsLDfCmpxnDhOstn1GTFlXhxiGglTaVdA3"/>
      <display value="Ellen (Other)"/>
    </item>
  </entry>
  <entry>
    <item>
      <reference value="FamilyMemberHistory/ePM.Qux3IY3tsLDfCmpxnDreCpIDxiDYvSwSdd2GZMl83"/>
      <display value="Lou (Other)"/>
    </item>
  </entry>
  <entry>
    <item>
      <reference value="FamilyMemberHistory/ePM.Qux3IY3tsLDfCmpxnDveFxWGXW2.MzRylZeKEzKw3"/>
      <display value="Half-sister"/>
    </item>
  </entry>
  <entry>
    <item>
      <reference value="FamilyMemberHistory/ePM.Qux3IY3tsLDfCmpxnDsX.rHMIGLfLICc9e8rjlNE3"/>
      <display value="Adoptive mother"/>
    </item>
  </entry>
  <entry>
    <item>
      <reference value="FamilyMemberHistory/ePM.Qux3IY3tsLDfCmpxnDktL26zVbCVSewma1pEizDg3"/>
      <display value="Adoptive father"/>
    </item>
  </entry>
</List>
```

10.1.33 List.Read (Hospital Problems) (R4)

File: site/apis/10-Clinical-General/1-R4/10148-List-Read-Hospital-Problems-R4.html
Epic API ID | 10148
FHIR version | R4
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | August 2021
HTTP method | GET
URL template | /api/FHIR/R4/List/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10148

Description

The List (Hospital Problem) API is used to represent the list of problems addressed during the associated hospital encounter.

It includes problems that are currently marked as hospital problems for that encounter, and it also includes problems that were previously marked as hospital problems for that encounter but that are no longer marked as such. Each entry includes a reference to the Condition (Problem List) resource instance for the problem, details about whether the problem is currently marked as a hospital problem, details about whether the problem is the principal problem for the encounter, and the problem's present on admission status.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | String | False | | | The List FHIR ID

Response elements

Name | Type | Optional | Array | Cardinality | Description
List | List | False | | | An instance of the R4 List FHIR resource.

List.code | CodeableConcept | False | | | Code “hospital-problem” in Epic’s “/list-type” definition.

List.encounter | Reference | False | | | The encounter that this hospital problem list is for

List.entry | Entry | True | Yes | | There is one entry for each hospital problem on the encounter’s hospital problem list.

List.entry.deleted | Boolean | True | | | If true, this problem is no longer marked as a hospital problem for this encounter, but it had been previously. This is omitted if the problem is currently marked as a hospital problem.

List.entry.extension (also: hospital-problem-present-on-admission) | Extension | False | Yes | | The Present on Admission status for this problem on this encounter. Default values include: Yes No Unknown Clinically Undetermined Exempt from POA reporting Epic organizations may define additional custom values.

List.entry.extension (also: is-principal-hospital-problem) | Extension | False | Yes | | If true, this problem is the principal problem for this hospital encounter.

List.entry.item | Reference | False | | | Reference to the Condition (Problem List Item) resource for this problem list item. The display name is set to the display name of the problem’s diagnosis.

List.mode | String | False | | | Always “working”

List.status | String | False | | | Always “current”

List.subject | Reference | False | | | The patient that this hospital problem list is for.

List.title | String | False | | | Always “Hospital Problem List”

Sample request

```
https://hostname/instance/api/FHIR/R4/List/eWmQViNOBS-
    laHgREP6mMLg2goK4z8icONOq6uQX9ulM7OKnxp8CJcKIVQRCWDcg3BhOJ2RFgetUI6l8CMmDZfQ3
```

Sample response


```
<List xmlns="http://hl7.org/fhir">
  <id value="eWmQViNOBS-laHgREP6mMLg2goK4z8icONOq6uQX9ulM7OKnxp8CJcKIVQRCWDcg3BhOJ2RFgetUI6l8CMmDZfQ3"/>
  <status value="current"/>
  <mode value="working"/>
  <title value="Hospital Problem List"/>
  <code>
    <coding>
      <system value="http://open.epic.com/FHIR/StructureDefinition/list-type"/>
      <code value="hospital-problems"/>
      <display value="Hospital Problem List"/>
    </coding>
    <text value="Hospital Problem List"/>
  </code>
  <subject>
    <reference value="Patient/emFzuaGzscRApD-FRiPcjzg3"/>
    <display value="Johnsen, Bruce Banner"/>
  </subject>
  <encounter>
    <reference value="Encounter/e7du6YOFXxZOXpdLadukGrrZe.p0.YpT8Xz.ywOuTJh03"/>
  </encounter>
  <entry>
    <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/is-principal-hospital-problem">
      <valueBoolean value="false"/>
    </extension>
    <deleted value="true"/>
    <item>
      <reference value="Condition/emmYRtD0ekHjEJHnndSisgILtK.-JGUaZdpiJsF9a-eg3"/>
      <display value="Acute bronchitis"/>
    </item>
  </entry>
  <entry>
    <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/is-principal-hospital-problem">
      <valueBoolean value="false"/>
    </extension>
    <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/hospital-problem-present-on-admission">
      <valueCoding>
        <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.10.726582.611"/>
        <code value="4"/>
        <display value="Clinically Undetermined"/>
      </valueCoding>
    </extension>
    <item>
      <reference value="Condition/e2zEi7GDsij3PMcZQHQGB8SUuSHbw3s8wVSveexq57eQ3"/>
      <display value="Acute recurrent streptococcal tonsillitis"/>
    </item>
  </entry>
  <entry>
    <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/is-principal-hospital-problem">
      <valueBoolean value="true"/>
    </extension>
    <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/hospital-problem-present-on-admission">
      <valueCoding>
        <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.10.726582.611"/>
        <code value="1"/>
        <display value="Yes"/>
      </valueCoding>
    </extension>
    <item>
      <reference value="Condition/eiKzn5BBwMqGuXQ2uswjZgk8emitbtKUX0DjzI0PjPko3"/>
      <display value="Abdominal pain"/>
    </item>
```

```
  </entry>
  <entry>
    <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/is-principal-hospital-problem">
      <valueBoolean value="false"/>
    </extension>
    <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/hospital-problem-present-on-admission">
      <valueCoding>
        <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.10.726582.611"/>
        <code value="1"/>
        <display value="Yes"/>
      </valueCoding>
    </extension>
    <item>
      <reference value="Condition/eHSNwOlK3jOy1mvo.wyJoCoFtlF.Cjv0jIQPzhNzUJ143"/>
      <display value="Open wound"/>
    </item>
  </entry>
  <entry>
    <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/is-principal-hospital-problem">
      <valueBoolean value="false"/>
    </extension>
    <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/hospital-problem-present-on-admission">
      <valueCoding>
        <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.10.726582.611"/>
        <code value="1"/>
        <display value="Yes"/>
      </valueCoding>
    </extension>
    <item>
      <reference value="Condition/ealoQatr7JcmEDBoZSgxWNl3r1veaBoRuNC.VP1sCF7M3"/>
      <display value="Effusion of right shoulder joint"/>
    </item>
  </entry>
  <entry>
    <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/is-principal-hospital-problem">
      <valueBoolean value="false"/>
    </extension>
    <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/hospital-problem-present-on-admission">
      <valueCoding>
        <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.10.726582.611"/>
        <code value="1"/>
        <display value="Yes"/>
      </valueCoding>
    </extension>
    <item>
      <reference value="Condition/e2MMZPHo548dhjYMKSjl9..B20ZrdDAozLznKqEGNQXk3"/>
      <display value="Sprain of left wrist"/>
    </item>
  </entry>
  <entry>
    <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/is-principal-hospital-problem">
      <valueBoolean value="false"/>
    </extension>
    <deleted value="true"/>
    <item>
      <reference value="Condition/eSAzHvrKQ0e1-Ekv.RDgXpy3mxbeubXNxsbta83LTXdY3"/>
      <display value="Difficulty controlling anger"/>
    </item>
  </entry>
  <entry>
    <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/is-principal-hospital-problem">
      <valueBoolean value="false"/>
    </extension>
    <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/hospital-problem-present-on-admission">
      <valueCoding>
        <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.10.726582.611"/>
        <code value="1"/>
        <display value="Yes"/>
      </valueCoding>
    </extension>
    <item>
      <reference value="Condition/e-UsEpEy3QQEKwOdfjCO0fzOuZEw-i4Af5BjQZ1IUfEo3"/>
      <display value="Burns involving less than 10% of body surface"/>
    </item>
  </entry>
</List>
```

10.1.34 List.Read (Problems) (R4)

File: site/apis/10-Clinical-General/1-R4/10144-List-Read-Problems-R4.html
Epic API ID | 10144
FHIR version | R4
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes

ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | August 2021
HTTP method | GET
URL template | /api/FHIR/R4/List/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10144

Description

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | String | False | | | The List FHIR ID

Response elements

Name | Type | Optional | Array | Cardinality | Description
List | List | False | | | An instance of the R4 List FHIR resource.
List.code | CodeableConcept | False | | | The code representing the type of list searched. Always " problems".
List.emptyReason | CodeableConcept | True | | | If the patient has no problems documented, the API returns an emptyReason of “notstarted”.
List.entry | Entry | False | Yes | | Each entry is a BackboneElement with the "item" property populated with a reference to the Condition resource and the "display" property populated with the problem name.
List.entry.item | Reference | False | | | List of FHIR IDs returned by the Condition (Problems) resource search, excluding external and deleted problems.
List.id | String | False | | | The encoded list ID.
List.mode | String | False | | | The processing mode that applies to this list. Always "working".
List.status | String | False | | | Status of the problem list. Always "current" because each list represents only the patient’s current problems.
List.subject | Reference | False | | | Reference to the Patient resource with which the List is associated.
List.title | String | False | | | The title representing this entity. Always “Problem List”.

Sample request

```
https://hostname/instance/api/FHIR/R4/List/eCyYT.pbw-dM8y2l-n4jHXwW3ec5p5L.WVuJc8oZ05J4ZpGWHv3N.zVVB48q9PYc83
```

Sample response

```
<List xmlns="http://hl7.org/fhir">
  <id value="eCyYT.pbw-dM8y2l-n4jHXwW3ec5p5L.WVuJc8oZ05J4ZpGWHv3N.zVVB48q9PYc83"/>
  <status value="current"/>
  <mode value="working"/>
  <title value="Problem List"/>
  <code>
    <coding>
      <system value="http://open.epic.com/FHIR/StructureDefinition/list-type"/>
      <code value="problems"/>
    </coding>
  </code>
  <subject>
    <reference value="Patient/ewHtJxSZtNcyaWTgdXk-d.w3"/>
    <display value="LastName, FirstName"/>
  </subject>
  <entry>
    <item>
      <reference value="Condition/e8fbejXht0MGZ4CSBur8j129s-6iyIrTb81ObB038JHk3"/>
      <display value="Diab d/t undrl cond w hyprosm w/o nonket hyprgly-hypros coma"/>
    </item>
  </entry>
  <entry>
    <item>
      <reference value="Condition/enTH1oSb0pSfafmTOaJbUK-bTm-8LOuX-c29KU9FZCbc3"/>
      <display value="Chronic cough"/>
    </item>
  </entry>
  <entry>
    <item>
      <reference value="Condition/eBT3MlFLZZH64HMMlucSk8vYv1GT0F32LioVHJGJUNAQ3"/>
      <display value="5-hydroxytryptamine-3 receptor antagonist overdose of undetermined intent"/>
    </item>
  </entry>
  <entry>
    <item>
      <reference value="Condition/ee5NJyzEOnrF3-M8WEYBnypp5EH.8KUehgNUwU0qqeMI3"/>
      <display value="AEW Term F"/>
    </item>
  </entry>
  <entry>
    <item>
      <reference value="Condition/eD7kRt4TBMComOcXjT43OSnhHt2eyPt9QQ4pbxHyaD2k3"/>
      <display value="Granuloma inguinale in male"/>
    </item>
```

```
  </entry>
  <entry>
    <item>
      <reference value="Condition/eNBMArwECYMFrWa8QJY6BsTvq2jfNPDcwgREwtur39jk3"/>
      <display value="Yunis-Varon Syndrome"/>
    </item>
  </entry>
  <entry>
    <item>
      <reference value="Condition/ek1fpHqucYtlTL3wZuNGKRwmJQMvyJGXlBmDY.Az2bPs3"/>
      <display value="Fecal impaction"/>
    </item>
  </entry>
  <entry>
    <item>
      <reference value="Condition/erB1I.nPTR-b3fgKgBhM908EHVZ69hrV-mntznjMDvBM3"/>
      <display value="Polyalgia"/>
    </item>
  </entry>
</List>
```

10.1.35 List.Search (Allergies) (R4)

File: site/apis/10-Clinical-General/1-R4/10147-List-Search-Allergies-R4.html
Epic API ID | 10147
FHIR version | R4
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | August 2021
HTTP method | GET;POST
URL template | /api/FHIR/R4/List?code={code}&patient={patient}&subject={subject}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10147

Description

The List (Allergies) API is used to represent the current allergies in a patient's Allergy list in their chart.

This API can also be used to review the history of the patient's allergy list by including AuditEvent data using the _revInclude parameter. To obtain more details about a specific allergy included in the patient's allergy list, the AllergyIntolerance resource can be used.

Remarks

AuditEvent

The following details are returned as part of the AuditEvent structure if AuditEvent data is included with this resource. For more information about AuditEvent, refer to the FHIR Search Parameters article in the Documentation section of this site.

Name | Description | Is Optional | Is Array
type (coding) | Hard-coded value "Review History" | false | false
recorded (instant) | The recorded instant | false | false
agent (backbone) | Contains requestor, who, and role | true | false
agent.Requestor (Boolean) | | true | false
agent.Who (reference) | | true | false
agent.Role (coding) | Hard-coded value "Reviewer" | true | false
source (backbone) | Contains observer | true | false
source.Observer (reference) | Who reviewed the information | true | false
entity (backbone) | Contains what and detail | true | false
entity.what (reference) | The List | false | false
entity.detail (backbone) | Contains type and valueString | true | false
entity.detail.type (Boolean) | Key/value pair list | false | false
entity.detail.valueString (reference) | Key/value pair list | false | false

Request parameters

Name | Type | Optional | Array | Cardinality | Description
code | String | False | | | Must be "allergies"
patient | String | Either patient or subject must be specified. | | | The patient with whom the list is associated.
subject | String | Either patient or subject must be specified. | | | The patient with whom the list is associated.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description

date | String | True | | | Unsupported
empty-reason | String | True | | | If the patient has no allergies documented, the API returns an empty reason of “notstarted”.
encounter | String | True | | | Unsupported
identifier | String | True | | | Unsupported
item | String | True | | | AllergyIntolerance FHIR ID
notes | String | True | | | Unsupported
source | String | True | | | Unsupported
status | String | True | | | Status of the allergy list. Always "current" because each list represents only the patient’s current allergies.
title | String | True | | | The title representing this entity. Always “Allergy List”.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | BackboneElement | False | Yes | | An array of entries included in this bundle.
Bundle.entry.resource | ResourceContainer | False | | | A container for a single FHIR resource.
Bundle.entry.resource.List | List | False | | | An instance of the R4 List FHIR resource.
Bundle.entry.resource.List.code | CodeableConcept | False | | | The code representing the type of list searched. Always "allergies".
Bundle.entry.resource.List.emptyReason | CodeableConcept | True | | | If the patient has no allergies documented, the API returns an empty reason of “notstarted”.
Bundle.entry.resource.List.entry | Entry | False | Yes | | A single entry in the list.
Bundle.entry.resource.List.entry.deleted | Boolean | True | | | Code representing whether the allergy is deleted.
Bundle.entry.resource.List.entry.item | Reference | False | | | List of FHIR IDs returned by the AllergyIntolerance resource search, excluding external allergies.
Bundle.entry.resource.List.id | String | False | | | FHIR identifier for this List resource instance.
Bundle.entry.resource.List.mode | String | False | | | The processing mode that applies to this list. Always "working".
Bundle.entry.resource.List.status | String | False | | | Status of the allergy list. Always "current" because each list represents only the patient’s current allergies.
Bundle.entry.resource.List.subject | Reference | False | | | Reference to the Patient resource with whom the List is associated.
Bundle.entry.resource.List.title | String | False | | | The title representing this entity. Always “Allergy List”.

Sample request

```
https://hostname/instance/api/FHIR/R4/List?patient=ej1cY-oWY4m.DqW8JzxfpnA3&code=allergies&_revinclude=AuditEvent:entity
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 1,
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/List?code=allergies&patient=ej1cY-
    oWY4m.DqW8JzxfpnA3&_revinclude=AuditEvent:entity"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url":
    "https://hostname/instance/api/FHIR/R4/List/ehyYsfQIl0V4MLni61iNl7Y1vM.JjR7Z4CWTKvcUmResxzros6bbLu2cTM2.T7Cpn3"
        }
      ],
      "fullUrl":
    "https://hostname/instance/api/FHIR/R4/List/ehyYsfQIl0V4MLni61iNl7Y1vM.JjR7Z4CWTKvcUmResxzros6bbLu2cTM2.T7Cpn3",
      "resource": {
        "resourceType": "List",
        "id": "ehyYsfQIl0V4MLni61iNl7Y1vM.JjR7Z4CWTKvcUmResxzros6bbLu2cTM2.T7Cpn3",
        "status": "current",
        "mode": "working",
        "title": "Allergy List",
        "code": {
          "coding": [
            {
              "system": "http://open.epic.com/FHIR/StructureDefinition/list-type",
              "code": "allergies"
            }
          ]
        },
        "subject": {
          "reference": "Patient/ej1cY-oWY4m.DqW8JzxfpnA3",
          "display": "Everett, Interrobang"
        },
        "entry": [
          {
            "item": {
              "reference": "AllergyIntolerance/etjkduCPTph32r0ps6dnsV8IttW7J7C9Jr8gMkuzxk0c3",
              "display": "SACCHARIN"
```

```
        }
      },
      {
        "item": {
          "reference": "AllergyIntolerance/eCOp53AN6NdmycU7JhWYBgjk9m2U1TEbc9V.sMGFkwek3",
          "display": "IBUPROFEN"
        }
      },
      {
        "item": {
          "reference": "AllergyIntolerance/eZSyM9qErwsMxO4SjlHRY2fn9bpopgf7HCzzeWKUutyo3",
          "display": "ACETAMINOPHEN"
        }
      },
      {
        "item": {
          "reference": "AllergyIntolerance/el3HQNl0791lTDNKvlYUciGnUS5K1F3vwB1zrBtj49Uo3",
          "display": "PENICILLINS"
        }
      }
    ]
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/AuditEvent/enAUSM6sfOxm5qtTslYGVhjv6DFEdjs0u6kRaBPWRN8CIsgQjdYKO
ByIm9Of1mKa2JDMkDih5gJPS6csu-1nOTJiv8.Yp8D6bqcu3yweGIQng3IH5n8IBcc84l1JTwnZj3"
    }
  ],
  "fullUrl": "https://hostname/instance/api/FHIR/R4/AuditEvent/enAUSM6sfOxm5qtTslYGVhjv6DFEdjs0u6kRaBPWRN8CIsgQjdYKO
ByIm9Of1mKa2JDMkDih5gJPS6csu-1nOTJiv8.Yp8D6bqcu3yweGIQng3IH5n8IBcc84l1JTwnZj3",
  "resource": {
    "resourceType": "AuditEvent",
    "id": "enAUSM6sfOxm5qtTslYGVhjv6DFEdjs0u6kRaBPWRN8CIsgQjdYKOByIm9Of1mKa2JDMkDih5gJPS6csu-
1nOTJiv8.Yp8D6bqcu3yweGIQng3IH5n8IBcc84l1JTwnZj3",
    "type": {
      "system": "http://open.epic.com/FHIR/StructureDefinition/audit-event-type",
      "code": "Review History",
      "display": "Review History"
    },
    "recorded": "2021-06-11T15:44:20Z",
    "agent": [
      {
        "role": [
          {
            "coding": [
              {
                "system": "http://open.epic.com/FHIR/StructureDefinition/audit-event-agent-role",
                "code": "Reviewer",
                "display": "Reviewer"
              }
            ]
          }
        ],
        "who": {
          "reference": "Practitioner/e3FxRT6I36jzw8BMhICeeHw3",
          "display": "Robert Sutton Persona Mr Md, MD"
        },
        "requestor": true
      }
    ],
    "source": {
      "observer": {
        "reference": "Practitioner/e3FxRT6I36jzw8BMhICeeHw3",
        "display": "Robert Sutton Persona Mr Md, MD"
      }
    },
    "entity": [
      {
        "what": {
          "reference": "List/ehyYsfQIl0V4MLni61iNl7Y1vM.JjR7Z4CWTKvcUmResxzros6bbLu2cTM2.T7Cpn3"
        },
        "detail": [
          {
            "type": "Review Status",
            "valueString": "Reviewed"
          }
        ]
      }
    ]
  },
  "search": {
    "mode": "include"
  }
```

```
    },
    {
      "link": [
        {
          "relation": "self",
          "url": "https://hostname/instance/api/FHIR/R4/AuditEvent/enAUSM6sfOxm5qtTslYGVhjv6DFEdjs0u6kRaBPWRN8C20H8mLqt4
    Q0KakNFtVxtpoH0JoAID3wPOGbrYlHM-sFzOem5djbhtt0zuU.bEFPHuMQEvBdSuGdo-bO9XsCzP3"
        }
      ],
      "fullUrl": "https://hostname/instance/api/FHIR/R4/AuditEvent/enAUSM6sfOxm5qtTslYGVhjv6DFEdjs0u6kRaBPWRN8C20H8mLqt4
    Q0KakNFtVxtpoH0JoAID3wPOGbrYlHM-sFzOem5djbhtt0zuU.bEFPHuMQEvBdSuGdo-bO9XsCzP3",
      "resource": {
        "resourceType": "AuditEvent",
        "id": "enAUSM6sfOxm5qtTslYGVhjv6DFEdjs0u6kRaBPWRN8C20H8mLqt4Q0KakNFtVxtpoH0JoAID3wPOGbrYlHM-
    sFzOem5djbhtt0zuU.bEFPHuMQEvBdSuGdo-bO9XsCzP3",
        "type": {
          "system": "http://open.epic.com/FHIR/StructureDefinition/audit-event-type",
          "code": "Review History",
          "display": "Review History"
        },
        "recorded": "2021-04-26T21:47:00Z",
        "agent": [
          {
            "role": [
              {
                "coding": [
                  {
                    "system": "http://open.epic.com/FHIR/StructureDefinition/audit-event-agent-role",
                    "code": "Reviewer",
                    "display": "Reviewer"
                  }
                ]
              }
            ],
            "who": {
              "reference": "Practitioner/e3FxRT6I36jzw8BMhICeeHw3",
              "display": "Robert Sutton Persona Mr Md, MD"
            },
            "requestor": true
          }
        ],
        "source": {
          "observer": {
            "reference": "Practitioner/e3FxRT6I36jzw8BMhICeeHw3",
            "display": "Robert Sutton Persona Mr Md, MD"
          }
        },
        "entity": [
          {
            "what": {
              "reference": "List/ehyYsfQIl0V4MLni61iNl7Y1vM.JjR7Z4CWTKvcUmResxzros6bbLu2cTM2.T7Cpn3"
            },
            "detail": [
              {
                "type": "Review Status",
                "valueString": "Reviewed"
              }
            ]
          }
        ]
      },
      "search": {
        "mode": "include"
      }
    }
  ]
}
```


10.1.36 List.Search (Family History) (R4)

File: site/apis/10-Clinical-General/1-R4/10151-List-Search-Family-History-R4.html
Epic API ID | 10151
FHIR version | R4
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | August 2021
HTTP method | GET
URL template | /api/FHIR/R4/List?code={code}&patient={patient}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10151

Description

The List (Family History) API defines the list family members that have been documented in a patient's family history, as well as patient-level information that is documented alongside the family history, including adoptive parents, fertility status, and notes about the family history as a whole, referred to as pedigree annotations. Each family member is returned as a FamilyMemberHistory resource.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
code | String | False | | | Type of list. Should be set to "pedigree-list" for this API.
patient | String | Either patient or subject must be specified | | | Reference to another resource.
subject | String | Either Patient or Subject must be specified | | | The Patient resource with which the List is associated.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
date | String | True | | | Unsupported
empty-reason | String | True | | | Unsupported
encounter | String | True | | | Unsupported
identifier | String | True | | | Unsupported
item | String | True | | | FamilyHistory FHIR ID Reference
notes | String | True | | | Unsupported
source | String | True | | | Unsupported
status | String | True | | | Unsupported
title | String | True | | | Unsupported

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | BackboneElement | False | Yes | | An array of entries included in this bundle.
Bundle.entry.resource | ResourceContainer | False | | | A container for a single FHIR resource.
Bundle.entry.resource.List | List | False | | | An instance of the R4 List FHIR resource.
Bundle.entry.resource.List.code | CodeableConcept | False | | | Code "pedigree-list" in Epic's "/list-type" definition
Bundle.entry.resource.List.entry | Entry | False | Yes | | FamilyMemberHistory references for patient's family history
Bundle.entry.resource.List.entry.item | Reference | False | | | FamilyHistory FHIR ID Reference
Bundle.entry.resource.List.id | String | False | | | FHIR identifier for this List resource instance.
Bundle.entry.resource.List.mode | String | False | | | Always "working"
Bundle.entry.resource.List.status | String | False | | | Always "current"

Sample request

```
https://hostname/instance/api/FHIR/R4/List?patient=eV6Ofju2.rCOp6TCunVM9Rg3&code=pedigree-list
```

Sample response

```
<Bundle xmlns="http://hl7.org/fhir">
  <type value="searchset"/>
  <total value="1"/>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/R4/List?patient=eV6Ofju2.rCOp6TCunVM9Rg3&amp;code=pedigree-list"/>
  </link>
  <entry>
    <link>
      <relation value="self"/>
      <url value="https://hostname/instance/api/FHIR/R4/List/epz8xX7jOTqs7.On.3GKrfupkrN3CHnG7JlM3DZRNJgQ3"/>
    </link>
    <fullUrl value="https://hostname/instance/api/FHIR/R4/List/epz8xX7jOTqs7.On.3GKrfupkrN3CHnG7JlM3DZRNJgQ3"/>
    <resource>
      <List>
        <id value="epz8xX7jOTqs7.On.3GKrfupkrN3CHnG7JlM3DZRNJgQ3"/>
        <status value="current"/>
        <mode value="working"/>
        <title value="Pedigree List"/>
        <code>
          <coding>
            <system value="http://open.epic.com/FHIR/StructureDefinition/list-type"/>
            <code value="pedigree-list"/>
          </coding>
        </code>
        <subject>
          <reference value="Patient/eV6Ofju2.rCOp6TCunVM9Rg3"/>
          <display value="Smith, Sally"/>
        </subject>
        <entry>
          <item>
            <reference value="FamilyMemberHistory/ePM.Qux3IY3tsLDfCmpxnDhhUJ4WWDI.DRtTk-LVsHD03"/>
```

```
            <display value="Paula (Mother)"/>
          </item>
        </entry>
        <entry>
          <item>
            <reference value="FamilyMemberHistory/ePM.Qux3IY3tsLDfCmpxnDiFB6Z7eKudu2cH4dK2ZLDg3"/>
            <display value="Joe (Father)"/>
          </item>
        </entry>
        <entry>
          <item>
            <reference value="FamilyMemberHistory/ePM.Qux3IY3tsLDfCmpxnDhOstn1GTFlXhxiGglTaVdA3"/>
            <display value="Ellen (Other)"/>
          </item>
        </entry>
        <entry>
          <item>
            <reference value="FamilyMemberHistory/ePM.Qux3IY3tsLDfCmpxnDreCpIDxiDYvSwSdd2GZMl83"/>
            <display value="Lou (Other)"/>
          </item>
        </entry>
        <entry>
          <item>
            <reference value="FamilyMemberHistory/ePM.Qux3IY3tsLDfCmpxnDveFxWGXW2.MzRylZeKEzKw3"/>
            <display value="Half-sister"/>
          </item>
        </entry>
        <entry>
          <item>
            <reference value="FamilyMemberHistory/ePM.Qux3IY3tsLDfCmpxnDsX.rHMIGLfLICc9e8rjlNE3"/>
            <display value="Adoptive mother"/>
          </item>
        </entry>
        <entry>
          <item>
            <reference value="FamilyMemberHistory/ePM.Qux3IY3tsLDfCmpxnDktL26zVbCVSewma1pEizDg3"/>
            <display value="Adoptive father"/>
          </item>
        </entry>
      </List>
    </resource>
    <search>
      <mode value="match"/>
    </search>
  </entry>
</Bundle>
```


10.1.37 List.Search (Hospital Problems) (R4)

File: site/apis/10-Clinical-General/1-R4/10149-List-Search-Hospital-Problems-R4.html
Epic API ID | 10149
FHIR version | R4
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | August 2021
HTTP method | GET;POST
URL template | /api/FHIR/R4/List?code={code}&encounter={encounter}&patient={patient}&subject={subject}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10149

Description

The List (Hospital Problem) API is used to represent the list of problems addressed during the associated hospital encounter.

It includes problems that are currently marked as hospital problems for that encounter, and it also includes problems that were previously marked as hospital problems for that encounter but that are no longer marked as such. Each entry includes a reference to the Condition (Problem List) resource instance for the problem, details about whether the problem is currently marked as a hospital problem, details about whether the problem is the principal problem for the encounter, and the problem's present on admission status.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
code | String | False | | | The code representing the type of list to search. Must include "hospital-problems" to be valid for a Hospital Problem List search.
encounter | String | True | | | FHIR Encounter ID(s). If this is given, then this search returns a List for each encounter searched that has hospital problems. This search parameter must be combined with either the patient or subject search parameter.

patient | String | Either patient or subject must be specified. | | | FHIR ID for the Patient resource with whom the List is associated.
subject | String | Either patient or subject must be specified. | | | FHIR ID for the Patient resource with whom the List is associated.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
date | String | True | | | Unsupported
empty-reason | String | True | | | Unsupported
identifier | String | True | | | Unsupported
item | String | True | | | Condition (Problem List Item) FHIR ID
notes | String | True | | | Unsupported
source | String | True | | | Unsupported
status | String | True | | | Always “current”
title | String | True | | | Always “Hospital Problem List”

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | BackboneElement | False | Yes | | An array of entries included in this bundle.
Bundle.entry.resource | ResourceContainer | False | | | A container for a single FHIR resource.
Bundle.entry.resource.List | List | False | | | An instance of the R4 List FHIR resource.
Bundle.entry.resource.List.code | CodeableConcept | False | | | Code “hospital-problem” in Epic’s “/list-type” definition
Bundle.entry.resource.List.encounter | Reference | False | | | The encounter that this hospital problem list is for
Bundle.entry.resource.List.entry | Entry | False | Yes | | There is one entry for each hospital problem on the encounter’s hospital problem list.
Bundle.entry.resource.List.entry.deleted | Boolean | True | | | If true, this problem is no longer marked as a hospital problem for this encounter, but it had been previously. This is omitted if the problem is currently marked as a hospital problem.
Bundle.entry.resource.List.entry.extension (also: hospital-problem-present-on-admission) | Extension | False | Yes | | The Present on Admission status for this problem on this encounter. Default values include: Yes No Unknown Clinically Undetermined Exempt from POA reporting Epic organizations may define additional custom values.
Bundle.entry.resource.List.entry.extension (also: is-principal-hospital-problem) | Extension | False | Yes | | If true, this problem is the principal problem for this hospital encounter.
Bundle.entry.resource.List.entry.item | Reference | False | | | Reference to the Condition (Problem List Item) resource for this problem list item. The display name is set to the display name of the problem’s diagnosis.
Bundle.entry.resource.List.mode | String | False | | | Always “working”
Bundle.entry.resource.List.status | String | False | | | Always “current”
Bundle.entry.resource.List.subject | Reference | False | | | The patient that this hospital problem list is for
Bundle.entry.resource.List.title | String | False | | | Always “Hospital Problem List”

Sample request

```
https://hostname/instance/api/FHIR/R4/List?patient=emFzuaGzscRApD-FRiPcjzg3&code=hospital-problems
```

Sample response

```
<Bundle xmlns="http://hl7.org/fhir">
    <type value="searchset" />
    <total value="2" />
    <link>
        <relation value="self" />
        <url value="https://hostname/instance/api/FHIR/R4/List?patient=emFzuaGzscRApD-FRiPcjzg3&amp;code=hospital-
    problems" />
    </link>
    <entry>
        <link>
            <relation value="self" />
            <url value="https://hostname/instance/api/FHIR/R4/List/eWmQViNOBS-
    laHgREP6mMLg2goK4z8icONOq6uQX9ulM7OKnxp8CJcKIVQRCWDcg3BhOJ2RFgetUI6l8CMmDZfQ3" />
        </link>
        <fullUrl value="https://hostname/instance/api/FHIR/R4/List/eWmQViNOBS-
    laHgREP6mMLg2goK4z8icONOq6uQX9ulM7OKnxp8CJcKIVQRCWDcg3BhOJ2RFgetUI6l8CMmDZfQ3" />
        <resource>
            <List>
                <id value="eWmQViNOBS-laHgREP6mMLg2goK4z8icONOq6uQX9ulM7OKnxp8CJcKIVQRCWDcg3BhOJ2RFgetUI6l8CMmDZfQ3" />
                <status value="current" />
                <mode value="working" />
                <title value="Hospital Problem List" />
                <code>
                    <coding>
                        <system value="http://open.epic.com/FHIR/StructureDefinition/list-type" />
                        <code value="hospital-problems" />
                        <display value="Hospital Problem List" />
                    </coding>
                    <text value="Hospital Problem List" />
                </code>
                <subject>
                    <reference value="Patient/emFzuaGzscRApD-FRiPcjzg3" />
                    <display value="Johnsen, Bruce Banner" />
                </subject>
                <encounter>
                    <reference value="Encounter/e7du6YOFXxZOXpdLadukGrrZe.p0.YpT8Xz.ywOuTJh03" />
```

```
            </encounter>
            <entry>
                <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/is-principal-hospital-
problem">
                    <valueBoolean value="false" />
                </extension>
                <deleted value="true" />
                <item>
                    <reference value="Condition/emmYRtD0ekHjEJHnndSisgILtK.-JGUaZdpiJsF9a-eg3" />
                    <display value="Acute bronchitis" />
                </item>
            </entry>
            <entry>
                <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/is-principal-hospital-
problem">
                    <valueBoolean value="false" />
                </extension>
                <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/hospital-problem-present-on-
admission">
                    <valueCoding>
                        <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.10.726582.611" />
                        <code value="4" />
                        <display value="Clinically Undetermined" />
                    </valueCoding>
                </extension>
                <item>
                    <reference value="Condition/e2zEi7GDsij3PMcZQHQGB8SUuSHbw3s8wVSveexq57eQ3" />
                    <display value="Acute recurrent streptococcal tonsillitis" />
                </item>
            </entry>
            <entry>
                <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/is-principal-hospital-
problem">
                    <valueBoolean value="true" />
                </extension>
                <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/hospital-problem-present-on-
admission">
                    <valueCoding>
                        <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.10.726582.611" />
                        <code value="1" />
                        <display value="Yes" />
                    </valueCoding>
                </extension>
                <item>
                    <reference value="Condition/eiKzn5BBwMqGuXQ2uswjZgk8emitbtKUX0DjzI0PjPko3" />
                    <display value="Abdominal pain" />
                </item>
            </entry>
            <entry>
                <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/is-principal-hospital-
problem">
                    <valueBoolean value="false" />
                </extension>
                <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/hospital-problem-present-on-
admission">
                    <valueCoding>
                        <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.10.726582.611" />
                        <code value="1" />
                        <display value="Yes" />
                    </valueCoding>
                </extension>
                <item>
                    <reference value="Condition/eHSNwOlK3jOy1mvo.wyJoCoFtlF.Cjv0jIQPzhNzUJ143" />
                    <display value="Open wound" />
                </item>
            </entry>
            <entry>
                <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/is-principal-hospital-
problem">
                    <valueBoolean value="false" />
                </extension>
                <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/hospital-problem-present-on-
admission">
                    <valueCoding>
                        <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.10.726582.611" />
                        <code value="1" />
                        <display value="Yes" />
                    </valueCoding>
                </extension>
                <item>
                    <reference value="Condition/ealoQatr7JcmEDBoZSgxWNl3r1veaBoRuNC.VP1sCF7M3" />
                    <display value="Effusion of right shoulder joint" />
                </item>
            </entry>
            <entry>
                <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/is-principal-hospital-
problem">
                    <valueBoolean value="false" />
                </extension>
```

```
                <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/hospital-problem-present-on-
admission">
                    <valueCoding>
                        <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.10.726582.611" />
                        <code value="1" />
                        <display value="Yes" />
                    </valueCoding>
                </extension>
                <item>
                    <reference value="Condition/e2MMZPHo548dhjYMKSjl9..B20ZrdDAozLznKqEGNQXk3" />
                    <display value="Sprain of left wrist" />
                </item>
            </entry>
            <entry>
                <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/is-principal-hospital-
problem">
                    <valueBoolean value="false" />
                </extension>
                <deleted value="true" />
                <item>
                    <reference value="Condition/eSAzHvrKQ0e1-Ekv.RDgXpy3mxbeubXNxsbta83LTXdY3" />
                    <display value="Difficulty controlling anger" />
                </item>
            </entry>
            <entry>
                <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/is-principal-hospital-
problem">
                    <valueBoolean value="false" />
                </extension>
                <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/hospital-problem-present-on-
admission">
                    <valueCoding>
                        <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.10.726582.611" />
                        <code value="1" />
                        <display value="Yes" />
                    </valueCoding>
                </extension>
                <item>
                    <reference value="Condition/e-UsEpEy3QQEKwOdfjCO0fzOuZEw-i4Af5BjQZ1IUfEo3" />
                    <display value="Burns involving less than 10% of body surface" />
                </item>
            </entry>
        </List>
    </resource>
    <search>
        <mode value="match" />
    </search>
</entry>
<entry>
    <link>
        <relation value="self" />
        <url value="https://hostname/instance/api/FHIR/R4/List/eWmQViNOBS-
laHgREP6mMLpDgGb2asXjOg1UdCpMFVvaQ1WvkgkRgF2SIhnasNgx3EJiUpAYRK-n91nYYZPh-1A3" />
    </link>
    <fullUrl value="https://hostname/instance/api/FHIR/R4/List/eWmQViNOBS-
laHgREP6mMLpDgGb2asXjOg1UdCpMFVvaQ1WvkgkRgF2SIhnasNgx3EJiUpAYRK-n91nYYZPh-1A3" />
    <resource>
        <List>
            <id value="eWmQViNOBS-laHgREP6mMLpDgGb2asXjOg1UdCpMFVvaQ1WvkgkRgF2SIhnasNgx3EJiUpAYRK-n91nYYZPh-1A3" />
            <status value="current" />
            <mode value="working" />
            <title value="Hospital Problem List" />
            <code>
                <coding>
                    <system value="http://open.epic.com/FHIR/StructureDefinition/list-type" />
                    <code value="hospital-problems" />
                    <display value="Hospital Problem List" />
                </coding>
                <text value="Hospital Problem List" />
            </code>
            <subject>
                <reference value="Patient/emFzuaGzscRApD-FRiPcjzg3" />
                <display value="Johnsen, Bruce Banner" />
            </subject>
            <encounter>
                <reference value="Encounter/e6U0xu8ayP4Lfx0z6X9RPeFMpip-BAgwFDnVbz4MHBMA3" />
            </encounter>
            <entry>
                <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/is-principal-hospital-
problem">
                    <valueBoolean value="true" />
                </extension>
                <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/hospital-problem-present-on-
admission">
                    <valueCoding>
                        <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.10.726582.611" />
                        <code value="1" />
                        <display value="Yes" />
                    </valueCoding>
```

```
                </extension>
                <item>
                    <reference value="Condition/eHSNwOlK3jOy1mvo.wyJoCoFtlF.Cjv0jIQPzhNzUJ143" />
                    <display value="Open wound" />
                </item>
            </entry>
            <entry>
                <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/is-principal-hospital-
    problem">
                    <valueBoolean value="false" />
                </extension>
                <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/hospital-problem-present-on-
    admission">
                    <valueCoding>
                        <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.10.726582.611" />
                        <code value="2" />
                        <display value="No" />
                    </valueCoding>
                </extension>
                <item>
                    <reference value="Condition/ealoQatr7JcmEDBoZSgxWNl3r1veaBoRuNC.VP1sCF7M3" />
                    <display value="Effusion of right shoulder joint" />
                </item>
            </entry>
            <entry>
                <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/is-principal-hospital-
    problem">
                    <valueBoolean value="false" />
                </extension>
                <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/hospital-problem-present-on-
    admission">
                    <valueCoding>
                        <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.10.726582.611" />
                        <code value="2" />
                        <display value="No" />
                    </valueCoding>
                </extension>
                <item>
                    <reference value="Condition/e2MMZPHo548dhjYMKSjl9..B20ZrdDAozLznKqEGNQXk3" />
                    <display value="Sprain of left wrist" />
                </item>
            </entry>
            <entry>
                <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/is-principal-hospital-
    problem">
                    <valueBoolean value="false" />
                </extension>
                <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/hospital-problem-present-on-
    admission">
                    <valueCoding>
                        <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.10.726582.611" />
                        <code value="3" />
                        <display value="Unknown" />
                    </valueCoding>
                </extension>
                <item>
                    <reference value="Condition/eSAzHvrKQ0e1-Ekv.RDgXpy3mxbeubXNxsbta83LTXdY3" />
                    <display value="Difficulty controlling anger" />
                </item>
            </entry>
            <entry>
                <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/is-principal-hospital-
    problem">
                    <valueBoolean value="false" />
                </extension>
                <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/hospital-problem-present-on-
    admission">
                    <valueCoding>
                        <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.10.726582.611" />
                        <code value="3" />
                        <display value="Unknown" />
                    </valueCoding>
                </extension>
                <item>
                    <reference value="Condition/eOX1InBdOth.sDapGbTH91bKnzc0XG5.dnGAd4HSDxD03" />
                    <display value="Difficulty controlling anger" />
                </item>
            </entry>
        </List>
    </resource>
    <search>
        <mode value="match" />
    </search>
</entry>
</Bundle
```

10.1.38 List.Search (Problems) (R4)

File: site/apis/10-Clinical-General/1-R4/10145-List-Search-Problems-R4.html
Epic API ID | 10145
FHIR version | R4
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | August 2021
HTTP method | GET;POST
URL template | /api/FHIR/R4/List?code={code}&patient={patient}&subject={subject}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10145

Description

The List (Problem List) API is used to represent a list of the patient's current problems and conditions.

This resource is commonly used to review the history of the patient's problems and conditions with AuditEvent. For more details about a specific problem or condition, you can use the Condition.Read (Problems) (R4) web service.

Remarks

AuditEvent

The following details are returned as part of the AuditEvent structure if AuditEvent data is included with this resource. For more information about AuditEvent, refer to the FHIR Search Parameters article in the Documentation section of this site.

Name | Description | Is Optional | Is Array
type (Coding) | Defines the audit event type. Always "Review History". | true | false
recorded (Instant) | The instant that the user reviewed the patient's problem list. | false | false
agent (Backbone) | Describes the user who reviewed the patient's problem list.
agent.requestor (Boolean) | Always "true". | true | false
agent.who (Reference) | The user requesting the information. | false | false
agent.role (Coding) | Defines the agent role. Always "Reviewer".
source (Backbone) | The user who reviewed the patient's problem list.
source.observer (Reference) | The user requesting the information. | true | false
entity (Backbone) | Describes the problem list.
entity.what (Reference) | The List | true | false

Request parameters

Name | Type | Optional | Array | Cardinality | Description
code | String | False | | | The type of list to search for. Must be set to "problems".
patient | String | Either patient or subject must be specified. | | | The Patient resource with which the List is associated.
subject | String | Either patient or subject must be specified. | | | The Patient resource with which the List is associated.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
date | String | True | | | Unsupported
empty-reason | String | True | | | If the patient has no problems documented, the API returns an emptyReason of "notstarted".
encounter | String | True | | | Unsupported
identifier | String | True | | | Unsupported
item | String | True | | | Condition (Problems) FHIR ID
notes | String | True | | | Unsupported
source | String | True | | | Unsupported
status | String | True | | | Status of the problem list. Always "current" because each list represents only the patient's current problems.
title | String | True | | | The title representing this entity. Always "Problem List".

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | BackboneElement | False | Yes | | An array of entries included in this bundle.
Bundle.entry.resource | ResourceContainer | False | | | A container for a single FHIR resource.
Bundle.entry.resource.List | List | False | | | An instance of the R4 List FHIR resource.
Bundle.entry.resource.List.code | CodeableConcept | True | | | The code representing the type of list searched. Always "problems".
Bundle.entry.resource.List.emptyReason | CodeableConcept | False | | | If the patient has no problems documented, the API returns an emptyReason of "notstarted".
Bundle.entry.resource.List.entry | Entry | True | Yes | | Each entry is a BackboneElement with the "item" property populated with a reference to the Condition resource and the "display" property populated with the problem name.

Bundle.entry.resource.List.entry.item | Reference | False | | | List of FHIR IDs returned by the Condition (Problems) resource search, excluding external and deleted problems.

Bundle.entry.resource.List.id | String | False | | | FHIR identifier for this List resource instance.

Bundle.entry.resource.List.mode | String | False | | | The processing mode that applies to this list. Always "working".

Bundle.entry.resource.List.status | String | False | | | Status of the problem list. Always "current" because each list represents only the patient's current problems.

Bundle.entry.resource.List.subject | Reference | True | | | Reference to the Patient resource with which the List is associated.

Bundle.entry.resource.List.title | String | False | | | The title representing this entity. Always “Problem List”.

Sample request

```
https://hostname/instance/api/FHIR/R4/List?patient=exsKF19Z.1GVmpD2e3d.RIQ3&code=problems&_revInclude=AuditEvent:entity
```

Sample response

```
<Bundle xmlns="http://hl7.org/fhir">
  <type value="searchset"/>
  <total value="1"/>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/R4/List?patient=exsKF19Z.1GVmpD2e3d.RIQ3&amp;code=problems&amp;_revIn
    clude=AuditEvent:entity"/>
  </link>
  <entry>
    <link>
      <relation value="self"/>
      <url
    value="https://hostname/instance/api/FHIR/R4/List/eIpSjjINTljmpiCA6WglPdM.uAFAZ2DWCZ98sPxJAOhYiIFQn.ki6sAgMyiZ5ZfsB3
    "/>
    </link>
    <fullUrl
    value="https://hostname/instance/api/FHIR/R4/List/eIpSjjINTljmpiCA6WglPdM.uAFAZ2DWCZ98sPxJAOhYiIFQn.ki6sAgMyiZ5ZfsB3
    "/>
    <resource>
      <List>
        <id value="eIpSjjINTljmpiCA6WglPdM.uAFAZ2DWCZ98sPxJAOhYiIFQn.ki6sAgMyiZ5ZfsB3"/>
        <status value="current"/>
        <mode value="working"/>
        <title value="Problem List"/>
        <code>
          <coding>
            <system value="http://open.epic.com/FHIR/StructureDefinition/list-type"/>
            <code value="problems"/>
          </coding>
        </code>
        <subject>
          <reference value="Patient/exsKF19Z.1GVmpD2e3d.RIQ3"/>
          <display value="LastName, FirstName"/>
        </subject>
        <entry>
          <item>
            <reference value="Condition/eNNIspYNV4t-uXXI2iqkW7TY49HZOmiWCLe0RkFNQjqs3"/>
            <display value="Generalized abdominal pain"/>
          </item>
        </entry>
        <entry>
          <item>
            <reference value="Condition/egP7rCnj4ekRiz2SVDRelhnhj7HCnPqplEjS.vxARCJE3"/>
            <display value="Acute non-ST elevation myocardial infarction (NSTEMI)"/>
          </item>
        </entry>
        <entry>
          <item>
            <reference value="Condition/eO9zPXhbkgk1ezY-NDhfUsUH0Y2JpLh.Ajc3BEy6Z-Is3"/>
            <display value="Gout"/>
          </item>
        </entry>
      </List>
    </resource>
    <search>
      <mode value="match"/>
    </search>
  </entry>
  <entry>
    <link>
      <relation value="self"/>
      <url value="https://hostname/instance/api/FHIR/R4/AuditEvent/e5CN.zQlbIHF7CK30ShFtPJsKcOf274tJZfiXA6Ilvw0tI-
    XDsOmjZD9T88nBuwUN4Bx2RZoW4d34E0vHsnowwoe8fazZHo5drQT8vOWmnowxVD5tqyfg17gXWjQDZbCh3"/>
    </link>
    <fullUrl value="https://hostname/instance/api/FHIR/R4/AuditEvent/e5CN.zQlbIHF7CK30ShFtPJsKcOf274tJZfiXA6Ilvw0tI-
    XDsOmjZD9T88nBuwUN4Bx2RZoW4d34E0vHsnowwoe8fazZHo5drQT8vOWmnowxVD5tqyfg17gXWjQDZbCh3"/>
    <resource>
      <AuditEvent>
        <id value="e5CN.zQlbIHF7CK30ShFtPJsKcOf274tJZfiXA6Ilvw0tI-
    XDsOmjZD9T88nBuwUN4Bx2RZoW4d34E0vHsnowwoe8fazZHo5drQT8vOWmnowxVD5tqyfg17gXWjQDZbCh3"/>
        <type>
          <system value="http://open.epic.com/FHIR/StructureDefinition/audit-event-type"/>
```

```
        <code value="Review History"/>
        <display value="Review History"/>
      </type>
      <recorded value="2020-06-22T17:07:43Z"/>
      <agent>
        <role>
          <coding>
            <system value="http://open.epic.com/FHIR/StructureDefinition/audit-event-agent-role"/>
            <code value="Reviewer"/>
            <display value="Reviewer"/>
          </coding>
        </role>
        <who>
          <reference value="Practitioner/e3h.ZuGTnpnoqzYu4LAcboA3"/>
          <display value="Physician LastName, MD"/>
        </who>
        <requestor value="true"/>
      </agent>
      <source>
        <observer>
          <reference value="Practitioner/e3h.ZuGTnpnoqzYu4LAcboA3"/>
          <display value="Physician LastName, MD"/>
        </observer>
      </source>
      <entity>
        <what>
          <reference value="List/eIpSjjINTljmpiCA6WglPdM.uAFAZ2DWCZ98sPxJAOhYiIFQn.ki6sAgMyiZ5ZfsB3"/>
        </what>
      </entity>
    </AuditEvent>
  </resource>
  <search>
    <mode value="include"/>
  </search>
</entry>
<entry>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/R4/AuditEvent/e5CN.zQlbIHF7CK30ShFtPJsKcOf274tJZfiXA6Ilvw3YpLB7HO5Q
  DKBWavAA9Qno6th4RDaioLTC7oOlUgVWI0ZCMnsBIgQpKauYvrsbOg39TJTK3Abng39HlsA5dtiA3"/>
  </link>
  <fullUrl value="https://hostname/instance/api/FHIR/R4/AuditEvent/e5CN.zQlbIHF7CK30ShFtPJsKcOf274tJZfiXA6Ilvw3YpLB7HO
  5QDKBWavAA9Qno6th4RDaioLTC7oOlUgVWI0ZCMnsBIgQpKauYvrsbOg39TJTK3Abng39HlsA5dtiA3"/>
  <resource>
    <AuditEvent>
      <id value="e5CN.zQlbIHF7CK30ShFtPJsKcOf274tJZfiXA6Ilvw3YpLB7HO5QDKBWavAA9Qno6th4RDaioLTC7oOlUgVWI0ZCMnsBIgQpKauY
  vrsbOg39TJTK3Abng39HlsA5dtiA3"/>
      <type>
        <system value="http://open.epic.com/FHIR/StructureDefinition/audit-event-type"/>
        <code value="Review History"/>
        <display value="Review History"/>
      </type>
      <recorded value="2020-12-15T20:02:09Z"/>
      <agent>
        <role>
          <coding>
            <system value="http://open.epic.com/FHIR/StructureDefinition/audit-event-agent-role"/>
            <code value="Reviewer"/>
            <display value="Reviewer"/>
          </coding>
        </role>
        <who>
          <reference value="Practitioner/e3h.ZuGTnpnoqzYu4LAcboA3"/>
          <display value="Physician LastName, MD"/>
        </who>
        <requestor value="true"/>
      </agent>
      <source>
        <observer>
          <reference value="Practitioner/e3h.ZuGTnpnoqzYu4LAcboA3"/>
          <display value="Physician LastName, MD"/>
        </observer>
      </source>
      <entity>
        <what>
          <reference value="List/eIpSjjINTljmpiCA6WglPdM.uAFAZ2DWCZ98sPxJAOhYiIFQn.ki6sAgMyiZ5ZfsB3"/>
        </what>
      </entity>
    </AuditEvent>
  </resource>
  <search>
    <mode value="include"/>
  </search>
</entry>
<entry>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/R4/AuditEvent/e5CN.zQlbIHF7CK30ShFtPJsKcOf274tJZfiXA6Ilvw1EZrW3ZPZl
  p08XoQrt0.GvKnnjJEIlb9dPf.aqvg58dGa2bP6Li8z0binoUjxgGtN17up.QrATZ-7wywjpXJHF3"/>
```

```
  </link>
  <fullUrl value="https://hostname/instance/api/FHIR/R4/AuditEvent/e5CN.zQlbIHF7CK30ShFtPJsKcOf274tJZfiXA6Ilvw1EZrW3ZP
  Zlp08XoQrt0.GvKnnjJEIlb9dPf.aqvg58dGa2bP6Li8z0binoUjxgGtN17up.QrATZ-7wywjpXJHF3"/>
  <resource>
    <AuditEvent>
      <id value="e5CN.zQlbIHF7CK30ShFtPJsKcOf274tJZfiXA6Ilvw1EZrW3ZPZlp08XoQrt0.GvKnnjJEIlb9dPf.aqvg58dGa2bP6Li8z0bino
  UjxgGtN17up.QrATZ-7wywjpXJHF3"/>
      <type>
        <system value="http://open.epic.com/FHIR/StructureDefinition/audit-event-type"/>
        <code value="Review History"/>
        <display value="Review History"/>
      </type>
      <recorded value="2020-12-15T20:03:25Z"/>
      <agent>
        <role>
          <coding>
            <system value="http://open.epic.com/FHIR/StructureDefinition/audit-event-agent-role"/>
            <code value="Reviewer"/>
            <display value="Reviewer"/>
          </coding>
        </role>
        <who>
          <reference value="Practitioner/e3h.ZuGTnpnoqzYu4LAcboA3"/>
          <display value="Physician LastName, MD"/>
        </who>
        <requestor value="true"/>
      </agent>
      <source>
        <observer>
          <reference value="Practitioner/e3h.ZuGTnpnoqzYu4LAcboA3"/>
          <display value="Physician LastName, MD"/>
        </observer>
      </source>
      <entity>
        <what>
          <reference value="List/eIpSjjINTljmpiCA6WglPdM.uAFAZ2DWCZ98sPxJAOhYiIFQn.ki6sAgMyiZ5ZfsB3"/>
        </what>
      </entity>
    </AuditEvent>
  </resource>
  <search>
    <mode value="include"/>
  </search>
</entry>
<entry>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/R4/AuditEvent/e5CN.zQlbIHF7CK30ShFtPJsKcOf274tJZfiXA6Ilvw0ncHsSAf3h
  1lFljoEMHoMUpr.SxffwqdpS1LkjS8UC8l.n6BL2OO7YJCH9j63pVjvtyyXjXUgWEEJSzSgDL3dP3"/>
  </link>
  <fullUrl value="https://hostname/instance/api/FHIR/R4/AuditEvent/e5CN.zQlbIHF7CK30ShFtPJsKcOf274tJZfiXA6Ilvw0ncHsSAf
  3h1lFljoEMHoMUpr.SxffwqdpS1LkjS8UC8l.n6BL2OO7YJCH9j63pVjvtyyXjXUgWEEJSzSgDL3dP3"/>
  <resource>
    <AuditEvent>
      <id value="e5CN.zQlbIHF7CK30ShFtPJsKcOf274tJZfiXA6Ilvw0ncHsSAf3h1lFljoEMHoMUpr.SxffwqdpS1LkjS8UC8l.n6BL2OO7YJCH9
  j63pVjvtyyXjXUgWEEJSzSgDL3dP3"/>
      <type>
        <system value="http://open.epic.com/FHIR/StructureDefinition/audit-event-type"/>
        <code value="Review History"/>
        <display value="Review History"/>
      </type>
      <recorded value="2020-12-15T20:05:30Z"/>
      <agent>
        <role>
          <coding>
            <system value="http://open.epic.com/FHIR/StructureDefinition/audit-event-agent-role"/>
            <code value="Reviewer"/>
            <display value="Reviewer"/>
          </coding>
        </role>
        <who>
          <reference value="Practitioner/e3h.ZuGTnpnoqzYu4LAcboA3"/>
          <display value="Physician LastName, MD"/>
        </who>
        <requestor value="true"/>
      </agent>
      <source>
        <observer>
          <reference value="Practitioner/e3h.ZuGTnpnoqzYu4LAcboA3"/>
          <display value="Physician LastName, MD"/>
        </observer>
      </source>
      <entity>
        <what>
          <reference value="List/eIpSjjINTljmpiCA6WglPdM.uAFAZ2DWCZ98sPxJAOhYiIFQn.ki6sAgMyiZ5ZfsB3"/>
        </what>
      </entity>
    </AuditEvent>
  </resource>
```

```
  <search>
    <mode value="include"/>
  </search>
</entry>
<entry>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/R4/AuditEvent/e5CN.zQlbIHF7CK30ShFtPJsKcOf274tJZfiXA6Ilvw2GHzdCVxW-
  9eKxVgMlEi0H3UxKg06zcvwil4IwtPlDxs5y2CFcK9ypt6yJbbV62mMplaW4Y0y0W2YmtrboZyS13"/>
  </link>
  <fullUrl value="https://hostname/instance/api/FHIR/R4/AuditEvent/e5CN.zQlbIHF7CK30ShFtPJsKcOf274tJZfiXA6Ilvw2GHzdCVx
  W-9eKxVgMlEi0H3UxKg06zcvwil4IwtPlDxs5y2CFcK9ypt6yJbbV62mMplaW4Y0y0W2YmtrboZyS13"/>
  <resource>
    <AuditEvent>
      <id value="e5CN.zQlbIHF7CK30ShFtPJsKcOf274tJZfiXA6Ilvw2GHzdCVxW-
  9eKxVgMlEi0H3UxKg06zcvwil4IwtPlDxs5y2CFcK9ypt6yJbbV62mMplaW4Y0y0W2YmtrboZyS13"/>
      <type>
        <system value="http://open.epic.com/FHIR/StructureDefinition/audit-event-type"/>
        <code value="Review History"/>
        <display value="Review History"/>
      </type>
      <recorded value="2021-04-05T15:28:06Z"/>
      <agent>
        <role>
          <coding>
            <system value="http://open.epic.com/FHIR/StructureDefinition/audit-event-agent-role"/>
            <code value="Reviewer"/>
            <display value="Reviewer"/>
          </coding>
        </role>
        <who>
          <reference value="Practitioner/e3FxRT6I36jzw8BMhICeeHw3"/>
          <display value="Robert Sutton, MD"/>
        </who>
        <requestor value="true"/>
      </agent>
      <source>
        <observer>
          <reference value="Practitioner/e3FxRT6I36jzw8BMhICeeHw3"/>
          <display value="Robert Sutton, MD"/>
        </observer>
      </source>
      <entity>
        <what>
          <reference value="List/eIpSjjINTljmpiCA6WglPdM.uAFAZ2DWCZ98sPxJAOhYiIFQn.ki6sAgMyiZ5ZfsB3"/>
        </what>
      </entity>
    </AuditEvent>
  </resource>
  <search>
    <mode value="include"/>
  </search>
</entry>
<entry>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/R4/AuditEvent/e5CN.zQlbIHF7CK30ShFtPJsKcOf274tJZfiXA6Ilvw17UTulfN4s
  MftllsyGL6g9d2vOpvSiEkh.9wI4AHudED2swuBbUjMEeR6XRQ-E7md8dMYkzJeckFaB.rlfqB9L3"/>
  </link>
  <fullUrl value="https://hostname/instance/api/FHIR/R4/AuditEvent/e5CN.zQlbIHF7CK30ShFtPJsKcOf274tJZfiXA6Ilvw17UTulfN
  4sMftllsyGL6g9d2vOpvSiEkh.9wI4AHudED2swuBbUjMEeR6XRQ-E7md8dMYkzJeckFaB.rlfqB9L3"/>
  <resource>
    <AuditEvent>
      <id value="e5CN.zQlbIHF7CK30ShFtPJsKcOf274tJZfiXA6Ilvw17UTulfN4sMftllsyGL6g9d2vOpvSiEkh.9wI4AHudED2swuBbUjMEeR6X
  RQ-E7md8dMYkzJeckFaB.rlfqB9L3"/>
      <type>
        <system value="http://open.epic.com/FHIR/StructureDefinition/audit-event-type"/>
        <code value="Review History"/>
        <display value="Review History"/>
      </type>
      <recorded value="2021-04-09T16:16:07Z"/>
      <agent>
        <role>
          <coding>
            <system value="http://open.epic.com/FHIR/StructureDefinition/audit-event-agent-role"/>
            <code value="Reviewer"/>
            <display value="Reviewer"/>
          </coding>
        </role>
        <who>
          <reference value="Practitioner/e3h.ZuGTnpnoqzYu4LAcboA3"/>
          <display value="Physician LastName, MD"/>
        </who>
        <requestor value="true"/>
      </agent>
      <source>
        <observer>
          <reference value="Practitioner/e3h.ZuGTnpnoqzYu4LAcboA3"/>
          <display value="Physician LastName, MD"/>
```

```
        </observer>
      </source>
      <entity>
        <what>
          <reference value="List/eIpSjjINTljmpiCA6WglPdM.uAFAZ2DWCZ98sPxJAOhYiIFQn.ki6sAgMyiZ5ZfsB3"/>
        </what>
      </entity>
    </AuditEvent>
  </resource>
  <search>
    <mode value="include"/>
  </search>
 </entry>
 <entry>
  <link>
    <relation value="self"/>
    <url
  value="https://hostname/instance/api/FHIR/R4/AuditEvent/e5CN.zQlbIHF7CK30ShFtPJsKcOf274tJZfiXA6Ilvw3nevUBN0Fgr2TKx-
  QRIPZl29aBZ2F59JzhLP59S.ZCSsjIsB4S2LkzvZ2oe9empnc4IIUzFoBTpcEJlTE537LI3"/>
  </link>
  <fullUrl
  value="https://hostname/instance/api/FHIR/R4/AuditEvent/e5CN.zQlbIHF7CK30ShFtPJsKcOf274tJZfiXA6Ilvw3nevUBN0Fgr2TKx-
  QRIPZl29aBZ2F59JzhLP59S.ZCSsjIsB4S2LkzvZ2oe9empnc4IIUzFoBTpcEJlTE537LI3"/>
  <resource>
    <AuditEvent>
      <id value="e5CN.zQlbIHF7CK30ShFtPJsKcOf274tJZfiXA6Ilvw3nevUBN0Fgr2TKx-
  QRIPZl29aBZ2F59JzhLP59S.ZCSsjIsB4S2LkzvZ2oe9empnc4IIUzFoBTpcEJlTE537LI3"/>
      <type>
        <system value="http://open.epic.com/FHIR/StructureDefinition/audit-event-type"/>
        <code value="Review History"/>
        <display value="Review History"/>
      </type>
      <recorded value="2021-06-02T19:55:40Z"/>
      <agent>
        <role>
          <coding>
            <system value="http://open.epic.com/FHIR/StructureDefinition/audit-event-agent-role"/>
            <code value="Reviewer"/>
            <display value="Reviewer"/>
          </coding>
        </role>
        <who>
          <reference value="Practitioner/e3h.ZuGTnpnoqzYu4LAcboA3"/>
          <display value="Physician LastName, MD"/>
        </who>
        <requestor value="true"/>
      </agent>
      <source>
        <observer>
          <reference value="Practitioner/e3h.ZuGTnpnoqzYu4LAcboA3"/>
          <display value="Physician LastName, MD"/>
        </observer>
      </source>
      <entity>
        <what>
          <reference value="List/eIpSjjINTljmpiCA6WglPdM.uAFAZ2DWCZ98sPxJAOhYiIFQn.ki6sAgMyiZ5ZfsB3"/>
        </what>
      </entity>
    </AuditEvent>
  </resource>
  <search>
    <mode value="include"/>
  </search>
 </entry>
</Bundle>
```


10.1.39 Observation.Read (Care Experience Preference) (R4)

File: site/apis/10-Clinical-General/1-R4/40298-Observation-Read-Care-Experience-Preference-R4.html
Epic API ID | 40298
FHIR version | R4
Categories | Clinical General
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Observation/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=40298

Description

Use this API to retrieve care experience preference observations. The observations can be one of the following:

- Acceptable blood products for pregnancy
- Anesthesia methods planned for pregnancy
- Birth control planned after pregnancy
- Circumcision requested on behalf of pregnancy
- Delivery method planned for pregnancy
- Expected delivery location for pregnancy
- Preferred pharmacy
- Support person for pregnancy

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | Required | | | The Observation FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Observation (R4) | Observation
Observation (R4).category | CodeableConcept | | Yes | 1..* | Always includes "care-experience-preference". Flowsheet-based observations can include additional categories.
Observation (R4).code | CodeableConcept | Required | | 1..1 | All resources export with 95541-9 from http://loinc.org. Some resources also export with an additional code: - Acceptable blood products for pregnancy: acceptable-blood-products with code system http://open.epic.com/FHIR/CodeSystem/care-experience-preference - Anesthesia methods planned for pregnancy: anesthesia-method-plan with code system http://open.epic.com/FHIR/CodeSystem/care-experience-preference - Birth control planned after pregnancy: planned-birth-control with code system http://open.epic.com/FHIR/CodeSystem/care-experience-preference - Circumcision requested on behalf of pregnancy: circumcision-requested with code system http://open.epic.com/FHIR/CodeSystem/care-experience-preference - Delivery method planned for pregnancy: delivery-method-plan with code system http://open.epic.com/FHIR/CodeSystem/care-experience-preference - Expected delivery location for pregnancy: 85712-8 from http://loinc.org. - Preferred pharmacy: 103980-9 from http://loinc.org. - Support person for pregnancy: 100309-4 from http://loinc.org. Flowsheet-based observations additionally include: Encoded value of the internal Epic flowsheet ID External concept mappings such as LOINC and SNOMED
Observation (R4).component | BackboneElement | | Yes | 0..* | Includes extra information about the Observation. Populated only for: Patient-Entered Flowsheet Metadata Represents strict metadata if the coding sub-element is populated. The following strict metadata can be returned: 1-UUID The unique reading ID 2-Blood Glucose Meal Times 1-Before meal 2-After meal 3-Insulin Delivery Reason 1-Basal 2-Bolus 5-VO2 Max Test Type 1-Max exercise test 2-Sub-max exercise prediction 3-Non exercise prediction 6-Manually Entered - Whether or not the patient manually entered this reading into the system (as opposed to automatically entered by a device). false true Represents loose metadata if only the text sub-element is populated. UUID, Blood Glucose Meal times, Insulin Delivery Reason, and VO2 Max Test Type must have their value specified in component.valueString. Manually Entered is specified as a valueBoolean.
Observation (R4).component.code | CodeableConcept | | | 1..1 | All resources that use this element export with 95541-9 from http://loinc.org. These resources also export with an additional code: - Acceptable blood products for pregnancy: acceptable-blood-products with code system http://open.epic.com/FHIR/CodeSystem/care-experience-preference - Anesthesia methods planned for pregnancy: anesthesia-method-plan with code system http://open.epic.com/FHIR/CodeSystem/care-experience-preference - Birth control planned after pregnancy: planned-birth-control with code system http://open.epic.com/FHIR/CodeSystem/care-experience-preference
Observation (R4).component.valueBoolean | boolean | | | 0..1 | Only used for Patient-Entered Flowsheet Metadata values. Valid values include: For Code = 6-Manually Entered false true
Observation (R4).component.valueCodeableConcept | CodeableConcept | Conditional | | 0..1 | This is the documented value: - For acceptable blood products for pregnancy observations, this is a blood product that is acceptable to the patient in the case of an emergency, particularly during labor and delivery. - For anesthesia methods planned for pregnancy observations, this is a type of anesthesia that a patient plans to request for labor and delivery; it is usually discussed and captured during prenatal care. - For birth control planned after pregnancy observations, this is the patient's planned postpartum birth control method.
Observation (R4).component.valueString | string | | | 0..1 | Values for the patient-entered flowsheet metadata.
Observation (R4).dataAbsentReason | CodeableConcept | | | 0..1 | Indicates that an Observation value is missing.
Observation (R4).effectiveDateTime (also: effectiveDateTime) | dateTime | | | 0..1 | The instant when the observation is made or verified. For preferred pharmacy, this is when pharmacy was last entered, updated, or verified by a clinical user or patient. This is skipped when it's updated by a background user. For flowsheet-based observations, the date and time the value was taken.
Observation (R4).effectivePeriod | Period | | | 0..1 | This is the date range for the pregnancy the data was documented on, when applicable. Pregnancy data always has a start date and might or might not have an end date. The date range reflects when the pregnancy was noted and resolved, and is likely to be based on when the patient received care rather than specifically being clinical.
Observation (R4).encounter | Reference | Conditional | | 0..1 | Reference to an Encounter resource that this Observation belongs to. This element is only used by flowsheet-based Observations.
Observation (R4).extension (supportingInfo) (also: extension (supportingInfo)) | Extension | Conditional | Yes | 0..* | If the measurement is linked to a note written by a clinician, this returns a reference to the DocumentReference resource. URL: http://hl7.org/fhir/StructureDefinition/workflow-supportingInfo
Observation (R4).extension (template-id) (also: extension (template-id)) | Extension | | Yes | 0..1 | This is the specific flowsheet template (a collection of flowsheet rows and groups) to which data was filed. URL: http://open.epic.com/FHIR/StructureDefinition/extension/template-id
Observation (R4).extension (template-id).valueIdentifier | Identifier | | | 1..1 | The system and value for the flowsheet template.
Observation (R4).extension (template-id).valueIdentifier.system | String | | | 1..1 | The OID for the Epic internal flowsheet template identifier. This varies based on the specific Epic environment.
Observation (R4).extension (template-id).valueIdentifier.value | String | | | 1..1 | The Epic internal identifier of the flowsheet template.
Observation (R4).focus | Reference | | Yes | 0..1 | This is the pregnancy on which the data has been documented, when applicable.
Observation (R4).id | id | | | 0..1 | FHIR ID of the resource.
Observation (R4).interpretation | CodeableConcept | | Yes | 0..* | Returns a system of http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation and value of "A" for flowsheet values with an abnormal status. Returns an

environment-specific OID system with a value of "SIG" for flowsheet values with a significant status.

Observation (R4).issued | instant | Conditional | | 0..1 | For flowsheet-based observations, the date and time the observation was documented.

Observation (R4).method | CodeableConcept | | | 0..1 | For patient-entered flowsheets, this element is returned with a text value of "Patient-entered".

Observation (R4).note | Annotation | | Yes | 0..1 | The comment associated with the value.

Observation (R4).performer | Reference | | Yes | 0..* | The provider or patient who entered the documentation. For preferred pharmacy, if it was entered by a background process, this element is skipped.

Observation (R4).performer.extension (performerFunction) (also: extension (performerFunction)) | Extension | | Yes | 0..1 | Only used by flowsheet-based Observations. Additional Practitioners associated with the Observation value, if applicable. Possible values include: "data entry person" for Practitioners who transcribed the Observation into the system "verifier" for Practitioners who cosigned the Observation URL: http://hl7.org/fhir/StructureDefinition/event-performerFunction

Observation (R4).status | code | | | 1..1 | For preferred pharmacy observations, this is final if the pharmacy is currently preferred. This is cancelled if the pharmacy is no longer preferred. For observations for acceptable blood products for pregnancy, anesthesia methods planned for pregnancy, birth control planned after pregnancy, circumcision requested on behalf of pregnancy, delivery method planned for pregnancy, expected delivery location for pregnancy, and support person for pregnancy, this is final while there is documentation on file and is cancelled otherwise. For flowsheet-based observations, this element has the following possible values: "preliminary" - value is pended "final" - value has been filed "amended" - value has been edited "entered-in-error" - value has been deleted

Observation (R4).subject | Reference | | | 1..1 | This is the patient.

Observation (R4).valueCodeableConcept | CodeableConcept | | | 0..1 | This is the documented value: - For circumcision requested on behalf of pregnancy observations, this is the parents' decision regarding circumcision after the baby is born. - For delivery method planned for pregnancy observations, this is the planned method of delivery as documented prior to labor. - For expected delivery location for pregnancy observations, this is the location where the pregnant woman plans to deliver. - For preferred pharmacy observations, this is the NPI code of the pharmacy. For flowsheet-based observations, this element is populated if the data type is a category or custom list. Only one of the value* elements is returned.

Observation (R4).valueDateTime | dateTime | | | 0..1 | For flowsheet-based observations, this element is populated if data type is date. Only one of the value* elements is returned.

Observation (R4).valueQuantity | Quantity | | | 0..1 | For flowsheet-based observations, this element is populated if a value has units or the data type is numeric. Only one of the value* elements is returned.

Observation (R4).valueString | string | | | 0..1 | This is the documented value: - For support person for pregnancy observations, this is the person who has been designated to be the patient's support person for the pregnancy. For flowsheet-based observations, this element is populated if a value doesn't have units or the data type is string. Only one of the value* elements is returned.

Observation (R4).valueTime | time | | | 0..1 | For flowsheet-based observations, this element is populated if data type is time. Only one of the value* elements is returned.

Sample request

```
GET https://hostname/instance/api/FHIR/R4/Observation/DHGMA6y2kEFx934CuY08qWnKDscNin.hiC3qxW.hDJbg6bJRuWkLBkNeaw5H94Mm
```

Sample response


```
{
  "resourceType": "Observation",
  "id": "DHGMA6y2kEFx934CuY08qWnKDscNin.hiC3qxW.hDJbg6bJRuWkLBkNeaw5H94Mm",
  "status": "final",
  "category": [
    {
      "coding": [
        {
          "system": "http://hl7.org/fhir/us/core/CodeSystem/us-core-category",
          "code": "care-experience-preference",
          "display": "Care Experience Preference"
        }
      ],
      "text": "Care Experience Preference"
    }
  ],
  "code": {
    "coding": [
      {
        "system": "http://loinc.org",
        "code": "95541-9",
        "display": "Care experience preference"
      }
    ],
    "text": "Care experience preference"
  },
  "subject": {
    "reference": "Patient/eOgS6zE7C8czUV.9e8lcHoQ3",
    "display": "Sam, Ham"
  },
  "valueCodeableConcept": {
    "coding": [
      {
        "system": "http://hl7.org/fhir/sid/us-npi",
        "code": "1013192590",
        "display": "PLEXUS PHARMACY - LANCASTER, SC - 703 N WHITE STREET"
      }
    ],
    "text": "PLEXUS PHARMACY - LANCASTER, SC - 703 N WHITE STREET"
  }
}
```


10.1.40 Observation.Read (Genomics) (R4)

File: site/apis/10-Clinical-General/1-R4/10304-Observation-Read-Genomics-R4.html
Epic API ID | 10304
FHIR version | R4
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Observation/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10304

Description

The Observation.Read (Genomics) resource returns genomic variant information. Genomic variants are alterations in the most common DNA sequence and are fundamentally different from traditional lab data. For example, a given CBC test is relevant immediately after it is resulted but is generally less relevant over time because a patient's blood count changes over time. In contrast, genomic variant information maintains its relevancy as time passes because a patient's DNA remains largely unchanged throughout their life.

When variant information is available discretely, external systems can use this variant information to drive clinical decision support and leverage it in clinical and organizational reporting. This reference can be queried by a patient ID.

Variant data responses are split into different types of genomic observations representing high-level findings and implications. These observations are variant, genotype, diagnostic implication, and therapeutic implication. The response also includes a grouper observation which contains references to all genomic observations that are part of the same piece of variant data. Each genomic observation contains unique components and metadata and can be accessed directly through a Read. If no code is specified in the request URL all genomic observation types will be returned.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | False | | | The Observation FHIR ID. If provided, other search criteria are bypassed.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Observation | Observation | False | | | An instance of the R4 Observation resource.
Observation.extension (also: Affected-exons) | Extension | False | Yes | | The reported exons for a variant.
Observation.extension (also: Affected-introns) | Extension | False | Yes | | The reported introns for a variant.
Observation.basedOn | Reference | False | Yes | | The procedure or service associated with this observation.
Observation.category | CodeableConcept | True | Yes | | The Observaction type. Always "genomics." A related "laboratory" observation is also returned.
Observation.code | CodeableConcept | False | | | The code of the observation.
Observation.component | BackboneElement | True | Yes | | Component information.
Observation.component.extension (also: Clonal-hematopoiesis) | Extension | False | Yes | | Returned only for Diagnostic Implication elements. Indicates the degree of confidence that the variant arises from clonal hematopoiesis rather than representing a primary tumor variant.
Observation.component.extension | Extension | False | Yes | | Returned only for Therapeutic Implication elements with a Medication-assessed (51963-7) component. The recommended alteration to the medication based on the reported drug-gene interactions.
Observation.component.extension (also: Interpretation-annotation) | Extension | False | Yes | | Returned only for Diagnostic Implication and Therapeutic Implication elements. The free-text classification narrative associated with the genomic interpretation. When the original note contained rich-text formatting (e.g., bold, tables, or images) that could not be preserved in plain text, one or more data-conversion-warning child extensions are included.
Observation.component.valueCodeableConcept | CodeableConcept | True | | | Allelic-read-depth (82121-5): The number of reads that identified the allele in question. Returned only for Variant observations. Allelic-state (53034-5): The relationship between alleles found at the same locus on different chromosomes. Returned only for Variant observations. Alt-allele (69551-0): The DNA string in the test sample that differs from the reference sequence. Returned only for Variant observations. Amino-acid-chg (48005-3):The amino acid change caused by the variant. Returned only for Variant observations. Associated-phenotype (81259-4): The phenotype associated with the variant. Returned only for Diagnostic Implication observations. Chromosome-identifier (48000-4): The chromosome where the variant is located. Returned only for Variant observations. Clinical-significance (53037-8): The clinical significance of the variant. Returned only for Diagnostic Implication observations. Copy-number (82155-3): The copy number of the large variant when applicable. Returned only for Diagnostic Implication observations. Cytogenetic-location (48001-2): The cytogenetic (chromosome) location of the variant. Returned only for Diagnostic Implication observations. Dna-chg (48004-6): The nucleotide change in the DNA. Returned only for Diagnostic Implication observations. Dna-chg-type (48019-4): The type of the DNA change reported for the variant. Returned only for Diagnostic Implication observations. Effect-medication-efficacy (51961-1): The predicted phenotype for the ability of a drug to bind to the intended site to deliver its intended effect. Returned only for Therapeutic Implication observations. Effect-medication-high-risk (83009-1): The predicted phenotype for risk of adverse drug reaction. Returned only for Therapeutic Implication observations. Effect-medication-metabolism (53040-2): The effect on drug metabolism for a pharmacogenomic variant. Returned only for Therapeutic Implication observations. Exact-start-end (81254-5): The starting and ending position of the variant's allele. Returned only for Variant observations. Functional-effect: The effect that the variant has on the function of the gene. Returned only for Diagnostic Implication observations. Gene-studied (48018-6): The gene where the variant is located. Returned only for Genotype and Variant observations. Genomic-dna-chg (81290-9): The nucleotide change that occurs in the DNA relative to the genomic reference sequence or chromosome. Returned only for Variant observations. Genomic-ref-seq (48013-7): The genomic reference sequence. Returned only for Variant observations. Genomic-source-class (48002-0): The source of the genomic variant. Defines whether the variant is germline or somatic. Returned only for Variant observations. Inner-start-end (81302-2): The inner start and end position of a region known to be affected by a variant. Returned only for Variant observations. Medication-assessed (51963-7):

The medication being assessed for usage based on drug-gene interactions. Returned only for Therapeutic Implication observations. Mode-of-inheritance: The mode of inheritance for the phenotype associated with the variant. Returned only for Diagnostic Implication observations. Molecular-consequence: The effect of the variant on a genomic, transcript, or protein sequence Outer-start-end (81301-4): The outer start and end position of a region known to be affected by a variant. Returned only for Variant observations. Ref-allele (69547-8): The DNA string in the reference allele with which the DNA string in the test sample differs. Returned only for Variant observations. Reference-sequence-assembly (62374-4): The reference genome used for the sequence. Returned only for Variant observations. Sample-allelic-frequency (81258-6): Reports the percentage of all the reads at this genomic location that were represented by the given allele. Returned only for Variant observations. Transcript-ref-seq (51958-7): The transcript reference sequence used to identify the variants. Returned only for Variant observations. Variant-inheritance: From whom the variant was inherited. Variation-code (81252-9): The definition of the variant. Contains the HGVS name and associated codes.

Observation.derivedFrom | Reference | False | Yes | | References the genotype observations on the same variant. Specific to therapeutic implications.

Observation.effectiveDateTime | dateTime | True | | | The prioritized instant of the associated order.

Observation.extension (also: Effect-medication-transport) | Extension | False | Yes | | Returned only for Therapeutic elements. The predicted phenotype for effect on drug transportation. LOINC 82309-6

Observation.extension | Extension | True | Yes | | Returned only for Variant elements. The protein reference sequence for this variant.

Observation.hasMember | Reference | True | Yes | | Contains resource IDs for sub-types. Only included if the code=grouper. Values include: hasMember: diagnostic-implication hasMember: variant hasMember: genotype hasMember: therapeutic-implication

Observation.issued | instant | True | | | The observation instant.

Observation.method | CodeableConcept | True | | | The method used to analyze the chromosome structural variation.

Observation.note | Annotation | True | Yes | | The clinical interpretation of the genomic variant.

Observation.note.extension (also: Oncogenicity) | Extension | True | Yes | | Returned only for Diagnostic Implication and Therapeutic Implication elements. The oncogenicity of the genomic variant

Observation.extension (also: Oncogenicity) | Extension | False | Yes | | Returned only for Diagnostic Implication and Therapeutic Implication elements. The oncogenicity of the genomic variant

Observation.extension (also: Repeat-pair) | Extension | False | Yes

Observation.status | code | False | | | The status of the lab order associated with the variant.

Observation.subject | Reference | True | | | The Patient FHIR ID linked to the variant information.

Observation.valueCodeableConcept | CodeableConcept | False | | | Allelic-read-depth (82121-5): The number of reads that identified the allele in question. Returned only for Variant elements. Allelic-state (53034-5): The relationship between alleles found at the same locus on different chromosomes. Returned only for Variant elements. Alt-allele (69551-0): The DNA string in the test sample that differs from the reference sequence. Returned only for Variant elements. Amino-acid-chg (48005-3):The amino acid change caused by the variant. Returned only for Variant elements. Associated-phenotype (81259-4): The phenotype associated with the variant. Returned only for Diagnostic Implication and Therapeutic Implication elements. Chromosome-identifier (48000-4): The chromosome where the variant is located. Returned only for Variant elements. Clinical-significance (53037-8): The clinical significance of the variant. Returned only for Diagnostic Implication and Therapeutic Implication elements. Copy-number (82155-3): The copy number of the large variant when applicable. Returned only for Diagnostic Implication elements. Cytogenetic-location (48001-2): The cytogenetic (chromosome) location of the variant. Returned only for Diagnostic Implication elements. Cytogenomic-nomenclature (81291-7): The reported ISCN name for Structural, Fusion, and Copy Number variants. Returned only for Variant elements. Dna-chg (48004-6): The nucleotide change in the DNA. Returned only for Diagnostic Implication elements. Diagnostic-evidence-level (93044-6): The diagnostic evidence level of the variant when applicable. Returned only for Diagnostic Implication observations. Dna-chg-type (48019-4): The type of the DNA change reported for the variant. Returned only for Diagnostic Implication elements. Effect-medication-efficacy (51961-1): The predicted phenotype for the ability of a drug to bind to the intended site to deliver its intended effect. Returned only for Therapeutic Implication elements. Effect-medication-high-risk (83009-1): The predicted phenotype for risk of adverse drug reaction. Returned only for Therapeutic Implication elements. Effect-medication-metabolism (53040-2): The effect on drug metabolism for a pharmacogenomic variant. Returned only for Therapeutic Implication elements. Exact-start-end (81254-5): The starting and ending position of the variant's allele. Returned only for Variant elements. Functional-effect: The effect that the variant has on the function of the gene. Returned only for Diagnostic Implication elements. Gene-studied (48018-6): The gene where the variant is located. Returned only for Genotype and Variant elements. Genomic-dna-chg (81290-9): The nucleotide change that occurs in the DNA relative to the genomic reference sequence or chromosome. Returned only for Variant elements. Genomic-ref-seq (48013-7): The genomic reference sequence. Returned only for Variant elements. Genomic-source-class (48002-0): The source of the genomic variant. Defines whether the variant is germline or somatic. Returned only for Variant elements. Inner-start-end (81302-2): The inner start and end position of a region known to be affected by a variant. Returned only for Variant elements. Medication-assessed (51963-7): The medication being assessed for usage based on drug-gene interactions. Returned only for Therapeutic Implication elements. Mode-of-inheritance: The mode of inheritance for the phenotype associated with the variant. Returned only for Diagnostic Implication and Therapeutic Implication elements. Molecular-consequence: The effect of the variant on a genomic, transcript, or protein sequence Outer-start-end (81301-4): The outer start and end position of a region known to be affected by a variant. Returned only for Variant elements. Prognostic-evidence-level (93044-6): The prognostic evidence level of the variant when applicable. Returned only for Diagnostic Implication observations. Ref-allele (69547-8): The DNA string in the reference allele with which the DNA string in the test sample differs. Returned only for Variant elements. Reference-sequence-assembly (62374-4): The reference genome used for the sequence. Returned only for Variant elements. Sample-allelic-frequency (81258-6): Reports the percentage of all the reads at this genomic location that were represented by the given allele. Returned only for Variant elements. Therapeutic-evidence-level (93044-6): The therapeutic evidence level of the variant when applicable. Returned only for Diagnostic Implication observations. Transcript-ref-seq (81258-6): The transcript reference sequence used to identify the variants. Returned only for Variant elements. Variant-inheritance: From whom the variant was inherited. Variation-code (81252-9): The definition of the variant. Contains the HGVS name and associated codes.

Sample request

```
https://hostname/instance/api/FHIR/R4/Observation/eUZ671fmZxtLjtto.i5tiC17grWlaHA2QYBL6pkBzrsg3
```

Sample response

```
<Observation xmlns="http://hl7.org/fhir">
  <id value="eUZ671fmZxtLjtto.i5tiC17grWlaHA2QYBL6pkBzrsg3"/>
  <basedOn>
    <reference value="ServiceRequest/eY0zM9XwMAsfr83odpCgcifyOxXr.2BKRPWcXghHs9b03"/>
  </basedOn>
  <status value="amended"/>
  <category>
    <coding>
      <system value="http://open.epic.com/FHIR/StructureDefinition/observation-category"/>
      <code value="genomics"/>
      <display value="Genomics Grouper"/>
    </coding>
```

```
    <text value="Genomics Grouper"/>
  </category>
  <category>
    <coding>
      <system value="http://terminology.hl7.org/CodeSystem/observation-category"/>
      <code value="laboratory"/>
      <display value="Laboratory"/>
    </coding>
    <text value="Laboratory"/>
  </category>
  <code>
    <coding>
      <system value="http://ncithesaurus-stage.nci.nih.gov"/>
      <code value="C43359"/>
      <display value="Group"/>
    </coding>
    <text value="Genomic Result Group"/>
  </code>
  <subject>
    <reference value="Patient/eyr87CEI8vy-nL6hL5yccppCd83J03x-7iMjZnGMSlrE3"/>
    <display value="Garcia Rodriguez, Hugo"/>
  </subject>
  <effectiveDateTime value="2021-12-17T06:00:00Z"/>
  <issued value="2021-12-17T15:37:17Z"/>
  <hasMember>
    <reference value="Observation/eUZ671fmZxtLjtto.i5tiC9BweAAcVFBDp6dJ.vN4VzU3"/>
    <display value="Variant"/>
  </hasMember>
  <hasMember>
    <reference value="Observation/eUZ671fmZxtLjtto.i5tiC3UzHp6paXUbJr9FMjMIDX43"/>
    <display value="Diagnostic Implication"/>
  </hasMember>
  <hasMember>
    <reference value="Observation/e4t0a8Yo6ZFhNUAbGJoR4P-DeSZGzoSLzEyxaQ0oD1gM3"/>
    <display value="Therapeutic Implication"/>
  </hasMember>
  <hasMember>
    <reference value="Observation/e4t0a8Yo6ZFhNUAbGJoR4P2m1Dsb.hyC8Oi7resT-R8I3"/>
    <display value="Genotype"/>
  </hasMember>
</Observation>
```

Change log

Epic Version | Description | Type
May 2026 | FHIR IDs for Observation (Genomics) (R4) resources will no longer be valid if generated by previous versions, due to a change in format. | Breaking change
August 2025 | Observation (Genomics) is updated such that repeat expansion number now supports quantity and range values. | Breaking change
August 2025 | For variant concepts 511- Allele Start/End, Structural Inner Start/End- 546, and Structural Outer Start/End 548, the system of http://hl7.org/fhir/uv/genomics-reporting/CodeSystem/tbd-codes-cs was updated to http://loinc.org because the returned codes are LOINC codes. | Breaking change

10.1.41 Observation.Read (Phenotype) (R4)

File: site/apis/10-Clinical-General/1-R4/11330-Observation-Read-Phenotype-R4.html
Epic API ID | 11330
FHIR version | R4
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Observation/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=11330

Description

This web service retrieves a phenotype from a patient's chart. Phenotypes represent observable characteristics of an individual and are used by laboratories to determine which genes to prioritize and which variants to include in the lab report.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | Required | | 1..1 | The Observation FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description

Observation (R4) | Observation
Observation (R4).category | CodeableConcept | | Yes | 1..1 | The Observation type. Always "phenotype". System: http://terminology.hl7.org/CodeSystem/observation-category Code: phenotype
Observation (R4).code | CodeableConcept | | | 1..1 | The code of the Observation. System: http://human-phenotype-ontology.org Code: HPO Code, for example: HP:0009805
Observation (R4).effectiveDateTime | dateTime | | | 1..1 | When the phenotype was first documented.
Observation (R4).id | id | | | 1..1 | The FHIR ID for the Observation.
Observation (R4).note | Annotation | | Yes | 0..1 | Patient specific comments about the phenotype.
Observation (R4).performer | Reference | | Yes | 0..* | A reference to the documenting provider or user.
Observation (R4).status | code | | | 1..1 | Always set to "final".
Observation (R4).subject | Reference | | | 1..1 | The FHIR ID for the patient.
Observation (R4).valueCodeableConcept | CodeableConcept | | | 1..1 | The LOINC code for presence or absence of a phenotype. The values come from LOINC Answer List LL1937-3. Positive and Resolved both map to "Present (LA9633-4)" Negative maps to "Absent (LA9634-2)"

Sample request

```
GET https://hostname/instance/api/FHIR/R4/Observation/fzcMkZ2CZ4RRxccgmgdWeHw4
```

Sample response


```
{
  "resourceType": "Observation",
  "id": "fzcMkZ2CZ4RRxccgmgdWeHw4",
  "status": "final",
  "category": [
    {
      "coding": [
        {
          "system": "http://open.epic.com/FHIR/StructureDefinition/observation-category",
          "code": "phenotype",
          "display": "Phenotype"
        }
      ],
      "text": "Phenotype"
    }
  ],
  "code": {
    "coding": [
      {
        "system": "http://human-phenotype-ontology.org",
        "code": "HP:0001251",
        "display": "Ataxia"
      }
    ],
    "text": "Ataxia"
  },
  "subject": {
    "reference": "Patient/eKswDJcNtbx94h39Rf33YUg3",
    "display": "Test, Patient"
  },
  "effectiveDateTime": "2025-07-23",
  "performer": [
    {
      "reference": "Practitioner/euJOUNeZyX6mofWELWG0Ibg3",
      "display": "Starter Provider"
    }
  ],
  "valueCodeableConcept": {
    "coding": [
      {
        "system": "http://loinc.org",
        "code": "LA9633-4",
        "display": "Present"
      }
    ],
    "text": "Positive"
  },
  "note": [
    {
      "text": "note content"
    }
  ]
}
```


10.1.42 Observation.Read (SDOH Assessments) (R4)

File: site/apis/10-Clinical-General/1-R4/11105-Observation-Read-SDOH-Assessments-R4.html
Epic API ID | 11105
FHIR version | R4
Categories | Clinical General
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes

Keywords | sdoh
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Observation/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=11105

Description

This web service retrieves assessment data from a patient's chart for only those assessments that relate to social drivers of health (SDOH) data. The same SDOH data is also returned in the Observation (Assessments) API, but this API is limited to include only SDOH data.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | False | | | The Observation FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Observation | Observation | False | | | Resource bundle. For more information, refer to the FHIR specification.
Observation.category | CodeableConcept | False | Yes | | Always returns "survey" and "sdoh".
Observation.code | CodeableConcept | False | | | The LOINC code that corresponds to the measured SDOH.
Observation.dataAbsentReason | CodeableConcept | True | | | Indicates why an Observation value is missing, if applicable.
Observation.derivedFrom | Reference | True | Yes | | This element displays any Observations that are related to the returned Observation.
Observation.effectiveDateTime | dateTime | False | | | The date and time the Observation was taken.
Observation.encounter | Reference | True | | | Encounter associated with this observation value, if applicable.
Observation.hasMember | Reference | True | Yes | | This element returns any panel members an Observation contains.
Observation.issued | instant | False | | | The date and time the observation was taken.
Observation.note | Annotation | True | Yes | | The comment.
Observation.performer | Reference | False | Yes | | Who filed this Observation.
Observation.status | code | False | | | Patient associated with the Observation.
Observation.subject | Reference | False | | | Patient associated with the Observation.
Observation.valueCodeableConcept | CodeableConcept | This element is populated if the data type is category or custom list. | | | This element is populated if the data type is category or custom list.
Observation.valueQuantity | Quantity | This element is populated if a value has units or the data type is numeric or a percentile. | | | This element is populated if a value has units or the data type is numeric or a percentile.
Observation.valueString | string | This element is populated if a value doesn't have units or the data type is string. | | | This element is populated if a value doesn't have units or the data type is string.
Observation.extension (also: workflow-supportingInfo) | Extension | True | Yes | | If the measurement is linked to a note written by a clinician, this is a reference to the DocumentReference resource.

Sample request

```
https://hostname/instance/api/FHIR/R4/Observation/efcc3Attawr2io1psQPAqb70IsFeXNeDXRiCG5IIx6wQszyW3dolmz2usKrtZAlDaQtZxA
    xfBEf5I0qRJDkRmvg3
```

Sample response


```
{
  "resourceType": "Observation",
  "id": "efcc3Attawr2io1psQPAqb70IsFeXNeDXRiCG5IIx6wQszyW3dolmz2usKrtZAlDaQtZxAxfBEf5I0qRJDkRmvg3",
  "extension": [
    {
      "valueReference": {
        "reference": "DocumentReference/eUnlJdBrqpaShs-0W7WVWWZ.xcSopy17JDplU0b993NI3"
      },
      "url": "http://hl7.org/fhir/StructureDefinition/workflow-supportingInfo"
    }
  ],
  "status": "amended",
  "category": [
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/observation-category",
          "code": "survey",
          "display": "Survey"
        }
      ]
    },
    {
      "coding": [
        {
          "system": "http://hl7.org/fhir/us/core/CodeSystem/us-core-category",
          "code": "sdoh",
          "display": "SDOH"
```

```
        }
      ]
    }
  ],
  "code": {
    "coding": [
      {
        "system": "http://open.epic.com/FHIR/StructureDefinition/observation-flowsheet-id",
        "code": "tijeKcVSPxJBt4VeWqyuskG7",
        "display": "(I/We) worried whether (my/our) food would run out before (I/we) got money to buy more.  "
      },
      {
        "system": "http://loinc.org",
        "code": "88122-7",
        "display": "(I/We) worried whether (my/our) food would run out before (I/we) got money to buy more.  "
      }
    ]
  },
  "subject": {
    "reference": "Patient/eDOEgtZkys6EIcn1-XIATQQ3",
    "display": "Smith, Patient Test"
  },
  "encounter": {
    "reference": "Encounter/eh0sTg.kAw5MItnpwiwnxh5WGO44DNwtsZgYnU05kLEU3",
    "identifier": {
      "use": "usual",
      "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.3.698084.8",
      "value": "10005526256"
    },
    "display": "Office Visit"
  },
  "effectiveDateTime": "2024-08-29T21:00:00Z",
  "issued": "2024-08-29T21:10:43Z",
  "performer": [
    {
      "reference": "Practitioner/e4ZKPXZ0ux4Z80UHywzHyHg3",
      "display": "Practitioner Test, MD"
    }
  ],
  "valueCodeableConcept": {
    "coding": [
      {
        "system": "http://loinc.org",
        "code": "LA28397-0",
        "display": "Often true"
      }
    ]
  },
  "note": [
    {
      "text": "Patient struggled to answer question"
    }
  ],
  "derivedFrom": [
    {
      "reference":
    "Observation/efcc3Attawr2io1psQPAqb70IsFeXNeDXRiCG5IIx6wQGiTqxqrzg1smuWXTuFiQqghuLqt31Ln0ktmGvI.iYrQ3"
    }
  ]
}
```


10.1.43 Observation.Read (Social History) (R4)

File: site/apis/10-Clinical-General/1-R4/967-Observation-Read-Social-History-R4.html
Epic API ID | 967
FHIR version | R4
Categories | Clinical General
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Keywords | Smoking; Social; Social history;Flowsheets
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | November 2019
HTTP method | GET
URL template | api/FHIR/R4/Observation/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=967

Description

The Observation (Social History) sub-resource retrieves one or more of the following details:

Type of data | Description
Smoking history | The status of a patient's tobacco use. This could be used to determine if the patient is a current or former tobacco user, what type of tobacco product is/was used, and the dates the patient started and, if a former user, stopped using tobacco. This includes passive exposure information in the code response element.
Pregnancy Status | The patient’s current pregnancy status.
Pregnancy Intention | The patient’s intentions of becoming pregnant in the next year. Available starting in the February 2024 version of Epic.
Lactation Status | The patient’s current lactation status.
Physical Findings of Uterus | Whether the patient has a uterus.
Estimated Delivery Date | The patient’s estimated date of delivery.
Last Menstrual Period | The patient’s last menstrual period.
Alcohol use history | The patient's alcohol use history and, if applicable, the type and amount of alcohol the patient consumes.
Drug use history | The patient's drug use history and, if applicable, the type and amount of drugs the patient consumes.
Living situation | The patient's current living situation.
Occupational history | The patient's occupational history, including start and end dates, industry, employer, and additional comments.
Social documentation | The social documentation notes stored in the patient's history.
Education history | The patient's education history.
Sexual orientation | The patient's sexual orientation. Available starting in the May 2024 version of Epic.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | Required | | | The Observation FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Observation (R4) | Observation | | | | This is the root
Observation (R4).category | CodeableConcept | | Yes | 0..* | This is always "social-history".
Observation (R4).code | CodeableConcept | | | 1..1 | The following value(s), depending on the type of data returned: Smoking history: LOINC code 72166-2 SNOMED code 365980008. Sexual Orientation: LOINC code 76690-7 Alcohol use history: LOINC code 11331-6 SNOMED code 228273003 Drug use history: LOINC code 11343-1 SNOMED code 228366006 Living situation: SNOMED code 365508006 Pregnancy status: LOINC code 82810-3 Pregnancy intention: LOINC code 86645-9 Lactation status: LOINC code 63895-7 Physical findings of uterus: LOINC code 32486-3 Estimated delivery date: LOINC code 11778-8 Menstrual status: LOINC code 3146-8 Last menstrual status: LOINC code 8665-2
Observation (R4).component | BackboneElement | | Yes | 0..* | Applies only to smoking history, alcohol use history, drug use history, education, and occupational history. Holds additional detail depending on the type of Observation: Smoking history: Can include one or more of the following code values, paired with either a valueCodeableConcept or a valueQuantity that includes additional detail: Tobacco product (LOINC 81228-9; SNOMED 53661000146106) Cigarette pack-years (SNOMED 401201003) Tobacco use duration (LOINC 88029-4) Packs per day (LOINC 8663-7; SNOMED 266918002) Alcohol use history: The amount of alcohol consumed per day (SNOMED 160573003), expressed in UCUM units as alcoholic_units/day. Drug use history: Can include one or more of the following code values, paired with either a valueCodeableConcept or a valueString that includes additional detail: Type of substance (LOINC 63643-1; SNOMED 410942007) Frequency of misuse (SNOMED 228390007) Education: Can include the following code value, paired with either a valueCodeableConcept or valueString that includes additional detail: The patient’s highest level of education (LOINC code 82589-3) Occupational: Can include one or more of the following code values, paired with either a valueCodeableConcept or valueString that includes additional detail: The patient’s occupation (LOINC code LP133724-7) The patient’s employer (LOINC code 80427-8) The patient’s occupation industry (LOINC code 86188-0)
Observation (R4).component.code | CodeableConcept | | | 1..1 | Often a LOINC or SNOMED code.
Observation (R4).component.valueCodeableConcept | CodeableConcept | | | 0..1 | CodeableConcept value.
Observation (R4).component.valueQuantity | Quantity | | | 0..1 | Quantity value.
Observation (R4).component.valueString | string | | | 0..1 | String value.
Observation (R4).dataAbsentReason | CodeableConcept | | | 0..1 | Populated if there is no last menstrual period start date but there is a last menstrual period start date comment or precision. Populated if there is no lactation status but there is a lactation status comment.
Observation (R4).derivedFrom | Reference | | Yes | 0..* | References the corresponding menstrual status Observation that the pregnancy status is derived from.
Observation (R4).effectiveDateTime | dateTime | | | 0..1 | The date of the encounter on which the Observation resource was documented for the following types of data: Pregnancy status Menstrual status Last menstrual status Lactation status Sexual Orientation Pregnancy intention
Observation (R4).effectivePeriod | Period | | | 0..1 | The start and end dates of the Observation for the following types of data: Smoking history For smokers, when the patient started and quit smoking (if applicable), and data absent extension if there are no dates on file. For non-smokers, the date range is the from the date the status was recorded to today. Occupation history
Observation (R4).id | id | | | 0..1 | The Observation FHIR ID.
Observation (R4).interpretation | CodeableConcept | | Yes | 0..* | Precision of the last menstrual period.
Observation (R4).issued | instant | | | 0..1 | The date when this status is documented. This is set to the date that the status was assessed, if available. Otherwise, this is set to one of the following values to indicate that the response is considered up-to-date as of the specified date: The date of the patient's most recent history encounter, if available. The current date. The issued date may not be populated for every Observation type.
Observation (R4).note | Annotation | | Yes | 0..* | The comments for smoking history, alcohol use history, drug use history, employment history, last menstrual period, or lactation status. For organizations in the Netherlands, this is sent in both provider and patient contexts. For all other organizations, this is sent only in provider context.
Observation (R4).performer | Reference | | Yes | 0..* | The person who documented the status. This is not populated if the status is never assessed.
Observation (R4).status | code | | | 1..1 | This value is always "final". More detailed status information, such as whether the patient’s smoking status is unknown, can be found by reviewing the values specified in the code element or the valueCodeableConcept element.
Observation (R4).subject | Reference | | | 0..1 | The Patient FHIR ID.
Observation (R4).valueBoolean | boolean | | | 0..1 | Yes/No answer to whether the patient has a uterus.
Observation (R4).valueCodeableConcept | CodeableConcept | | | 0..1 | The SNOMED code or other value pertaining to smoking status, alcohol use status, drug use status, living situation, pregnancy status, lactation status, or pregnancy intention. Default Epic-to-SNOMED mappings are listed below. This value can also be modified by organization-specific mapping. Smoking history supported values: 449868002-Smokes tobacco daily (finding) 428041000124106-Current some day smoker 230059006-Occasional cigarette smoker (only for organizations in the Netherlands)

77176002-Smoker, current status unknown 8517006-Former smoker 266919005-Never smoker 43381005-Passive smoker (only for organizations in the Netherlands) 266927001-Unknown if ever smoker 266919005-Never smoker 266927001-Unknown if ever smoker 428071000124103-Heavy tobacco smoker 428061000124105-Light tobacco smoker Alcohol use history supported values: 219006-Current drinker of alcohol (finding) 105542008-Current non-drinker of alcohol (finding) OTH-Other 82581004-Ex-drinker (finding) 228274009-Lifetime non-drinker (finding) Drug use history supported values: 361055000-Misuses drugs (finding) 228367002-Does not misuse drugs (finding) OTH-Other 44870007-Misused drugs in past (finding) 228368007-Has never misused drugs Living situation has no default mapping, but can be customized by the Epic organization. Pregnancy status supported values: 77386006-Pregnancy (finding) UNK-Unknown 60001007-Not pregnant (finding) Lactation status supported values: 373066001-Yes 373067005-No 261665006-Unknown Pregnancy intention supported values: 454411000124108-Wants to become pregnant (finding) 454391000124108-Ambivalent about becoming pregnant (finding) 454401000124105-No desire to become pregnant (finding) 454381000124105-Not sure of desire to become pregnant (finding)

Observation (R4).valueDateTime | dateTime | | | 0..1 | Estimated delivery date or the start date of the last menstrual period, if applicable.

Observation (R4).valueString | string | | | 0..1 | The text of the social documentation notes. Applies only to social documentation.

Sample request

```
GET https://hostname/instance/api/FHIR/R4/Observation/euvO8iq4aYWriTr-MA7S-IkCxqCXJDmc.rRo7MsHFYNG0HdfUGgXpHY.GcKccA1fi3
```

Sample response


```
{
  "resourceType": "Observation",
  "id": "euvO8iq4aYWriTr-MA7S-IkCxqCXJDmc.rRo7MsHFYNG0HdfUGgXpHY.GcKccA1fi3",
  "status": "final",
  "category": [
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/observation-category",
          "code": "social-history",
          "display": "Social History"
        }
      ],
      "text": "Social History"
    }
  ],
  "code": {
    "coding": [
      {
        "system": "http://loinc.org",
        "code": "72166-2",
        "display": "Tobacco smoking status"
      },
      {
        "system": "http://snomed.info/sct",
        "code": "365980008",
        "display": "Finding of tobacco use and exposure (finding)"
      }
    ],
    "text": "Smoking History"
  },
  "subject": {
    "reference": "Patient/eeiBNKuRR84uD0nIFn73BGQ3",
    "display": "Test, Patient FHIR"
  },
  "effectivePeriod": {
    "start": "2009-01-01"
  },
  "issued": "2021-05-11T05:00:00Z",
  "performer": [
    {
      "reference": "Practitioner/ekk17eu0BAPwHQ6s2NDWYEQ3",
      "display": "Dr. FHIR"
    }
  ],
  "valueCodeableConcept": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "449868002",
        "display": "Smokes tobacco daily"
      }
    ],
    "text": "Every Day"
  },
  "component": [
    {
      "code": {
        "coding": [
          {
            "system": "http://loinc.org",
            "code": "81228-9",
            "display": "Tobacco product"
          },
          {
            "system": "http://snomed.info/sct",
            "code": "53661000146106",
            "display": "Type of tobacco used (observable entity)"
```

```
      }
    ],
    "text": "Type"
  },
  "valueCodeableConcept": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "65568007",
        "display": "Cigarette Smoker"
      }
    ],
    "text": "Cigarettes"
  }
},
{
  "code": {
    "coding": [
      {
        "system": "http://loinc.org",
        "code": "81228-9",
        "display": "Tobacco product"
      },
      {
        "system": "http://snomed.info/sct",
        "code": "53661000146106",
        "display": "Type of tobacco used (observable entity)"
      }
    ],
    "text": "Type"
  },
  "valueCodeableConcept": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "81703003",
        "display": "Chews Tobacco"
      }
    ],
    "text": "Chew"
  }
},
{
  "code": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "401201003",
        "display": "Cigarette pack-years (observable entity)"
      }
    ],
    "text": "Cigarette pack-years (observable entity)"
  },
  "valueQuantity": {
    "value": 3.72,
    "unit": "{PackYears}",
    "system": "http://unitsofmeasure.org",
    "code": "{PackYears}"
  }
},
{
  "code": {
    "coding": [
      {
        "system": "http://loinc.org",
        "code": "88029-4",
        "display": "Tobacco use duration"
      }
    ],
    "text": "Years Used"
  },
  "valueQuantity": {
    "value": 12.4,
    "unit": "Years Used",
    "system": "http://unitsofmeasure.org",
    "code": "a"
  }
},
{
  "code": {
    "coding": [
      {
        "system": "http://loinc.org",
        "code": "8663-7",
        "display": "Cigarettes smoked current (pack per day) - Reported"
      },
      {
        "system": "http://snomed.info/sct",
        "code": "266918002",
```

```
            "display": "Tobacco smoking consumption (observable entity)"
          }
        ],
        "text": "Packs/Day"
      },
      "valueQuantity": {
        "value": 0.3,
        "unit": "Packs/Day",
        "system": "http://snomed.info/sct",
        "code": "228963008"
      }
    }
  ]
}
```


Change log

Epic Version | Description | Type

February 2024 | Updated the mappings for the “82810-3 Pregnancy Status” to return with SNOMED values more specific to Pregnancy Status. Be sure to update any functionality that relies on these codes. Also added the “86645-9 Pregnancy Intention” LOINC code which returns values for the patient’s willingness to become pregnant in the next year. | Breaking Change

November 2022 | Fixed a bug that caused the "32486-3 Physical Findings of Uterus" LOINC code return with a SNOMED system value (http://snomed.info/sct) instead of a LOINC system (http://loinc.org). If you use this LOINC code, make sure you aren't relying on seeing a SNOMED system. | Breaking Change

10.1.44 Observation.Search (Care Experience Preference) (R4)

File: site/apis/10-Clinical-General/1-R4/40301-Observation-Search-Care-Experience-Preference-R4.html

Epic API ID | 40301

FHIR version | R4

Categories | Clinical General

API type | USCDI

A la carte | Industry-Standard Level 1 API

USCDI | Yes

Available in inpatient and other contexts | Yes

HTTP method | GET

URL template | /api/FHIR/R4/Observation?based-on={based-on}&category={category}&code={code}&date={date}&focus={focus}&issued={issued}&patient={patient}&subject={subject}&_count={_count}&_id={_id}&_include={_include}&_lastupdated={_lastupdated}&_revinclude={_revinclude}

Supports OAuth 2.0 | True

Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients

Performs scope validation | False

Source | https://fhir.epic.com/Specifications?api=40301

Description

Use this API to retrieve care experience preference observations. The observations can be one of the following:

- Acceptable blood products for pregnancy
- Anesthesia methods planned for pregnancy
- Birth control planned after pregnancy
- Circumcision requested on behalf of pregnancy
- Delivery method planned for pregnancy
- Expected delivery location for pregnancy
- Preferred pharmacy
- Support person for pregnancy

Request parameters

Name | Type | Optional | Array | Cardinality | Description

_count | number | Optional | Yes | | Number of results per page.

_id | token | Optional | Yes | | FHIR resource IDs for the desired resources. Can be used in combination with _count, _include, and _revInclude. All others will be ignored.

_include | string | Optional | Yes | | Used to indicate which referenced resources to include in the search bundle.

_revinclude | string | Optional | Yes | | Used to include Provenance or AuditEvent resources that refer to search results, when available.

category | token | Conditional | Yes | | This is fixed to care-experience-preference. This is required if code is not provided. Note: Category assignments are subjective and can differ across organizations.

code | token | Conditional | Yes | | Required if category is not provided. Formatted as <system>|<code>. Searching for 95541-9 returns all care experience preference observations. All resources are returned with 95541-9 from http://loinc.org. Some resources are also returned with an additional code: - Acceptable blood products for pregnancy: acceptable-blood-products with code system http://open.epic.com/FHIR/CodeSystem/care-experience-preference - Anesthesia methods planned for pregnancy: anesthesia-method-plan with code system http://open.epic.com/FHIR/CodeSystem/care-experience-preference - Birth control planned after pregnancy: planned-birth-control with code system http://open.epic.com/FHIR/CodeSystem/care-experience-preference - Circumcision requested on behalf of pregnancy: circumcision-requested with code system http://open.epic.com/FHIR/CodeSystem/care-experience-preference - Delivery method planned for pregnancy: delivery-method-plan with code system http://open.epic.com/FHIR/CodeSystem/care-experience-preference - Expected delivery location for pregnancy: 85712-8 from http://loinc.org. - Preferred pharmacy: 103980-9 from http://loinc.org. - Support person for pregnancy: 100309-4 from

http://loinc.org. For flowsheet-based observations, valid codes and their systems include: LOINC Code System Value: http://loinc.org or no system value. Epic Encoded Flowsheet Identifier System Value: http://open.epic.com/FHIR/StructureDefinition/observation-flowsheet-id Note: The encoded flowsheet identifier must be generated by IT admins at an Epic organization.

date | date | Optional | Yes | | The date range of the observations. For preferred pharmacy, the date is when any pharmacy preference was last modified/verified/confirmed by a user or patient. For pregnancy-related observations, pregnancies are considered active between their noted and resolution dates, which are not clinical concepts. Data on pregnancies qualify for export when the pregnancy is active at any time during the specified date range. For flowsheet-based observations, the date and time the value was taken.

patient | reference | Conditional | Yes | | This or the subject is required to specify a patient.

subject | reference | Conditional | Yes | | This or the patient is required to specify a patient.

Response elements

Name | Type | Optional | Array | Cardinality | Description

Observation (R4) | Observation

Observation (R4).category | CodeableConcept | | Yes | 1..* | Always includes "care-experience-preference". Flowsheet-based observations can include additional categories.

Observation (R4).code | CodeableConcept | Required | | 1..1 | All resources export with 95541-9 from http://loinc.org. Some resources also export with an additional code: - Acceptable blood products for pregnancy: acceptable-blood-products with code system http://open.epic.com/FHIR/CodeSystem/care-experience-preference - Anesthesia methods planned for pregnancy: anesthesia-method-plan with code system http://open.epic.com/FHIR/CodeSystem/care-experience-preference - Birth control planned after pregnancy: planned-birth-control with code system http://open.epic.com/FHIR/CodeSystem/care-experience-preference - Circumcision requested on behalf of pregnancy: circumcision-requested with code system http://open.epic.com/FHIR/CodeSystem/care-experience-preference - Delivery method planned for pregnancy: delivery-method-plan with code system http://open.epic.com/FHIR/CodeSystem/care-experience-preference - Expected delivery location for pregnancy: 85712-8 from http://loinc.org. - Preferred pharmacy: 103980-9 from http://loinc.org. - Support person for pregnancy: 100309-4 from http://loinc.org. Flowsheet-based observations additionally include: Encoded value of the internal Epic flowsheet ID External concept mappings such as LOINC and SNOMED

Observation (R4).component | BackboneElement | | Yes | 0..* | Includes extra information about the Observation. Populated only for: Patient-Entered Flowsheet Metadata Represents strict metadata if the coding sub-element is populated. The following strict metadata can be returned: 1-UUID The unique reading ID 2-Blood Glucose Meal Times 1-Before meal 2-After meal 3-Insulin Delivery Reason 1-Basal 2-Bolus 5-VO2 Max Test Type 1-Max exercise test 2-Sub-max exercise prediction 3-Non exercise prediction 6-Manually Entered - Whether or not the patient manually entered this reading into the system (as opposed to automatically entered by a device). false true Represents loose metadata if only the text sub-element is populated. UUID, Blood Glucose Meal times, Insulin Delivery Reason, and VO2 Max Test Type must have their value specified in component.valueString. Manually Entered is specified as a valueBoolean.

Observation (R4).component.code | CodeableConcept | | | 1..1 | All resources that use this element export with 95541-9 from http://loinc.org. These resources also export with an additional code: - Acceptable blood products for pregnancy: acceptable-blood-products with code system http://open.epic.com/FHIR/CodeSystem/care-experience-preference - Anesthesia methods planned for pregnancy: anesthesia-method-plan with code system http://open.epic.com/FHIR/CodeSystem/care-experience-preference - Birth control planned after pregnancy: planned-birth-control with code system http://open.epic.com/FHIR/CodeSystem/care-experience-preference

Observation (R4).component.valueBoolean | boolean | | | 0..1 | Only used for Patient-Entered Flowsheet Metadata values. Valid values include: For Code = 6-Manually Entered false true

Observation (R4).component.valueCodeableConcept | CodeableConcept | Conditional | | 0..1 | This is the documented value: - For acceptable blood products for pregnancy observations, this is a blood product that is acceptable to the patient in the case of an emergency, particularly during labor and delivery. - For anesthesia methods planned for pregnancy observations, this is a type of anesthesia that a patient plans to request for labor and delivery; it is usually discussed and captured during prenatal care. - For birth control planned after pregnancy observations, this is the patient's planned postpartum birth control method.

Observation (R4).component.valueString | string | | | 0..1 | Values for the patient-entered flowsheet metadata.

Observation (R4).dataAbsentReason | CodeableConcept | | | 0..1 | Indicates that an Observation value is missing.

Observation (R4).effectiveDateTime (also: effectiveDateTime) | dateTime | | | 0..1 | The instant when the observation is made or verified. For preferred pharmacy, this is when pharmacy was last entered, updated, or verified by a clinical user or patient. This is skipped when it's updated by a background user. For flowsheet-based observations, the date and time the value was taken.

Observation (R4).effectivePeriod | Period | | | 0..1 | This is the date range for the pregnancy the data was documented on, when applicable. Pregnancy data always has a start date and might or might not have an end date. The date range reflects when the pregnancy was noted and resolved, and is likely to be based on when the patient received care rather than specifically being clinical.

Observation (R4).encounter | Reference | Conditional | | 0..1 | Reference to an Encounter resource that this Observation belongs to. This element is only used by flowsheet-based Observations.

Observation (R4).extension (supportingInfo) (also: extension (supportingInfo)) | Extension | Conditional | Yes | 0..* | If the measurement is linked to a note written by a clinician, this returns a reference to the DocumentReference resource. URL: http://hl7.org/fhir/StructureDefinition/workflow-supportingInfo

Observation (R4).extension (template-id) (also: extension (template-id)) | Extension | | Yes | 0..1 | This is the specific flowsheet template (a collection of flowsheet rows and groups) to which data was filed. URL: http://open.epic.com/FHIR/StructureDefinition/extension/template-id

Observation (R4).extension (template-id).valueIdentifier | Identifier | | | 1..1 | The system and value for the flowsheet template.

Observation (R4).extension (template-id).valueIdentifier.system | String | | | 1..1 | The OID for the Epic internal flowsheet template identifier. This varies based on the specific Epic environment.

Observation (R4).extension (template-id).valueIdentifier.value | String | | | 1..1 | The Epic internal identifier of the flowsheet template.

Observation (R4).focus | Reference | | Yes | 0..1 | This is the pregnancy on which the data has been documented, when applicable.

Observation (R4).id | id | | | 0..1 | FHIR ID of the resource.

Observation (R4).interpretation | CodeableConcept | | Yes | 0..* | Returns a system of http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation and value of "A" for flowsheet values with an abnormal status. Returns an environment-specific OID system with a value of "SIG" for flowsheet values with a significant status.

Observation (R4).issued | instant | Conditional | | 0..1 | For flowsheet-based observations, the date and time the observation was documented.

Observation (R4).method | CodeableConcept | | | 0..1 | For patient-entered flowsheets, this element is returned with a text value of "Patient-entered".

Observation (R4).note | Annotation | | Yes | 0..1 | The comment associated with the value.

Observation (R4).performer | Reference | | Yes | 0..* | The provider or patient who entered the documentation. For preferred pharmacy, if it was entered by a background process, this element is skipped.

Observation (R4).performer.extension (performerFunction) (also: extension (performerFunction)) | Extension | | Yes | 0..1 | Only used by flowsheet-based Observations. Additional Practitioners associated with the Observation value, if applicable. Possible values include: "data entry person" for Practitioners who transcribed the Observation into the system "verifier" for Practitioners who cosigned the Observation URL:

http://hl7.org/fhir/StructureDefinition/event-performerFunction

Observation (R4).status | code | | | 1..1 | For preferred pharmacy observations, this is final if the pharmacy is currently preferred. This is cancelled if the pharmacy is no longer preferred. For observations for acceptable blood products for pregnancy, anesthesia methods planned for pregnancy, birth control planned after pregnancy, circumcision requested on behalf of pregnancy, delivery method planned for pregnancy, expected delivery location for pregnancy, and support person for pregnancy, this is final while there is documentation on file and is cancelled otherwise. For flowsheet-based observations, this element has the following possible values: "preliminary" - value is pended "final" - value has been filed "amended" - value has been edited "entered-in-error" - value has been deleted

Observation (R4).subject | Reference | | | 1..1 | This is the patient.

Observation (R4).valueCodeableConcept | CodeableConcept | | | 0..1 | This is the documented value: - For circumcision requested on behalf of pregnancy observations, this is the parents' decision regarding circumcision after the baby is born. - For delivery method planned for pregnancy observations, this is the planned method of delivery as documented prior to labor. - For expected delivery location for pregnancy observations, this is the location where the pregnant woman plans to deliver. - For preferred pharmacy observations, this is the NPI code of the pharmacy. For flowsheet-based observations, this element is populated if the data type is a category or custom list. Only one of the value* elements is returned.

Observation (R4).valueDateTime | dateTime | | | 0..1 | For flowsheet-based observations, this element is populated if data type is date. Only one of the value* elements is returned.

Observation (R4).valueQuantity | Quantity | | | 0..1 | For flowsheet-based observations, this element is populated if a value has units or the data type is numeric. Only one of the value* elements is returned.

Observation (R4).valueString | string | | | 0..1 | This is the documented value: - For support person for pregnancy observations, this is the person who has been designated to be the patient's support person for the pregnancy. For flowsheet-based observations, this element is populated if a value doesn't have units or the data type is string. Only one of the value* elements is returned.

Observation (R4).valueTime | time | | | 0..1 | For flowsheet-based observations, this element is populated if data type is time. Only one of the value* elements is returned.

Sample request

```
GET https://hostname/instance/api/FHIR/R4/Observation?patient=eOgS6zE7C8czUV.9e8lcHoQ3&category=care-experience-
    preference
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 3,
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Observation?category=care-experience-
    preference&patient=eOgS6zE7C8czUV.9e8lcHoQ3"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url":
    "https://hostname/instance/api/FHIR/R4/Observation/DHGMA6y2kEFx934CuY08qWnKDscNin.hiC3qxW.hDJbg6bJRuWkLBkNeaw5H94Mm"
        }
      ],
      "fullUrl":
    "https://hostname/instance/api/FHIR/R4/Observation/DHGMA6y2kEFx934CuY08qWnKDscNin.hiC3qxW.hDJbg6bJRuWkLBkNeaw5H94Mm"
    ,
      "resource": {
        "resourceType": "Observation",
        "id": "DHGMA6y2kEFx934CuY08qWnKDscNin.hiC3qxW.hDJbg6bJRuWkLBkNeaw5H94Mm",
        "status": "final",
        "category": [
          {
            "coding": [
              {
                "system": "http://hl7.org/fhir/us/core/CodeSystem/us-core-category",
                "code": "care-experience-preference",
                "display": "Care Experience Preference"
              }
            ],
            "text": "Care Experience Preference"
          }
        ],
        "code": {
          "coding": [
            {
              "system": "http://loinc.org",
              "code": "95541-9",
              "display": "Care experience preference"
            }
          ],
          "text": "Care experience preference"
        },
        "subject": {
          "reference": "Patient/eOgS6zE7C8czUV.9e8lcHoQ3",
          "display": "Sam, Ham"
        },
        "valueCodeableConcept": {
          "coding": [
```

```
        {
          "system": "http://hl7.org/fhir/sid/us-npi",
          "code": "1013192590",
          "display": "PLEXUS PHARMACY - LANCASTER, SC - 703 N WHITE STREET"
        }
      ],
      "text": "PLEXUS PHARMACY - LANCASTER, SC - 703 N WHITE STREET"
    }
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url": "https://vs-icx18-25/Interconnect-FHIR-CDE-
Unsecured/api/FHIR/R4/Observation/DHGMA6y2kEFx934CuY08qdVirWEuhnRJNrxY0xjx1nJq54k3xTtKeOXLYFopAEYX"
    }
  ],
  "fullUrl": "https://vs-icx18-25/Interconnect-FHIR-CDE-
Unsecured/api/FHIR/R4/Observation/DHGMA6y2kEFx934CuY08qdVirWEuhnRJNrxY0xjx1nJq54k3xTtKeOXLYFopAEYX",
  "resource": {
    "resourceType": "Observation",
    "id": "DHGMA6y2kEFx934CuY08qdVirWEuhnRJNrxY0xjx1nJq54k3xTtKeOXLYFopAEYX",
    "status": "final",
    "category": [
      {
        "coding": [
          {
            "system": "http://hl7.org/fhir/us/core/CodeSystem/us-core-category",
            "code": "care-experience-preference",
            "display": "Care Experience Preference"
          }
        ],
        "text": "Care Experience Preference"
      }
    ],
    "code": {
      "coding": [
        {
          "system": "http://loinc.org",
          "code": "95541-9",
          "display": "Care experience preference"
        }
      ],
      "text": "Care experience preference"
    },
    "subject": {
      "reference": "Patient/eOgS6zE7C8czUV.9e8lcHoQ3",
      "display": "Sam, Ham"
    },
    "valueCodeableConcept": {
      "text": "MAIN PHARMACY"
    }
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url": "https://vs-icx18-25/Interconnect-FHIR-CDE-
Unsecured/api/FHIR/R4/Observation/DHGMA6y2kEFx934CuY08qdhSrPRGYAW6ZuexuPeQcu64RbWyfztQa-N9bIdGXRm2"
    }
  ],
  "fullUrl": "https://vs-icx18-25/Interconnect-FHIR-CDE-
Unsecured/api/FHIR/R4/Observation/DHGMA6y2kEFx934CuY08qdhSrPRGYAW6ZuexuPeQcu64RbWyfztQa-N9bIdGXRm2",
  "resource": {
    "resourceType": "Observation",
    "id": "DHGMA6y2kEFx934CuY08qdhSrPRGYAW6ZuexuPeQcu64RbWyfztQa-N9bIdGXRm2",
    "status": "final",
    "category": [
      {
        "coding": [
          {
            "system": "http://hl7.org/fhir/us/core/CodeSystem/us-core-category",
            "code": "care-experience-preference",
            "display": "Care Experience Preference"
          }
        ],
        "text": "Care Experience Preference"
      }
    ],
    "code": {
      "coding": [
```

```
            {
              "system": "http://loinc.org",
              "code": "95541-9",
              "display": "Care experience preference"
            }
          ],
          "text": "Care experience preference"
        },
        "subject": {
          "reference": "Patient/eOgS6zE7C8czUV.9e8lcHoQ3",
          "display": "Sam, Ham"
        },
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://hl7.org/fhir/sid/us-npi",
              "code": "1023268489",
              "display": "WALGREENS DRUG STORE #12099 - TALLADEGA, AL - 503 BATTLE ST E AT SEC OF JOHNSON ST & E BATTLE
    ST"
            }
          ],
          "text": "WALGREENS DRUG STORE #12099 - TALLADEGA, AL - 503 BATTLE ST E AT SEC OF JOHNSON ST & E BATTLE ST"
        }
      },
      "search": {
        "mode": "match"
      }
    }
  ]
}
```


10.1.45 Observation.Search (Genomics) (R4)

File: site/apis/10-Clinical-General/1-R4/10298-Observation-Search-Genomics-R4.html
Epic API ID | 10298
FHIR version | R4
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Observation?patient={patient}&category=genomics&code={code}&based-on={based-on}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10298

Description

The Observation.Search (Genomics) resource returns genomic variant information. Genomic variants are alterations in the most common DNA sequence and are fundamentally different from traditional lab data. For example, a given CBC test is relevant immediately after it is resulted but is generally less relevant over time because a patient's blood count changes over time. In contrast, genomic variant information maintains its relevancy as time passes because a patient's DNA remains largely unchanged throughout their life.

When variant information is available discretely, external systems can use this variant information to drive clinical decision support and leverage it in clinical and organizational reporting. This reference can be queried by a patient ID.

Variant data responses are split into different types of genomic observations representing high-level findings and implications. These observations are variant, genotype, diagnostic implication, and therapeutic implication. The response also includes a grouper observation which contains references to all genomic observations that are part of the same piece of variant data. Each genomic observation contains unique components and metadata and can be accessed directly through a Read. If no code is specified in the request URL all genomic observation types will be returned.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
based-on | String | True | Yes | | FHIR ID for service result or order related to the result. The reference must be to a FHIR resource that is contained within the same FHIR server. If ServiceRequest is supplied, it must resolve to an order on the referenced patient.
category | token | True | Yes | | Use "genomics" to search for genomic variants.
code | token | True | Yes | | The observation identifier. The following values can be used to search for each particular genomic observation: Grouper: “C43359” Variant: “69548-6” Diagnostic Implication: “diagnostic-implication” Genotype: “84413-4” Therapeutic Implication: “therapeutic-implication” The observation identifier. The following values can be used to search for each particular genomic observation: Grouper: “C43359” Variant: “69548-6” Diagnostic Implication: “diagnostic-implication” Genotype: “84413-4” Therapeutic Implication: “therapeutic-implication”
combo-code | token | True | Yes | | Unsupported
combo-data-absent-reason | token | True | Yes | | Unsupported
combo-value-concept | token | True | Yes | | Unsupported
component-code | token | True | Yes | | The component code
component-data-absent-reason | token | True | Yes | | Unsupported

component-value-concept | token | True | Yes | | The value of the component observation, if the value is a CodeableConcept.
data-absent-reason | token | True | Yes | | Unsupported
date | date | True | Yes | | Obtained date/time. If the obtained element is a period, a date that falls in the period.
derived-from | String | True | Yes | | Related measurements the observation is made from. References the genotype findings.
device | String | True | Yes | | Unsupported
encounter | reference | True | Yes | | Unsupported
focus | reference | True | Yes | | Unsupported
has-member | reference | True | Yes | | Related resource that belongs to the Observation group
identifier | token | True | Yes | | Unsupported
issued | date | True | Yes | | Date/Time the observation was made available
method | token | True | Yes | | The method used to analyze the chromosome structural variation.
part-of | String | True | Yes | | Unsupported
patient | reference | True | Yes | | Reference to a patient resource. If both patient and subject are provided, they must match.
performer | reference | True | Yes | | Unsupported
specimen | String | True | Yes | | Unsupported
status | token | True | Yes | | The status of the lab order associated with the variant.
subject | reference | True | Yes | | Reference to a patient resource. If both patient and subject are provided, they must match.
value-concept | token | True | Yes | | Unsupported
value-date | date | True | Yes | | Unsupported
value-string | string | True | Yes | | The value of the genomic variant

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | BackboneElement | False | Yes | | An array of entries included in this bundle.
Bundle.entry.resource | ResourceContainer | False | | | A container for a single FHIR resource.
Bundle.entry.resource.Observation | Observation | False | | | An instance of the R4 Observation resource.
Bundle.entry.resource.Observation.extension (also: Allele-name) | Extension | False | Yes | | Returned only on the Variant observation. The allele name of the variant. LOINC 84414-2
Bundle.entry.resource.Observation.extension (also: amplification) | Extension | False | Yes | | Returned only on the Diagnostic Implication observation. Indicates whether a copy number variation meets the amplification threshold.
Bundle.entry.resource.Observation.basedOn | Reference | False | Yes | | The procedure or service associated with this observation.
Bundle.entry.resource.Observation.extension (also: Basis-for-allelic-phase) | Extension | False | Yes | | Returned only for Variant elements. The evidential basis for the allelic phase. LOINC 82309-6
Bundle.entry.resource.Observation.category | CodeableConcept | False | Yes | | The Observation type. Always "genomics". A second "laboratory" category is also returned.
Bundle.entry.resource.Observation.code | CodeableConcept | False | | | The observation identifier. The following values can be used to search for a particular genomic sub-category: Grouper: “C43359” Variant: “69548-6” Diagnostic Implication: “diagnostic-implication” Genotype: “84413-4” Therapeutic Implication: “therapeutic-implication”
Bundle.entry.resource.Observation.component | BackboneElement | False | Yes | | Component information.
Bundle.entry.resource.Observation.component.extension (also: Clonal-hematopoiesis) | Extension | False | Yes | | Returned only for Diagnostic Implication elements. Indicates the degree of confidence that the variant arises from clonal hematopoiesis rather than representing a primary tumor variant.
Bundle.entry.resource.Observation.component.extension | Extension | False | Yes | | Returned as an extension of the Gene-studied (48018-6) component. Whether the studied gene is part of a genetic fusion.
Bundle.entry.resource.Observation.component.extension (also: Interpretation-annotation) | Extension | False | Yes | | Returned only for Diagnostic Implication and Therapeutic Implication elements. The free-text classification narrative associated with the genomic interpretation. When the original note contained rich-text formatting (e.g., bold, tables, or images) that could not be preserved in plain text, one or more data-conversion-warning child extensions are included.
Bundle.entry.resource.Observation.component.extension (also: Medication-usage-suggestion) | Extension | False | Yes | | Returned only on Therapeutic Implication observations with a Medication-assessed (51963-7) component. The recommended alteration to the medication based on the reported drug-gene interactions.
Bundle.entry.resource.Observation.component.extension (also: Medication-usage-suggestion-type) | Extension | False | Yes | | Returned only for Therapeutic Implication elements with a Medication-assessed (51963-7) component. The predicted phenotype for drug metabolism. LOINC 82116-5
Bundle.entry.resource.Observation.component.valueCodeableConcept | CodeableConcept | False | | | Allelic-read-depth (82121-5): The number of reads that identified the allele in question. Returned only for Variant elements. Allelic-state (53034-5): The relationship between alleles found at the same locus on different chromosomes. Returned only for Variant elements. Alt-allele (69551-0): The DNA string in the test sample that differs from the reference sequence. Returned only for Variant elements. Amino-acid-chg (48005-3):The amino acid change caused by the variant. Returned only for Variant elements. Associated-phenotype (81259-4): The phenotype associated with the variant. Returned only for Diagnostic Implication elements. Chromosome-identifier (48000-4): The chromosome where the variant is located. Returned only for Variant elements. Clinical-significance (53037-8): The clinical significance of the variant. Returned only for Diagnostic Implication elements. Copy-number (82155-3): The copy number of the large variant when applicable. Returned only for Diagnostic Implication elements. Cytogenetic-location (48001-2): The cytogenetic (chromosome) location of the variant. Returned only for Diagnostic Implication elements. Dna-chg (48004-6): The nucleotide change in the DNA. Returned only for Diagnostic Implication elements. Dna-chg-type (48019-4): The type of the DNA change reported for the variant. Returned only for Diagnostic Implication elements. Effect-medication-efficacy (51961-1): The predicted phenotype for the ability of a drug to bind to the intended site to deliver its intended effect. Returned only for Therapeutic Implication elements. Effect-medication-high-risk (83009-1): The predicted phenotype for risk of adverse drug reaction. Returned only for Therapeutic Implication elements. Effect-medication-metabolism (53040-2): The effect on drug metabolism for a pharmacogenomic variant. Returned only for Therapeutic Implication elements. Exact-start-end (81254-5): The starting and ending position of the variant’s allele. Returned only for Variant elements. Functional-effect: The effect that the variant has on the function of the gene. Returned only for Diagnostic Implication elements. Gene-studied (48018-6): The gene where the variant is located. Returned only for Genotype and Variant elements. Genomic-dna-chg (81290-9): The nucleotide change that occurs in the DNA relative to the genomic reference sequence or chromosome. Returned only for Variant elements. Genomic-ref-seq (48013-7): The genomic reference sequence. Returned only for Variant elements. Genomic-source-class (48002-0): The source of the genomic variant. Defines whether the variant is germline or somatic. Returned only for Variant elements. Inner-start-end (81302-2): The inner start and end position of a region known to be affected by a variant. Returned only for Variant elements. Medication-assessed (51963-7): The medication being assessed for usage based on drug-gene interactions. Returned only for

Therapeutic Implication elements. Mode-of-inheritance: The mode of inheritance for the phenotype associated with the variant. Returned only for Diagnostic Implication elements. Molecular-consequence: The effect of the variant on a genomic, transcript, or protein sequence Outer-start-end (81301-4): The outer start and end position of a region known to be affected by a variant. Returned only for Variant elements. Ref-allele (69547-8): The DNA string in the reference allele with which the DNA string in the test sample differs. Returned only for Variant elements. Reference-sequence-assembly (62374-4): The reference genome used for the sequence. Returned only for Variant elements. Sample-allelic-frequency (81258-6): Reports the percentage of all the reads at this genomic location that were represented by the given allele. Returned only for Variant elements. Transcript-ref-seq (81258-6): The transcript reference sequence used to identify the variants. Returned only for Variant elements. Variant-inheritance: From whom the variant was inherited. Variation-code (81252-9): The definition of the variant. Contains the HGVS name and associated codes.

Bundle.entry.resource.Observation.derivedFrom | Reference | False | Yes | | References the genotype findings. Specific to therapeutic implications and diagnostic implications.

Bundle.entry.resource.Observation.extension (also: Dna-region) | Extension | False | Yes | | Returned only for Variant elements. The region within the gene where the variant is located. LOINC 47999-8

Bundle.entry.resource.Observation.effectiveDateTime | dateTime | False | | | The prioritized instant of the associated order.

Bundle.entry.resource.Observation.extension (also: Effect-medication-transport) | Extension | False | Yes | | Returned only on Therapeutic Implication observations. The predicted phenotype for effect on drug transportation.

Bundle.entry.resource.Observation.extension | Extension | False | Yes | | Returned only for Variant elements. The protein reference sequence for this variant.

Bundle.entry.resource.Observation.hasMember | Reference | False | Yes | | Contains resource IDs for sub-types. Included only if the code=C43359.

Bundle.entry.resource.Observation.issued | instant | False | | | The observation instant.

Bundle.entry.resource.Observation.method | CodeableConcept | False | | | The method used to analyze the chromosome structural variation.

Bundle.entry.resource.Observation.extension (also: Mosaicism) | Extension | False | Yes | | Returned only for Diagnostic Implication elements. Indicates whether mosaicism is detected. LOINC 62363-7

Bundle.entry.resource.Observation.note | Annotation | False | Yes | | Returned only for Genotype and Variant elements. The clinical interpretation of the genomic variant.

Bundle.entry.resource.Observation.extension (also: Oncogenicity) | Extension | False | Yes | | Returned only for Diagnostic Implication and Therapeutic Implication elements. The oncogenicity of the genomic variant

Bundle.entry.resource.Observation.extension (also: Penetrance) | Extension | True | Yes | | Returned only for Diagnostic Implication and Therapeutic Implication elements. The percentage of individuals with a given genotype who exhibit the phenotype associated with that genotype.

Bundle.entry.resource.Observation.extension (also: Phenotype-description) | Extension | True | Yes | | Returned only for Diagnostic Implication and Therapeutic Implication elements. The description of phenotypes possibly associated with the variants.

Bundle.entry.resource.Observation.extension (also: Precision-of-boundaries) | Extension | False | Yes | | Returned only for Variant elements. The boundary position of the variant.

Bundle.entry.resource.Observation.extension (also: Repeat-pair) | Extension | False | Yes

Bundle.entry.resource.Observation.extension (also: Reported-arrCGH-ratio) | Extension | False | Yes | | Returned only for Variant elements. The reported aCGH ratio for the structural variant.

Bundle.entry.resource.Observation.status | code | False | | | The status of the lab order associated with the variant.

Bundle.entry.resource.Observation.subject | Reference | False | | | The Patient FHIR ID linked to the variant information.

Bundle.entry.resource.Observation.valueCodeableConcept | CodeableConcept | False | | | Allelic-read-depth (82121-5): The number of reads that identified the allele in question. Returned only for Variant elements. Allelic-state (53034-5): The relationship between alleles found at the same locus on different chromosomes. Returned only for Variant elements. Alt-allele (69551-0): The DNA string in the test sample that differs from the reference sequence. Returned only for Variant elements. Amino-acid-chg (48005-3):The amino acid change caused by the variant. Returned only for Variant elements. Associated-phenotype (81259-4): The phenotype associated with the variant. Returned only for Diagnostic Implication and Therapeutic Implication elements. Chromosome-identifier (48000-4): The chromosome where the variant is located. Returned only for Variant elements. Clinical-significance (53037-8): The clinical significance of the variant. Returned only for Diagnostic Implication and Therapeutic Implication elements. Copy-number (82155-3): The copy number of the large variant when applicable. Returned only for Diagnostic Implication elements. Cytogenetic-location (48001-2): The cytogenetic (chromosome) location of the variant. Returned only for Diagnostic Implication elements. Diagnostic-evidence-level (93044-6): The diagnostic evidence level of the variant when applicable. Returned only for Diagnostic Implication elements. Cytogenomic-nomenclature (81291-7): The reported ISCN name for Structural, Fusion, and Copy Number variants. Returned only for Variant elements. Dna-chg (48004-6): The nucleotide change in the DNA. Returned only for Diagnostic Implication elements. Dna-chg-type (48019-4): The type of the DNA change reported for the variant. Returned only for Diagnostic Implication elements. Effect-medication-efficacy (51961-1): The predicted phenotype for the ability of a drug to bind to the intended site to deliver its intended effect. Returned only for Therapeutic Implication elements. Effect-medication-high-risk (83009-1): The predicted phenotype for risk of adverse drug reaction. Returned only for Therapeutic Implication elements. Effect-medication-metabolism (53040-2): The effect on drug metabolism for a pharmacogenomic variant. Returned only for Therapeutic Implication elements. Exact-start-end (81254-5): The starting and ending position of the variant's allele. Returned only for Variant elements. Functional-effect: The effect that the variant has on the function of the gene. Returned only for Diagnostic Implication elements. Gene-studied (48018-6): The gene where the variant is located. Returned only for Genotype and Variant elements. Genomic-dna-chg (81290-9): The nucleotide change that occurs in the DNA relative to the genomic reference sequence or chromosome. Returned only for Variant elements. Genomic-ref-seq (48013-7): The genomic reference sequence. Returned only for Variant elements. Genomic-source-class (48002-0): The source of the genomic variant. Defines whether the variant is germline or somatic. Returned only for Variant elements. Inner-start-end (81302-2): The inner start and end position of a region known to be affected by a variant. Returned only for Variant elements. Medication-assessed (51963-7): The medication being assessed for usage based on drug-gene interactions. Returned only for Therapeutic Implication elements. Mode-of-inheritance: The mode of inheritance for the phenotype associated with the variant. Returned only for Diagnostic Implication and Therapeutic Implication elements. Molecular-consequence: The effect of the variant on a genomic, transcript, or protein sequence Outer-start-end (81301-4): The outer start and end position of a region known to be affected by a variant. Returned only for Variant elements. Prognostic-evidence-level (93044-6): The prognostic evidence level of the variant when applicable. Returned only for Diagnostic Implication elements. Ref-allele (69547-8): The DNA string in the reference allele with which the DNA string in the test sample differs. Returned only for Variant elements. Reference-sequence-assembly (62374-4): The reference genome used for the sequence. Returned only for Variant elements. Sample-allelic-frequency (81258-6): Reports the percentage of all the reads at this genomic location that were represented by the given allele. Returned only for Variant elements. Therapeutic-evidence-level (93044-6): The therapeutic evidence level of the variant when applicable. Returned only for Diagnostic Implication elements. Transcript-ref-seq (81258-6): The transcript reference sequence used to identify the variants. Returned only for Variant elements. Variant-inheritance: From whom the variant was inherited. Variation-code (81252-9): The definition of the variant. Contains the HGVS name and associated codes.

Sample request

```
https://hostname/instance/api/FHIR/R4/Observation?patient=eyr87CEI8vy-
    nL6hL5yccppCd83J03x-7iMjZnGMSlrE3&category=genomics
```

Sample response


```
<Bundle xmlns="http://hl7.org/fhir">
 <type value="searchset" />
 <total value="5" />
 <link>
 <relation value="self" />
 <url value="https://hostname/instance/api/FHIR/R4/Observation?patient=eyr87CEI8vy-
    nL6hL5yccppCd83J03x-7iMjZnGMSlrE3&amp;category=genomics" />
 </link>
 <entry>
 <link>
 <relation value="self" />
 <url value="https://hostname/instance/api/FHIR/R4/Observation/egcZd1Jrr7564EkduqejavL06fkCEirtUlwrD44f6W-Y3" />
 </link>
 <fullUrl value="https://hostname/instance/api/FHIR/R4/Observation/egcZd1Jrr7564EkduqejavL06fkCEirtUlwrD44f6W-Y3" />
 <resource>
 <Observation>
 <id value="egcZd1Jrr7564EkduqejavL06fkCEirtUlwrD44f6W-Y3" />
 <basedOn>
 <reference value="ServiceRequest/eY0zM9XwMAsfr83odpCgcifyOxXr.2BKRPWcXghHs9b03" />
 </basedOn>
 <status value="amended" />
 <category>
 <coding>
 <system value="http://open.epic.com/FHIR/StructureDefinition/observation-category" />
 <code value="genomics" />
 <display value="Genomics Grouper" />
 </coding>
 <text value="Genomics Grouper" />
 </category>
 <category>
 <coding>
 <system value="http://terminology.hl7.org/CodeSystem/observation-category" />
 <code value="laboratory" />
 <display value="Laboratory" />
 </coding>
 <text value="Laboratory" />
 </category>
 <code>
 <coding>
 <system value="http://ncithesaurus-stage.nci.nih.gov" />
 <code value="C43359" />
 <display value="Group" />
 </coding>
 <text value="Genomic Result Group" />
 </code>
 <subject>
 <reference value="Patient/eyr87CEI8vy-nL6hL5yccppCd83J03x-7iMjZnGMSlrE3" />
 <display value="Garcia Rodriguez, Hugo" />
 </subject>
 <effectiveDateTime value="2021-12-17T06:00:00Z" />
 <issued value="2021-12-17T15:37:17Z" />
 <hasMember>
 <reference value="Observation/eUZ671fmZxtLjtto.i5tiC9BweAAcVFBDp6dJ.vN4VzU3" />
 <display value="Variant" />
 </hasMember>
 <hasMember>
 <reference value="Observation/eUZ671fmZxtLjtto.i5tiC3UzHp6paXUbJr9FMjMIDX43" />
 <display value="Diagnostic Implication" />
 </hasMember>
 <hasMember>
 <reference value="Observation/e4t0a8Yo6ZFhNUAbGJoR4P-DeSZGzoSLzEyxaQ0oD1gM3" />
 <display value="Therapeutic Implication" />
 </hasMember>
 <hasMember>
 <reference value="Observation/e4t0a8Yo6ZFhNUAbGJoR4P2m1Dsb.hyC8Oi7resT-R8I3" />
 <display value="Genotype" />
 </hasMember>
 </Observation>
 </resource>
 <search>
 <mode value="match" />
 </search>
 </entry>
 <entry>
 <link>
 <relation value="self" />
 <url value="https://hostname/instance/api/FHIR/R4/Observation/eUZ671fmZxtLjtto.i5tiC9BweAAcVFBDp6dJ.vN4VzU3" />
 </link>
 <fullUrl value="https://hostname/instance/api/FHIR/R4/Observation/eUZ671fmZxtLjtto.i5tiC9BweAAcVFBDp6dJ.vN4VzU3" />
 <resource>
 <Observation>
 <id value="eUZ671fmZxtLjtto.i5tiC9BweAAcVFBDp6dJ.vN4VzU3" />
 <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/variant-type">
 <valueCodeableConcept>
 <coding>
 <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.4.866582.35" />
 <code value="1" />
```

```
<display value="Simple" />
</coding>
</valueCodeableConcept>
</extension>
<extension url="http://open.epic.com/FHIR/StructureDefinition/extension/dna-region">
<extension url="code">
<valueCodeableConcept>
<coding>
<system value="http://loinc.org" />
<code value="47999-8" />
</coding>
<text value="dna-region" />
</valueCodeableConcept>
</extension>
<extension url="value">
<valueString value="DNA Region" />
</extension>
</extension>
<extension url="http://open.epic.com/FHIR/StructureDefinition/extension/protein-reference-sequence">
<valueString value="Protein Ref Seq" />
</extension>
<extension url="http://open.epic.com/FHIR/StructureDefinition/extension/basis-for-allelic-phase">
<extension url="code">
<valueCodeableConcept>
<coding>
<system value="http://loinc.org" />
<code value="82309-6" />
</coding>
<text value="basis-for-allelic-phase" />
</valueCodeableConcept>
</extension>
<extension url="value">
<valueCodeableConcept>
<coding>
<system value="urn:oid:1.2.840.114350.1.13.5325.1.7.4.866582.1680" />
<code value="2" />
<display value="Family DNA" />
</coding>
</valueCodeableConcept>
</extension>
</extension>
<extension url="http://open.epic.com/FHIR/StructureDefinition/extension/reported-arrcgh-ratio">
<valueString value="0.32" />
</extension>
<extension url="http://open.epic.com/FHIR/StructureDefinition/extension/allele-name">
<extension url="code">
<valueCodeableConcept>
<coding>
<system value="http://loinc.org" />
<code value="84414-2" />
</coding>
<text value="allele-name" />
</valueCodeableConcept>
</extension>
<extension url="value">
<valueString value="Simple VAR Allele" />
</extension>
</extension>
<extension url="http://open.epic.com/FHIR/StructureDefinition/extension/precision-of-boundaries">
<valueString value="Simple VAR Precision of Boundaries" />
</extension>
<basedOn>
<reference value="ServiceRequest/eY0zM9XwMAsfr83odpCgcifyOxXr.2BKRPWcXghHs9b03" />
</basedOn>
<status value="amended" />
<category>
<coding>
<system value="http://open.epic.com/FHIR/StructureDefinition/observation-category" />
<code value="genomics" />
<display value="Variant Data" />
</coding>
<text value="Variant Data" />
</category>
<category>
<coding>
<system value="http://terminology.hl7.org/CodeSystem/observation-category" />
<code value="laboratory" />
<display value="Laboratory" />
</coding>
<text value="Laboratory" />
</category>
<code>
<coding>
<system value="http://loinc.org" />
<code value="69548-6" />
<display value="Genetic variant assessment" />
</coding>
<text value="Assessment of Simple VAR" />
</code>
```

```
<subject>
<reference value="Patient/eyr87CEI8vy-nL6hL5yccppCd83J03x-7iMjZnGMSlrE3" />
<display value="Garcia Rodriguez, Hugo" />
</subject>
<effectiveDateTime value="2021-12-17T06:00:00Z" />
<issued value="2021-12-17T15:37:17Z" />
<valueCodeableConcept>
<coding>
<system value="urn:oid:1.2.840.114350.1.13.5325.1.7.4.866582.5000" />
<code value="1" />
<display value="Detected" />
</coding>
</valueCodeableConcept>
<component>
<code>
<coding>
<system value="http://loinc.org" />
<code value="48018-6" />
</coding>
<text value="gene-studied" />
</code>
<valueCodeableConcept>
<coding>
<system value="http://www.genenames.org/data/gene-symbol-report/" />
<code value="60" />
</coding>
<text value="ABCC9" />
</valueCodeableConcept>
</component>
<component>
<code>
<coding>
<system value="http://loinc.org" />
<code value="48001-2" />
</coding>
<text value="cytogenetic-location" />
</code>
<valueCodeableConcept>
<coding>
<system value="http://www.ncbi.nlm.nih.gov/genemap99" />
<code value="Ex Cyto Location" />
</coding>
<text value="Ex Cyto Location" />
</valueCodeableConcept>
</component>
<component>
<code>
<coding>
<system value="http://loinc.org" />
<code value="62374-4" />
</coding>
<text value="reference-sequence-assembly" />
</code>
<valueCodeableConcept>
<coding>
<system value="urn:oid:1.2.840.114350.1.13.5325.1.7.4.866582.1040" />
<code value="2" />
<display value="NCBI36" />
</coding>
</valueCodeableConcept>
</component>
<component>
<code>
<coding>
<system value="http://loinc.org" />
<code value="48004-6" />
</coding>
<text value="dna-chg" />
</code>
<valueCodeableConcept>
<coding>
<system value="http://varnomen.hgvs.org" />
<code value="Simple VAR DNA Change" />
</coding>
<text value="Simple VAR DNA Change" />
</valueCodeableConcept>
</component>
<component>
<code>
<coding>
<system value="http://loinc.org" />
<code value="48019-4" />
</coding>
<text value="dna-chg-type" />
</code>
<valueCodeableConcept>
<coding>
<system value="urn:oid:1.2.840.114350.1.13.5325.1.7.4.866582.1240" />
<code value="6" />
```

```
<display value="Inversion" />
</coding>
</valueCodeableConcept>
</component>
<component>
<code>
<coding>
<system value="http://hl7.org/fhir/uv/genomics-reporting/CodeSystem/tbd-codes-cs" />
<code value="molecular-consequence" />
</coding>
<text value="molecular-consequence" />
</code>
<valueCodeableConcept>
<coding>
<system value="urn:oid:1.2.840.114350.1.13.5325.1.7.4.866582.1260" />
<code value="4" />
<display value="Frameshift Variant" />
</coding>
</valueCodeableConcept>
</component>
<component>
<code>
<coding>
<system value="http://loinc.org" />
<code value="81252-9" />
</coding>
<text value="variation-code" />
</code>
<valueCodeableConcept>
<coding>
<system value="https://www.ncbi.nlm.nih.gov/clinvar/" />
<version value="Example VAR Coding System" />
<code value="Example Variant Identifier" />
<display value="Example HGVS Name" />
</coding>
</valueCodeableConcept>
</component>
<component>
<code>
<coding>
<system value="http://loinc.org" />
<code value="81290-9" />
</coding>
<text value="genomic-dna-chg" />
</code>
<valueCodeableConcept>
<coding>
<system value="http://varnomen.hgvs.org" />
<code value="Example GEN DNA CHG" />
</coding>
<text value="Example GEN DNA CHG" />
</valueCodeableConcept>
</component>
<component>
<code>
<coding>
<system value="http://loinc.org" />
<code value="48002-0" />
</coding>
<text value="genomic-source-class" />
</code>
<valueCodeableConcept>
<coding>
<system value="urn:oid:1.2.840.114350.1.13.5325.1.7.4.866582.1650" />
<code value="1" />
<display value="Germline" />
</coding>
</valueCodeableConcept>
</component>
<component>
<code>
<coding>
<system value="http://loinc.org" />
<code value="48005-3" />
</coding>
<text value="amino-acid-chg" />
</code>
<valueCodeableConcept>
<coding>
<system value="http://varnomen.hgvs.org" />
<code value="Example AA CHNG" />
</coding>
<text value="Example AA CHNG" />
</valueCodeableConcept>
</component>
<component>
<code>
<coding>
<system value="http://loinc.org" />
```

```
<code value="51958-7" />
</coding>
<text value="transcript-ref-seq" />
</code>
<valueCodeableConcept>
<coding>
<system value="http://www.ncbi.nlm.nih.gov/refseq" />
<code value="Example TRAN REF SEQ" />
</coding>
<text value="Example TRAN REF SEQ" />
</valueCodeableConcept>
</component>
<component>
<code>
<coding>
<system value="http://loinc.org" />
<code value="48013-7" />
</coding>
<text value="genomic-ref-seq" />
</code>
<valueCodeableConcept>
<coding>
<code value="Genomic Ref Seq" />
</coding>
<text value="Genomic Ref Seq" />
</valueCodeableConcept>
</component>
<component>
<code>
<coding>
<system value="http://loinc.org" />
<code value="81258-6" />
</coding>
<text value="sample-allelic-frequency" />
</code>
<valueQuantity>
<value value="12345" />
</valueQuantity>
</component>
<component>
<code>
<coding>
<system value="http://loinc.org" />
<code value="82121-5" />
</coding>
<text value="allelic-read-depth" />
</code>
<valueQuantity>
<value value="67890" />
</valueQuantity>
</component>
<component>
<code>
<coding>
<system value="http://loinc.org" />
<code value="53034-5" />
</coding>
<text value="allelic-state" />
</code>
<valueCodeableConcept>
<coding>
<system value="urn:oid:1.2.840.114350.1.13.5325.1.7.4.866582.1600" />
<code value="1" />
<display value="Heteroplasmic" />
</coding>
</valueCodeableConcept>
</component>
<component>
<code>
<coding>
<system value="http://loinc.org" />
<code value="82155-3" />
</coding>
<text value="copy-number" />
</code>
<valueQuantity>
<value value="5" />
</valueQuantity>
</component>
<component>
<code>
<coding>
<system value="http://loinc.org" />
<code value="69547-8" />
</coding>
<text value="ref-allele" />
</code>
<valueString value="Example Reference Allele" />
</component>
```

```
<component>
<code>
<coding>
<system value="http://loinc.org" />
<code value="69551-0" />
</coding>
<text value="alt-allele" />
</code>
<valueString value="Example Observed Allele" />
</component>
<component>
<code>
<coding>
<system value="http://hl7.org/fhir/uv/genomics-reporting/CodeSystem/tbd-codes-cs" />
<code value="81302-2" />
</coding>
<text value="inner-start-end" />
</code>
<valueRange>
<low>
<value value="3" />
</low>
<high>
<value value="7" />
</high>
</valueRange>
</component>
<component>
<code>
<coding>
<system value="http://hl7.org/fhir/uv/genomics-reporting/CodeSystem/tbd-codes-cs" />
<code value="81301-4" />
</coding>
<text value="outer-start-end" />
</code>
<valueRange>
<low>
<value value="1" />
</low>
<high>
<value value="10" />
</high>
</valueRange>
</component>
<component>
<code>
<coding>
<system value="http://loinc.org" />
<code value="48000-4" />
</coding>
<text value="chromosome-identifier" />
</code>
<valueCodeableConcept>
<coding>
<system value="urn:oid:1.2.840.114350.1.13.5325.1.7.4.866582.1050" />
<code value="2" />
<display value="2 " />
</coding>
</valueCodeableConcept>
</component>
<code>
<coding>
<system value="http://loinc.org" />
<code value="81291-7" />
</coding>
<text value="cytogenomic-nomenclature" />
</code>
<valueString value="arr[GRCh37] 22q11.21(18919528_21460594)x1" />
</component>
<component>
<code>
<coding>
<system value="http://loinc.org" />
<code value="82120-7" />
</coding>
<text value="variant-inheritance" />
</code>
<valueCodeableConcept>
<coding>
<system value="urn:oid:1.2.840.114350.1.13.5325.1.7.4.866582.1670" />
<code value="9" />
<display value="Other" />
</coding>
</valueCodeableConcept>
</component>
</Observation>
</resource>
<search>
<mode value="match" />
```

```
</search>
</entry>
<entry>
<link>
<relation value="self" />
<url value="https://hostname/instance/api/FHIR/R4/Observation/eUZ671fmZxtLjtto.i5tiC3UzHp6paXUbJr9FMjMIDX43" />
</link>
<fullUrl value="https://hostname/instance/api/FHIR/R4/Observation/eUZ671fmZxtLjtto.i5tiC3UzHp6paXUbJr9FMjMIDX43" />
<resource>
<Observation>
<id value="eUZ671fmZxtLjtto.i5tiC3UzHp6paXUbJr9FMjMIDX43" />
<extension url="http://open.epic.com/FHIR/StructureDefinition/extension/penetrance">
<valueString value=".87" />
</extension>
<extension url="http://open.epic.com/FHIR/StructureDefinition/extension/mosaicism">
<extension url="code">
<valueCodeableConcept>
<coding>
<system value="http://loinc.org" />
<code value="62363-7" />
</coding>
<text value="mosaicism" />
</valueCodeableConcept>
</extension>
<extension url="value">
<valueCodeableConcept>
<coding>
<system value="urn:oid:1.2.840.114350.1.13.5325.1.7.4.866582.1690" />
<code value="50" />
<display value="Not detected" />
</coding>
</valueCodeableConcept>
</extension>
</extension>
<extension url="http://open.epic.com/FHIR/StructureDefinition/extension/amplification">
<valueCodeableConcept>
<coding>
<system value="urn:oid:1.2.840.114350.1.13.5325.1.7.4.866582.3080" />
<code value="1" />
<display value="Yes" />
</coding>
</valueCodeableConcept>
</extension>
<extension url="http://open.epic.com/FHIR/StructureDefinition/extension/phenotype-description">
<valueString value="Simple VAR Phenotype Description" />
</extension>
<basedOn>
<reference value="ServiceRequest/eY0zM9XwMAsfr83odpCgcifyOxXr.2BKRPWcXghHs9b03" />
</basedOn>
<status value="amended" />
<category>
<coding>
<system value="http://open.epic.com/FHIR/StructureDefinition/observation-category" />
<code value="genomics" />
<display value="Diagnostic Implication" />
</coding>
<text value="Diagnostic Implication" />
</category>
<category>
<coding>
<system value="http://terminology.hl7.org/CodeSystem/observation-category" />
<code value="laboratory" />
<display value="Laboratory" />
</coding>
<text value="Laboratory" />
</category>
<code>
<coding>
<system value="http://hl7.org/fhir/uv/genomics-reporting/CodeSystem/TbdCodes" />
<code value="diagnostic-implication" />
<display value="Diagnostic Implication" />
</coding>
<text value="Diagnostic Implication" />
</code>
<subject>
<reference value="Patient/eyr87CEI8vy-nL6hL5yccppCd83J03x-7iMjZnGMSlrE3" />
<display value="Garcia Rodriguez, Hugo" />
</subject>
<effectiveDateTime value="2021-12-17T06:00:00Z" />
<issued value="2021-12-17T15:37:17Z" />
<derivedFrom>
<reference value="Observation/eUZ671fmZxtLjtto.i5tiC9BweAAcVFBDp6dJ.vN4VzU3" />
<display value="Variant" />
</derivedFrom>
<component>
<code>
<coding>
<system value="http://loinc.org" />
<code value="53037-8" />
```

```
</coding>
<text value="clinical-significance" />
</code>
<valueCodeableConcept>
<coding>
<system value="urn:oid:1.2.840.114350.1.13.5325.1.7.4.866582.5010" />
<code value="20" />
<display value="Risk allele" />
</coding>
</valueCodeableConcept>
</component>
<component>
<code>
<coding>
<system value="http://hl7.org/fhir/uv/genomics-reporting/CodeSystem/tbd-codes-cs" />
<code value="functional-effect" />
</coding>
<text value="functional-effect" />
</code>
<valueCodeableConcept>
<coding>
<system value="urn:oid:1.2.840.114350.1.13.5325.1.7.4.866582.1290" />
<code value="20" />
<display value="Gain of Function" />
</coding>
</valueCodeableConcept>
</component>
<component>
<code>
<coding>
<system value="http://loinc.org" />
<code value="81259-4" />
</coding>
<text value="associated-phenotype" />
</code>
<valueCodeableConcept>
<coding>
<system value="http://hl7.org/fhir/sid/icd-9-cm/diagnosis" />
<code value="22959" />
</coding>
<text value="Simple VAR Phenotype Name" />
</valueCodeableConcept>
</component>
<component>
<code>
<coding>
<system value="http://hl7.org/fhir/uv/genomics-reporting/CodeSystem/tbd-codes-cs" />
<code value="condition-inheritance" />
</coding>
<text value="mode-of-inheritance" />
</code>
<valueCodeableConcept>
<coding>
<system value="urn:oid:1.2.840.114350.1.13.5325.1.7.4.866582.1904" />
<code value="170" />
<display value="X-linked recessive inheritance" />
</coding>
</valueCodeableConcept>
</component>
</Observation>
</resource>
<search>
<mode value="match" />
</search>
</entry>
<entry>
<link>
<relation value="self" />
<url value="https://hostname/instance/api/FHIR/R4/Observation/e4t0a8Yo6ZFhNUAbGJoR4P-DeSZGzoSLzEyxaQ0oD1gM3" />
</link>
<fullUrl value="https://hostname/instance/api/FHIR/R4/Observation/e4t0a8Yo6ZFhNUAbGJoR4P-DeSZGzoSLzEyxaQ0oD1gM3" />
<resource>
<Observation>
<id value="e4t0a8Yo6ZFhNUAbGJoR4P-DeSZGzoSLzEyxaQ0oD1gM3" />
<basedOn>
<reference value="ServiceRequest/eY0zM9XwMAsfr83odpCgcifyOxXr.2BKRPWcXghHs9b03" />
</basedOn>
<status value="amended" />
<category>
<coding>
<system value="http://open.epic.com/FHIR/StructureDefinition/observation-category" />
<code value="genomics" />
<display value="Therapeutic Implication" />
</coding>
<text value="Therapeutic Implication" />
</category>
<category>
<coding>
<system value="http://terminology.hl7.org/CodeSystem/observation-category" />
```

<code value="laboratory" />
<display value="Laboratory" />
</coding>
<text value="Laboratory" />
</category>
<code>
<coding>
<system value="http://hl7.org/fhir/uv/genomics-reporting/CodeSystem/TbdCodes" />
<code value="therapeutic-implication" />
<display value="Therapeutic Implication" />
</coding>
<text value="Therapeutic Implication" />
</code>
<subject>
<reference value="Patient/eyr87CEI8vy-nL6hL5yccppCd83J03x-7iMjZnGMSlrE3" />
<display value="Garcia Rodriguez, Hugo" />
</subject>
<effectiveDateTime value="2021-12-17T06:00:00Z" />
<issued value="2021-12-17T15:37:17Z" />
<derivedFrom>
<reference value="Observation/e4t0a8Yo6ZFhNUAbGJoR4P2m1Dsb.hyC8Oi7resT-R8I3" />
<display value="Genotype" />
</derivedFrom>
<component>
<code>
<coding>
<system value="http://loinc.org" />
<code value="53040-2" />
</coding>
<text value="Genetic variation's effect on drug metabolism" />
</code>
<valueCodeableConcept>
<coding>
<system value="urn:oid:1.2.840.114350.1.13.5325.1.7.4.866582.6010" />
<code value="5" />
<display value="Poor metabolizer" />
</coding>
</valueCodeableConcept>
</component>
<component>
<code>
<coding>
<system value="http://loinc.org" />
<code value="51961-1" />
</coding>
<text value="Genetic variation's effect on drug efficacy" />
</code>
<valueCodeableConcept>
<coding>
<system value="urn:oid:1.2.840.114350.1.13.5325.1.7.4.866582.6011" />
<code value="3" />
<display value="Presumed resistant" />
</coding>
</valueCodeableConcept>
</component>
<component>
<code>
<coding>
<system value="http://loinc.org" />
<code value="83009-1" />
</coding>
<text value="Genetic variation's effect on high-risk allele" />
</code>
<valueCodeableConcept>
<coding>
<system value="urn:oid:1.2.840.114350.1.13.5325.1.7.4.866582.6012" />
<code value="1" />
<display value="Low risk" />
</coding>
</valueCodeableConcept>
</component>
<component>
<extension url="http://open.epic.com/FHIR/StructureDefinition/extension/medication-usage-suggestion-type">
<extension url="code">
<valueCodeableConcept>
<coding>
<system value="http://loinc.org" />
<code value="82116-5" />
</coding>
<text value="Medication usage suggestion type" />
</valueCodeableConcept>
</extension>
<extension url="value" /></extension>
<extension url="http://open.epic.com/FHIR/StructureDefinition/extension/medication-usage-suggestion-narrative">
<extension url="code">
<valueCodeableConcept>
<coding>
<system value="http://loinc.org" />
<code value="83010-9" />

```
</coding>
<text value="Medication usage suggestion narrative" />
</valueCodeableConcept>
</extension>
<extension url="value">
<valueString value="PGX VAR (Single) Medication Usage Suggestion" />
</extension>
</extension>
<code>
<coding>
<system value="http://loinc.org" />
<code value="51963-7" />
</coding>
<text value="Medication Assessed" />
</code>
<valueCodeableConcept>
<coding>
<system value="https://www.nlm.nih.gov/research/umls/rxnorm" />
<code value="31669" />
</coding>
<text value="PGX VAR (Single) Medication Assessed" />
</valueCodeableConcept>
</component>
</Observation>
</resource>
<search>
<mode value="match" />
</search>
</entry>
<entry>
<link>
<relation value="self" />
<url value="https://hostname/instance/api/FHIR/R4/Observation/e4t0a8Yo6ZFhNUAbGJoR4P2m1Dsb.hyC8Oi7resT-R8I3" />
</link>
<fullUrl value="https://hostname/instance/api/FHIR/R4/Observation/e4t0a8Yo6ZFhNUAbGJoR4P2m1Dsb.hyC8Oi7resT-R8I3" />
<resource>
<Observation>
<id value="e4t0a8Yo6ZFhNUAbGJoR4P2m1Dsb.hyC8Oi7resT-R8I3" />
<extension url="http://open.epic.com/FHIR/StructureDefinition/extension/variant-type">
<valueCodeableConcept>
<coding>
<system value="urn:oid:1.2.840.114350.1.13.5325.1.7.4.866582.35" />
<code value="50" />
<display value="Pharmacogenomic Genotype" />
</coding>
</valueCodeableConcept>
</extension>
<extension url=”http://open.epic.com/FHIR/StructureDefinition/extension/affected-exons” />
<valueRange>
<low>
<value value="2" />
</low>
<high>
<value value="15" />
</high>
</valueRange>
</extension>
<extension url=”http://open.epic.com/FHIR/StructureDefinition/extension/affected-introns” />
<valueRange>
<low>
<value value="4" />
</low>
<high>
<value value="10" />
</high>
</valueRange>
</extension>
<basedOn>
<reference value="ServiceRequest/eY0zM9XwMAsfr83odpCgcifyOxXr.2BKRPWcXghHs9b03" />
</basedOn>
<status value="amended" />
<category>
<coding>
<system value="http://open.epic.com/FHIR/StructureDefinition/observation-category" />
<code value="genomics" />
<display value="Genotype" />
</coding>
<text value="Genotype" />
</category>
<category>
<coding>
<system value="http://terminology.hl7.org/CodeSystem/observation-category" />
<code value="laboratory" />
<display value="Laboratory" />
</coding>
<text value="Laboratory" />
</category>
<code>
<coding>
```

```
<system value="http://loinc.org" />
<code value="84413-4" />
<display value="Genotype" />
</coding>
<text value="Genotype" />
</code>
<subject>
<reference value="Patient/eyr87CEI8vy-nL6hL5yccppCd83J03x-7iMjZnGMSlrE3" />
<display value="Garcia Rodriguez, Hugo" />
</subject>
<effectiveDateTime value="2021-12-17T06:00:00Z" />
<issued value="2021-12-17T15:37:17Z" />
<valueCodeableConcept>
<text value="ABCA 10 Genotype" />
</valueCodeableConcept>
<component>
<extension url="http://open.epic.com/FHIR/StructureDefinition/extension/is-fused">
<valueBoolean value="true" />
</extension>
<code>
<coding>
<system value="http://loinc.org" />
<code value="48018-6" />
</coding>
<text value="gene-studied" />
</code>
<valueCodeableConcept>
<coding>
<system value="http://www.genenames.org/data/gene-symbol-report/" />
<code value="30" />
</coding>
<text value="ABCA10" />
</valueCodeableConcept>
</component>
</Observation>
</resource>
<search>
<mode value="match" />
</search>
</entry>
</Bundle>
```


Change log

Epic Version | Description | Type
May 2026 | FHIR IDs for Observation (Genomics) (R4) resources will no longer be valid if generated by previous versions, due to a change in format. | Breaking change
August 2025 | Observation (Genomics) is updated such that repeat expansion number now supports quantity and range values. | Breaking change
August 2025 | For variant concepts 511- Allele Start/End, Structural Inner Start/End- 546, and Structural Outer Start/End 548, the system of http://hl7.org/fhir/uv/genomics-reporting/CodeSystem/tbd-codes-cs was updated to http://loinc.org because the returned codes are LOINC codes. | Breaking change

10.1.46 Observation.Search (Phenotype) (R4)

File: site/apis/10-Clinical-General/1-R4/11328-Observation-Search-Phenotype-R4.html
Epic API ID | 11328
FHIR version | R4
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Observation?based-on={based-on}&category={category}&code={code}&date={date}&focus={focus}&issued={issued}&patient={patient}&subject={subject}&_count={_count}&_id={_id}&_include={_include}&_revinclude={_revinclude}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=11328

Description

This web service retrieve phenotypes from a patient's chart. Phenotypes represent observable characteristics of an individual and are used by laboratories to determine which genes should be focused on and which variants to include in the lab report.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
_count | number | Optional | Yes | | Number of results per page.
_id | token | Optional | Yes | | FHIR resource IDs for the desired resources. Can be used in combination with _count, _include, and _revInclude. All others are ignored.
_include | string | Optional | Yes | | Used to indicate which referenced resources to include in the search bundle.

_revinclude | string | Optional | Yes | | Used to include Provenance or AuditEvent resources that refer to search results, when available.
category | token | Conditional | Yes | | Always "phenotype" for this API. Either category or code must be provided.
code | token | Conditional | Yes | | The code of the Observation. System: http://human-phenotype-ontology.org Code: HPO Code, for example: HP:0009805 Either category or code must be provided.
patient | reference | Conditional | Yes | | The FHIR ID for the patient associated with the Observation. Either patient or subject must be provided.
subject | reference | Conditional | Yes | | The FHIR ID for the patient associated with the Observation. Either patient or subject must be provided.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
date | date | Optional | Yes | | The date the phenotype was recorded.
performer | reference | Optional | Yes | | A reference to the Practitioner FHIR ID for the resulting agency director. Format for use in post-filtering is performer=Practitioner/<Lab Director Practitioner FHIR ID>

Response elements

Name | Type | Optional | Array | Cardinality | Description
Observation (R4) | Observation
Observation (R4).category | CodeableConcept | | Yes | 1..1 | The Observation type. Always "phenotype". System: http://terminology.hl7.org/CodeSystem/observation-category Code: phenotype
Observation (R4).code | CodeableConcept | | | 1..1 | The code of the Observation. System: http://human-phenotype-ontology.org Code: HPO Code, for example: HP:0009805
Observation (R4).effectiveDateTime | dateTime | | | 1..1 | When the phenotype was first documented.
Observation (R4).id | id | | | 1..1 | The FHIR ID for the Observation.
Observation (R4).note | Annotation | | Yes | 0..1 | Patient specific comments about the phenotype.
Observation (R4).performer | Reference | | Yes | 0..* | A reference to the documenting provider or user.
Observation (R4).status | code | | | 1..1 | Always set to "final".
Observation (R4).subject | Reference | | | 1..1 | The FHIR ID for the patient.
Observation (R4).valueCodeableConcept | CodeableConcept | | | 1..1 | The LOINC code for presence or absence of a phenotype. The values come from LOINC Answer List LL1937-3. Positive and Resolved both map to "Present (LA9633-4)" Negative maps to "Absent (LA9634-2)"

Sample request

```
GET https://hostname/instance/api/FHIR/R4/Observation?patient=ej4nfxzysjaysjmuv4rtbna3&category=phenotype
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 2,
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Observation?category=phenotype&patient=ej4nfxzysjaysjmuv4rtbna3"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url": "https://hostname/instance/api/FHIR/R4/Observation/fzcMkZ2CZ4RRxccgmgdWeHw4"
        }
      ],
      "fullUrl": "https://hostname/instance/api/FHIR/R4/Observation/fzcMkZ2CZ4RRxccgmgdWeHw4",
      "resource": {
        "resourceType": "Observation",
        "id": "fzcMkZ2CZ4RRxccgmgdWeHw4",
        "status": "final",
        "category": [
          {
            "coding": [
              {
                "system": "http://open.epic.com/FHIR/StructureDefinition/observation-category",
                "code": "phenotype",
                "display": "Phenotype"
              }
            ],
            "text": "Phenotype"
          }
        ],
        "code": {
          "coding": [
            {
              "system": "http://human-phenotype-ontology.org",
              "code": "HP:0001251",
              "display": "Ataxia"
            }
          ],
          "text": "Ataxia"
        },
        "subject": {
```

```
      "reference": "Patient/ej4nfxzysjaysjmuv4rtbna3",
      "display": "Test, Patient"
    },
    "effectiveDateTime": "2025-07-23",
    "performer": [
      {
        "reference": "Practitioner/euJOUNeZyX6mofWELWG0Ibg3",
        "display": "Starter Provider"
      }
    ],
    "valueCodeableConcept": {
      "coding": [
        {
          "system": "http://loinc.org",
          "code": "LA9633-4",
          "display": "Present"
        }
      ],
      "text": "Positive"
    },
    "note": [
      {
        "text": "note text"
      }
    ]
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Observation/fTnG87aSZt4W9AOj9lArr.w4"
    }
  ],
  "fullUrl": "https://hostname/instance/api/FHIR/R4/Observation/fTnG87aSZt4W9AOj9lArr.w4",
  "resource": {
    "resourceType": "Observation",
    "id": "fTnG87aSZt4W9AOj9lArr.w4",
    "status": "final",
    "category": [
      {
        "coding": [
          {
            "system": "http://open.epic.com/FHIR/StructureDefinition/observation-category",
            "code": "phenotype",
            "display": "Phenotype"
          }
        ],
        "text": "Phenotype"
      }
    ],
    "code": {
      "coding": [
        {
          "system": "http://human-phenotype-ontology.org",
          "code": "HP:0001626",
          "display": "Abnormality of the cardiovascular system"
        }
      ],
      "text": "Abnormality of the cardiovascular system"
    },
    "subject": {
      "reference": "Patient/ej4nfxzysjaysjmuv4rtbna3",
      "display": "Test, Patient"
    },
    "effectiveDateTime": "2025-07-28",
    "performer": [
      {
        "reference": "Practitioner/euJOUNeZyX6mofWELWG0Ibg3",
        "display": "Starter Provider"
      }
    ],
    "valueCodeableConcept": {
      "coding": [
        {
          "system": "http://loinc.org",
          "code": "LA9633-4",
          "display": "Present"
        }
      ],
      "text": "Positive"
    }
  },
  "search": {
    "mode": "match"
  }
```

```
    }
  ]
}
```

10.1.47 Observation.Search (SDOH Assessments) (R4)

File: site/apis/10-Clinical-General/1-R4/11104-Observation-Search-SDOH-Assessments-R4.html
Epic API ID | 11104
FHIR version | R4
Categories | Clinical General
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Keywords | sdoh
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Observation?category={category}&code={code}&date={date}& patient={patient}&subject={subject}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=11104

Description

This web service retrieves assessment data from a patient's chart for only those assessments that relate to social drivers of health (SDOH) data. The same SDOH data is also returned in the Observation (Assessments) API, but this API is limited to include only SDOH data.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
category | token | This parameter is optional if the code parameter is used in the search request. | | | Use "sdoh" or "survey" to search for SDOH observations.
code | token | This parameter is optional if the category parameter is used in the search request. | | | The LOINC code that corresponds to the measured SDOH Assessment
date | date | True | | | The date or date range when the SDOH assessment was taken
patient | reference | False | | | Target patient for this search
subject | reference | True | | | Target patient for this search.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
based-on | String | True | | | Unsupported
combo-code | token | True | | | The LOINC code that corresponds to the measured assessment.
combo-data-absent-reason | token | True | | | The reason an observation value is missing.
combo-value-concept | token | True | | | CodeableConcept value associated with the assessment
component-code | token | True | | | Unsupported
component-data-absent-reason | token | True | | | Unsupported
component-value-concept | token | False | | | Unsupported
data-absent-reason | token | True | | | The reason an observation value is missing.
derived-from | String | True | | | An observation that is related to the returned observation.
device | String | True | | | Unsupported
encounter | reference | True | | | Unsupported
focus | reference | True | | | Unsupported
has-member | reference | True | | | Unsupported
identifier | token | True | | | Unsupported
issued | date | True | | | Unsupported
method | token | True | | | Unsupported
part-of | String | True | | | Unsupported
performer | reference | True | | | Reference to whomever performed the observation.
specimen | String | False | | | Unsupported
status | token | True | | | The status of the observation. Will be either "final" or "amended" depending on whether the field has been corrected.
value-concept | token | True | | | CodeableConcept value associated with the assessment
value-date | date | True | | | Unsupported
value-string | string | True | | | String value associated with the assessment

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | BackboneElement | False | Yes | | An array of entries included in this bundle.
Bundle.entry.resource | ResourceContainer | False | | | A container for a single FHIR resource.
Bundle.entry.resource.Observation | Observation | False | | | An instance of the R4 Observation resource.
Bundle.entry.resource.Observation.category | CodeableConcept | False | Yes | | Always returns "survey" and "sdoh".

Bundle.entry.resource.Observation.code | CodeableConcept | False | | | The LOINC code that corresponds to the measured SDOH assessment.
Bundle.entry.resource.Observation.dataAbsentReason | CodeableConcept | True | | | Indicates why an observation value is missing, if applicable.
Bundle.entry.resource.Observation.derivedFrom | Reference | True | Yes | | This element displays any Observations that are related to the returned Observation.
Bundle.entry.resource.Observation.effectiveDateTime | dateTime | False | | | The date and time the Observation was taken.
Bundle.entry.resource.Observation.encounter | Reference | True | | | Encounter associated with this observation value, if applicable.
Bundle.entry.resource.Observation.hasMember | Reference | True | Yes | | This element returns any panel members an Observation contains.
Bundle.entry.resource.Observation.issued | instant | False | | | The date and time the observation was taken.
Bundle.entry.resource.Observation.note | Annotation | True | Yes | | The comment
Bundle.entry.resource.Observation.performer | Reference | False | Yes | | Who filed this Observation.
Bundle.entry.resource.Observation.status | code | False | | | The status of the Observation. This is set to amended or final depending on whether the field has been corrected.
Bundle.entry.resource.Observation.subject | Reference | False | | | Patient associated with the Observation.
Bundle.entry.resource.Observation.valueCodeableConcept | CodeableConcept | This element is populated if the data type is category or custom list. | | | This element is populated if the data type is category or custom list.
Bundle.entry.resource.Observation.valueQuantity | Quantity | This element is populated if a value has units or the data type is numeric or a percentile. | | | This element is populated if a value has units or the data type is numeric or a percentile.
Bundle.entry.resource.Observation.valueString | string | This element is populated if a value doesn't have units or the data type is string. | | | This element is populated if a value doesn't have units or the data type is string.
Bundle.entry.resource.Observation.extension (also: workflow-supportingInfo) | Extension | True | Yes | | If the measurement is linked to a note written by a clinician, this is a reference to the DocumentReference resource.
OperationOutcome | OperationOutcome | False

Sample request

```
https://hostname/instance/api/FHIR/R4/Observation?patient=egEK7QBIy5r1ipD2Za3zkmw3&category=sdoh
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 2,
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Observation?category=sdoh&patient=egEK7QBIy5r1ipD2Za3zkmw3"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url": "https://hostname/instance/api/FHIR/R4/Observation/etlVC9LG0xZ-
    PP5marhsAN7eeR8kxbjl8fVk81OodXj-o2fyIFMidJCiiKR8xfEr8-UVXmInQckQbJ-7676.ccg3"
        }
      ],
      "fullUrl": "https://hostname/instance/api/FHIR/R4/Observation/etlVC9LG0xZ-
    PP5marhsAN7eeR8kxbjl8fVk81OodXj-o2fyIFMidJCiiKR8xfEr8-UVXmInQckQbJ-7676.ccg3",
      "resource": {
        "resourceType": "Observation",
        "id": "etlVC9LG0xZ-PP5marhsAN7eeR8kxbjl8fVk81OodXj-o2fyIFMidJCiiKR8xfEr8-UVXmInQckQbJ-7676.ccg3",
        "status": "final",
        "category": [
          {
            "coding": [
              {
                "system": "http://terminology.hl7.org/CodeSystem/observation-category",
                "code": "survey",
                "display": "Survey"
              }
            ]
          },
          {
            "coding": [
              {
                "system": "http://hl7.org/fhir/us/core/CodeSystem/us-core-category",
                "code": "sdoh",
                "display": "SDOH"
              }
            ]
          }
        ],
        "code": {
          "coding": [
            {
              "system": "http://loinc.org",
              "code": "98975-6",
              "display": "Children's Health Watch Housing Stability Vital Signs"
            },
            {
              "system": "http://loinc.org",
              "code": "98976-4",
```

```
          "display": "In the past 12 months, was there a time when you were not able to pay the mortgage or rent on
time?"
        },
        {
          "system": "http://loinc.org",
          "code": "98977-2",
          "display": "In the past 12 months, how many times have you moved where you were living?"
        },
        {
          "system": "http://loinc.org",
          "code": "98978-0",
          "display": "At any time in the past 12 months, were you homeless or living in shelter [including now]?"
        }
      ]
    },
    "subject": {
      "reference": "Patient/egEK7QBIy5r1ipD2Za3zkmw3",
      "display": "Grey, Eärl LNS"
    },
    "encounter": {
      "reference": "Encounter/eh0sTg.kAw5MItnpwiwnxh5WGO44DNwtsZgYnU05kLEU3",
      "identifier": {
        "use": "usual",
        "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.3.698084.8",
        "value": "10005526256"
      },
      "display": "Office Visit"
    },
    "effectiveDateTime": "2024-09-05T14:00:00Z",
    "issued": "2024-09-05T14:05:43Z",
    "performer": [
      {
        "reference": "Practitioner/e4ZKPXZ0ux4Z80UHywzHyHg3",
        "display": "Practitioner Test, MD"
      }
    ],
    "hasMember": [
      {
        "reference": "Observation/etlVC9LG0xZ-PP5marhsAN7eeR8kxbjl8fVk81OodXj-
VuOvfOrfwoQO6aiKSG5I4aWGV4.CNSIlwFSP.qMWSbg3"
      }
    ]
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Observation/etlVC9LG0xZ-PP5marhsAN7eeR8kxbjl8fVk81OodXj-
VuOvfOrfwoQO6aiKSG5I4aWGV4.CNSIlwFSP.qMWSbg3"
    }
  ],
  "fullUrl": "https://hostname/instance/api/FHIR/R4/Observation/etlVC9LG0xZ-PP5marhsAN7eeR8kxbjl8fVk81OodXj-
VuOvfOrfwoQO6aiKSG5I4aWGV4.CNSIlwFSP.qMWSbg3",
  "resource": {
    "resourceType": "Observation",
    "id": "etlVC9LG0xZ-PP5marhsAN7eeR8kxbjl8fVk81OodXj-VuOvfOrfwoQO6aiKSG5I4aWGV4.CNSIlwFSP.qMWSbg3",
    "status": "final",
    "category": [
      {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/observation-category",
            "code": "survey",
            "display": "Survey"
          }
        ]
      },
      {
        "coding": [
          {
            "system": "http://hl7.org/fhir/us/core/CodeSystem/us-core-category",
            "code": "sdoh",
            "display": "SDOH"
          }
        ]
      }
    ],
    "code": {
      "coding": [
        {
          "system": "http://loinc.org",
          "code": "98976-4",
          "display": "In the past 12 months, was there a time when you were not able to pay the mortgage or rent on
time?"
        }
```

```
          ]
        },
        "subject": {
          "reference": "Patient/egEK7QBIy5r1ipD2Za3zkmw3",
          "display": "Grey, Eärl LNS"
        },
        "encounter": {
          "reference": "Encounter/eh0sTg.kAw5MItnpwiwnxh5WGO44DNwtsZgYnU05kLEU3",
          "identifier": {
            "use": "usual",
            "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.3.698084.8",
            "value": "10005526256"
          },
          "display": "Office Visit"
        },
        "effectiveDateTime": "2024-09-05T14:00:00Z",
        "issued": "2024-09-05T14:05:32Z",
        "performer": [
          {
            "reference": "Practitioner/e4ZKPXZ0ux4Z80UHywzHyHg3",
            "display": "Practitioner Test, MD"
          }
        ],
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://loinc.org",
              "code": "LA32-8",
              "display": "No"
            }
          ]
        },
        "derivedFrom": [
          {
            "reference": "Observation/etlVC9LG0xZ-
    PP5marhsAN7eeR8kxbjl8fVk81OodXj-o2fyIFMidJCiiKR8xfEr8-UVXmInQckQbJ-7676.ccg3"
          }
        ]
      },
      "search": {
        "mode": "match"
      }
    }
  ]
}
```


10.1.48 Observation.Search (Social History) (R4)

File: site/apis/10-Clinical-General/1-R4/972-Observation-Search-Social-History-R4.html
Epic API ID | 972
FHIR version | R4
Categories | Clinical General
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Keywords | Smoking; Social; Social history;Flowsheets
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | November 2019
HTTP method | GET
URL template | api/FHIR/R4/Observation?patient={patient}&subject={subject}&category={category}&code={code}&issued={issued}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=972

Description

The Observation (Social History) sub-resource searches for patient observations which include any of the following types of data:

Type of data | Description
Smoking history | The status of a patient's tobacco use. This could be used to determine if the patient is a current or former tobacco user, what type of tobacco product is/was used, and the dates the patient started and, if a former user, stopped using tobacco. This includes passive exposure information in the code response element.
Pregnancy Status | The patient’s current pregnancy status.
Pregnancy Intention | The patient’s intentions of becoming pregnant in the next year. Available starting in the February 2024 version of Epic.
Lactation Status | The patient’s current lactation status.
Physical Findings of Uterus | Whether the patient has a uterus.
Estimated Delivery Date | The patient’s estimated date of delivery.
Last Menstrual Period | The patient’s last menstrual period.
Alcohol use history | The patient's alcohol use history and, if applicable, the type and amount of alcohol the patient consumes.

Drug use history | The patient's drug use history and, if applicable, the type and amount of drugs the patient consumes.
Living situation | The patient's current living situation.
Occupational history | The patient's occupational history, including start and end dates, industry, employer, and additional comments.
Social documentation | The social documentation notes stored in the patient's history.
Education history | The patient's education history.
Sexual orientation | The patient's sexual orientation. Available starting in the May 2024 version of Epic.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
_count | number | Optional | Yes | | Number of results per page.
_id | token | Optional | Yes | | FHIR resource IDs for the desired resources. Can be used in combination with _count, _include, and _revInclude. All others will be ignored.
_include | string | Optional | Yes | | Used to indicate which referenced resources to include in the search bundle.
_revinclude | string | Optional | Yes | | Used to include Provenance or AuditEvent resources that refer to search results, when available.
category | token | Conditional | Yes | | This is always "social-history". Either a category or a code must be provided.
code | token | Conditional | Yes | | Either the LOINC or SNOMED code for a particular type of Observation may be used. This may include any of the following values: Smoking history: LOINC code 72166-2 or SNOMED code 365980008. Passive exposure: SNOMED code 43381005 Alcohol use history: LOINC code 11331-6 or SNOMED code 228273003 Drug use history: LOINC code 11343-1 or SNOMED code 228366006 Living situation: SNOMED code 365508006 Education: LOINC code LG41797-1 Employment history: LOINC code 11341-5 Social documentation: LOINC code 29762-2 Pregnancy status: LOINC code 82810-3 Uterus findings: LOINC code 32486-3 Lactation Status: LOINC code 63895-7 Menstrual status: LOINC code 3146-8 Last menstrual period start-date: LOINC code 8665-2 Estimated Delivery Date: LOINC code 11778-8 Pregnancy Intention: LOINC code 86645-9 Breastfeeding status: LOINC code 63895-7 Sexual orientation: LOINC code 76690-7 Either a category or a code must be provided.
date | date | Optional | Yes | | Date when the Observation occurred.
patient | reference | Conditional | Yes | | Reference to a Patient resource. Either patient or subject is required.
subject | reference | Conditional | Yes | | Reference to a Patient resource. Either patient or subject is required.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
combo-code | token | Optional | Yes | | Matches to the code value returned in the code or component.code elements, if present, such as a SNOMED code. Refer to the code and component.code response elements for more information.
combo-value-concept | token | Optional | Yes | | The SNOMED code or other code value pertaining to smoking status, alcohol use status, drug use status, living situation, or other social history information. This value might be modified by organization-specific mapping. Refer to the valueCodeableConcept or component.valueCodeableConcept response element for more information.
component-code | token | Optional | Yes | | Matches to the code value returned in component.code, if present, such as a SNOMED code. Refer to the component.code response element for more information.
component-value-concept | token | Optional | Yes | | The SNOMED code or other code value pertaining to smoking status, alcohol use status, drug use status, living situation, or other social history information. This value might be modified by organization-specific mapping. Refer to the component.valueCodeableConcept response element for more information.
issued | date | Optional | Yes | | The date range the Observation is obtained.
performer | reference | Optional | Yes | | The party responsible for documenting smoking history status. Does not apply to other types of social history data.
status | token | Optional | Yes | | If specified, should always be "final" when searching for social history data.
value-concept | token | Optional | Yes | | The SNOMED code or other value pertaining to smoking status, alcohol use status, drug use status, living situation, or other social history information. This value might be modified by organization-specific mapping. Refer to the valueCodeableConcept response element for more information.
value-string | string | Optional | Yes | | The text in valueString for this Observation, such as the text of the social documentation note.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Observation (R4) | Observation
Observation (R4).category | CodeableConcept | | Yes | 0..* | This is always "social-history".
Observation (R4).code | CodeableConcept | | | 1..1 | The following value(s), depending on the type of data returned: Smoking history: LOINC code 72166-2 SNOMED code 365980008. Sexual Orientation: LOINC code 76690-7 Alcohol use history: LOINC code 11331-6 SNOMED code 228273003 Drug use history: LOINC code 11343-1 SNOMED code 228366006 Living situation: SNOMED code 365508006 Pregnancy status: LOINC code 82810-3 Pregnancy intention: LOINC code 86645-9 Lactation status: LOINC code 63895-7 Physical findings of uterus: LOINC code 32486-3 Estimated delivery date: LOINC code 11778-8 Menstrual status: LOINC code 3146-8 Last menstrual status: LOINC code 8665-2
Observation (R4).component | BackboneElement | | Yes | 0..* | Applies only to smoking history, alcohol use history, drug use history, education, and occupational history. Holds additional detail depending on the type of Observation: Smoking history: Can include one or more of the following code values, paired with either a valueCodeableConcept or a valueQuantity that includes additional detail: Tobacco product (LOINC 81228-9; SNOMED 53661000146106) Cigarette pack-years (SNOMED 401201003) Tobacco use duration (LOINC 88029-4) Packs per day (LOINC 8663-7; SNOMED 266918002) Alcohol use history: The amount of alcohol consumed per day (SNOMED 160573003), expressed in UCUM units as alcoholic_units/day. Drug use history: Can include one or more of the following code values, paired with either a valueCodeableConcept or a valueString that includes additional detail: Type of substance (LOINC 63643-1; SNOMED 410942007) Frequency of misuse (SNOMED 228390007) Education: Can include the following code value, paired with either a valueCodeableConcept or valueString that includes additional detail: The patient’s highest level of education (LOINC code 82589-3) Occupational: Can include one or more of the following code values, paired with either a valueCodeableConcept or valueString that includes additional detail: The patient’s occupation (LOINC code LP133724-7) The patient’s employer (LOINC code 80427-8) The patient’s occupation industry (LOINC code 86188-0)
Observation (R4).component.code | CodeableConcept | | | 1..1 | Often a LOINC or SNOMED code.
Observation (R4).component.valueCodeableConcept | CodeableConcept | | | 0..1 | CodeableConcept value.
Observation (R4).component.valueQuantity | Quantity | | | 0..1 | Quantity value.
Observation (R4).component.valueString | string | | | 0..1 | String value.

Observation (R4).dataAbsentReason | CodeableConcept | | | 0..1 | Populated if there is no last menstrual period start date but there is a last menstrual period start date comment or precision. Populated if there is no lactation status but there is a lactation status comment.

Observation (R4).derivedFrom | Reference | | Yes | 0..* | References the corresponding menstrual status Observation that the pregnancy status is derived from.

Observation (R4).effectiveDateTime | dateTime | | | 0..1 | The date of the encounter on which the Observation resource was documented for the following types of data: Pregnancy status Menstrual status Last menstrual status Lactation status Sexual Orientation Pregnancy intention

Observation (R4).effectivePeriod | Period | | | 0..1 | The start and end dates of the Observation for the following types of data: Smoking history For smokers, when the patient started and quit smoking (if applicable), and data absent extension if there are no dates on file. For non-smokers, the date range is the from the date the status was recorded to today. Occupation history

Observation (R4).id | id | | | 0..1 | The Observation FHIR ID.

Observation (R4).interpretation | CodeableConcept | | Yes | 0..* | Precision of the last menstrual period.

Observation (R4).issued | instant | | | 0..1 | The date when this status is documented. This is set to the date that the status was assessed, if available. Otherwise, this is set to one of the following values to indicate that the response is considered up-to-date as of the specified date: The date of the patient's most recent history encounter, if available. The current date. The issued date may not be populated for every Observation type.

Observation (R4).note | Annotation | | Yes | 0..* | The comments for smoking history, alcohol use history, drug use history, employment history, last menstrual period, or lactation status. For organizations in the Netherlands, this is sent in both provider and patient contexts. For all other organizations, this is sent only in provider context.

Observation (R4).performer | Reference | | Yes | 0..* | The person who documented the status. This is not populated if the status is never assessed.

Observation (R4).status | code | | | 1..1 | This value is always "final". More detailed status information, such as whether the patient's smoking status is unknown, can be found by reviewing the values specified in the code element or the valueCodeableConcept element.

Observation (R4).subject | Reference | | | 0..1 | The Patient FHIR ID.

Observation (R4).valueBoolean | boolean | | | 0..1 | Yes/No answer to whether the patient has a uterus.

Observation (R4).valueCodeableConcept | CodeableConcept | | | 0..1 | The SNOMED code or other value pertaining to smoking status, alcohol use status, drug use status, living situation, pregnancy status, lactation status, or pregnancy intention. Default Epic-to-SNOMED mappings are listed below. This value can also be modified by organization-specific mapping. Smoking history supported values: 449868002-Smokes tobacco daily (finding) 428041000124106-Current some day smoker 230059006-Occasional cigarette smoker (only for organizations in the Netherlands) 77176002-Smoker, current status unknown 8517006-Former smoker 266919005-Never smoker 43381005-Passive smoker (only for organizations in the Netherlands) 266927001-Unknown if ever smoker 266919005-Never smoker 266927001-Unknown if ever smoker 428071000124103-Heavy tobacco smoker 428061000124105-Light tobacco smoker Alcohol use history supported values: 219006-Current drinker of alcohol (finding) 105542008-Current non-drinker of alcohol (finding) OTH-Other 82581004-Ex-drinker (finding) 228274009-Lifetime non-drinker (finding) Drug use history supported values: 361055000-Misuses drugs (finding) 228367002-Does not misuse drugs (finding) OTH-Other 44870007-Misused drugs in past (finding) 228368007-Has never misused drugs Living situation has no default mapping, but can be customized by the Epic organization. Pregnancy status supported values: 77386006-Pregnancy (finding) UNK-Unknown 60001007-Not pregnant (finding) Lactation status supported values: 373066001-Yes 373067005-No 261665006-Unknown Pregnancy intention supported values: 454411000124108-Wants to become pregnant (finding) 454391000124108-Ambivalent about becoming pregnant (finding) 454401000124105-No desire to become pregnant (finding) 454381000124105-Not sure of desire to become pregnant (finding)

Observation (R4).valueDateTime | dateTime | | | 0..1 | Estimated delivery date or the start date of the last menstrual period, if applicable.

Observation (R4).valueString | string | | | 0..1 | The text of the social documentation notes. Applies only to social documentation.

Sample request

```
https://hostname/instance/api/FHIR/R4/Observation?patient=eeiBNKuRR84uD0nIFn73BGQ3&category=social-history
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 2,
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Observation?patient=eeiBNKuRR84uD0nIFn73BGQ3&category=social-
    history"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url": "https://hostname/instance/api/FHIR/R4/Observation/euvO8iq4aYWriTr-
    MA7S-IkCxqCXJDmc.rRo7MsHFYNG0HdfUGgXpHY.GcKccA1fi3"
        }
      ],
      "fullUrl": "https://hostname/instance/api/FHIR/R4/Observation/euvO8iq4aYWriTr-
    MA7S-IkCxqCXJDmc.rRo7MsHFYNG0HdfUGgXpHY.GcKccA1fi3",
      "resource": {
        "resourceType": "Observation",
        "id": "euvO8iq4aYWriTr-MA7S-IkCxqCXJDmc.rRo7MsHFYNG0HdfUGgXpHY.GcKccA1fi3",
        "status": "final",
        "category": [
          {
            "coding": [
              {
                "system": "http://terminology.hl7.org/CodeSystem/observation-category",
                "code": "social-history",
                "display": "Social History"
              }
            ],
            "text": "Social History"
          }
```

```
],
"code": {
  "coding": [
    {
      "system": "http://loinc.org",
      "code": "72166-2",
      "display": "Tobacco smoking status"
    },
    {
      "system": "http://snomed.info/sct",
      "code": "365980008",
      "display": "Finding of tobacco use and exposure (finding)"
    }
  ],
  "text": "Smoking History"
},
"subject": {
  "reference": "Patient/eeiBNKuRR84uD0nIFn73BGQ3",
  "display": "Test, Patient FHIR"
},
"effectivePeriod": {
  "start": "2009-01-01"
},
"issued": "2021-05-11T05:00:00Z",
"performer": [
  {
    "reference": "Practitioner/ekk17eu0BAPwHQ6s2NDWYEQ3",
    "display": "Test FHIR, MD"
  }
],
"valueCodeableConcept": {
  "coding": [
    {
      "system": "http://snomed.info/sct",
      "code": "449868002",
      "display": "Smokes tobacco daily"
    }
  ],
  "text": "Every Day"
},
"component": [
  {
    "code": {
      "coding": [
        {
          "system": "http://loinc.org",
          "code": "81228-9",
          "display": "Tobacco product"
        },
        {
          "system": "http://snomed.info/sct",
          "code": "53661000146106",
          "display": "Type of tobacco used (observable entity)"
        }
      ],
      "text": "Type"
    },
    "valueCodeableConcept": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "65568007",
          "display": "Cigarette Smoker"
        }
      ],
      "text": "Cigarettes"
    }
  },
  {
    "code": {
      "coding": [
        {
          "system": "http://loinc.org",
          "code": "81228-9",
          "display": "Tobacco product"
        },
        {
          "system": "http://snomed.info/sct",
          "code": "53661000146106",
          "display": "Type of tobacco used (observable entity)"
        }
      ],
      "text": "Type"

    },
    "valueCodeableConcept": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "81703003",
```

```
              "display": "Chews Tobacco"
            }
          ],
          "text": "Chew"
        }
      },
      {
        "code": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "401201003",
              "display": "Cigarette pack-years (observable entity)"
            }
          ],
          "text": "Cigarette pack-years (observable entity)"
        },
        "valueQuantity": {
          "value": 3.72,
          "unit": "{PackYears}",
          "system": "http://unitsofmeasure.org",
          "code": "{PackYears}"
        }
      },
      {
        "code": {
          "coding": [
            {
              "system": "http://loinc.org",
              "code": "88029-4",
              "display": "Tobacco use duration"
            }
          ],
          "text": "Years Used"
        },
        "valueQuantity": {
          "value": 12.4,
          "unit": "Years Used",
          "system": "http://unitsofmeasure.org",
          "code": "a"
        }
      },
      {
        "code": {
          "coding": [
            {
              "system": "http://loinc.org",
              "code": "8663-7",
              "display": "Cigarettes smoked current (pack per day) - Reported"
            },
            {
              "system": "http://snomed.info/sct",
              "code": "266918002",
              "display": "Tobacco smoking consumption (observable entity)"
            }
          ],
          "text": "Packs/Day"
        },
        "valueQuantity": {
          "value": 0.3,
          "unit": "Packs/Day",
          "system": "http://snomed.info/sct",
          "code": "228963008"
        }
      }
    ]
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Observation/euvO8iq4aYWriTr-
MA7S-IkCxqCXJDmc.rRo7MsHFYNH7Qg8BxrXz6rsT-bDm9pqY3"
    }
  ],
  "fullUrl": "https://hostname/instance/api/FHIR/R4/Observation/euvO8iq4aYWriTr-
MA7S-IkCxqCXJDmc.rRo7MsHFYNH7Qg8BxrXz6rsT-bDm9pqY3",
  "resource": {
    "resourceType": "Observation",
    "id": "euvO8iq4aYWriTr-MA7S-IkCxqCXJDmc.rRo7MsHFYNH7Qg8BxrXz6rsT-bDm9pqY3",
    "status": "final",
    "category": [
      {
        "coding": [
          {
```

```
              "system": "http://terminology.hl7.org/CodeSystem/observation-category",
              "code": "social-history",
              "display": "Social History"
            }
          ],
          "text": "Social History"
        }
      ],
      "code": {
        "coding": [
          {
            "system": "http://loinc.org",
            "code": "11331-6",
            "display": "Alcohol Use History"
          },
          {
            "system": "http://snomed.info/sct",
            "code": "228273003",
            "display": "Finding relating to alcohol drinking behavior (finding)"
          }
        ],
        "text": "Alcohol Use History"
      },
      "subject": {
        "reference": "Patient/eeiBNKuRR84uD0nIFn73BGQ3",
        "display": "Test, Patient FHIR"
      },
      "issued": "2021-05-11T05:00:00Z",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "219006",
            "display": "Current drinker of alcohol (finding)"
          }
        ],
        "text": "Yes"
      },
      "note": [
        {
          "text": "Patient noted that they battle with addiction dependency"
        }
      ],
      "component": [
        {
          "code": {
            "coding": [
              {
                "system": "http://snomed.info/sct",
                "code": "160573003",
                "display": "Alcohol intake (observable entity)"
              }
            ],
            "text": "Alcohol intake (observable entity)"
          },
          "valueQuantity": {
            "value": 4.79,
            "unit": "alcoholic_units/day",
            "system": "http://unitsofmeasure.org",
            "code": "{alcoholic_units}/d"
          }
        }
      ]
    },
    "search": {
      "mode": "match"
    }
  }
 ]
}
```


Change log

Epic Version	Description	Type
November 2024	A query for Observation resources without passing in a date parameter now returns all resources instead of only the most recent.	Breaking Change
February 2024	Added industry and start/end date information under Occupational history in Social History search and read results.	
February 2024	Updated the mappings for the “82810-3 Pregnancy Status” to return with SNOMED values more specific to Pregnancy Status. Be sure to update any functionality that relies on these codes. Also added the “86645-9 Pregnancy Intention” LOINC code which returns values for the patient’s willingness to become pregnant in the next year.	Breaking Change
November 2022	Fixed a bug that caused the "32486-3 Physical Findings of Uterus" LOINC code return with a SNOMED system value (http://snomed.info/sct) instead of a LOINC system (http://loinc.org). If you use this LOINC code, make sure you aren't relying on seeing a SNOMED system.	Breaking Change

10.1.49 Procedure.Read (Nursing Intervention) (R4)

File: site/apis/10-Clinical-General/1-R4/10470-Procedure-Read-Nursing-Intervention-R4.html
Epic API ID | 10470
FHIR version | R4
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Procedure/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10470

Description

This resource can be used only by healthcare organizations in the Netherlands. This resource cannot be used in any other locales at this time.

The FHIR Procedure resource defines an activity performed on or with a patient as part of the provision of care. The Procedure (Intervention) API is designed to summarize any care or treatment carried out by a nurse in response to problems that arise in relation to a patient's care plan goals. This resource returns information about inpatient care plan interventions only. The read interaction enables the lookup of a procedure resource by a constant server ID. The read interaction allows clients to store only the server ID and with a single request retrieve the most up-to-date information about a procedure. Read interactions begin with a client having previously established a relationship, typically when following a reference to this resource from another resource or through querying for procedures through the search interaction.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | False | | | The Procedure FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Procedure | Procedure | False | | | The Procedure FHIR ID.
Procedure.category | CodeableConcept | False | | | A code that classifies the procedure. This will always be SNOMED code 9632001 (nursing procedure) for the Procedure (Intervention) API.
Procedure.code | CodeableConcept | False | | | External codes associated with the procedure.
Procedure.extension (also: Frequency) | Extension | False | Yes | | How often an intervention was performed.
Procedure.note | Annotation | True | Yes | | Comment on the nursing intervention.
Procedure.performedPeriod | Period | True | | | The period of time over which the procedure was performed.
Procedure.performer | BackboneElement | True | Yes | | Who performed the procedure.
Procedure.performer.actor | Reference | True | | | Reference to the practitioner that performed the procedure. The reading physician or, if unavailable, the resulting physician.
Procedure.extension (also: Requester) | Extension | False | Yes | | The user who requested the intervention.
Procedure.status | code | False | | | The status of the Procedure. This describes how the status of the intervention maps to the status of a procedure in this resource. Intervention Status Procedure Status in-progress in-progress completed completed cancelled entered-in-error
Intervention Status | Procedure Status
in-progress | in-progress
completed | completed
cancelled | entered-in-error
Procedure.subject | Reference | False | | | Reference to the patient resource for whom this procedure resource is relevant.
Procedure.extension (also: TreatmentObjective) | Extension | False | Yes | | The description of the treatment goal that the intervention decision is based on. Uses the "pertains to goal" code system.

Sample request

```
https://hostname/instance/api/FHIR/R4/Procedure/enUqBOgIgnH0uYkGqnsBQsdFw8XZSd-axvzJVJ9ybs6E3
```

Sample response


```
{
  "resourceType": "Procedure",
  "id": "enUqBOgIgnH0uYkGqnsBQsdFw8XZSd-axvzJVJ9ybs6E3",
  "extension": [
    {
      "valueReference": {
        "reference": "Goal/eFj3HGT8DXPwINZezNND31g3",
        "display": "IPQA CS Goal"
      },
      "url": "http://hl7.org/fhir/StructureDefinition/resource-pertainsToGoal"
    },
    {
      "valueTiming": {
        "repeat": {
          "frequency": 2,
```

```
          "period": 1,
          "periodUnit": "d"
        },
        "code": {
          "coding": [
            {
              "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.2.697081",
              "code": "200006",
              "display": "BID"
            }
          ],
          "text": "BID"
        }
      },
      "url": "http://hl7.org/fhir/StructureDefinition/procedure-schedule"
    },
    {
      "valueReference": {
        "reference": "Practitioner/e24exSNO1wPT5FJn2sKyTaQ3",
        "display": "John Doe, MD"
      },
      "url": "http://nictiz.nl/fhir/StructureDefinition/zib-NursingIntervention-Requester"
    }
  ],
  "status": "entered-in-error",
  "category": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "9632001",
        "display": "Nursing procedure (procedure)"
      }
    ],
    "text": "Nursing Intervention"
  },
  "code": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "416393001",
        "display": "Procedure time out (procedure)"
      },
      {
        "system": "http://snomed.info/sct",
        "code": "707000009",
        "display": "SNOMED CT parsable string (foundation metadata concept)"
      }
    ],
    "text": "IPQA CS Goal Int"
  },
  "subject": {
    "reference": "Patient/eGmXCJpemueo3cQT8TXd13A3",
    "display": "Smith, Josiah"
  },
  "performedPeriod": {
    "end": "2022-05-03T15:39:15Z"
  },
  "performer": [
    {
      "actor": {
        "reference": "Practitioner/e24exSNO1wPT5FJn2sKyTaQ3",
        "type": "Practitioner",
        "display": "John Doe, MD"
      }
    }
  ],
  "note": [
    {
      "authorReference": {
        "reference": "Practitioner/e24exSNO1wPT5FJn2sKyTaQ3",
        "display": "John Doe, MD"
      },
      "time": "2022-05-03T15:35:59Z",
      "text": "Patient was asleep."
    }
  ]
}
```


Change log

10.1.50 Procedure.Read (Orders) (R4)

File: site/apis/10-Clinical-General/1-R4/975-Procedure-Read-Orders-R4.html
Epic API ID | 975
FHIR version | R4
Categories | Clinical General
API type | USCDI

A la carte | Industry-Standard Level 1 API
USCDI | Yes
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | November 2019
HTTP method | GET
URL template | api/FHIR/R4/Procedure/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=975

Description

The FHIR Procedure resource defines an activity performed on or with a patient as part of the provision of care. It corresponds with surgeries and procedures performed, including endoscopies and biopsies, as well as less invasive actions like counseling and physiotherapy. Because terminology codes for procedures are often billing related, some procedures might not have a terminology code, especially when billing activities are not performed within Epic.

The Procedure (Orders) API is designed to summarize any performed procedures for a patient for which an order has been created. Other types of procedures may be returned in other APIs, such as the Procedure (Surgeries) API for procedures documented as part of a surgery. Future upcoming procedures that have not been completed can't be read by this API.

The read interaction enables the lookup of a Procedure resource by a constant server ID. The read interaction allows clients to store only the server ID, and with a single request, retrieve the most up-to-date information about a procedure. Read interactions begin with a client having previously established a relationship, typically when following a reference to this resource from another resource or through querying for procedures through the search interaction.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | Required | | | The Procedure FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Procedure (R4) | Procedure
Procedure (R4).asserter | Reference | | | 0..1 | Reference to the Practitioner resource who asserts this procedure.
Procedure (R4).basedOn | Reference | | Yes | 0..1 | Reference to the ServiceRequest resource, which contains details of the request for the procedure.
Procedure (R4).category | CodeableConcept | | | 0..1 | A code that classifies the procedure. This is always SNOMED code 103693007 (Diagnostic procedure) for the Procedure (Orders) resource.
Procedure (R4).code | CodeableConcept | | | 1..1 | External codes associated with the procedure.
Procedure (R4).complication | CodeableConcept | | Yes | 0..* | Complication tracking event.
Procedure (R4).encounter | Reference | | | 0..1 | Reference to an Encounter resource associated with the Procedure.
Procedure (R4).id | id | | | 1..1 | The Procedure FHIR ID.
Procedure (R4).identifier | Identifier | | Yes | 1..* | System-specific ID for the procedure and its order.
Procedure (R4).performedDateTime | dateTime | | | 1..1 | When the procedure was performed.
Procedure (R4).performedPeriod | Period | | | 0..1 | The period of time in which the procedure was performed.
Procedure (R4).performer | BackboneElement | | Yes | 0..* | Who performed the procedure.
Procedure (R4).performer.actor | Reference | | | 0..1 | Reference to the Practitioner that performed the procedure. The reading physicians or, if unavailable, the resulting physician.
Procedure (R4).performer.onBehalfOf | Reference | | | 0..1 | The resulting agency or lab.
Procedure (R4).reasonCode | CodeableConcept | | Yes | 0..* | External codes for diagnosis records associated with the procedure.
Procedure (R4).recorder | Reference | | | 0..1 | Reference to the Practitioner resource who recorded the procedure.
Procedure (R4).report | Reference | | Yes | 0..1 | Reference to the DiagnosticReport resource for the procedure's order.
Procedure (R4).status | code | | | 1..1 | The status of the Procedure. Epic Value FHIR Event Status Value cancelled stopped active in-progress completed completed aborted stopped
Epic Value | FHIR Event Status Value
cancelled | stopped
active | in-progress
completed | completed
aborted | stopped
Procedure (R4).subject | Reference | | | 1..1 | The Patient FHIR ID for whom this Procedure resource is relevant.

Sample request

```
GET https://hostname/instance/api/FHIR/R4/Procedure/edKALtDcuTHWnRKIEr3Jo0pNpyvR27ZfzRHNVtzFTSVk3
```

Sample response


```
{
  "resourceType": "Procedure",
  "id": "edKALtDcuTHWnRKIEr3Jo0pNpyvR27ZfzRHNVtzFTSVk3",
  "identifier": [
    {
```

```
      "use": "usual",
      "type": {
        "text": "ORD"
      },
      "system": "urn:oid:1.2.840.114350.1.13.861.1.7.2.798268",
      "value": "3326502"
    },
    {
      "use": "usual",
      "type": {
        "text": "EAP"
      },
      "system": "urn:oid:1.2.840.114350.1.13.861.1.7.2.696580",
      "value": "2096"
    }
  ],
  "status": "completed",
  "category": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "103693007",
        "display": "Diagnostic procedure"
      }
    ],
    "text": "Ordered Procedures"
  },
  "code": {
    "coding": [
      {
        "system": "http://www.ama-assn.org/go/cpt",
        "code": "29405",
        "display": "APPLY SHORT LEG CAST"
      }
    ],
    "text": "APPLY SHORT LEG CAST"
  },
  "subject": {
    "reference": "Patient/ej4nFxzySJAysjMuv4rTBNA3",
    "display": "zzFhir, Fhirgirl"
  },
  "encounter": {
    "reference": "Encounter/ej5R7X.4RdaFNR5vWdsTpVTtccmkRkWJcKixc5XVY4cQ3"
  },
  "performedDateTime": "2019-03-04T22:31:27Z"
}
```


10.1.51 Procedure.Read (Patient-Reported Surgical History) (R4)

File: site/apis/10-Clinical-General/1-R4/10029-Procedure-Read-Patient-Reported-Surgical-History-R4.html
Epic API ID | 10029
FHIR version | R4
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | February 2021
HTTP method | GET
URL template | /api/FHIR/R4/Procedure/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10029

Description

The FHIR Procedure resource defines an activity performed on or with a patient as part of the provision of care. It corresponds with surgeries and procedures performed, including endoscopies and biopsies, as well as less invasive actions like counseling and physiotherapy. Because terminology codes for procedures are often billing related, some procedures might not have a terminology code, especially when billing activities are not performed within Epic.

The Procedure (Patient-Reported Surgical History) resource is designed to summarize historical surgical procedures a patient has had, as well as pertinent negatives (indicating a patient has not had a particular surgery). Other types of procedures may be returned in other resources, such as the Procedure (Orders) resource for procedures created from orders. In the May 2021 version of Epic and earlier versions, be aware when using the Procedure resource that each unique procedure could appear twice for patients because it can appear as both a Surgical History entry and a Surgical Procedure entry.

The read interaction enables the lookup of a Surgical History resource by a constant server ID, typically retrieved from a different resource. Read interactions begin with a client having previously established a relationship, typically when following a reference to this resource from another resource or through querying for procedures through the search interaction.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | False | | | The Procedure FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Procedure | Procedure | False
Procedure.bodySite | CodeableConcept | True | Yes | | Where on the body the procedure was performed.
Procedure.bodySite.extension (also: surgicalhistory-laterality) | Extension | True | Yes | | Laterality options using SNOMED codes and names: 7771000 (Left) 24028007 (Right) 51440002 (Right and left) 385432009 (Not applicable) 276727009 (Null)
Procedure.category | CodeableConcept | True | | | A code that classifies the procedure. This will always be SNOMED code 387713003 (Surgical procedure) for the Procedure (Surgical History) API.
Procedure.code | CodeableConcept | True | | | External codes associated with the procedure.
Procedure.note | Annotation | True | Yes | | Comments about the surgical history or pertinent negative.
Procedure.performedDateTime | dateTime | True | | | When the procedure was performed.
Procedure.status | code | False | | | The status of the Procedure. FHIR Event Status Returned when... completed positive procedure not-done pertinent negative *Note: Omitted FHIR event status equivalent values are not returned by this search
FHIR Event Status | Returned when...
completed | positive procedure
not-done | pertinent negative
Procedure.subject | Reference | False | | | Reference to the Patient resource for whom this Procedure resource is relevant.
Procedure.extension (also: surgical-history-laterality) | Extension | True | Yes | | Laterality options using SNOMED codes and names: 7771000 (Left) 24028007 (Right) 51440002 (Right and left) 385432009 (Not applicable) 276727009 (Null)
Procedure.extension (also: surgical-history-source) | Extension | True | Yes | | Free-text description of the source of the information. For example “Provider” or “Patient.”

Sample request

```
https://hostname/instance/api/FHIR/R4/Procedure/edKALtDcuTHWnRKIEr3Jo0pNpyvR27ZfzRHNVtzFTSVk3
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 1,
  "link": [
    {
      "relation": "self",
      "url":
    "https://hostname/instance/api/FHIR/R4/Procedure?patient=eyb0wFCjScO4c54vJCAvBHg3&_revInclude=Provenance:target"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url": "https://hostname/instance/api/FHIR/R4/Procedure/esDSRiB-AywSL88jYw-ZM6sEJnn0d9jBxCvBT-8Rk599J-rX-
    iEFP39skqRH79sgl3"
        }
      ],
      "fullUrl": "https://hostname/instance/api/FHIR/R4/Procedure/esDSRiB-AywSL88jYw-ZM6sEJnn0d9jBxCvBT-8Rk599J-rX-
    iEFP39skqRH79sgl3",
      "resource": {
        "resourceType": "Procedure",
        "id": "esDSRiB-AywSL88jYw-ZM6sEJnn0d9jBxCvBT-8Rk599J-rX-iEFP39skqRH79sgl3",
        "extension": [
          {
            "valueString": "Provider",
            "url": "http://open.epic.com/FHIR/StructureDefinition/extension/surgical-history-source"
          }
        ],
        "status": "completed",
        "category": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "387713003",
              "display": "Surgical procedure"
            }
          ],
          "text": "Surgical History"
        },
        "code": {
          "coding": [
            {
              "system": "http://www.ama-assn.org/go/cpt",
              "code": "36778483",
              "display": "LEG AMPUTATION"
            }
          ],
```

```
        "text": "LEG AMPUTATION"
      },
      "subject": {
        "reference": "Patient/eyb0wFCjScO4c54vJCAvBHg3",
        "display": "Name, Shroud, PhD"
      },
      "performedDateTime": "2020-11-03",
      "bodySite": [
        {
          "extension": [
            {
              "valueCodeableConcept": {
                "coding": [
                  {
                    "system": "http://snomed.info/sct",
                    "code": "447561005",
                    "display": "SNOMED CT source code to target map code correlation not specified"
                  }
                ],
                "text": "Left"
              },
              "url": "http://open.epic.com/FHIR/StructureDefinition/extension/surgical-history-laterality"
            }
          ]
        }
      ]
    },
    "search": {
      "mode": "match"
    }
  },
  {
    "link": [
      {
        "relation": "self",
        "url": "https://hostname/instance/api/FHIR/R4/Provenance/eoUS7LVLK31M9n0hyriZ3GDp53Rr1Fj5FscqJecPglPJkyb2Kp0D6
  X2foU.QAO7yUMqFAco53Civ.HbpWTrOcoEblNfOQ.nq0zbIRwNflHag3"
      }
    ],
    "fullUrl": "https://hostname/instance/api/FHIR/R4/Provenance/eoUS7LVLK31M9n0hyriZ3GDp53Rr1Fj5FscqJecPglPJkyb2Kp0D6
  X2foU.QAO7yUMqFAco53Civ.HbpWTrOcoEblNfOQ.nq0zbIRwNflHag3",
    "resource": {
      "resourceType": "Provenance",
      "id":
  "eoUS7LVLK31M9n0hyriZ3GDp53Rr1Fj5FscqJecPglPJkyb2Kp0D6X2foU.QAO7yUMqFAco53Civ.HbpWTrOcoEblNfOQ.nq0zbIRwNflHag3",
      "target": [
        {
          "reference": "Procedure/esDSRiB-AywSL88jYw-ZM6sEJnn0d9jBxCvBT-8Rk599J-rX-iEFP39skqRH79sgl3"
        }
      ],
      "recorded": "2020-12-03T16:33:50Z",
      "agent": [
        {
          "type": {
            "coding": [
              {
                "system": "http://terminology.hl7.org/CodeSystem/provenance-participant-type",
                "code": "author",
                "display": "Author"
              }
            ],
            "text": "Author"
          },
          "who": {
            "display": "Development Environment "
          }
        },
        {
          "type": {
            "coding": [
              {
                "system": "http://hl7.org/fhir/us/core/CodeSystem/us-core-provenance-participant-type",
                "code": "transmitter",
                "display": "Transmitter"
              }
            ],
            "text": "Transmitter"
          },
          "who": {
            "display": "Development Environment"
          }
        }
      ]
    },
    "search": {
      "mode": "include"
    }
  }
]
```

}

Change log

Epic Version | Description | Type
May 2023 (Special Updates to February 2023) | Code systems for the following code sets now return a URL in place of an OID in the following elements: Procedure.bodySite, Procedure.bodySite.extension(BodySite-Qualifier).valueCodeableConceptProcedure System URL OID SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 LOINC http://loinc.org 2.16.840.1.113883.6.1 | Breaking change
System | URL | OID
SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96
LOINC | http://loinc.org | 2.16.840.1.113883.6.1

10.1.52 Procedure.Read (Restricting Intervention) (R4)

File: site/apis/10-Clinical-General/1-R4/10474-Procedure-Read-Restricting-Intervention-R4.html
Epic API ID | 10474
FHIR version | R4
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Procedure/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10474

Description

This resource is available only for healthcare organizations in the Netherlands.

The FHIR Procedure resource defines an activity performed on or with a patient as part of the provision of care.

The Procedure (Restriction Intervention) API is designed to record interventions that are used to implement freedom restricting measures on the patient.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | False | | | The Procedure FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Procedure | Procedure | False
Procedure.extension (also: Assent) | Extension | True | Yes | | Records whether a patient or their legal representative provided consent.
Procedure.category | CodeableConcept | False | | | A code that classifies the procedure. Always includes SNOMED code “225317005” and the custom Epic category “freedom-restricting-intervention” for this resource.
Procedure.code | CodeableConcept | False | | | External codes associated with the procedure. Bound to TypeOfInterventionCodelist.
Procedure.encounter | Reference | True | | | Reference to the encounter in which the procedure took place.
Procedure.id | id | False | | | The Procedure FHIR ID.
Procedure.extension (also: legallyCapable) | Extension | False | Yes | | Whether a patient is legally capable. Contained within a parent zib-FreedomRestrictingIntervention-LegallyCapable extension.
Procedure.extension (also: legallyCapableComment) | Extension | True | Yes | | Includes free-text instruction information on a patient’s legal capability. Contained within a parent zib-FreedomRestrictingIntervention-LegallyCapable extension.
Procedure.performedPeriod | Period | False | | | The period of time over which the procedure was performed.
Procedure.reasonCode | CodeableConcept | True | Yes | | The reason for use that provides the justification for applying involuntary care. Mapped to ReasonForUseCodelist.
Procedure.status | code | False | | | The status of the procedure order. The table below describes how the status of the intervention procedure order maps to the status returned by this resource. Order Status FHIR Status Pending, Sent, Holding for Referral Preparation Verified, Pending Verify In-Progress Canceled or Discontinued (without documentation) Stopped Denied Approval Not-Done Suspend Suspended Resulted, Completed, Dispensed, Discontinued or Canceled (with documentation) Completed
Order Status | FHIR Status
Pending, Sent, Holding for Referral | Preparation
Verified, Pending Verify | In-Progress
Canceled or Discontinued (without documentation) | Stopped
Denied Approval | Not-Done
Suspend | Suspended
Resulted, Completed, Dispensed, Discontinued or Canceled (with documentation) | Completed
Procedure.subject | Reference | False | | | Reference to the Patient resource for whom this Procedure resource is relevant.

Sample request

```
https://hostname/instance/api/FHIR/R4/Procedure/fFK-tRE1AmmsrTnHfWRlXZofEtsqrIGnyBvpRohoGPuM4
```

Sample response


```
{
  "resourceType": "Procedure",
  "id": "fFK-tRE1AmmsrTnHfWRlXZofEtsqrIGnyBvpRohoGPuM4",
  "extension": [
    {
      "extension": [
        {
          "valueBoolean": false,
          "url": "legallyCapable"
        },
        {
          "valueString": "Patient's legal representative provided consent for this action",
          "url": "legallyCapableComment"
        }
      ],
      "url": "http://nictiz.nl/fhir/StructureDefinition/zib-FreedomRestrictingIntervention-LegallyCapable"
    },
    {
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "urn:oid:2.16.840.1.113883.2.4.3.11.60.40.4.6.1",
            "code": "4",
            "display": "Instemming wettelijke vertegenwoordiger"
          }
        ],
        "text": "Instemming wettelijke vertegenwoordiger"
      },
      "url": "http://nictiz.nl/fhir/StructureDefinition/zib-FreedomRestrictingIntervention-Assent"
    }
  ],
  "status": "in-progress",
  "category": {
    "coding": [
      {
        "system": "http://open.epic.com/FHIR/StructureDefinition/Procedure-category",
        "code": "freedom-restricting-intervention",
        "display": "Freedom Restricting Intervention"
      },
      {
        "system": "http://snomed.info/sct",
        "code": "225317005",
        "display": "Restriction of movement"
      }
    ],
    "text": "Restriction of movement"
  },
  "code": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "225210009",
        "display": "LOCKED DOOR PROCEDURE (PROCEDURE)"
      }
    ],
    "text": "LOCKED DOOR PROCEDURE (PROCEDURE)"
  },
  "subject": {
    "reference": "Patient/eojz6maS66J54Bwze3hFhDg3",
    "display": "Henderson, T."
  },
  "encounter": {
    "reference": "Encounter/eT5I-WeoUShAaaQqwGiwpyg3",
    "identifier": {
      "use": "usual",
      "system": "urn:oid:1.2.840.114350.1.13.260.1.7.3.698084.8",
      "value": "239103"
    },
    "display": "Hospital Encounter"
  },
  "performedPeriod": {
    "start": "2022-07-14T20:21:00Z"
  },
  "reasonCode": [
    {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "121481000146105",
          "display": "Diversion of situation that could result in self-harm or harm to others"
        }
      ],
      "text": "Diversion of situation that could result in self-harm or harm to others"
    }
  ]
}
```

10.1.53 Procedure.Read (SDOH Intervention) (R4)

File: site/apis/10-Clinical-General/1-R4/11055-Procedure-Read-SDOH-Intervention-R4.html
Epic API ID | 11055
FHIR version | R4
Categories | Clinical General
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Procedure/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=11055

Description

Procedure.Read (SDOH Intervention) allows a client to look up current and historic social determinants of health (SDOH) interventions completed in Epic from a user sending a request to a community-based organization to address an at-risk domain for a patient.

This resource includes general information about the intervention, including the patient to whom it applies and who made the request.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | False | | | The Procedure FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Procedure | Procedure | False
Procedure.basedOn | Reference | False | Yes | | Sub-element description.
Procedure.category | CodeableConcept | False | | | The FHIR codeable concept: hardcoded to SNOMED code – SCTID 410606002: Social service procedure (procedure).
Procedure.code | CodeableConcept | False | | | The FHIR codeable concept: hardcoded to SNOMED code – SCTID 306206005: Referral to service.
Procedure.encounter | Reference | False | | | Reference to an encounter resource.
Procedure.performedDateTime | dateTime | False | | | The date the procedure was performed.
Procedure.recorder | Reference | False | | | Reference to a practitioner resource.
Procedure.status | code | False | | | The status of the procedure. Always “completed” for this API.
Procedure.subject | Reference | False | | | Reference to patient resource.

Sample request

```
https://hostname/instance/api/FHIR/R4/Procedure/fRCKdJNwYvxZZwcKp-dszPg4
```

Sample response


```
{
  "resourceType": "Procedure",
  "id": "fRCKdJNwYvxZZwcKp-dszPg4",
  "basedOn": [
    {
      "reference": "ServiceRequest/eNaYhHi6xequPT3aBgleofl-gWkv0Gsb7m5WFgsWpPOh8u1J-JnCQVhISmBkbWcmv3",
      "type": "ServiceRequest"
    }
  ],
  "status": "completed",
  "category": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "410606002",
        "display": "Social service procedure (procedure)"
      }
    ],
    "text": "Social service procedure (procedure)"
  },
  "code": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "306206005",
        "display": "Referral to service"
      }
    ],
    "text": "Referral to service"
```

```
  },
  "subject": {
    "reference": "Patient/e8OeNeovb7izjGIErw9MUUg3",
    "type": "Patient",
    "display": "Smith, Janice"
  },
  "encounter": {
    "reference": "Encounter/eRsROYaTxtaNmMib-Sh0DUke14oCfJBsQ3daJGXeTFQo3",
    "identifier": {
      "use": "usual",
      "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.3.698084.8",
      "value": "10003597712"
    },
    "display": "Office Visit"
  },
  "performedDateTime": "2024-04-17T15:19:52Z",
  "recorder": {
    "reference": "Practitioner/eeLsHlvm0KvPn-y4wW-AuzQ3",
    "type": "Practitioner",
    "display": "Social Worker Janisko, MSW"
  }
}
```


10.1.54 Procedure.Read (Surgeries) (R4)

File: site/apis/10-Clinical-General/1-R4/10041-Procedure-Read-Surgeries-R4.html
Epic API ID | 10041
FHIR version | R4
Categories | Clinical General
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | November 2020
HTTP method | GET
URL template | /api/FHIR/R4/Procedure/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10041

Description

The FHIR Procedure resource defines an activity performed on or with a patient as part of the provision of care. It corresponds with surgeries and procedures performed, including endoscopies and biopsies, as well as less invasive actions like counseling and physiotherapy. Because terminology codes for procedures are often billing related, some procedures might not have a terminology code, especially when billing activities are not performed within Epic.

The Procedure (Surgeries) API is designed to summarize any performed surgical procedures for a patient. Other types of procedures may be returned in other APIs, such as the Procedure (Orders) API for procedures created from orders.

To gather a complete list of surgical procedure information for a patient, three different Procedure APIs must be consulted. While a surgical procedure may be documented in multiple places in Epic, it will be returned only by one Procedure FHIR API, in the following order of precedence:

* Procedure (Orders): Includes any procedure that is ordered by a clinician and completed during an encounter. If a procedure is returned by this API, it won't be returned by the Procedure (Surgeries) or Procedure (Surgical History) APIs.

* Procedure (Surgeries): Includes surgical procedures attached to a surgical log. If a procedure is returned by this API, it won't be returned by the Procedure (Surgical History) API.

* Procedure (Surgical History): Includes any manually documented surgical history or pertinent negatives that were not documented on a surgical log or ordered within Epic.

This resource is designed for a high-level summarization around the occurrence of a procedure, and not for specific procedure log documentation - a concept that does not yet have a defined FHIR Resource.

The read interaction enables the lookup of a Procedure resource by a constant server ID. Read interactions begin with a client having previously established a relationship, typically when following a reference to this resource from another resource or through querying for procedures through the search interaction.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | Required | | | The Procedure FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description

Procedure (R4) | Procedure

Procedure (R4).basedOn | Reference | | Yes | 0..* | Reference to the ServiceRequest resource that the procedure log was created from.

Procedure (R4).bodySite | CodeableConcept | | Yes | 0..* | External code for the operative region for the procedure.

Procedure (R4).bodySite.extension (BodySite-Qualifier) (also: extension (BodySite-Qualifier)) | Extension | | Yes | 0..1 | Laterality for the body site. URL: http://nictiz.nl/fhir/StructureDefinition/BodySite-Qualifier

Procedure (R4).category | CodeableConcept | | | 0..1 | A code that classifies the procedure. This will always be SNOMED code 387713003 (Surgical procedure) for the Procedure (Surgeries) API.

Procedure (R4).code | CodeableConcept | | | 1..1 | External codes associated with the procedure. Can include code types specified in the PROCEDURE_CODE_TYPE (1825) profile variable if those codes are available on the target record.

Procedure (R4).complication | CodeableConcept | | Yes | 0..* | Complication tracking event.

Procedure (R4).encounter | Reference | | | 0..1 | Reference to an Encounter resource associated with the Procedure.

Procedure (R4).extension (data-absent-reason) (also: extension (data-absent-reason)) | Extension | | Yes | 0..* | The reason why normally expected content of the data element is missing. Set to "not-performed" for surgeries without a performed time or period. URL:http://hl7.org/fhir/StructureDefinition/data-absent-reason

Procedure (R4).extension (log-level-procedure-codes) (also: extension (log-level-procedure-codes)) | Extension | | Yes | 0..* | Additional external codes for diagnosis records not associated with a specific procedure. Includes a list of one or more valueCodeableConcept codings, and the text value "Additional Procedure Codes". URL: http://open.epic.com/FHIR/StructureDefinition/extension/log-level-procedure-codes

Procedure (R4).focalDevice | BackboneElement | | Yes | 0..* | Reference to a Device resource marked as implanted, explanted, or adjusted during the procedure.

Procedure (R4).focalDevice.action | CodeableConcept | | | 1..1 | Implant action.

Procedure (R4).focalDevice.manipulated | Reference | | | 0..1 | Implant ID.

Procedure (R4).id | id | | | 1..1 | The Procedure FHIR ID.

Procedure (R4).identifier | Identifier | | Yes | 1..* | System-specific ID for the procedure order record.

Procedure (R4).location | Reference | | | 0..1 | Reference to the Location resource where the procedure was performed.

Procedure (R4).outcome | CodeableConcept | | | 0..1 | Discharge disposition from the linked ADT contact.

Procedure (R4).performedDateTime | dateTime | | | 1..1 | When the procedure was performed.

Procedure (R4).performedPeriod | Period | | | 1..1 | Period of time over which the procedure was performed.

Procedure (R4).performer | BackboneElement | | Yes | 0..* | Who performed the procedure.

Procedure (R4).performer.actor | Reference | | | 0..1 | Reference to the Practitioner that performed the procedure.

Procedure (R4).reasonCode | CodeableConcept | | Yes | 0..* | The coded reason that the procedure was performed.

Procedure (R4).reasonReference | Reference | | Yes | 0..* | Applicable only to organizations in the Netherlands. Reference to a Condition resource associated with the DBC episode for the procedure.

Procedure (R4).recorder | Reference | | | 0..1 | Audit user.

Procedure (R4).report | Reference | | Yes | 0..* | Reference to the DocumentReference resource that contains notes from the surgical log.

Procedure (R4).status | code | | | 1..1 | The status of the Procedure. Epic Value FHIR Event Status cancelled stopped active in-progress completed completed aborted stopped

Epic Value | FHIR Event Status

cancelled | stopped

active | in-progress

completed | completed

aborted | stopped

Procedure (R4).statusReason | CodeableConcept | | | 0..1 | The reason that a procedure was not performed, which is sent only when the status is Stopped.

Procedure (R4).subject | Reference | | | 1..1 | Reference to the Patient resource for whom this Procedure resource is relevant.

Procedure (R4).usedReference | Reference | | Yes | 0..* | Reference to the Device resource marked as a wasted implant that was used during the procedure.

Sample request

```
GET https://hostname/instance/api/FHIR/R4/Procedure/edkaltdcuthwnrkier3jo0pnpyvr27zfzrhnvtzftsvk3
```

Sample response


```
{
  "resourceType": "Procedure",
  "id": "fv4NzEJK7mo6EzLIHD7ZAYaLJVXNQExFkm55pWMZHX-M4",
  "identifier": [
    {
      "use": "usual",
      "type": {
        "text": "ORL"
      },
      "system": "urn:oid:1.2.840.114350.1.13.861.1.7.2.798276",
      "value": "178124"
    },
    {
      "use": "usual",
      "type": {
        "text": "OPE"
      },
      "system": "urn:oid:1.2.840.114350.1.13.861.1.7.2.798069",
      "value": "207340"
    }
  ],
  "status": "completed",
  "category": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
```

```
        "code": "387713003",
        "display": "Surgical procedure"
      }
    ],
    "text": "Surgical Procedures"
  },
  "code": {
    "text": "Appendectomy"
  },
  "subject": {
    "reference": "Patient/e1uV557J2LZDVPSZfxOtXyw3",
    "display": "OpTime, Fhirman"
  },
  "encounter": {
    "reference": "Encounter/el8BJ6-udqPGspHLpTl8hFw64lqOLXJt8mqYBGbqaKRo3",
    "identifier": {
      "use": "usual",
      "system": "urn:oid:1.2.840.114350.1.13.861.1.7.3.698084.8",
      "value": "10008467329"
    },
    "display": "Surgery"
  },
  "performedPeriod": {
    "start": "2024-04-30T19:22:00Z",
    "end": "2024-05-01T20:35:00Z"
  },
  "recorder": {
    "reference": "Practitioner/eMPvR2m5RxFtz6hEZ-B6hiA3",
    "type": "Practitioner",
    "display": "Dave Cv Surgeon"
  },
  "location": {
    "reference": "Location/ejlzkA8sKOIVzg8NCPMp-Cw3",
    "display": "WI UHM MAIN OR"
  },
  "bodySite": [
    {
      "extension": [
        {
          "valueCodeableConcept": {
            "coding": [
              {
                "system": "urn:oid:1.2.840.114350.1.13.861.1.7.4.798276.2010",
                "code": "4",
                "display": "N/A"
              }
            ],
            "text": "N/A"
          },
          "url": "http://nictiz.nl/fhir/StructureDefinition/BodySite-Qualifier"
        }
      ],
      "coding": [
        {
          "system": "urn:oid:1.2.840.114350.1.13.861.1.7.4.798276.2015",
          "code": "1",
          "display": "Abdomen"
        }
      ],
      "text": "Abdomen"
    }
  ],
  "report": [
    {
      "reference": "DocumentReference/fsgKwTeUQ60.Qcy4G4cc7foABkyhWF4J.fMMEWP3aCX44"
    },
    {
      "reference": "DocumentReference/fpuurmPyzd-tI3JVTZ8XELyROIm9Vw4wdZAOMbOKDD2U4"
    },
    {
      "reference": "DocumentReference/fhWV6qPPhmBrStr.hD5J3PpjUAYFAjv5c2Ry5KczoqfU4"
    }
  ]
}
```


Change log

Epic Version | Description | Type
May 2023 (Special Updates to February 2023) | Code systems for the following code sets now return a URL in place of an OID in the following elements: Procedure.bodySite, Procedure.bodySite.extension(BodySite-Qualifier).valueCodeableConceptProcedure System URL OID SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 LOINC http://loinc.org 2.16.840.1.113883.6.1 | Breaking change
System | URL | OID
SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96
LOINC | http://loinc.org | 2.16.840.1.113883.6.1

10.1.55 Procedure.Search (Nursing Intervention) (R4)

File: site/apis/10-Clinical-General/1-R4/10479-Procedure-Search-Nursing-Intervention-R4.html
Epic API ID | 10479
FHIR version | R4
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Procedure?category={category}&date={date}&patient={patient}&subject={subject}&_id={_id}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10479

Description

This resource can be used only by healthcare organizations in the Netherlands. This resource cannot be used in any other locales at this time.

The FHIR Procedure resource defines an activity performed on or with a patient as part of the provision of care.

The Procedure (Intervention) API is designed to summarize any care or treatment carried out by a nurse in response to problems that arise in relation to a patient's care plan goals.

This resource returns information about inpatient care plan interventions only.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
category | token | True | | | A code that classifies the procedure. This will always be SNOMED code 9632001 (Nursing procedure) for the Procedure (Nursing Intervention) API.
patient | reference | False | | | The Patient FHIR resource ID.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
based-on | reference | True | | | Unsupported
code | token | True | | | A code to identify the procedure.
date | date | True | | | Unsupported
encounter | reference | True | | | Unsupported
identifier | token | True | | | Unsupported
instantiates-canonical | String | True | | | Unsupported
location | reference | True | | | Unsupported
part-of | reference | True | | | Unsupported
performer | reference | True | | | The practitioner performing the procedure.
reason-code | token | True | | | Unsupported
reason-reference | reference | True | | | Unsupported
status | token | True | | | The status of the procedure. Refer to the response element for more information.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | BackboneElement | False | Yes | | An array of entries included in this bundle.
Bundle.entry.resource | ResourceContainer | False | | | A container for a single FHIR resource.
Bundle.entry.resource.Procedure | Procedure | False
Bundle.entry.resource.Procedure.category | CodeableConcept | False | | | A code that classifies the procedure. This will always be SNOMED code 9632001 (Nursing procedure) for the Procedure (Intervention) API.
Bundle.entry.resource.Procedure.code | CodeableConcept | False | | | External codes associated with the procedure.
Bundle.entry.resource.Procedure.extension (also: Frequency) | Extension | False | Yes | | How often an intervention was performed.
Bundle.entry.resource.Procedure.note | Annotation | True | Yes | | Comment on the nursing intervention.
Bundle.entry.resource.Procedure.performedPeriod | Period | True | | | The period of time over which the procedure was performed.
Bundle.entry.resource.Procedure.performer | BackboneElement | True | Yes | | Who performed the procedure.
Bundle.entry.resource.Procedure.extension (also: Requester) | Extension | False | Yes | | The user who requested the intervention.
Bundle.entry.resource.Procedure.status | code | False | | | The status of the procedure. Intervention Status Procedure Status in-progress in-progress completed completed cancelled entered-in-error
Intervention Status | Procedure Status
in-progress | in-progress
completed | completed
cancelled | entered-in-error
Bundle.entry.resource.Procedure.subject | Reference | False | | | Reference to the Patient resource for whom this Procedure resource is relevant.
Bundle.entry.resource.Procedure.extension (also: TreatmentObjective) | Extension | False | Yes | | The description of the treatment goal that the intervention decision is based on. Uses the "pertains to goal" code system.
Bundle.total | Int64 | True | | | Total number of resources that matched the search criteria.

Sample request

```
https://hostname/instance/api/FHIR/R4/Procedure?category=9632001&patient=eHS2.Rr5rdHLZwVFIApIfhA3
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 3,
  "link": [
    {
      "relation": "self",
      "url": "https://vs-icx18-11.epic.com/Interconnect-Netherlands-Stage1-nl-
    NL/api/FHIR/R4/Procedure?category=9632001&patient=eHS2.Rr5rdHLZwVFIApIfhA3"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url": "https://vs-icx18-11.epic.com/Interconnect-Netherlands-Stage1-nl-
    NL/api/FHIR/R4/Procedure/fFWD0Fm.sK6CNRVFiULGAZSGeNMizP9quMcaHkAxaNQY4"
        }
      ],
      "fullUrl": "https://vs-icx18-11.epic.com/Interconnect-Netherlands-Stage1-nl-
    NL/api/FHIR/R4/Procedure/fFWD0Fm.sK6CNRVFiULGAZSGeNMizP9quMcaHkAxaNQY4",
      "resource": {
        "resourceType": "Procedure",
        "id": "fFWD0Fm.sK6CNRVFiULGAZSGeNMizP9quMcaHkAxaNQY4",
        "extension": [
          {
            "valueReference": {
              "reference": "Goal/ezIkQ5v86zJzyR-YnMpXiMg3",
              "display": "Comfort level"
            },
            "url": "http://hl7.org/fhir/StructureDefinition/resource-pertainsToGoal"
          },
          {
            "valueTiming": {
              "code": {
                "coding": [
                  {
                    "system": "urn:oid:1.2.840.114350.1.13.260.1.7.2.697081",
                    "code": "2000083",
                    "display": "1/2H AC"
                  }
                ],
                "text": "1/2H AC"
              }
            },
            "url": "http://hl7.org/fhir/StructureDefinition/procedure-schedule"
          },
          {
            "valueReference": {
              "reference": "Practitioner/erQJyGdrWyxRqS1YlqSKI.w3",
              "display": "Doctor. Sample"
            },
            "url": "http://nictiz.nl/fhir/StructureDefinition/zib-NursingIntervention-Requester"
          }
        ],
        "status": "in-progress",
        "category": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "9632001",
              "display": "Nursing procedure (procedure)"
            }
          ],
          "text": "Nursing Intervention"
        },
        "code": {
          "text": "Activity as tolerated"
        },
        "subject": {
          "reference": "Patient/eHS2.Rr5rdHLZwVFIApIfhA3",
          "display": "Sample, M."
        },
        "performedPeriod": {
          "start": "2022-06-20"
        },
        "performer": [
          {
            "actor": {
              "reference": "Practitioner/erQJyGdrWyxRqS1YlqSKI.w3",
              "type": "Practitioner",
              "display": "Doctor. Sample"
```

```
        }
      }
    ]
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url": "https://vs-icx18-11.epic.com/Interconnect-Netherlands-Stage1-nl-
NL/api/FHIR/R4/Procedure/fEp1Ah3HzsTmD0sdnEm1lxoiQJKDnvptI5dcdagLhFLc4"
    }
  ],
  "fullUrl": "https://vs-icx18-11.epic.com/Interconnect-Netherlands-Stage1-nl-
NL/api/FHIR/R4/Procedure/fEp1Ah3HzsTmD0sdnEm1lxoiQJKDnvptI5dcdagLhFLc4",
  "resource": {
    "resourceType": "Procedure",
    "id": "fEp1Ah3HzsTmD0sdnEm1lxoiQJKDnvptI5dcdagLhFLc4",
    "extension": [
      {
        "valueReference": {
          "reference": "Goal/esSMozX6MnLp66QUNi9PiIw3",
          "display": "MAXIMIZE RESPIRATORY FUNCTION"
        },
        "url": "http://hl7.org/fhir/StructureDefinition/resource-pertainsToGoal"
      },
      {
        "valueTiming": {
          "code": {
            "coding": [
              {
                "system": "urn:oid:1.2.840.114350.1.13.260.1.7.2.697081",
                "code": "2000092",
                "display": "1 x/day"
              }
            ],
            "text": "1 x/day"
          }
        },
        "url": "http://hl7.org/fhir/StructureDefinition/procedure-schedule"
      },
      {
        "valueReference": {
          "reference": "Practitioner/erQJyGdrWyxRqS1YlqSKI.w3",
          "display": "Doctor. Sample"
        },
        "url": "http://nictiz.nl/fhir/StructureDefinition/zib-NursingIntervention-Requester"
      }
    ],
    "status": "in-progress",
    "category": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "9632001",
          "display": "Nursing procedure (procedure)"
        }
      ],
      "text": "Nursing Intervention"
    },
    "code": {
      "text": "RESPIRATORY MUSCLES, IMPROVE STRENGTH/ENDURANCE"
    },
    "subject": {
      "reference": "Patient/eHS2.Rr5rdHLZwVFIApIfhA3",
      "display": "Sample, M."
    },
    "performedPeriod": {
      "start": "2022-06-20"
    },
    "performer": [
      {
        "actor": {
          "reference": "Practitioner/erQJyGdrWyxRqS1YlqSKI.w3",
          "type": "Practitioner",
          "display": "Doctor. Sample"
        }
      }
    ]
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
```

```
      {
        "relation": "self",
        "url": "https://vs-icx18-11.epic.com/Interconnect-Netherlands-Stage1-nl-
    NL/api/FHIR/R4/Procedure/f.hWsPbjolCumnN9yKjf8bjKZbWXKoTKhR4IVJdz5HBA4"
      }
    ],
    "fullUrl": "https://vs-icx18-11.epic.com/Interconnect-Netherlands-Stage1-nl-
    NL/api/FHIR/R4/Procedure/f.hWsPbjolCumnN9yKjf8bjKZbWXKoTKhR4IVJdz5HBA4",
    "resource": {
      "resourceType": "Procedure",
      "id": "f.hWsPbjolCumnN9yKjf8bjKZbWXKoTKhR4IVJdz5HBA4",
      "extension": [
        {
          "valueReference": {
            "reference": "Goal/ePXw.u7DWbg2eToGcXJN.YA3",
            "display": "Mobility level"
          },
          "url": "http://hl7.org/fhir/StructureDefinition/resource-pertainsToGoal"
        },
        {
          "valueTiming": {
            "code": {
              "coding": [
                {
                  "system": "urn:oid:1.2.840.114350.1.13.260.1.7.2.697081",
                  "code": "900000002",
                  "display": "BID"
                }
              ],
              "text": "BID"
            }
          },
          "url": "http://hl7.org/fhir/StructureDefinition/procedure-schedule"
        },
        {
          "valueReference": {
            "reference": "Practitioner/erQJyGdrWyxRqS1YlqSKI.w3",
            "display": "Doctor. Sample"
          },
          "url": "http://nictiz.nl/fhir/StructureDefinition/zib-NursingIntervention-Requester"
        }
      ],
      "status": "in-progress",
      "category": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "9632001",
            "display": "Nursing procedure (procedure)"
          }
        ],
        "text": "Nursing Intervention"
      },
      "code": {
        "text": "EXERCISE THERAPY - AMBULATION"
      },
      "subject": {
        "reference": "Patient/eHS2.Rr5rdHLZwVFIApIfhA3",
        "display": "Sample, M."
      },
      "performedPeriod": {
        "start": "2022-06-20"
      },
      "performer": [
        {
          "actor": {
            "reference": "Practitioner/erQJyGdrWyxRqS1YlqSKI.w3",
            "type": "Practitioner",
            "display": "Doctor. Sample"
          }
        }
      ],
      "note": [
        {
          "authorReference": {
            "reference": "Practitioner/erQJyGdrWyxRqS1YlqSKI.w3",
            "display": "Doctor. Sample"
          },
          "time": "2022-06-20T14:22:19Z",
          "text": "Here is a note about exercise therapy ambulation\r\n"
        }
      ]
    },
    "search": {
      "mode": "match"
    }
  }
 ]
}
```

10.1.56 Procedure.Search (Orders) (R4)

File: site/apis/10-Clinical-General/1-R4/976-Procedure-Search-Orders-R4.html
Epic API ID | 976
FHIR version | R4
Categories | Clinical General
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | November 2019
HTTP method | GET
URL template | api/FHIR/R4/Procedure?date={date}&patient={patient}&subject={subject}&category={category}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=976

Description

The FHIR Procedure resource defines an activity performed on or with a patient as part of the provision of care. It corresponds with surgeries and procedures performed, including endoscopies and biopsies, as well as less invasive actions like counseling and physiotherapy.

The Procedure (Orders) API is designed to summarize any performed procedures for a patient for which an order has been created. Other types of procedures may be returned in other APIs, such as the Procedure (Surgeries) API for procedures documented as part of a surgery. Upcoming future orders that have not been completed aren't returned by this API.

Organizations can have differences when determining which performable activities are classified as procedures, so the types of activities included in FHIR Procedure search can vary between organizations.

Because terminology codes for procedures are often billing related, some procedures might not have a standard code (for example, CPT), especially when billing activities are not performed within Epic. For organizations that do perform billing activities within Epic, procedures might not have a CPT code until the charges are generated for the procedure.

When searching, only completed procedures are returned.

To gather a complete list of surgical procedure information for a patient, three different Procedure APIs must be consulted. While a surgical procedure may be documented in multiple places in Epic, it will be returned only by one Procedure FHIR API, in the following order of precedence:

* Procedure (Orders): Includes any procedure that is ordered by a clinician and completed during an encounter. If a procedure is returned by this API, it won't be returned by the Procedure (Surgeries) or Procedure (Surgical History) APIs.

* Procedure (Surgeries): Includes surgical procedures attached to a surgical log. If a procedure is returned by this API, it won't be returned by the Procedure (Surgical History) API.

* Procedure (Surgical History): Includes any manually documented surgical history or pertinent negatives that were not documented on a surgical log or ordered within Epic.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
_count | number | Optional | Yes | | Number of results per page.
_id | token | Optional | Yes | | FHIR resource IDs for the desired resources. Can be used in combination with _count, _include, and _revInclude. All others will be ignored.
_include | string | Optional | Yes | | Used to indicate which referenced resources to include in the search bundle.
_revinclude | string | Optional | Yes | | Used to include Provenance or AuditEvent resources that refer to search results, when available.
category | token | Optional | Yes | | A code that classifies the procedure. To limit the search to only the Orders sub-resource, this should be set to SNOMED code 103693007 (Diagnostic procedure).
date | date | Optional | Yes | | Date or period that the procedure was performed, using the FHIR date parameter format.
patient | reference | Conditional | Yes | | The Patient FHIR resource ID. Either subject or patient must be specified. If both are specified, they must reference the same patient.
subject | reference | Conditional | Yes | | Subject of the Procedure. Either subject or patient must be specified. If both are specified, they must reference the same patient.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
based-on | reference | Optional | Yes | | The ServiceRequest resource which contains details of the request for the procedure
code | token | Optional | Yes | | External codes associated with the procedure.
encounter | reference | Optional | Yes | | Reference to an Encounter resource associated with the Procedure.
performer | reference | Optional | Yes | | The FHIR ID of the user who performed the procedure.
reason-code | token | Optional | Yes | | External codes for diagnosis records associated with the procedure.
status | token | Optional | Yes | | The status of the Procedure. See the response element description for detailed values.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Procedure (R4) | Procedure
Procedure (R4).asserter | Reference | | | 0..1 | Reference to the Practitioner resource who asserts this procedure.
Procedure (R4).basedOn | Reference | | Yes | 0..1 | Reference to the ServiceRequest resource, which contains details of the request for the procedure.
Procedure (R4).category | CodeableConcept | | | 0..1 | A code that classifies the procedure. This is always SNOMED code 103693007 (Diagnostic procedure) for the Procedure (Orders) resource.
Procedure (R4).code | CodeableConcept | | | 1..1 | External codes associated with the procedure.
Procedure (R4).complication | CodeableConcept | | Yes | 0..* | Complication tracking event.
Procedure (R4).encounter | Reference | | | 0..1 | Reference to an Encounter resource associated with the Procedure.
Procedure (R4).id | id | | | 1..1 | The Procedure FHIR ID.
Procedure (R4).identifier | Identifier | | Yes | 1..* | System-specific ID for the procedure and its order.
Procedure (R4).performedDateTime | dateTime | | | 1..1 | When the procedure was performed.
Procedure (R4).performedPeriod | Period | | | 0..1 | The period of time in which the procedure was performed.
Procedure (R4).performer | BackboneElement | | Yes | 0..* | Who performed the procedure.
Procedure (R4).performer.actor | Reference | | | 0..1 | Reference to the Practitioner that performed the procedure. The reading physicians or, if unavailable, the resulting physician.
Procedure (R4).performer.onBehalfOf | Reference | | | 0..1 | The resulting agency or lab.
Procedure (R4).reasonCode | CodeableConcept | | Yes | 0..* | External codes for diagnosis records associated with the procedure.
Procedure (R4).recorder | Reference | | | 0..1 | Reference to the Practitioner resource who recorded the procedure.
Procedure (R4).report | Reference | | Yes | 0..1 | Reference to the DiagnosticReport resource for the procedure's order.
Procedure (R4).status | code | | | 1..1 | The status of the Procedure. Epic Value FHIR Event Status Value cancelled stopped active in-progress completed completed aborted stopped
Epic Value | FHIR Event Status Value
cancelled | stopped
active | in-progress
completed | completed
aborted | stopped
Procedure (R4).subject | Reference | | | 1..1 | The Patient FHIR ID for whom this Procedure resource is relevant.

Sample request

```
GET https://hostname/instance/api/FHIR/R4/Procedure?patient=ej4nFxzySJAysjMuv4rTBNA3&date=gt2019-01-01&date=lt2019-02-20
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 1,
  "link": [
    {
      "relation": "self",
      "url":
    "https://hostname/instance/api/FHIR/R4/Procedure?patient=ej4nFxzySJAysjMuv4rTBNA3&date=gt2019-01-01&date=lt2019-02-
    20"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url": "https://hostname/instance/api/FHIR/R4/Procedure/eMm0.ONP9yY6gU1plJr8Cloh9xVrrivStBjA6GTod-Ag3"
        }
      ],
      "fullUrl": "https://hostname/instance/api/FHIR/R4/Procedure/eMm0.ONP9yY6gU1plJr8Cloh9xVrrivStBjA6GTod-Ag3",
      "resource": {
        "resourceType": "Procedure",
        "id": "eMm0.ONP9yY6gU1plJr8Cloh9xVrrivStBjA6GTod-Ag3",
        "identifier": [
          {
            "use": "usual",
            "type": {
              "text": "ORD"
            },
            "system": "urn:oid:1.2.840.114350.1.13.861.1.7.2.798268",
            "value": "3287774"
          },
          {
            "use": "usual",
            "type": {
              "text": "EAP"
            },
            "system": "urn:oid:1.2.840.114350.1.13.861.1.7.2.696580",
            "value": "307"
          }
        ],
        "status": "completed",
        "category": {
```

```
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "103693007",
              "display": "Diagnostic procedure"
            }
          ],
          "text": "Ordered Procedures"
        },
        "code": {
          "coding": [
            {
              "system": "http://www.ama-assn.org/go/cpt",
              "code": "11100",
              "display": "BIOPSY OF SKIN LESION"
            },
            {
              "system": "http://snomed.info/sct",
              "code": "129902007",
              "display": "BIOPSY OF SKIN LESION"
            },
            {
              "system": "http://loinc.org",
              "code": "10002-4",
              "display": "BIOPSY OF SKIN LESION"
            }
          ],
          "text": "BIOPSY OF SKIN LESION"
        },
        "subject": {
          "reference": "Patient/ej4nFxzySJAysjMuv4rTBNA3",
          "display": "zzFhir,Fhirgirl"
        },
        "encounter": {
          "reference": "Encounter/ej5R7X.4RdaFNR5vWdsTpVTtccmkRkWJcKixc5XVY4cQ3"
        },
        "performedDateTime": "2019-02-15",
        "reasonCode": [
          {
            "coding": [
              {
                "system": "http://snomed.info/sct",
                "code": "400097005",
                "display": "Ingrowing nail"
              }
            ],
            "text": "Ingrowing nail"
          }
        ]
      },
      "search": {
        "mode": "match"
      }
    }
  ]
}
```


10.1.57 Procedure.Search (Patient-Reported Surgical History) (R4)

File: site/apis/10-Clinical-General/1-R4/10030-Procedure-Search-Patient-Reported-Surgical-History-R4.html
Epic API ID | 10030
FHIR version | R4
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | February 2021
HTTP method | GET
URL template | /api/FHIR/R4/Procedure?category=387713003&patient={patient}&subject={subject}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10030

Description

The FHIR Procedure resource defines an activity performed on or with a patient as part of the provision of care. It corresponds with surgeries and procedures performed, including endoscopies and biopsies, as well as less invasive actions like counseling and physiotherapy.

The Procedure (Patient-Reported Surgical History) API is designed to summarize historical surgical procedures a patient has had, as well as pertinent negatives (indicating a patient has not had a particular surgery). Other types of procedures may be

returned in other APIs, such as the Procedure (Orders) API for procedures created from orders. In Epic version May 2021 and earlier, be aware when using the Procedure resource that each unique procedure could appear twice for patients because it can appear as both a Surgical History entry and a Surgical Procedure entry.

To gather a complete list of surgical procedure information for a patient , three different Procedure APIs must be consulted. While a surgical procedure may be documented in multiple places in Epic, it will be returned only by one Procedure FHIR API, in the following order of precedence:

* Procedure (Orders): Includes any procedure that is ordered by a clinician and completed during an encounter. If a procedure is returned by this API, it won't be returned by the Procedure (Surgeries) or Procedure (Surgical History) APIs.

* Procedure (Surgeries): Includes surgical procedures attached to a surgical log. If a procedure is returned by this API, it won't be returned by the Procedure (Surgical History) API.

* Procedure (Surgical History): Includes any manually documented surgical history or pertinent negatives that were not documented on a surgical log or ordered within Epic.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
category | token | True | | | A code that classifies the procedure. This will always be SNOMED code 387713003 (Surgical procedure) for the Procedure (Surgical history) API.
patient | reference | Either subject or patient must be specified. If both are specified, they must reference the same patient. | | | The Patient FHIR resource ID.
subject | reference | Either subject or patient must be specified. If both are specified, they must reference the same patient. | | | Subject of the Procedure.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
based-on | reference | True | | | Unsupported
code | token | True | | | External codes associated with the procedure.
date | date | True | | | When the procedure was performed
encounter | reference | True | | | Unsupported
identifier | token | True | | | Unsupported
instantiates-canonical | String | True | | | Unsupported
location | reference | True | | | Unsupported
part-of | reference | True | | | Unsupported
performer | reference | True | | | Unsupported
reason-code | token | True | | | Unsupported
reason-reference | reference | True | | | Unsupported
status | token | True | | | The status of the Procedure. Refer to the response element for more information.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | BackboneElement | False | Yes | | An array of entries included in this bundle.
Bundle.entry.resource | ResourceContainer | False | | | A container for a single FHIR resource.
Bundle.entry.resource.Procedure | Procedure | False
Bundle.entry.resource.Procedure.bodySite | CodeableConcept | True | Yes | | Where on the body the procedure was performed.
Bundle.entry.resource.Procedure.bodySite.extension (also: surgical-history-laterality) | Extension | True | Yes | | Laterality options using SNOMED codes and names: 7771000 (Left) 24028007 (Right) 51440002 (Right and left) 385432009 (Not applicable) 276727009 (Null)
Bundle.entry.resource.Procedure.category | CodeableConcept | True | | | A code that classifies the procedure. This will always be SNOMED code 387713003 (Surgical procedure) for the Procedure (Surgical History) API.
Bundle.entry.resource.Procedure.code | CodeableConcept | True | | | External codes associated with the procedure.
Bundle.entry.resource.Procedure.note | Annotation | True | Yes | | Comments about the surgical history or pertinent negative.
Bundle.entry.resource.Procedure.performedDateTime | dateTime | False | | | When the procedure was performed.
Bundle.entry.resource.Procedure.status | code | False | | | The status of the Procedure. FHIR Event Status Returned when... completed positive procedure not-done pertinent negative *Note: Omitted FHIR event status equivalent values are not returned by this search
FHIR Event Status | Returned when...
completed | positive procedure
not-done | pertinent negative
Bundle.entry.resource.Procedure.subject | Reference | False | | | Reference to the Patient resource for whom this Procedure resource is relevant.
Bundle.entry.resource.Procedure.extension (also: surgical-history-laterality) | Extension | True | Yes | | Laterality options using SNOMED codes and names: 7771000 (Left) 24028007 (Right) 51440002 (Right and left) 385432009 (Not applicable) 276727009 (Null)
Bundle.entry.resource.Procedure.extension (also: surgical-history-source) | Extension | True | Yes | | Free-text description of the source of the information. For example "Provider" or "Patient."
Bundle.total | Int64 | True | | | Total number of resources that matched the search criteria.

Sample request

```
https://hostname/instance/api/FHIR/R4/Procedure?category=387713003&patient=ej4nFxzySJAysjMuv4rTBNA3
```

Sample response

```
{
```

```
"resourceType": "Bundle",
"type": "searchset",
"total": 2,
"link": [
  {
    "relation": "self",
    "url":
  "https://hostname/instance/api/FHIR/R4/Procedure?patient=eyb0wFCjScO4c54vJCAvBHg3&_revInclude=Provenance:target"
  }
],
"entry": [
  {
    "link": [
      {
        "relation": "self",
        "url": "https://hostname/instance/api/FHIR/R4/Procedure/esDSRiB-AywSL88jYw-
  ZM6pLL8uiKV5X2-iw.L7W5GCA4h8IoUDnwhxnDSiTbbo1A3"
      }
    ],
    "fullUrl": "https://hostname/instance/api/FHIR/R4/Procedure/esDSRiB-AywSL88jYw-
  ZM6pLL8uiKV5X2-iw.L7W5GCA4h8IoUDnwhxnDSiTbbo1A3",
    "resource": {
      "resourceType": "Procedure",
      "id": "esDSRiB-AywSL88jYw-ZM6pLL8uiKV5X2-iw.L7W5GCA4h8IoUDnwhxnDSiTbbo1A3",
      "extension": [
        {
          "valueString": "Provider",
          "url": "http://open.epic.com/FHIR/StructureDefinition/extension/surgical-history-source"
        }
      ],
      "status": "completed",
      "category": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "387713003",
            "display": "Surgical procedure"
          }
        ],
        "text": "Surgical History"
      },
      "code": {
        "coding": [
          {
            "system": "http://www.ama-assn.org/go/cpt",
            "code": "36778483",
            "display": "LEG AMPUTATION"
          }
        ],
        "text": "LEG AMPUTATION"
      },
      "subject": {
        "reference": "Patient/eyb0wFCjScO4c54vJCAvBHg3",
        "display": "Name, Shroud, PhD"
      },
      "performedDateTime": "2020-11-03",
      "bodySite": [
        {
          "extension": [
            {
              "valueCodeableConcept": {
                "coding": [
                  {
                    "system": "http://snomed.info/sct",
                    "code": "447561005",
                    "display": "SNOMED CT source code to target map code correlation not specified"
                  }
                ],
                "text": "Left"
              },
              "url": "http://open.epic.com/FHIR/StructureDefinition/extension/surgical-history-laterality"
            }
          ]
        }
      ]
    },
    "search": {
      "mode": "match"
    }
  },
  {
    "link": [
      {
        "relation": "self",
        "url": "https://hostname/instance/api/FHIR/R4/Provenance/eoUS7LVLK31M9n0hyriZ3GIx.1HdOLq.isIM.eyvrjg241T8RhUuD
  4oMPm4ShCnjCTiVBagLsJxdat6dsSqaEcOMaA3j7Va.i0R35vkRBY7g3"
      }
    ],
    "fullUrl": "https://hostname/instance/api/FHIR/R4/Provenance/eoUS7LVLK31M9n0hyriZ3GIx.1HdOLq.isIM.eyvrjg241T8RhUuD
```

```
4oMPm4ShCnjCTiVBagLsJxdat6dsSqaEcOMaA3j7Va.i0R35vkRBY7g3",
  "resource": {
    "resourceType": "Provenance",
    "id":
"eoUS7LVLK31M9n0hyriZ3GIx.1HdOLq.isIM.eyvrjg241T8RhUuD4oMPm4ShCnjCTiVBagLsJxdat6dsSqaEcOMaA3j7Va.i0R35vkRBY7g3",
    "target": [
      {
        "reference": "Procedure/esDSRiB-AywSL88jYw-ZM6pLL8uiKV5X2-iw.L7W5GCA4h8IoUDnwhxnDSiTbbo1A3"
      }
    ],
    "recorded": "2020-12-11T06:00:00Z",
    "agent": [
      {
        "type": {
          "coding": [
            {
              "system": "http://terminology.hl7.org/CodeSystem/provenance-participant-type",
              "code": "author",
              "display": "Author"
            }
          ],
          "text": "Author"
        },
        "who": {
          "reference": "Practitioner/eDYyJA9FNDKf7rD4dvhzgjQ3",
          "display": "User Name"
        },
        "onBehalfOf": {
          "display": "Current Development Environment (CDE) - Smaug"
        }
      },
      {
        "type": {
          "coding": [
            {
              "system": "http://hl7.org/fhir/us/core/CodeSystem/us-core-provenance-participant-type",
              "code": "transmitter",
              "display": "Transmitter"
            }
          ],
          "text": "Transmitter"
        },
        "who": {
          "display": "Current Development Environment (CDE) - Smaug"
        }
      }
    ]
  },
  "search": {
    "mode": "include"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Procedure/esDSRiB-AywSL88jYw-
ZM6pLL8uiKV5X2-iw.L7W5GCDnI0kXxoI0dD5BGO9mOkxp3"
    }
  ],
  "fullUrl": "https://hostname/instance/api/FHIR/R4/Procedure/esDSRiB-AywSL88jYw-
ZM6pLL8uiKV5X2-iw.L7W5GCDnI0kXxoI0dD5BGO9mOkxp3",
  "resource": {
    "resourceType": "Procedure",
    "id": "esDSRiB-AywSL88jYw-ZM6pLL8uiKV5X2-iw.L7W5GCDnI0kXxoI0dD5BGO9mOkxp3",
    "extension": [
      {
        "valueString": "Provider",
        "url": "http://open.epic.com/FHIR/StructureDefinition/extension/surgical-history-source"
      }
    ],
    "status": "not-done",
    "category": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "387713003",
          "display": "Surgical procedure"
        }
      ],
      "text": "Surgical History"
    },
    "code": {
      "coding": [
        {
          "system": "http://www.ama-assn.org/go/cpt",
          "code": "15836",
          "display": "EXCISE EXCESS SKIN TISSUE,ARM"
        }
```

```
          ],
          "text": "EXCISE EXCESS SKIN TISSUE,ARM"
        },
        "subject": {
          "reference": "Patient/eyb0wFCjScO4c54vJCAvBHg3",
          "display": "Name, Shroud, PhD"
        },
        "performedDateTime": "2020-12-11",
        "note": [
          {
            "text": "This is pertinent negative"
          }
        ]
      },
      "search": {
        "mode": "match"
      }
    },
    {
      "link": [
        {
          "relation": "self",
          "url": "https://hostname/instance/api/FHIR/R4/Provenance/eoUS7LVLK31M9n0hyriZ3GIx.1HdOLq.isIM.eyvrjg241T8RhUuD
    4oMPm4ShCnjCb27JrzQw8Cps8cy52AKJ6MofAZ7eN1zXUWfyw66jF9A3"
        }
      ],
      "fullUrl": "https://hostname/instance/api/FHIR/R4/Provenance/eoUS7LVLK31M9n0hyriZ3GIx.1HdOLq.isIM.eyvrjg241T8RhUuD
    4oMPm4ShCnjCb27JrzQw8Cps8cy52AKJ6MofAZ7eN1zXUWfyw66jF9A3",
      "resource": {
        "resourceType": "Provenance",
        "id":
    "eoUS7LVLK31M9n0hyriZ3GIx.1HdOLq.isIM.eyvrjg241T8RhUuD4oMPm4ShCnjCb27JrzQw8Cps8cy52AKJ6MofAZ7eN1zXUWfyw66jF9A3",
        "target": [
          {
            "reference": "Procedure/esDSRiB-AywSL88jYw-ZM6pLL8uiKV5X2-iw.L7W5GCDnI0kXxoI0dD5BGO9mOkxp3"
          }
        ],
        "recorded": "2020-12-11T06:00:00Z",
        "agent": [
          {
            "type": {
              "coding": [
                {
                  "system": "http://terminology.hl7.org/CodeSystem/provenance-participant-type",
                  "code": "author",
                  "display": "Author"
                }
              ],
              "text": "Author"
            },
            "who": {
              "reference": "Practitioner/eDYyJA9FNDKf7rD4dvhzgjQ3",
              "display": "User Name"
            },
            "onBehalfOf": {
              "display": "Current Development Environment (CDE) - Smaug"
            }
          },
          {
            "type": {
              "coding": [
                {
                  "system": "http://hl7.org/fhir/us/core/CodeSystem/us-core-provenance-participant-type",
                  "code": "transmitter",
                  "display": "Transmitter"
                }
              ],
              "text": "Transmitter"
            },
            "who": {
              "display": "Current Development Environment (CDE) - Smaug"
            }
          }
        ]
      },
      "search": {
        "mode": "include"
      }
    }
  ]
}
```


Change log

Epic Version | Description | Type

May 2023 (Special Updates to February 2023) | Code systems for the following code sets now return a URL in place of an OID in the following elements: Procedure.bodySite, Procedure.bodySite.extension(BodySite-Qualifier).valueCodeableConceptProcedure System URL OID SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 LOINC http://loinc.org 2.16.840.1.113883.6.1 | Breaking change

System | URL | OID
SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96
LOINC | http://loinc.org | 2.16.840.1.113883.6.1

10.1.58 Procedure.Search (Restricting Intervention) (R4)

File: site/apis/10-Clinical-General/1-R4/10472-Procedure-Search-Restricting-Intervention-R4.html
Epic API ID | 10472
FHIR version | R4
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Procedure?category={category}&patient={patient}&subject={subject}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10472

Description

This resource is available only for healthcare organizations in the Netherlands.

The FHIR Procedure resource defines an activity performed on or with a patient as part of the provision of care.

The Procedure (Restriction Intervention) API is designed to record interventions that are used to implement freedom restricting measures on the patient.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
category | token | True | | | A code that classifies the procedure. Use SNOMED code “225317005” or custom Epic category value “freedom-restricting-intervention” to search for restricting interventions.
patient | reference | Either "patient" or "subject" is required. If both are specified, they must match. | | | A Patient FHIR resource ID representing the subject of the intervention.
subject | reference | Either "patient" or "subject" is required. If both are specified, they must match. | | | A Patient FHIR resource ID representing the subject of the intervention.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
based-on | reference | True | | | Unsupported
code | token | True | | | External codes associated with the procedure. Bound to TypeOfInterventionCodelist.
date | date | True | | | The date when the procedure was performed.
encounter | reference | True | | | Reference to the encounter in which the procedure took place.
identifier | token | True | | | Unsupported
instantiates-canonical | String | True | | | Unsupported
location | reference | True | | | Unsupported
part-of | reference | True | | | Unsupported
performer | reference | True | | | Unsupported
reason-code | token | True | | | The reason for use that provides the justification for applying involuntary care. Mapped to ReasonForUseCodelist.
reason-reference | reference | True | | | Unsupported
status | token | True | | | The status of the procedure order. See the table in the response elements for specific values.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | BackboneElement | False | Yes | | An array of entries included in this bundle.
Bundle.entry.resource | ResourceContainer | False | | | A container for a single FHIR resource.
Bundle.entry.resource.Procedure | Procedure | False
Bundle.entry.resource.Procedure.extension (also: Assent) | Extension | True | Yes | | Records whether a patient or their legal representative provided consent.
Bundle.entry.resource.Procedure.category | CodeableConcept | False | | | A code that classifies the procedure. Always includes SNOMED code “225317005” and the custom Epic category “freedom-restricting-intervention” for this resource.
Bundle.entry.resource.Procedure.code | CodeableConcept | False | | | External codes associated with the procedure. Bound to TypeOfInterventionCodelist.
Bundle.entry.resource.Procedure.encounter | Reference | True | | | Reference to the encounter in which the procedure took place.
Bundle.entry.resource.Procedure.extension (also: legallyCapable) | Extension | False | Yes | | Whether a patient is legally capable. Contained within a parent zib-FreedomRestrictingIntervention-LegallyCapable extension.
Bundle.entry.resource.Procedure.extension (also: legallyCapableComment) | Extension | True | Yes | | Includes free-text instruction information on a patient’s legal capability. Contained within a parent zib-FreedomRestrictingIntervention-LegallyCapable extension.
Bundle.entry.resource.Procedure.performedPeriod | Period | False | | | The period of time over which the procedure was performed.

Bundle.entry.resource.Procedure.reasonCode | CodeableConcept | True | Yes | | The reason for use that provides the justification for applying involuntary care. Mapped to ReasonForUseCodelist.
Bundle.entry.resource.Procedure.status | code | False | | | The status of the procedure order. The table below describes how the status of the intervention procedure order maps to the status returned by this resource. Order Status FHIR Status Pending, Sent, Holding for Referral Preparation Verified, Pending Verify In-Progress Canceled or Discontinued (without documentation) Stopped Denied Approval Not-Done Suspend Suspended Resulted, Completed, Dispensed, Discontinued or Canceled (with documentation) Completed
Order Status | FHIR Status
Pending, Sent, Holding for Referral | Preparation
Verified, Pending Verify | In-Progress
Canceled or Discontinued (without documentation) | Stopped
Denied Approval | Not-Done
Suspend | Suspended
Resulted, Completed, Dispensed, Discontinued or Canceled (with documentation) | Completed
Bundle.entry.resource.Procedure.subject | Reference | False | | | Reference to the Patient resource for whom this Procedure resource is relevant.

Sample request

```
https://hostname/instance/api/FHIR/R4/Procedure?category=freedom-restricting-
    intervention&subject=eojz6maS66J54Bwze3hFhDg3
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 1,
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Procedure?category=freedom-restricting-
    intervention&subject=eojz6maS66J54Bwze3hFhDg3"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url": "https://hostname/instance/api/FHIR/R4/Procedure/fFK-tRE1AmmsrTnHfWRlXZofEtsqrIGnyBvpRohoGPuM4"
        }
      ],
      "fullUrl": "https://hostname/instance/api/FHIR/R4/Procedure/fFK-tRE1AmmsrTnHfWRlXZofEtsqrIGnyBvpRohoGPuM4",
      "resource": {
        "resourceType": "Procedure",
        "id": "fFK-tRE1AmmsrTnHfWRlXZofEtsqrIGnyBvpRohoGPuM4",
        "extension": [
          {
            "extension": [
              {
                "valueBoolean": false,
                "url": "legallyCapable"
              },
              {
                "valueString": "Patient's legal representative provided consent for this action",
                "url": "legallyCapableComment"
              }
            ],
            "url": "http://nictiz.nl/fhir/StructureDefinition/zib-FreedomRestrictingIntervention-LegallyCapable"
          },
          {
            "valueCodeableConcept": {
              "coding": [
                {
                  "system": "urn:oid:2.16.840.1.113883.2.4.3.11.60.40.4.6.1",
                  "code": "4",
                  "display": "Instemming wettelijke vertegenwoordiger"
                }
              ],
              "text": "Instemming wettelijke vertegenwoordiger"
            },
            "url": "http://nictiz.nl/fhir/StructureDefinition/zib-FreedomRestrictingIntervention-Assent"
          }
        ],
        "status": "in-progress",
        "category": {
          "coding": [
            {
              "system": "http://open.epic.com/FHIR/StructureDefinition/Procedure-category",
              "code": "freedom-restricting-intervention",
              "display": "Freedom Restricting Intervention"
            },
            {
              "system": "http://snomed.info/sct",
              "code": "225317005",
              "display": "Restriction of movement"
            }
```

```
          ],
          "text": "Restriction of movement"
        },
        "code": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "225210009",
              "display": "LOCKED DOOR PROCEDURE (PROCEDURE)"
            }
          ],
          "text": "LOCKED DOOR PROCEDURE (PROCEDURE)"
        },
        "subject": {
          "reference": "Patient/eojz6maS66J54Bwze3hFhDg3",
          "display": "Henderson, T."
        },
        "encounter": {
          "reference": "Encounter/eT5I-WeoUShAaaQqwGiwpyg3",
          "identifier": {
            "use": "usual",
            "system": "urn:oid:1.2.840.114350.1.13.260.1.7.3.698084.8",
            "value": "239103"
          },
          "display": "Hospital Encounter"
        },
        "performedPeriod": {
          "start": "2022-07-14T20:21:00Z"
        },
        "reasonCode": [
          {
            "coding": [
              {
                "system": "http://snomed.info/sct",
                "code": "121481000146105",
                "display": "Diversion of situation that could result in self-harm or harm to others"
              }
            ],
            "text": "Diversion of situation that could result in self-harm or harm to others"
          }
        ]
      },
      "search": {
        "mode": "match"
      }
    }
  ]
}
```


10.1.59 Procedure.Search (SDOH Intervention) (R4)

File: site/apis/10-Clinical-General/1-R4/11038-Procedure-Search-SDOH-Intervention-R4.html
Epic API ID | 11038
FHIR version | R4
Categories | Clinical General
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Procedure?category={category}&code={code}&date={date}&patient={patient}&subject={subject}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=11038

Description

This resource includes general information about the intervention, including the patient to whom it applies and who made the request.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
category | token | True | | | Set category to the following token construction: system | code, where: * system: http://snomed.info/sct * code: 410606002 The code SCTID: 410606002 represents the “‘Social service procedure (procedure)”’ to find SDOH intervention type of Procedure.
code | token | True | | | Set code to the following token construction: system | code, where: * system: http://snomed.info/sct * code: 306206005 The code SCTID: 306206005 represents the ‘Referral to service’ to find SDOH intervention type of Procedure.
date | date | True | | | The date the Procedure was completed.
patient | reference | Either patient or subject must be provided. | | | Search for Procedure resources for a specified patient ID. You can use both "patient" and "subject" equivalently, but if you use both they must look to the same value.

subject | reference | Either patient or subject must be provided | | | Search for Procedure resources for a specified subject. You can use both "patient" and "subject" equivalently, but if you use both they must look to the same value.

Response elements

Name | Type | Optional | Array | Cardinality | Description

Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.

Bundle.entry | BackboneElement | False | Yes | | An array of entries included in this bundle.

Bundle.entry.resource | ResourceContainer | False | | | A container for a single FHIR resource.

Bundle.entry.resource.Procedure | Procedure | False | | | A single instance of an R4 Procedure FHIR resource.

Bundle.entry.resource.Procedure.basedOn | Reference | False | Yes | | Reference to the ServiceRequest that represents the request metadata.

Bundle.entry.resource.Procedure.category | CodeableConcept | False | | | The FHIR codeable concept: always SNOMED code – SCTID 410606002: Social service procedure (procedure) for this API.

Bundle.entry.resource.Procedure.code | CodeableConcept | False | | | The FHIR codeable concept: always SNOMED code – SCTID 306206005: Referral to service for this API.

Bundle.entry.resource.Procedure.encounter | Reference | False | | | Reference to an Encounter resource.

Bundle.entry.resource.Procedure.performedDateTime | dateTime | False | | | The date the procedure was performed.

Bundle.entry.resource.Procedure.recorder | Reference | False | | | Reference to a Practitioner resource.

Bundle.entry.resource.Procedure.status | code | False | | | The status of the procedure. Always “completed” for this API.

Bundle.entry.resource.Procedure.subject | Reference | False | | | Reference to Patient resource.

Sample request

```
https://hostname/instance/api/FHIR/R4/Procedure?patient=e8OeNeovb7izjGIErw9MUUg3&category=http://snomed.info/sct|4106060
    02
```

Sample response

```
<Bundle xmlns="http://hl7.org/fhir">
  <type value="searchset"/>
  <total value="1"/>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/R4/Procedure?category=http://snomed.info/sct|410606002&amp;patient=e8
    OeNeovb7izjGIErw9MUUg3"/>
  </link>
  <entry>
    <link>
      <relation value="self"/>
      <url value="https://hostname/instance/api/FHIR/R4/Procedure/fRCKdJNwYvxZZwcKp-dszPg4"/>
    </link>
    <fullUrl value="https://hostname/instance/api/FHIR/R4/Procedure/fRCKdJNwYvxZZwcKp-dszPg4"/>
    <resource>
      <Procedure>
        <id value="fRCKdJNwYvxZZwcKp-dszPg4"/>
        <basedOn>
          <reference value="ServiceRequest/eNaYhHi6xequPT3aBgleofl-gWkv0Gsb7m5WFgsWpPOh8u1J-JnCQVhISmBkbWcmv3"/>
          <type value="ServiceRequest"/>
        </basedOn>
        <status value="completed"/>
        <category>
          <coding>
            <system value="http://snomed.info/sct"/>
            <code value="410606002"/>
            <display value="Social service procedure (procedure)"/>
          </coding>
          <text value="Social service procedure (procedure)"/>
        </category>
        <code>
          <coding>
            <system value="http://snomed.info/sct"/>
            <code value="306206005"/>
            <display value="Referral to service"/>
          </coding>
          <text value="Referral to service"/>
        </code>
        <subject>
          <reference value="Patient/e8OeNeovb7izjGIErw9MUUg3"/>
          <type value="Patient"/>
          <display value="Smith, Janice "/>
        </subject>
        <encounter>
          <reference value="Encounter/eRsROYaTxtaNmMib-Sh0DUke14oCfJBsQ3daJGXeTFQo3"/>
          <identifier>
            <use value="usual"/>
            <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.3.698084.8"/>
            <value value="10003597712"/>
          </identifier>
          <display value="Office Visit"/>
        </encounter>
        <performedDateTime value="2024-04-17T15:19:52Z"/>
        <recorder>
          <reference value="Practitioner/eeLsHlvm0KvPn-y4wW-AuzQ3"/>
          <type value="Practitioner"/>
          <display value="Social Worker Janisko, MSW"/>
```

```
        </recorder>
      </Procedure>
    </resource>
    <search>
      <mode value="match"/>
    </search>
  </entry>
</Bundle>
```

10.1.60 Procedure.Search (Surgeries) (R4)

File: site/apis/10-Clinical-General/1-R4/10042-Procedure-Search-Surgeries-R4.html
Epic API ID | 10042
FHIR version | R4
Categories | Clinical General
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | November 2020
HTTP method | GET
URL template | /api/FHIR/R4/Procedure?date={date}&patient={patient}&subject={subject}&category=387713003
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10042

Description

The FHIR Procedure resource defines an activity performed on or with a patient as part of the provision of care. It corresponds with surgeries and procedures performed, including endoscopies and biopsies, as well as less invasive actions like counseling and physiotherapy.

The Procedure (Surgeries) API is designed to summarize any performed surgical procedures for a patient. Other types of procedures may be returned in other APIs, such as the Procedure (Orders) API for procedures created from orders.

Organizations can have differences when determining which performable activities are classified as procedures, so the types of activities included in FHIR Procedure search can vary between organizations.

Because terminology codes for procedures are often billing related, some procedures might not have a standard code (for example, CPT), especially when billing activities are not performed within Epic. For organizations that do perform billing activities within Epic, procedures might not have a CPT code until the charges are generated for the procedure.

This resource is designed for a high-level summarization around the occurrence of a procedure, and not for specific procedure log documentation - a concept that does not yet have a defined FHIR Resource. When searching, only completed procedures are returned.

To gather a complete list of surgical procedure information for a patient, three different Procedure APIs must be consulted. While a surgical procedure may be documented in multiple places in Epic, it will be returned only by one Procedure FHIR API, in the following order of precedence:

* Procedure (Orders): Includes any procedure that is ordered by a clinician and completed during an encounter. If a procedure is returned by this API, it won't be returned by the Procedure (Surgeries) or Procedure (Surgical History) APIs.

* Procedure (Surgeries): Includes surgical procedures attached to a surgical log. If a procedure is returned by this API, it won't be returned by the Procedure (Surgical History) API.

* Procedure (Surgical History): Includes any manually documented surgical history or pertinent negatives that were not documented on a surgical log or ordered within Epic.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
_count | number | Optional | Yes | | Number of results per page.
_id | token | Optional | Yes | | FHIR resource IDs for the desired resources. Can be used in combination with _count, _include, and _revInclude. All others will be ignored.
_include | string | Optional | Yes | | Used to indicate which referenced resources to include in the search bundle.
_revinclude | string | Optional | Yes | | Used to include Provenance or AuditEvent resources that refer to search results, when available.
category | token | Optional | Yes | | A code that classifies the procedure. A code that classifies the procedure. This will always be SNOMED code 387713003 (Surgical procedure) for the Procedure (Surgeries) API.
date | date | Optional | Yes | | Date or period that the procedure was performed, using the FHIR date parameter format.
patient | reference | Conditional | Yes | | The Patient FHIR resource ID.
subject | reference | Conditional | Yes | | The FHIR ID for the patient associated with the Procedure. Either patient or subject must be provided.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description

based-on | reference | Optional | Yes | | The ServiceRequest resource that the procedure log was created from.
code | token | Optional | Yes | | External codes associated with the procedure.
encounter | reference | Optional | Yes | | Reference to an Encounter resource associated with the Procedure.
identifier | token | Optional | Yes | | System-specific ID for the procedure order record.
location | reference | Optional | Yes | | The Location resource where the procedure was performed.
performer | reference | Optional | Yes | | The Practitioner that performed the procedure.
reason-code | token | Optional | Yes | | Diagnosis codes associated with the procedure.
reason-reference | reference | Optional | Yes | | Applicable only to organizations in the Netherlands. Reference to a Condition associated with the DBC episode for the procedure.
status | token | Optional | Yes | | The status of the Procedure. See the element in the response section for specific values.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Procedure (R4) | Procedure
Procedure (R4).basedOn | Reference | | Yes | 0..* | Reference to the ServiceRequest resource that the procedure log was created from.
Procedure (R4).bodySite | CodeableConcept | | Yes | 0..* | External code for the operative region for the procedure.
Procedure (R4).bodySite.extension (BodySite-Qualifier) (also: extension (BodySite-Qualifier)) | Extension | | Yes | 0..1 | Laterality for the body site. URL: http://nictiz.nl/fhir/StructureDefinition/BodySite-Qualifier
Procedure (R4).category | CodeableConcept | | | 0..1 | A code that classifies the procedure. This will always be SNOMED code 387713003 (Surgical procedure) for the Procedure (Surgeries) API.
Procedure (R4).code | CodeableConcept | | | 1..1 | External codes associated with the procedure. Can include code types specified in the PROCEDURE_CODE_TYPE (1825) profile variable if those codes are available on the target record.
Procedure (R4).complication | CodeableConcept | | Yes | 0..* | Complication tracking event.
Procedure (R4).encounter | Reference | | | 0..1 | Reference to an Encounter resource associated with the Procedure.
Procedure (R4).extension (data-absent-reason) (also: extension (data-absent-reason)) | Extension | | Yes | 0..* | The reason why normally expected content of the data element is missing. Set to "not-performed" for surgeries without a performed time or period. URL:http://hl7.org/fhir/StructureDefinition/data-absent-reason
Procedure (R4).extension (log-level-procedure-codes) (also: extension (log-level-procedure-codes)) | Extension | | Yes | 0..* | Additional external codes for diagnosis records not associated with a specific procedure. Includes a list of one or more valueCodeableConcept codings, and the text value "Additional Procedure Codes". URL: http://open.epic.com/FHIR/StructureDefinition/extension/log-level-procedure-codes
Procedure (R4).focalDevice | BackboneElement | | Yes | 0..* | Reference to a Device resource marked as implanted, explanted, or adjusted during the procedure.
Procedure (R4).focalDevice.action | CodeableConcept | | | 1..1 | Implant action.
Procedure (R4).focalDevice.manipulated | Reference | | | 0..1 | Implant ID.
Procedure (R4).id | id | | | 1..1 | The Procedure FHIR ID.
Procedure (R4).identifier | Identifier | | Yes | 1..* | System-specific ID for the procedure order record.
Procedure (R4).location | Reference | | | 0..1 | Reference to the Location resource where the procedure was performed.
Procedure (R4).outcome | CodeableConcept | | | 0..1 | Discharge disposition from the linked ADT contact.
Procedure (R4).performedDateTime | dateTime | | | 1..1 | When the procedure was performed.
Procedure (R4).performedPeriod | Period | | | 1..1 | Period of time over which the procedure was performed.
Procedure (R4).performer | BackboneElement | | Yes | 0..* | Who performed the procedure.
Procedure (R4).performer.actor | Reference | | | 0..1 | Reference to the Practitioner that performed the procedure.
Procedure (R4).reasonCode | CodeableConcept | | Yes | 0..* | The coded reason that the procedure was performed.
Procedure (R4).reasonReference | Reference | | Yes | 0..* | Applicable only to organizations in the Netherlands. Reference to a Condition resource associated with the DBC episode for the procedure.
Procedure (R4).recorder | Reference | | | 0..1 | Audit user.
Procedure (R4).report | Reference | | Yes | 0..* | Reference to the DocumentReference resource that contains notes from the surgical log.
Procedure (R4).status | code | | | 1..1 | The status of the Procedure. Epic Value FHIR Event Status cancelled stopped active in-progress completed completed aborted stopped
Epic Value | FHIR Event Status
cancelled | stopped
active | in-progress
completed | completed
aborted | stopped
Procedure (R4).statusReason | CodeableConcept | | | 0..1 | The reason that a procedure was not performed, which is sent only when the status is Stopped.
Procedure (R4).subject | Reference | | | 1..1 | Reference to the Patient resource for whom this Procedure resource is relevant.
Procedure (R4).usedReference | Reference | | Yes | 0..* | Reference to the Device resource marked as a wasted implant that was used during the procedure.

Sample request

```
GET https://hostname/instance/api/FHIR/R4/Procedure?category=387713003&patient=e1uV557J2LZDVPSZfxOtXyw3
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 1,
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Procedure?category=387713003&patient=e1uV557J2LZDVPSZfxOtXyw3"
    }
```

```
],
"entry": [
  {
    "link": [
      {
        "relation": "self",
        "url": "https://hostname/instance/api/FHIR/R4/Procedure/fv4NzEJK7mo6EzLIHD7ZAYaLJVXNQExFkm55pWMZHX-M4"
      }
    ],
    "fullUrl": "https://hostname/instance/api/FHIR/R4/Procedure/fv4NzEJK7mo6EzLIHD7ZAYaLJVXNQExFkm55pWMZHX-M4",
    "resource": {
      "resourceType": "Procedure",
      "id": "fv4NzEJK7mo6EzLIHD7ZAYaLJVXNQExFkm55pWMZHX-M4",
      "identifier": [
        {
          "use": "usual",
          "type": {
            "text": "ORL"
          },
          "system": "urn:oid:1.2.840.114350.1.13.861.1.7.2.798276",
          "value": "178124"
        },
        {
          "use": "usual",
          "type": {
            "text": "OPE"
          },
          "system": "urn:oid:1.2.840.114350.1.13.861.1.7.2.798069",
          "value": "207340"
        }
      ],
      "status": "completed",
      "category": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "387713003",
            "display": "Surgical procedure"
          }
        ],
        "text": "Surgical Procedures"
      },
      "code": {
        "text": "Appendectomy"
      },
      "subject": {
        "reference": "Patient/e1uV557J2LZDVPSZfxOtXyw3",
        "display": "OpTime, Fhirman"
      },
      "encounter": {
        "reference": "Encounter/el8BJ6-udqPGspHLpTl8hFw64lqOLXJt8mqYBGbqaKRo3",
        "identifier": {
          "use": "usual",
          "system": "urn:oid:1.2.840.114350.1.13.861.1.7.3.698084.8",
          "value": "10008467329"
        },
        "display": "Surgery"
      },
      "performedPeriod": {
        "start": "2024-04-30T19:22:00Z",
        "end": "2024-05-01T20:35:00Z"
      },
      "recorder": {
        "reference": "Practitioner/eMPvR2m5RxFtz6hEZ-B6hiA3",
        "type": "Practitioner",
        "display": "Dave Cv Surgeon"
      },
      "location": {
        "reference": "Location/ejlzkA8sKOIVzg8NCPMp-Cw3",
        "display": "WI UHM MAIN OR"
      },
      "bodySite": [
        {
          "extension": [
            {
              "valueCodeableConcept": {
                "coding": [
                  {
                    "system": "urn:oid:1.2.840.114350.1.13.861.1.7.4.798276.2010",
                    "code": "4",
                    "display": "N/A"
                  }
                ],
                "text": "N/A"
              },
              "url": "http://nictiz.nl/fhir/StructureDefinition/BodySite-Qualifier"
            }
          ],
          "coding": [
```

```
              {
                "system": "urn:oid:1.2.840.114350.1.13.861.1.7.4.798276.2015",
                "code": "1",
                "display": "Abdomen"
              }
            ],
            "text": "Abdomen"
          }
        ],
        "report": [
          {
            "reference": "DocumentReference/fsgKwTeUQ60.Qcy4G4cc7foABkyhWF4J.fMMEWP3aCX44"
          },
          {
            "reference": "DocumentReference/fpuurmPyzd-tI3JVTZ8XELyROIm9Vw4wdZAOMbOKDD2U4"
          },
          {
            "reference": "DocumentReference/fhWV6qPPhmBrStr.hD5J3PpjUAYFAjv5c2Ry5KczoqfU4"
          }
        ]
      },
      "search": {
        "mode": "match"
      }
    }
  ]
}
```


Change log

Epic Version | Description | Type
May 2023 (Special Updates to February 2023) | Code systems for the following code sets now return a URL in place of an OID in the following elements: Procedure.bodySite, Procedure.bodySite.extension(BodySite-Qualifier).valueCodeableConceptProcedure System URL OID SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 LOINC http://loinc.org 2.16.840.1.113883.6.1 | Breaking change
System | URL | OID
SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96
LOINC | http://loinc.org | 2.16.840.1.113883.6.1

10.1.61 ResearchStudy.Read (R4)

File: site/apis/10-Clinical-General/1-R4/1002-ResearchStudy-Read-R4.html
Epic API ID | 1002
FHIR version | R4
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | February 2020
HTTP method | GET
URL template | api/FHIR/R4/ResearchStudy/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=1002

Description

The ResearchStudy resource includes general information about the research studies tracked in Epic, including the title, status, identifier, and principal investigator. These studies focus on the safety, efficacy, comparative effectiveness, and other information about medications, devices, therapies, and other intervention and investigative techniques intended to increase the field of healthcare-related knowledge.

This resource requires additional setup by the organization if they have custom category list values for research study status.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | String | False | | | The ResearchStudy FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
ResearchStudy | ResearchStudy | False
ResearchStudy.arm | Arm | True | Yes | | Available starting in the August 2021 Epic version. A group or subgroup of a research study.
ResearchStudy.arm.description | String | True | | | Short explanation of the study path.
ResearchStudy.arm.name | String | False | | | Label for the study arm.
ResearchStudy.description | String | True | | | Available starting in the August 2021 Epic version. What this study is doing.

ResearchStudy.id | String | False | | | The ResearchStudy FHIR ID.
ResearchStudy.identifier | Identifier | True | Yes | | Business identifier or study code. The National Clinical Trial (NCT) code is represented as an identifier with the identifier.system of http://clinicaltrials.gov. Other organization-specific codes will have organizational-specific local identifier systems.
ResearchStudy.principalInvestigator | Reference | True | | | Researcher who oversees multiple aspects of the study.
ResearchStudy.status | String | False | | | The current state of the study.
ResearchStudy.title | String | True | | | The name of the study.

Sample request

```
https://hostname/instance/api/FHIR/R4/ResearchStudy/erXhPENfYk8OTVVy9R5bleA3
```

Sample response


```
{
  "resourceType": "ResearchStudy",
  "id": "erXhPENfYk8OTVVy9R5bleA3",
  "identifier": [
    {
      "use": "official",
      "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.3.828372.100",
      "value": "101"
    },
    {
      "use": "usual",
      "system": "http://clinicaltrials.gov",
      "value": " NCT04302"
    }
  ],
  "title": "Pain Management Study",
  "status": "active",
  "description": "The description of the study.",
  "principalInvestigator": {
    "reference": "Practitioner/eflbSK90VWBZBmt0qRlZxaQ3",
    "display": "Physician Beacon, MD"
  },
  "arm": [
    {
      "name": "A",
      "description": "Study Branch A"
    },
    {
      "name": "B",
      "description": "Study Branch B"
    }
  ]
}
```


10.1.62 ResearchStudy.Search (R4)

File: site/apis/10-Clinical-General/1-R4/1003-ResearchStudy-Search-R4.html
Epic API ID | 1003
FHIR version | R4
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | February 2020
HTTP method | GET
URL template | api/FHIR/R4/ResearchStudy?identifier={identifier}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=1003

Description

The ResearchStudy resource includes general information about the research studies tracked in Epic, including the title, status, identifier, and principal investigator. These studies focus on the safety, efficacy, comparative effectiveness, and other information about medications, devices, therapies, and other intervention and investigative techniques intended to increase the field of healthcare-related knowledge.

This resource requires additional setup by the organization if they have custom category list values for research study status.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
identifier | String | False | | | The study code or, starting in the May 2024 Epic version, the study's National Clinical Trial (NCT) number. The NCT code is represented as an identifier with the identifier.system of http://clinicaltrials.gov. For example: http://clinicaltrials.gov|NCT12345678
status | String | True | | | The current state of the study.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
category | String | True | | | Unsupported
date | String | True | | | Unsupported
focus | String | True | | | Unsupported
keyword | String | True | | | Unsupported
location | String | True | | | Unsupported
partof | String | True | | | Unsupported
principalinvestigator | String | True | | | Researcher who oversees multiple aspects of the study
protocol | String | True | | | Unsupported
site | String | True | | | Unsupported
sponsor | String | True | | | Unsupported
title | String | False | | | Name of the study

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | BackboneElement | False | Yes | | An array of entries included in this bundle.
Bundle.entry.resource | ResourceContainer | False | | | A container for a single FHIR resource.
Bundle.entry.resource.ResearchStudy | ResearchStudy | False
Bundle.entry.resource.ResearchStudy.arm | Arm | True | Yes | | Available starting in the August 2021 Epic version. A group or subgroup of a research study.
Bundle.entry.resource.ResearchStudy.arm.description | String | True | | | Short explanation of the study path.
Bundle.entry.resource.ResearchStudy.arm.name | String | True | | | Label for the study arm.
Bundle.entry.resource.ResearchStudy.description | String | False | | | Available starting in the August 2021 Epic version. What this study is doing.
Bundle.entry.resource.ResearchStudy.id | String | False | | | The ResearchStudy FHIR ID.
Bundle.entry.resource.ResearchStudy.identifier | Identifier | True | Yes | | Business identifier or study code. The business identifier is guaranteed to be unique and use is official. The National Clinical Trial (NCT) code is represented as an identifier with the identifier.system of http://clinicaltrials.gov. Other organization-specific codes have organizational-specific local identifier systems.
Bundle.entry.resource.ResearchStudy.identifier.value | String | True | | | Available starting in the August 2021 Epic version. The unique identifier.
Bundle.entry.resource.ResearchStudy.principalInvestigator | Reference | True | | | Researcher who oversees multiple aspects of the study.
Bundle.entry.resource.ResearchStudy.status | String | False | | | The current state of the study.
Bundle.entry.resource.ResearchStudy.title | String | True | | | The name of the study.
Bundle.total | Int64 | True | | | Total number of resources that matched the search criteria.

Sample request

```
https://hostname/instance/api/FHIR/R4/ResearchStudy?identifier=101
```

Sample response


```
{
  "type": "searchset",
  "total": 1,
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/ResearchStudy?identifier=101"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url": "https://hostname/instance/api/FHIR/R4/ResearchStudy/erXhPENfYk8OTVVy9R5bleA3"
        }
      ],
      "fullUrl": "https://hostname/instance/api/FHIR/R4/ResearchStudy/erXhPENfYk8OTVVy9R5bleA3",
      "resource": {
        "resourceType": "ResearchStudy",
        "id": "erXhPENfYk8OTVVy9R5bleA3",
        "identifier": [
          {
            "use": "official",
            "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.3.828372.100",
            "value": "101"
          },
          {
            "use": "usual",
            "system": "http://clinicaltrials.gov",
            "value": "NCT04302"
          }
        ],
        "title": "Pain Management Study",
        "status": "active",
        "description": "The description of the study.",
        "principalInvestigator": {
```

```
          "reference": "Practitioner/eflbSK90VWBZBmt0qRlZxaQ3",
          "display": "Physician Beacon, MD"
        },
        "arm": [
          {
            "name": "A",
            "description": "Study Branch A"
          },
          {
            "name": "B",
            "description": "Study Branch B"
          }
        ]
      },
      "search": {
        "mode": "match"
      }
    }
  ]
}
```

10.1.63 ResearchSubject.Read (R4)

File: site/apis/10-Clinical-General/1-R4/10142-ResearchSubject-Read-R4.html
Epic API ID | 10142
FHIR version | R4
Categories | Clinical General
API type | Industry-Standard
USCDI | No
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | August 2021
HTTP method | GET
URL template | /api/FHIR/R4/ResearchSubject/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10142

Description

The ResearchSubject resource includes general information about the research subject tracked in Epic, including study, study arm, consent status, and other key items. Epic tracks human beings only for research associations. A Patient resource is traceable to a specific person for our cases.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | String | False | | | The research study enrollment record FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
ResearchSubject | ResearchSubject | False | | | An instance of the R4 ResearchSubject resource.
ResearchSubject.assignedArm | String | True | | | The study branch the subject is part of.
ResearchSubject.id | String | False | | | The FHIR ID
ResearchSubject.identifier | Identifier | True | Yes | | Business identifier for research subject in a study.
ResearchSubject.individual | Reference | False | | | The subject's patient ID.
ResearchSubject.extension (also: ParticipantID) | Extension | True | Yes | | The patient's alias for their enrollment in the research study.
ResearchSubject.period | Period | True | | | Start and end of the subject's participation in the study.
ResearchSubject.status | String | False | | | The progression of a study subject through a study. Contains values from the research subject status value set. Mapping might vary from one Epic organization to another.
ResearchSubject.study | Reference | False | | | The study the subject is part of.

Sample request

```
https://hostname/instance/api/FHIR/R4/ResearchSubject/eWizkki6sxC5mUveh-qf06g3
```

Sample response

```
{
  "resourceType": "ResearchSubject",
  "id": "eWizkki6sxC5mUveh-qf06g3",
  "extension": [
    {
      "valueString": "2021PMAAr",
      "url": "http://open.epic.com/FHIR/StructureDefinition/research-subject-participant-Id"
    }
  ],
  "identifier": [
    {
```

```
      "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.2.766582",
      "value": "978"
    }
  ],
  "status": "on-study",
  "period": {
    "start": "2021-03-10"
  },
  "study": {
    "reference": "ResearchStudy/erXhPENfYk8OTVVy9R5bleA3",
    "display": "Pain Management Study"
  },
  "individual": {
    "reference": "Patient/e9I.Nc8WJLGlJVyorSs6SZQ3",
    "display": "Ptlastn, Ptfirstn"
  },
  "assignedArm": "Study Arm B"
}
```


10.1.64 ResearchSubject.Search (R4)

File: site/apis/10-Clinical-General/1-R4/10143-ResearchSubject-Search-R4.html
Epic API ID | 10143
FHIR version | R4
Categories | Clinical General
API type | Industry-Standard
USCDI | No
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | August 2021
HTTP method | GET;POST
URL template | /api/FHIR/R4/ResearchSubject?patient={patient}&status={status}&study={study}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10143

Description

The ResearchSubject resource includes general information about the research subject tracked in Epic, including study, study arm, consent status, and other key items. Epic tracks human beings only for research associations. A Patient resource is traceable to a specific person for our cases.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
patient | String | False | | | The subject's FHIR ID
status | String | True | | | The progression of a study subject through a study.
study | String | True | | | The study FHIR ID

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
date | String | True | | | Start and end of participation
identifier | String | True | | | Business Identifier for research subject in a study

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | BackboneElement | False | Yes | | An array of entries included in this bundle.
Bundle.entry.resource | ResourceContainer | False | | | A container for a single FHIR resource.
Bundle.entry.resource.ResearchSubject | ResearchSubject | False | | | An instance of the R4 ResearchSubject resource.
Bundle.entry.resource.ResearchSubject.assignedArm | String | True | | | The study branch the subject is part of
Bundle.entry.resource.ResearchSubject.id | String | False | | | The ResearchSubject FHIR ID
Bundle.entry.resource.ResearchSubject.identifier | Identifier | True | Yes | | Business identifier for research subject in a study
Bundle.entry.resource.ResearchSubject.individual | Reference | False | | | The subject's patient ID
Bundle.entry.resource.ResearchSubject.period | Period | True | | | Start and end of the subject's participation in the study
Bundle.entry.resource.ResearchSubject.status | String | False | | | The progression of a study subject through a study. Contains values from the research subject status value set. Mapping might vary from one Epic organization to another.
Bundle.entry.resource.ResearchSubject.study | Reference | False | | | The study the subject is part of
OperationOutcome | OperationOutcome | False
OperationOutcome.extension (also: participantID) | Extension | True | Yes | | The patient's alias for their enrollment in the research study.

Sample request

```
https://hostname/instance/api/FHIR/R4/ResearchSubject?patient=e9I.Nc8WJLGlJVyorSs6SZQ3&study=erXhPENfYk8OTVVy9R5bleA3&st
    atus=on-study
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 1,
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/ResearchSubject?patient=e9I.Nc8WJLGlJVyorSs6SZQ3&study=erXhPENfYk8OT
    VVy9R5bleA3&status=on-study"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url": "https://hostname/instance/api/FHIR/R4/ResearchSubject/eWizkki6sxC5mUveh-qf06g3"
        }
      ],
      "fullUrl": "https://hostname/instance/api/FHIR/R4/ResearchSubject/eWizkki6sxC5mUveh-qf06g3",
      "resource": {
        "resourceType": "ResearchSubject",
        "id": "eWizkki6sxC5mUveh-qf06g3",
        "extension": [
          {
            "valueString": "2021PMAAr",
            "url": "http://open.epic.com/FHIR/StructureDefinition/research-subject-participant-Id"
          }
        ],
        "identifier": [
          {
            "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.2.766582",
            "value": "978"
          }
        ],
        "status": "on-study",
        "period": {
          "start": "2021-03-10"
        },
        "study": {
          "reference": "ResearchStudy/erXhPENfYk8OTVVy9R5bleA3",
          "display": "Pain Management Study"
        },
        "individual": {
          "reference": "Patient/e9I.Nc8WJLGlJVyorSs6SZQ3",
          "display": "Ptlastn, Ptfirstn"
        },
        "assignedArm": "B"
      },
      "search": {
        "mode": "match"
      }
    }
  ]
}
```


10.1.65 Substance.Read (R4)

File: site/apis/10-Clinical-General/1-R4/10075-Substance-Read-R4.html
Epic API ID | 10075
FHIR version | R4
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | May 2021
HTTP method | GET
URL template | /api/FHIR/R4/Substance/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10075

Description

This resource allows a client to look up Substance details by providing a specific Substance ID. Read interactions typically begin with a client having previously established a relationship, often through querying for Medications through the search interaction and retrieving the read ID of the substance through the Medication.Ingredient element.

This resource returns details of a substance, such as an electrolyte ion component of a total parenteral nutrition (TPN) medication.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | String | False | | | FHIR identifier for a Substance resource.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Substance | Substance | False | | | An instance of the R4 Substance resource.
Substance.code | CodeableConcept | False | | | A code or set of codes that identify this substance.

Sample request

```
https://hostname/instance/api/FHIR/R4/Substance/evktNAKUKJz-13W.ew.bJCw3
```

Sample response

```
<Substance xmlns="http://hl7.org/fhir">
  <id value="evktNAKUKJz-13W.ew.bJCw3"/>
  <code>
    <coding>
      <system value="urn:oid:1.2.840.114350.1.13.861.1.7.4.698288.571"/>
      <code value="1"/>
      <display value="Sodium"/>
    </coding>
    <text value="Sodium"/>
  </code>
</Substance>
```

10.1.66 Substance.Search (R4)

File: site/apis/10-Clinical-General/1-R4/10076-Substance-Search-R4.html
Epic API ID | 10076
FHIR version | R4
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | May 2021
HTTP method | GET
URL template | /api/FHIR/R4/Substance?_id={_id}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10076

Description

This resource allows a client to look up Substance details by searching by a specific Substance ID. Because only the _id search parameter is currently supported, this Search interaction is similar to Substance.Read.

This resource returns details of a substance, such as an electrolyte ion component of a total parenteral nutrition (TPN) medication.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | BackboneElement | False | Yes | | An array of entries included in this bundle.
Bundle.entry.resource | ResourceContainer | False | | | A container for a single FHIR resource.
Bundle.entry.resource.Substance | Substance | False | | | An instance of the R4 Substance resource.
Bundle.entry.resource.Substance.code | CodeableConcept | False | | | A code or set of codes that identify this substance.

Sample request

```
https://hostname/instance/api/FHIR/R4/Substance?_id=evktNAKUKJz-13W.ew.bJCw3
```

Sample response

```
<Substance xmlns="http://hl7.org/fhir">
  <id value="evktNAKUKJz-13W.ew.bJCw3"/>
  <code>
    <coding>
      <system value="urn:oid:1.2.840.114350.1.13.861.1.7.4.698288.571"/>
      <code value="1"/>
```

```
      <display value="Sodium"/>
    </coding>
    <text value="Sodium"/>
  </code>
</Substance>
```

10.2 Clinical General: STU3

File: site/apis/10-Clinical-General/2-STU3/index.html

* AllergyIntolerance.Create (STU3)
* AllergyIntolerance.Read (STU3)
* AllergyIntolerance.Search (STU3)
* Condition.Create (Problems) (STU3)
* Condition.Read (Encounter Diagnosis, Problems) (STU3)
* Condition.Search (Encounter Diagnosis, Problems) (STU3)
* Flag.Read (Health Concern) (STU3)
* Flag.Read (Infection) (STU3)
* Flag.Search (Health Concern) (STU3)
* Flag.Search (Infection) (STU3)
* Observation.$lastn (Social History) (STU3)
* Observation.Read (Social History) (STU3)
* Observation.Search (Social History) (STU3)
* Procedure.Read (Nursing Intervention) (STU3)
* Procedure.Read (Orders, Surgeries) (STU3)
* Procedure.Read (Restricting Intervention) (STU3)
* Procedure.Read (Surgical History) (STU3)
* Procedure.Search (Nursing Intervention) (STU3)
* Procedure.Search (Orders, Surgeries) (STU3)
* Procedure.Search (Restricting Intervention) (STU3)
* Procedure.Search (Surgical History) (STU3)

10.2.1 AllergyIntolerance.Create (STU3)

File: site/apis/10-Clinical-General/2-STU3/463-AllergyIntolerance-Create-STU3.html
Epic API ID | 463
FHIR version | STU3
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | Epic 2018
HTTP method | POST
URL template | api/FHIR/STU3/AllergyIntolerance
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=463

Description

The AllergyIntolerance resource defines clinical information about a patient's allergic response to a substance. It defines the substance that elicited the response, as well as when the reaction occurred, the severity, and the type of reaction noted.

The AllergyIntolerance resource accommodates creating allergies by adding them to the "holding tank" that drives the EpicCare Reconcile Outside Data Activity. Information in the holding tank is not yet in the patient's chart until reconciled by a clinician. Once a clinician reconciles the allergy, the unconfirmed AllergyIntolerance in the "holding tank" will no longer appear in AllergyIntolerance.Search results. A new AllergyIntolerance resource associated with the reconciled allergy will be available in the normal AllergyIntolerance.Search API

Request parameters

Name | Type | Optional | Array | Cardinality | Description
AllergyIntolerance | AllergyIntolerance | False | | | A single AllergyIntolerance resource.
AllergyIntolerance.clinicalStatus | code | True | | | The current status of the allergy. Only a value of "active" is supported. We do not support creating inactive allergies.

AllergyIntolerance.code | CodeableConcept | False | | | The allergen name and codes.
AllergyIntolerance.code.coding | Coding | False | Yes | | The allergen name and codes.
AllergyIntolerance.criticality | code | True | | | Refers to the risk of clinical harm of a reaction to the substance. Possible values are: unable-to-assess high low
AllergyIntolerance.note | Annotation | True | Yes | | Any comments associated with the allergy.
AllergyIntolerance.note.text | String | False | | | Any comments associated with the allergy.
AllergyIntolerance.onsetDateTime | dateTime | True | | | The date the allergy was identified.
AllergyIntolerance.onsetPeriod | Period | True | | | The date range when the allergy was identified.
AllergyIntolerance.onsetPeriod.end | String | True | | | The date the allergy was resolved.
AllergyIntolerance.onsetPeriod.start | String | True | | | The date the allergy was identified.
AllergyIntolerance.patient | Reference | False | | | Reference to a patient resource for whom the allergy is relevant.
AllergyIntolerance.reaction | BackboneElement | True | Yes | | List of known reactions the patient had to the substance.
AllergyIntolerance.reaction.description | string | True | | | The name of the reaction.
AllergyIntolerance.reaction.manifestation | CodeableConcept | True | Yes | | A single reaction the patient had to the substance.
AllergyIntolerance.reaction.manifestation.coding | Coding | True | Yes | | Coding value
AllergyIntolerance.reaction.manifestation.text | String | True
AllergyIntolerance.type | code | True | | | Possible values include: allergy intolerance
AllergyIntolerance.verificationStatus | code | True | | | How certain it is that the patient has an allergy to the identified substance. Only a value of "unconfirmed" is supported. Allergies must be reconciled into the chart to be considered confirmed.

Sample request

```
<AllergyIntolerance xmlns="http://hl7.org/fhir">
  <clinicalStatus value="active"/>
  <verificationStatus value="unconfirmed"/>
  <type value="allergy"/>
  <category value="medication"/>
  <category value="food"/>
  <category value="environment"/>
  <criticality value="high"/>
  <code>
    <coding>
      <system value="2.16.840.1.113883.6.88"/>
      <code value="18235"/>
    </coding>
    <text value="PEANUT OIL"/>
  </code>
  <patient>
    <reference value="ei3VxpFrMgB.FFYYDTYpsNd00jxZZpsKYvvBIJrv.I7Q3"/>
  </patient>
  <onsetDateTime value="2009-05-04T16:44:39Z"/>
  <note>
    <text value=" More notes "/>
  </note>
  <note>
    <text value="Test comments&#13;&#10;Comments from other org."/>
  </note>
  <reaction>
    <manifestation>
      <coding>
        <system value="2.16.840.1.113883.6.96"/>
        <code value="126485001"/>
      </coding>
      <text value="Hives"/>
    </manifestation>
    <description value="Hives"/>
  </reaction>
</AllergyIntolerance>
```

Sample response

```
<AllergyIntolerance xmlns="http://hl7.org/fhir">
  <id value="e.NZCBNU8HwWk71Gs0ZMlc.M.pgwbPhy1SQHkcLzv4ZJhtGsqhlqTtAWI0uQ19GjsiyMHv1tVhBP9lEXJlycIQg3"/>
  <verificationStatus value="unconfirmed"/>
  <criticality value="high"/>
  <code>
    <coding>
      <system value="http://www.nlm.nih.gov/research/umls/rxnorm"/>
      <code value="18235"/>
      <display value="PEANUT OIL"/>
    </coding>
    <text value="PEANUT OIL"/>
  </code>
  <patient>
    <reference value="https://hostname/instance/api/FHIR/STU3/Patient/eE0Uoh0kTfU6iiX3-pI94vg3"/>
    <display value="Jjs, Allergies"/>
  </patient>
  <assertedDate value="2009-05-04"/>
  <note>
    <text value="Test comments&#13;&#10;Comments from other org. More notes"/>
  </note>
  <reaction>
    <manifestation>
      <coding>
```

```
        <system value="http://snomed.info/sct"/>
        <code value="126485001"/>
      </coding>
      <text value="Hives"/>
    </manifestation>
    <description value="Hives"/>
    <severity value="severe"/>
  </reaction>
</AllergyIntolerance>
```

Change log

Epic Version | Description | Type
August 2023 | AllergyIntolerance.code.system will return URLs in place of an OID for the following code systems: System URL OID RxNorm http://www.nlm.nih.gov/research/umls/rxnorm 2.16.840.1.113883.6.88 | Breaking change
System | URL | OID
RxNorm | http://www.nlm.nih.gov/research/umls/rxnorm | 2.16.840.1.113883.6.88
November 2022 | Code systems for the following code sets now return a URL in place of an OID: ActCode, ICD-10-CM, ICD-10-CA, ICD-10-NL, SNOMED System URL OID ActCode STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode 2.16.840.1.113883.5.4 ICD-10-CM http://hl7.org/fhir/sid/icd-10-cm 2.16.840.1.113883.6.90 ICD-10-CA http://hl7.org/fhir/sid/icd-10-ca 2.16.840.1.113883.6.94 ICD-10-NL http://hl7.org/fhir/sid/icd-10-nl 2.16.840.1.113883.6.3.2 SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 | Breaking change
System | URL | OID
ActCode | STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode | 2.16.840.1.113883.5.4
ICD-10-CM | http://hl7.org/fhir/sid/icd-10-cm | 2.16.840.1.113883.6.90
ICD-10-CA | http://hl7.org/fhir/sid/icd-10-ca | 2.16.840.1.113883.6.94
ICD-10-NL | http://hl7.org/fhir/sid/icd-10-nl | 2.16.840.1.113883.6.3.2
SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96
November 2022 | Epic has updated the behavior of certain modifier elements in FHIR write APIs to ensure the data in the incoming request body matches the data filed to the database and to comply with the HL7 FHIR specification. Previously, the modifier element was ignored in certain situations, and an undocumented default value was saved in its place. Now, if the modifier element is included and is set to an unaccepted value, a fatal error logs, preventing the request from filing to the database. The FHIR request returns an OperationOutcome with the invalid element and what value is accepted for that element. Element Accepted Incoming Value Default Value clinicalStatus active active verificationStatus unconfirmed unconfirmed | Substantive change
Element | Accepted Incoming Value | Default Value
clinicalStatus | active | active
verificationStatus | unconfirmed | unconfirmed

10.2.2 AllergyIntolerance.Read (STU3)

File: site/apis/10-Clinical-General/2-STU3/464-AllergyIntolerance-Read-STU3.html
Epic API ID | 464
FHIR version | STU3
Categories | Clinical General
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Keywords | Allergies
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | Epic 2018
HTTP method | GET
URL template | api/FHIR/STU3/AllergyIntolerance/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=464

Description

The AllergyIntolerance resource defines clinical information about a patient's allergic response to a substance. It defines the substance that elicited the response, as well as when the reaction occurred, the severity, and the type of reaction noted.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | Required | | 1..1 | The AllergyIntolerance FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
AllergyIntolerance (STU3) | AllergyIntolerance | | | | A single AllergyIntolerance resource.
AllergyIntolerance (STU3).assertedDate | dateTime | | | 0..1 | The date the allergy was recorded in Epic.
AllergyIntolerance (STU3).category | code | | Yes | 0..* | The category of the allergy. The following default values are supported, but additional values can be configured by each Epic organization: food medication environment biologic
AllergyIntolerance (STU3).clinicalStatus | code | | | 1..1 | The current status of the allergy. Possible values include: active resolved inactive

AllergyIntolerance (STU3).code | CodeableConcept | | | 1..* | The allergen name and codes.

AllergyIntolerance (STU3).criticality | code | | | 1..1 | Refers to the risk of clinical harm of a reaction to the substance. Possible values are: unable-to-assess high low

AllergyIntolerance (STU3).id | id | | | 1..1 | The AllergyIntolerance FHIR ID.

AllergyIntolerance (STU3).meta | Meta | | | 0..* | The document metadata

AllergyIntolerance (STU3).meta.profile | String | | Yes | 0..1 | Available in Netherlands environments only. This environment shows that this API conforms to the profile connected to the following canonical url: http://nictiz.nl/fhir/StructureDefinition/zib-AllergyIntolerance

AllergyIntolerance (STU3).note | Annotation | | Yes | 0..1 | Additional free text details about the allergy.

AllergyIntolerance (STU3).onsetDateTime | dateTime | | | 0..1 | If an allergy has a noted date, that date is specified here. This element is returned in the following locales dependent on Epic version: In the August 2023 version of Epic and earlier: returned in Dutch locales. In the November 2024 version of Epic and later: returned in all locales.

AllergyIntolerance (STU3).onsetPeriod | Period | | | 0..1 | If an allergy has a noted date, that date or date range for non-specific dates is specified here. This element is returned in the following locales dependent on Epic version: May 2023 and earlier: returned in all locales. August 2023 – August 2024: returned in non-Dutch locales only. November 2024 and later: no longer returned in any locale.

AllergyIntolerance (STU3).patient | Reference | | | 1..1 | Reference to a patient resource for whom the allergy is relevant.

AllergyIntolerance (STU3).reaction | BackboneElement | | Yes | 0..* | List of known reactions the patient had to the substance.

AllergyIntolerance (STU3).reaction.description | string | | | 0..1 | The name of the reaction.

AllergyIntolerance (STU3).reaction.manifestation | CodeableConcept | | Yes | 0..1 | A single reaction the patient had to the substance.

AllergyIntolerance (STU3).recorder | Reference | | | 0..1 | Reference to the practitioner who updated the allergy.

AllergyIntolerance (STU3).type | code | | | 0..1 | Description of the underlying type of the AllergyIntolerance. This is either "Allergy" or "Intolerance".

AllergyIntolerance (STU3).verificationStatus | code | | | 0..1 | If the allergy is on the patient's chart, this element specifies "confirmed." If it is a newly created allergy and still needs to be reconciled, it is "unconfirmed."

Sample request

```
GET https://hostname/instance/api/FHIR/STU3/AllergyIntolerance/eqDAzB9FdPh06vJ6ePbrn7vc0.sbJrCJdSWXPRTaDf5I3
```

Sample response


```
{
  "resourceType": "AllergyIntolerance",
  "id": "eqDAzB9FdPh06vJ6ePbrn7vc0.sbJrCJdSWXPRTaDf5I3",
  "clinicalStatus": "active",
  "verificationStatus": "confirmed",
  "category": [
    "medication"
  ],
  "criticality": "high",
  "code": {
    "coding": [
      {
        "system": "http://www.nlm.nih.gov/research/umls/rxnorm",
        "code": "76469",
        "display": "BACILLUS CALMETTE GUERIN LIVE"
      },
      {
        "system": "http://www.nlm.nih.gov/research/umls/rxnorm",
        "code": "221050"
      },
      {
        "system": "http://www.nlm.nih.gov/research/umls/rxnorm",
        "code": "314513"
      },
      {
        "system": "urn:oid:2.16.840.1.113883.4.9",
        "code": "0BQ62LY3KF"
      },
      {
        "system": "urn:oid:2.16.840.1.113883.4.9",
        "code": "2XQ558L16Z"
      },
      {
        "system": "urn:oid:2.16.840.1.113883.4.9",
        "code": "4CGF0IDN3W"
      },
      {
        "system": "urn:oid:2.16.840.1.113883.4.9",
        "code": "5H5854UBMZ"
      },
      {
        "system": "urn:oid:2.16.840.1.113883.4.9",
        "code": "68H4YH3TKU"
      },
      {
        "system": "urn:oid:2.16.840.1.113883.4.9",
        "code": "8VJE55B0VG"
      }
    ],
    "text": "BACILLUS CALMETTE GUERIN LIVE"
  },
  "patient": {
    "reference": "https://hostname/instance/api/FHIR/STU3/Patient/eiyYG1XjqctfG3K.zQvSlWw3",
    "display": "FHIR, Test Patient"
  },
```

```
  "onsetDateTime": "2024-11-24",
  "assertedDate": "2024-11-24T19:44:38Z",
  "recorder": {
    "reference": "Practitioner/eodo7Bj9dk-LtJSthMwARpw3",
    "type": "Practitioner",
    "display": "Fhirdoc Testdoctor"
  },
  "reaction": [
    {
      "manifestation": [
        {
          "text": "Hives"
        }
      ],
      "description": "Hives"
    }
  ]
}
```


Change log

Epic Version | Description | Type
November 2024 | AllergyIntolerance now returns onsetDateTime. The onsetPeriod element is no longer used. | Breaking change
August 2023 | AllergyIntolerance now returns onsetDateTime in Dutch locales and returns onsetPeriod as a range for all other locales. | Breaking change
August 2023 | AllergyIntolerance.code.system will return URLs in place of an OID for the following code systems: System URL OID RxNorm http://www.nlm.nih.gov/research/umls/rxnorm 2.16.840.1.113883.6.88 SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 | Breaking change
System | URL | OID
RxNorm | http://www.nlm.nih.gov/research/umls/rxnorm | 2.16.840.1.113883.6.88
SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96
November 2022 | AllergyIntolerance no longer returns reaction-level severities for an allergen documented in Epic. Instead, allergen-level severity is now returned as the criticality element. | Breaking change

10.2.3 AllergyIntolerance.Search (STU3)

File: site/apis/10-Clinical-General/2-STU3/465-AllergyIntolerance-Search-STU3.html
Epic API ID | 465
FHIR version | STU3
Categories | Clinical General
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Keywords | Allergies
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | Epic 2018
HTTP method | GET
URL template | /api/FHIR/STU3/AllergyIntolerance?patient={patient}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=465

Description

The AllergyIntolerance resource defines clinical information about a patient's allergic response to a substance. It defines the substance that elicited the response, as well as when the reaction occurred, the severity, and the type of reaction noted. The AllergyIntolerance resource can also accommodate search by ID and by patient, allowing it to return a list of allergies. If a patient has no active allergies, an AllergyIntolerance resource is returned indicating whether the patient's allergies have never been reviewed (are not on file), or if they have been reviewed and it has been determined that they have no known allergies.

This resource can also return unconfirmed AllergyIntolerance resources in the "holding tank" that have been sent from other organizations. Note - once a clinician reconciles an allergy, a new AllergyIntolerance resource associated with the reconciled allergy is included in the normal AllergyIntolerance.Search results, and the unconfirmed AllergyIntolerance in the "holding tank" does not appear in the AllergyIntolerance.Search results.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
_count | number | Optional | Yes | | Number of results per page.
_id | token | Optional | Yes | | FHIR resource IDs for the desired resources. Can be used in combination with _count and _include. All others will be ignored.
_include | string | Optional | Yes | | Used to indicate which referenced resources to include in the search bundle.
patient | reference | Required | Yes | | Reference to a patient resource.

Response elements

Name | Type | Optional | Array | Cardinality | Description
AllergyIntolerance (STU3) | AllergyIntolerance | | | | A single AllergyIntolerance resource.
AllergyIntolerance (STU3).assertedDate | dateTime | | | 0..1 | The date the allergy was recorded in Epic.
AllergyIntolerance (STU3).category | code | | Yes | 0..* | The category of the allergy. The following default values are supported, but additional values can be configured by each Epic organization: food medication environment biologic
AllergyIntolerance (STU3).clinicalStatus | code | | | 1..1 | The current status of the allergy. Possible values include: active resolved inactive
AllergyIntolerance (STU3).code | CodeableConcept | | | 1..* | The allergen name and codes.
AllergyIntolerance (STU3).criticality | code | | | 1..1 | Refers to the risk of clinical harm of a reaction to the substance. Possible values are: unable-to-assess high low
AllergyIntolerance (STU3).id | id | | | 1..1 | The AllergyIntolerance FHIR ID.
AllergyIntolerance (STU3).meta | Meta | | | 0..* | The document metadata
AllergyIntolerance (STU3).meta.profile | String | | Yes | 0..1 | Available in Netherlands environments only. This environment shows that this API conforms to the profile connected to the following canonical url: http://nictiz.nl/fhir/StructureDefinition/zib-AllergyIntolerance
AllergyIntolerance (STU3).note | Annotation | | Yes | 0..1 | Additional free text details about the allergy.
AllergyIntolerance (STU3).onsetDateTime | dateTime | | | 0..1 | If an allergy has a noted date, that date is specified here. This element is returned in the following locales dependent on Epic version: In the August 2023 version of Epic and earlier: returned in Dutch locales. In the November 2024 version of Epic and later: returned in all locales.
AllergyIntolerance (STU3).onsetPeriod | Period | | | 0..1 | If an allergy has a noted date, that date or date range for non-specific dates is specified here. This element is returned in the following locales dependent on Epic version: May 2023 and earlier: returned in all locales. August 2023 – August 2024: returned in non-Dutch locales only. November 2024 and later: no longer returned in any locale.
AllergyIntolerance (STU3).patient | Reference | | | 1..1 | Reference to a patient resource for whom the allergy is relevant.
AllergyIntolerance (STU3).reaction | BackboneElement | | Yes | 0..* | List of known reactions the patient had to the substance.
AllergyIntolerance (STU3).reaction.description | string | | | 0..1 | The name of the reaction.
AllergyIntolerance (STU3).reaction.manifestation | CodeableConcept | | Yes | 0..1 | A single reaction the patient had to the substance.
AllergyIntolerance (STU3).recorder | Reference | | | 0..1 | Reference to the practitioner who updated the allergy.
AllergyIntolerance (STU3).type | code | | | 0..1 | Description of the underlying type of the AllergyIntolerance. This is either "Allergy" or "Intolerance".
AllergyIntolerance (STU3).verificationStatus | code | | | 0..1 | If the allergy is on the patient's chart, this element specifies "confirmed." If it is a newly created allergy and still needs to be reconciled, it is "unconfirmed."

Sample request

```
GET https://hostname/instance/api/FHIR/STU3/AllergyIntolerance?patient=e1KjWgfIPfH21PGMzj3hD.g3
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 3,
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/STU3/AllergyIntolerance?patient=e1KjWgfIPfH21PGMzj3hD.g3"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url": "https://hostname/instance/api/FHIR/STU3/AllergyIntolerance/eg-
    BOctCzGjPpnCNsvANQ3Ltj0xNlS6z3DsGN4QBniYQ3"
        }
      ],
      "fullUrl": "https://hostname/instance/api/FHIR/STU3/AllergyIntolerance/eg-
    BOctCzGjPpnCNsvANQ3Ltj0xNlS6z3DsGN4QBniYQ3",
      "resource": {
        "resourceType": "AllergyIntolerance",
        "id": "eg-BOctCzGjPpnCNsvANQ3Ltj0xNlS6z3DsGN4QBniYQ3",
        "clinicalStatus": "active",
        "verificationStatus": "confirmed",
        "category": [
          "medication"
        ],
        "criticality": "high",
        "code": {
          "coding": [
            {
              "system": "http://www.nlm.nih.gov/research/umls/rxnorm",
              "code": "190521",
              "display": "ABACAVIR"
            },
            {
              "system": "http://www.nlm.nih.gov/research/umls/rxnorm",
              "code": "221052"
            },
            {
              "system": "urn:oid:2.16.840.1.113883.4.9",
              "code": "1PPL256S5A"
            },
            {
              "system": "urn:oid:2.16.840.1.113883.4.9",
              "code": "GZP7A66C3C"
```

```
        },
        {
          "system": "urn:oid:2.16.840.1.113883.4.9",
          "code": "J220T4J9Q2"
        },
        {
          "system": "urn:oid:2.16.840.1.113883.4.9",
          "code": "PMA9EH08PM"
        },
        {
          "system": "urn:oid:2.16.840.1.113883.4.9",
          "code": "RP0T719KX7"
        },
        {
          "system": "urn:oid:2.16.840.1.113883.4.9",
          "code": "WR2TIP26VS"
        },
        {
          "system": "http://snomed.info/sct",
          "code": "387005008",
          "display": "Abacavir (substance)"
        },
        {
          "system": "http://snomed.info/sct",
          "code": "412074007",
          "display": "Abacavir sulfate (substance)"
        },
        {
          "system": "http://snomed.info/sct",
          "code": "1306846000",
          "display": "Abacavir hydrochloride (substance)"
        }
      ],
      "text": "ABACAVIR"
    },
    "patient": {
      "reference": "https://hostname/instance/api/FHIR/STU3/Patient/e1KjWgfIPfH21PGMzj3hD.g3",
      "display": "FHIR, Test Patient"
    },
    "onsetDateTime": "2025-09-11",
    "assertedDate": "2025-09-11T18:27:29Z",
    "reaction": [
      {
        "manifestation": [
          {
            "text": "Anxiety"
          }
        ],
        "description": "Anxiety"
      }
    ]
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url":
"https://hostname/instance/api/FHIR/STU3/AllergyIntolerance/e5EZ4uFzMLbuvUMVvpG52UMkVPm5cDfNbn2pZoeXnS6c3"
    }
  ],
  "fullUrl":
"https://hostname/instance/api/FHIR/STU3/AllergyIntolerance/e5EZ4uFzMLbuvUMVvpG52UMkVPm5cDfNbn2pZoeXnS6c3",
  "resource": {
    "resourceType": "AllergyIntolerance",
    "id": "e5EZ4uFzMLbuvUMVvpG52UMkVPm5cDfNbn2pZoeXnS6c3",
    "clinicalStatus": "active",
    "verificationStatus": "confirmed",
    "category": [
      "medication"
    ],
    "criticality": "high",
    "code": {
      "coding": [
        {
          "system": "http://www.nlm.nih.gov/research/umls/rxnorm",
          "code": "1310550",
          "display": "BUTANE"
        },
        {
          "system": "urn:oid:2.16.840.1.113883.4.9",
          "code": "6LV4FOR43R"
        },
        {
          "system": "http://snomed.info/sct",
          "code": "259139006",
```

```
          "display": "Butane (substance)"
        },
        {
          "system": "http://snomed.info/sct",
          "code": "300000003",
          "display": "n-Butane (substance)"
        }
      ],
      "text": "BUTANE"
    },
    "patient": {
      "reference": "https://hostname/instance/api/FHIR/STU3/Patient/e1KjWgfIPfH21PGMzj3hD.g3",
      "display": "FHIR, Test Patient"
    },
    "onsetDateTime": "2025-09-19",
    "assertedDate": "2025-09-19T13:46:22Z",
    "reaction": [
      {
        "manifestation": [
          {
            "text": "Cough"
          }
        ],
        "description": "Cough"
      }
    ]
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url":
"https://hostname/instance/api/FHIR/STU3/AllergyIntolerance/eansrKZFkQX7hTqyXuprf.CO2vF5WCYZUrT2WgBGMnLU3"
    }
  ],
  "fullUrl":
"https://hostname/instance/api/FHIR/STU3/AllergyIntolerance/eansrKZFkQX7hTqyXuprf.CO2vF5WCYZUrT2WgBGMnLU3",
  "resource": {
    "resourceType": "AllergyIntolerance",
    "id": "eansrKZFkQX7hTqyXuprf.CO2vF5WCYZUrT2WgBGMnLU3",
    "clinicalStatus": "active",
    "verificationStatus": "confirmed",
    "category": [
      "medication"
    ],
    "criticality": "low",
    "code": {
      "coding": [
        {
          "system": "http://www.nlm.nih.gov/research/umls/rxnorm",
          "code": "611",
          "display": "ASPIRIN"
        },
        {
          "system": "http://www.nlm.nih.gov/research/umls/rxnorm",
          "code": "1191"
        },
        {
          "system": "http://www.nlm.nih.gov/research/umls/rxnorm",
          "code": "314293"
        },
        {
          "system": "urn:oid:2.16.840.1.113883.4.9",
          "code": "2JJ274J145"
        },
        {
          "system": "urn:oid:2.16.840.1.113883.4.9",
          "code": "E33TS05V6B"
        },
        {
          "system": "urn:oid:2.16.840.1.113883.4.9",
          "code": "E62HT5S2E9"
        },
        {
          "system": "urn:oid:2.16.840.1.113883.4.9",
          "code": "R16CO5Y76E"
        },
        {
          "system": "urn:oid:2.16.840.1.113883.4.9",
          "code": "WOD7W0DGZS"
        },
        {
          "system": "urn:oid:2.16.840.1.113883.4.9",
          "code": "XAN4V337CI"
        },
```

```
            {
              "system": "http://snomed.info/sct",
              "code": "25796002",
              "display": "Aluminium aspirin (substance)"
            },
            {
              "system": "http://snomed.info/sct",
              "code": "387458008",
              "display": "Aspirin (substance)"
            },
            {
              "system": "http://snomed.info/sct",
              "code": "735135008",
              "display": "Aspirin DL-lysine (substance)"
            }
          ],
          "text": "ASPIRIN"
        },
        "patient": {
          "reference": "https://hostname/instance/api/FHIR/STU3/Patient/e1KjWgfIPfH21PGMzj3hD.g3",
          "display": "FHIR, Test Patient"
        },
        "onsetDateTime": "2025-10-09",
        "assertedDate": "2025-10-09T20:53:53Z",
        "recorder": {
          "reference": "Practitioner/eodo7Bj9dk-LtJSthMwARpw3",
          "type": "Practitioner",
          "display": "Fhirdoc Testdoctor"
        },
        "reaction": [
          {
            "manifestation": [
              {
                "text": "Drowsy"
              }
            ],
            "description": "Drowsy"
          }
        ]
      },
      "search": {
        "mode": "match"
      }
    }
  ]
}
```


Change log

Epic Version | Description | Type
November 2024 | AllergyIntolerance now returns onsetDateTime. The onsetPeriod element is no longer used. | Breaking change
August 2023 | AllergyIntolerance now returns onsetDateTime in Dutch locales and returns onsetPeriod as a range for all other locales. | Breaking change
August 2023 | AllergyIntolerance.code.system will return URLs in place of an OID for the following code systems: /p> System URL OID RxNorm http://www.nlm.nih.gov/research/umls/rxnorm 2.16.840.1.113883.6.88 SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 | Breaking change
System | URL | OID
RxNorm | http://www.nlm.nih.gov/research/umls/rxnorm | 2.16.840.1.113883.6.88
SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96
November 2022 | AllergyIntolerance no longer returns reaction-level severities for an allergen documented in Epic. Instead, allergen-level severity is now returned as the criticality element. | Breaking change

10.2.4 Condition.Create (Problems) (STU3)

File: site/apis/10-Clinical-General/2-STU3/467-Condition-Create-Problems-STU3.html
Epic API ID | 467
FHIR version | STU3
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Keywords | Health Concerns;Problems;Problem List
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | Epic 2017
HTTP method | POST
URL template | api/FHIR/STU3/Condition
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=467

Description

This web service adds problems to a patient's problem list. A clinician must reconcile the problem in Epic before it is officially accepted into the patient Chart.

The Condition resource accommodates creating problems by adding them to the "holding tank" that drives the EpicCare Reconcile Outside Data Activity. Information in the holding tank is not yet in the patient's chart until reconciled by a clinician. Once a clinician reconciles the problem, the confirmed problem appears in Condition.Search results with a new FHIR ID. The pre-reconciled condition continues to be available via Read only and remains at a status of "unconfirmed" so you should not use the pre-reconciled Condition FHIR ID as a source of truth.

Remarks

None

Request parameters

Name | Type | Optional | Array | Cardinality | Description

Condition | Condition | False

Condition.clinicalStatus | String | True | | | The clinical status of the condition. Only values of "active" or "resolved" are supported. We do not support creating inactive problems.

Condition.code | CodeableConcept | False | | | The identifier for the condition.

Condition.code.coding | Coding | False | Yes | | Coding value

Condition.code.coding.code | String | You must include either this element or comments in the text element. | | | The code for the condition. This can be an ICD-10 or SNOMED code.

Condition.code.text | String | You must include either this element or the code in the code element. | | | The name of the problem.

Condition.note | Annotation | False | Yes | | The comments.

Condition.note.text | String | False

Condition.onsetDateTime | String | You can use either this element or onsetPeriod start. | | | The date the problem began.

Condition.onsetPeriod | Period | False | | | The start and end dates for a condition.

Condition.onsetPeriod.end | String | True | | | When the condition was resolved.

Condition.onsetPeriod.start | String | You can use either this element or onsetDateTime. | | | The date and time when the condition began. You can use either this element or onsetDateTime.

Condition.subject | Reference | False | | | The patient who has the condition.

Condition.verificationStatus | String | True | | | The verification status that supports the clinical status of the condition. Only a value of "provisional" is supported. Problems must be reconciled into the chart before they can be verified.

Sample request

```
{
  "resourceType": "Condition",
  "clinicalStatus": "active",
  "verificationStatus": "provisional",
  "assertedDate": "1998-01-10",
  "category": [
    {
      "text": "Problem List Item",
      "coding": [
        {
          "system": "http://hl7.org/fhir/condition-category",
          "code": "problem-list-item",
          "display": "Problem List Item"
        }
      ]
    }
  ],
  "severity": {
    "text": "High"
  },
  "code": {
    "text": "Black lung disease",
    "coding": [
      {
        "system": "http://hl7.org/fhir/sid/icd-10-cm",
        "code": "J60"
      },
      {
        "system": "http://snomed.info/sct",
        "code": "29422001"
      }
    ]
  },
  "subject": {
    "reference": "https://hostname/instance/api/FHIR/STU3/Patient/e3YS8ClDzw13DmI18v1fpRQ3",
    "display": "Jjs, Conditions A"
  },
  "note": [
    {
      "text": "This is the text of the note."
    }
  ]
}
```

Sample response

```
HTTP/1.1 201
Cache-Control: no-cache
Pragma: no-cache
Content-Type: application/fhir+json; charset=utf-8
Expires: -1
Location: https://hostname/instance/api/FHIR/STU3/Condition/eogxwIG1txVZNCM3dHEOvFowQX2QFI0kgC8kbjCEIA13gv1KXV-
    KkUMsNeEHnym4NflRgODZ7mPOUPfun4KNHiQ3
Server: Microsoft-IIS/8.5
X-AspNet-Version: 4.0.30319
X-Powered-By: ASP.NET
Date: Wed, 10 Jan 2018 20:25:11 GMT
Content-Length: 0
```

Change log

Epic Version | Description | Type
November 2022 | Code systems for the following code sets now return a URL in place of an OID: ActCode, ICD-10-CM, ICD-10-CA, ICD-10-NL, SNOMED System URL OID ActCode STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode 2.16.840.1.113883.5.4 ICD-10-CM http://hl7.org/fhir/sid/icd-10-cm 2.16.840.1.113883.6.90 ICD-10-CA http://hl7.org/fhir/sid/icd-10-ca 2.16.840.1.113883.6.94 ICD-10-NL http://hl7.org/fhir/sid/icd-10-nl 2.16.840.1.113883.6.3.2 SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 | Breaking change
System | URL | OID
ActCode | STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode | 2.16.840.1.113883.5.4
ICD-10-CM | http://hl7.org/fhir/sid/icd-10-cm | 2.16.840.1.113883.6.90
ICD-10-CA | http://hl7.org/fhir/sid/icd-10-ca | 2.16.840.1.113883.6.94
ICD-10-NL | http://hl7.org/fhir/sid/icd-10-nl | 2.16.840.1.113883.6.3.2
SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96
November 2022 | Epic has updated the behavior of certain modifier elements in FHIR write APIs to ensure the data in the incoming request body matches the data filed to the database and to comply with the HL7 FHIR specification. Previously, the modifier element was ignored in certain situations, and an undocumented default value was saved in its place. Now, if the modifier element is included and is set to an unaccepted value, a fatal error logs, preventing the request from filing to the database. The FHIR request returns an OperationOutcome with the invalid element and what value is accepted for that element. Element Accepted Incoming Value Default Value clinicalStatus active active verificationStatus provisional provisional | Substantive change
Element | Accepted Incoming Value | Default Value
clinicalStatus | active | active
verificationStatus | provisional | provisional

10.2.5 Condition.Read (Encounter Diagnosis, Problems) (STU3)

File: site/apis/10-Clinical-General/2-STU3/468-Condition-Read-Encounter-Diagnosis-Problems-STU3.html
Epic API ID | 468
FHIR version | STU3
Categories | Clinical General
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Keywords | Health Concerns;Problems;Problem List
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | Epic 2017
HTTP method | GET
URL template | api/FHIR/STU3/Condition/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=468

Description

This web service will retrieve condition (diagnosis) data from a patient's chart. This includes any data found in the patient's problem list and diagnoses across all encounters. This resource can be queried by a combination of patient ID and status.

Note that this resource retrieves only data stored in problem list records. As a result, medical history data documented outside of a patient's problem list isn't available to applications using this service unless that data is retrieved using another method.

Starting in February 2019, this resource also returns encounter-level diagnoses.

Starting in February 2020, this resource also returns health-concern category data, which includes certain problems in the problem list and the patient's health status.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | String | False | | | The Condition FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description

Condition | Condition | False

Condition.abatementDateTime | String | True | | | Starting in the February 2024 version of Epic, the date the problem was resolved.

Condition.abatementPeriod | Period | True | | | In the November 2023 and earlier versions of Epic, the date the condition was resolved.

Condition.assertedDate | String | True | | | The date the condition was noted.

Condition.category | CodeableConcept | True | Yes | | For encounter-level diagnoses, this is "encounter-diagnosis". For problems, this is "problem-list-item". For health concerns, this is "health-concern". Starting in the May 2021 version of Epic, can also include additional category codes based on diagnosis groupers defined by the Epic organization.

Condition.clinicalStatus | String | Optional for encounter-level diagnoses. | | | The status of the problem. Can be active, inactive, or resolved.

Condition.code | CodeableConcept | False | | | External codings for the condition. If a list of codes is sent, they're listed in this element. Otherwise, this element uses the problem name from the name element. Starting in the February 2024 version of Epic or with special updates to November 2023, August 2023, or May 2023 version of Epic, each code also sends its display name and the provider-entered problem name is sent if available.

Condition.context | Reference | True | | | Reference to an Encounter resource.

Condition.id | String | False | | | The FHIR ID for this Condition.

Condition.meta | Meta | True | | | The document metadata.

Condition.meta.profile | String | True | | | Available in Netherlands environments only. This environment shows that this API conforms to the profile connected to the following canonical url: http://nictiz.nl/fhir/StructureDefinition/zib-Problem

Condition.note | Annotation | True | Yes | | Free text comments for the note. Can be an Overview note or, starting in the February 2025 version of Epic, an Assessment & Plan note.

Condition.note.extension (also: annotationType) | Extension | False | Yes | | This extension is returned starting in the following Epic versions, dependent on note type: May 2025: Problem overview notes February 2025: Assessment & Plan notes Always contains a url of http://hl7.org/fhir/StructureDefinition/annotationType and one of the following valueCodeableConcept LOINC codings: Assessment & Plan notes: LOINC code 51847-2 (Evaluation + Plan note) Problem overview notes: LOINC code 68608-9 (Summary note)

Condition.note.authorReference | Reference | False | | | Who wrote the note. This element is returned starting in the following Epic versions, dependent on note type: May 2025: Problem overview notes February 2025: Assessment & Plan notes

Condition.note.extension (also: data-conversion-warning) | Extension | False | Yes | | This extension is returned starting in the following Epic versions, dependent on note type: May 2025: Problem overview notes February 2025: Assessment & Plan notes One or more of the following warnings can be returned depending on RTF to plan text conversion: Reason for warning Warning text Formatting was lost due to unicode characters or tables being present in the original note. Formatting of this note is different from the original. The original note was RTF. Formatting of this note might be different from the original. There were one or more images in the original note. Images from the original note were not included. There were one or more media attachments in the original note. Attached media from the original note were not included.

Reason for warning | Warning text

Formatting was lost due to unicode characters or tables being present in the original note. | Formatting of this note is different from the original.

The original note was RTF. | Formatting of this note might be different from the original.

There were one or more images in the original note. | Images from the original note were not included.

There were one or more media attachments in the original note. | Attached media from the original note were not included.

Condition.note.text | String | False | | | The content of the note. In the February 2025 version of Epic and earlier versions, this note text can include prepended data conversion warnings similar to those documented in the “extension (data-conversion-warning)” response element.

Condition.note.time | String | False | | | The time that the note was signed. This element is returned starting in the following Epic versions, dependent on note type: May 2025: Problem overview notes February 2025: Assessment & Plan notes

Condition.onsetDateTime | String | True | | | Starting in the February 2024 version of Epic, if a condition has a noted date, it appears here.

Condition.onsetPeriod | Period | True | | | In the November 2023 and earlier versions of Epic, if a condition has a noted date, it appears here.

Condition.severity | CodeableConcept | True | | | The priority of the problem.

Condition.subject | Reference | False | | | Reference to the Patient associated with the Condition.

Condition.verificationStatus | String | For organizations in the Netherlands, this value can be omitted for problems on the problem list | | | If this is on the patient's local chart, the value is "confirmed". If it is outside data, it is "unconfirmed".

Sample request

```
https://hostname/instance/api/FHIR/STU3/Condition/eCnLY4-dmArLHMh-h27PWTJ5ZInvHkej8DD7gLkd4rlg3
```

Sample response


```
{
  "resourceType": "Condition",
  "id": "eCnLY4-dmArLHMh-h27PWTJ5ZInvHkej8DD7gLkd4rlg3",
  "clinicalStatus": "active",
  "verificationStatus": "confirmed",
  "category": [
    {
      "coding": [
        {
          "system": "http://hl7.org/fhir/condition-category",
          "code": "problem-list-item",
          "display": "Problem List Item"
        }
      ],
      "text": "Problem List Item"
    }
  ],
  "code": {
    "coding": [
      {
        "system": "http://hl7.org/fhir/sid/icd-10-cm",
        "code": "L70.8",
        "display": "Other acne"
      },
      {
```

```
        "system": "http://snomed.info/sct",
        "code": "11381005",
        "display": "Acne"
      }
    ],
    "text": "Other acne"
  },
  "subject": {
    "reference": "https://vs-icx18-25.epic.com/Interconnect-FHIR-Current-Stage1-Primary-
    Unsecured/api/FHIR/STU3/Patient/eviuA2ieGBzz-m-3VB6H3HA3",
    "display": "Allergies, IPS"
  },
  "onsetDateTime": "2024-11-05",
  "assertedDate": "2024-11-05",
  "note": [
    {
      "extension": [
        {
          "valueString": "Formatting of this note might be different from the original.",
          "url": "http://open.epic.com/FHIR/StructureDefinition/extension/data-conversion-warning"
        },
        {
          "valueCodeableConcept": {
            "coding": [
              {
                "system": "http://loinc.org",
                "code": "51847-2",
                "display": "Evaluation + Plan note"
              }
            ],
            "text": "Assessment & Plan"
          },
          "url": "http://hl7.org/fhir/StructureDefinition/annotationType"
        }
      ],
      "authorReference": {
        "reference": "https://vs-icx18-25.epic.com/Interconnect-FHIR-Current-Stage1-Primary-
    Unsecured/api/FHIR/STU3/Practitioner/erHkfsagip4sPeSc22fVgrg3",
        "display": "Grace Practitioner, MD"
      },
      "time": "2024-11-05T21:53:54Z",
      "text": "Assessment & Plan Note\r\n"
    }
  ]
}
```


Change log

Epic Version | Description | Type

February 2024 | The USCDIv3 specification for Condition will move the Condition's start date and time from onsetPeriod to onsetDatetime and end date and time from abatementPeriod to abatementDatetime fields. This could impact the evaluation of a Condition if a Condition consumer is not prepared for the transition. | Breaking change

May 2023 (Special Updates to February 2023) | Code systems in the Condition.category element now return a URL in place of an OID. Typical code systems include: System URL OID SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 LOINC http://loinc.org 2.16.840.1.113883.6.1 | Breaking change

System | URL | OID

SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96

LOINC | http://loinc.org | 2.16.840.1.113883.6.1

November 2022 | Code systems in the Condition.code element for the following code sets now return a URL in place of an OID: ActCode, ICD-10-CM, ICD-10-CA, ICD-10-NL, SNOMED System URL OID ActCode STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode 2.16.840.1.113883.5.4 ICD-10-CM http://hl7.org/fhir/sid/icd-10-cm 2.16.840.1.113883.6.90 ICD-10-CA http://hl7.org/fhir/sid/icd-10-ca 2.16.840.1.113883.6.94 ICD-10-NL http://hl7.org/fhir/sid/icd-10-nl 2.16.840.1.113883.6.3.2 SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 | Breaking change

System | URL | OID

ActCode | STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode | 2.16.840.1.113883.5.4

ICD-10-CM | http://hl7.org/fhir/sid/icd-10-cm | 2.16.840.1.113883.6.90

ICD-10-CA | http://hl7.org/fhir/sid/icd-10-ca | 2.16.840.1.113883.6.94

ICD-10-NL | http://hl7.org/fhir/sid/icd-10-nl | 2.16.840.1.113883.6.3.2

SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96

10.2.6 Condition.Search (Encounter Diagnosis, Problems) (STU3)

File: site/apis/10-Clinical-General/2-STU3/469-Condition-Search-Encounter-Diagnosis-Problems-STU3.html

Epic API ID | 469

FHIR version | STU3

Categories | Clinical General

API type | USCDI

A la carte | Industry-Standard Level 1 API

USCDI | Yes

Keywords | Health Concerns;Problems;Problem List

Available in inpatient and other contexts | Yes

ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base

ONC certification versions | Epic 2017
HTTP method | GET
URL template |
/api/FHIR/STU3/Condition?subject={subject}&patient={patient}&clinical-status={clinical-status}&category={category}&encounter={encounter}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=469

Description

This web service will retrieve condition (diagnosis) data from a patient's chart. This includes any data found in the patient's problem list across all encounters. This resource can be queried by a combination of patient ID and status.

Note that this resource retrieves only data stored in problem list (LPL) records. As a result, medical history data documented outside of a patient's problem list isn't available to applications using this service unless that data is retrieved using another method.

This resource does not return unconfirmed Condition resources in the "holding tank" that drives the EpicCare Reconcile Outside Data Activity. Note - once a clinician reconciles an problem, a new Condition resource associated with the reconciled problem will be available in the normal Condition.Search results.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
category | String | True | | | This is "problem-list-item", "encounter-diagnosis", or "health-concern". The default is to return problem list conditions.
clinical-status | String | True | | | Status of the condition FHIR Status Epic Status active active conditions resolved resolved conditions inactive deleted conditions In non-Netherlands environments, if no status is specified, all active (confirmed) conditions are returned. In Netherlands environments, if no status is specified, all conditions (active, resolved, and inactive) are returned. *Note: Any FHIR Status not listed in the above table is not returned.
FHIR Status | Epic Status
active | active conditions
resolved | resolved conditions
inactive | deleted conditions
encounter | String | True | | | A reference to an Encounter resource that the diagnosis is asserted. If not provided, only problem list conditions will be returned.
patient | String | Either patient or subject must be specified. If both are specified, they must reference the same patient. | | | The Patient FHIR resource ID.
subject | String | Either patient or subject must be specified. If both are specified, they must reference the same patient. | | | Subject of the Condition.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | Entry | False | Yes | | A single entry in the resource bundle.
Bundle.entry.resource | ResourceContainer | False | | | A single resource returned as part of this bundle.
Bundle.entry.resource.Condition | Condition | False
Bundle.entry.resource.Condition.abatementDateTime | String | True | | | Starting in the February 2024 version of Epic, the date the problem was resolved.
Bundle.entry.resource.Condition.abatementPeriod | Period | True | | | In the November 2023 and earlier versions of Epic, the date the problem was resolved.
Bundle.entry.resource.Condition.assertedDate | String | True | | | Date the problem was noted.
Bundle.entry.resource.Condition.category | CodeableConcept | True | Yes | | For encounter-level diagnoses, this is "encounter-diagnosis". For problems, this is "problem-list-item". For health concerns, this is "health-concern". Starting in the May 2021 version of Epic, can also include additional category codes based on diagnosis groupers defined by the Epic organization.
Bundle.entry.resource.Condition.clinicalStatus | String | Optional for encounter-level diagnoses. | | | The status of the problem. Can be active, inactive, or resolved.
Bundle.entry.resource.Condition.code | CodeableConcept | False | | | The external codes associated with the problem. If a list of codes is sent, they're listed in this element. Otherwise, this element uses the problem name from the name element. Starting in the February 2024 version of Epic or with special updates to November 2023, August 2023, or May 2023 version of Epic, each code also sends its display name and the provider-entered problem name is sent if available.
Bundle.entry.resource.Condition.id | String | False
Bundle.entry.resource.Condition.meta | Meta | True | | | The document metadata.
Bundle.entry.resource.Condition.meta.profile | String | True | | | Available in Netherlands environments only. This environment shows that this API conforms to the profile connected to the following canonical url: http://nictiz.nl/fhir/StructureDefinition/zib-Problem
Bundle.entry.resource.Condition.note | Annotation | True | Yes | | Free text comments for the note. Can be an Overview note or, starting in the February 2025 version of Epic, an Assessment & Plan note.
Bundle.entry.resource.Condition.note.extension (also: annotationType) | Extension | False | Yes | | This extension is returned starting in the following Epic versions, dependent on note type: May 2025: Problem overview notes February 2025: Assessment & Plan notes Always contains a url of http://hl7.org/fhir/StructureDefinition/annotationType and one of the following valueCodeableConcept LOINC codings: Assessment & Plan notes: LOINC code 51847-2 (Evaluation + Plan note) Problem overview notes: LOINC code 68608-9 (Summary note)
Bundle.entry.resource.Condition.note.authorReference | Reference | False | | | Who wrote the note. This element is returned starting in the following Epic versions, dependent on note type: May 2025: Problem overview notes February 2025: Assessment & Plan notes
Bundle.entry.resource.Condition.note.extension (also: data-conversion-warning) | Extension | False | Yes | | This extension is returned starting in the following Epic versions, dependent on note type: May 2025: Problem overview notes February 2025: Assessment & Plan notes One or more of the following warnings can be returned depending on RTF to plan text conversion: Reason for warning Warning text Formatting was lost due to unicode characters or tables being present in the original note. Formatting of this note is different from the original. The original note was RTF. Formatting of

this note might be different from the original. There were one or more images in the original note. Images from the original note were not included. There were one or more media attachments in the original note. Attached media from the original note were not included.

Reason for warning | Warning text

Formatting was lost due to unicode characters or tables being present in the original note. | Formatting of this note is different from the original.

The original note was RTF. | Formatting of this note might be different from the original.

There were one or more images in the original note. | Images from the original note were not included.

There were one or more media attachments in the original note. | Attached media from the original note were not included.

Bundle.entry.resource.Condition.note.text | String | False | | | The content of the note. In the February 2025 version of Epic and earlier versions, this note text can include prepended data conversion warnings similar to those documented in the “extension (data-conversion-warning)” response element.

Bundle.entry.resource.Condition.note.time | String | False | | | The time that the note was signed. This element is returned starting in the following Epic versions, dependent on note type: May 2025: Problem overview notes February 2025: Assessment & Plan notes

Bundle.entry.resource.Condition.onsetDateTime | String | True | | | Starting in the February 2024 version of Epic, if a condition has a noted date, it appears here.

Bundle.entry.resource.Condition.onsetPeriod | Period | True | | | In the November 2023 and earlier versions of Epic, if a condition has a noted date, it appears here.

Bundle.entry.resource.Condition.severity | CodeableConcept | True | | | The priority of the problem.

Bundle.entry.resource.Condition.subject | Reference | False | | | Reference to the Patient associated with the Condition.

Bundle.entry.resource.Condition.verificationStatus | String | For organizations in the Netherlands, this value can be omitted for problems on the problem list. | | | If this is on the patient's local chart, the value is "confirmed". If it is outside data, it is "unconfirmed". If it is deleted, it is "entered-in-error".

Sample request

```
https://hostname/instance/api/FHIR/STU3/Condition?patient=eviuA2ieGBzz-m-3VB6H3HA3
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 2,
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/STU3/Condition?Patient=eviuA2ieGBzz-m-3VB6H3HA3"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url": "https://hostname/instance/api/FHIR/STU3/Condition/eCnLY4-dmArLHMh-h27PWTJ5ZInvHkej8DD7gLkd4rlg3"
        }
      ],
      "fullUrl": "https://hostname/instance/api/FHIR/STU3/Condition/eCnLY4-dmArLHMh-h27PWTJ5ZInvHkej8DD7gLkd4rlg3",
      "resource": {
        "resourceType": "Condition",
        "id": "eCnLY4-dmArLHMh-h27PWTJ5ZInvHkej8DD7gLkd4rlg3",
        "clinicalStatus": "active",
        "verificationStatus": "confirmed",
        "category": [
          {
            "coding": [
              {
                "system": "http://hl7.org/fhir/condition-category",
                "code": "problem-list-item",
                "display": "Problem List Item"
              }
            ],
            "text": "Problem List Item"
          }
        ],
        "code": {
          "coding": [
            {
              "system": "http://hl7.org/fhir/sid/icd-10-cm",
              "code": "L70.8",
              "display": "Other acne"
            },
            {
              "system": "http://snomed.info/sct",
              "code": "11381005",
              "display": "Acne"
            }
          ],
          "text": "Other acne"
        },
        "subject": {
          "reference": "https://hostname/instance/api/FHIR/STU3/Patient/eviuA2ieGBzz-m-3VB6H3HA3",
          "display": "Allergies, IPS"
        },
        "onsetDateTime": "2024-11-05",
        "assertedDate": "2024-11-05",
```

```
    "note": [
      {
        "extension": [
          {
            "valueString": "Formatting of this note might be different from the original.",
            "url": "http://open.epic.com/FHIR/StructureDefinition/extension/data-conversion-warning"
          },
          {
            "valueCodeableConcept": {
              "coding": [
                {
                  "system": "http://loinc.org",
                  "code": "51847-2",
                  "display": "Evaluation + Plan note"
                }
              ],
              "text": "Assessment & Plan"
            },
            "url": "http://hl7.org/fhir/StructureDefinition/annotationType"
          }
        ],
        "authorReference": {
          "reference": "https://hostname/instance/api/FHIR/STU3/Practitioner/erHkfsagip4sPeSc22fVgrg3",
          "display": "Grace Practitioner, MD"
        },
        "time": "2024-11-05T21:53:54Z",
        "text": "Assessment & Plan Note\r\n"
      }
    ]
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/STU3/Condition/eOehPddWAhK6IiU-t5zKD4q6FWpPmZWHicXuxueVMGYs3"
    }
  ],
  "fullUrl": "https://hostname/instance/api/FHIR/STU3/Condition/eOehPddWAhK6IiU-t5zKD4q6FWpPmZWHicXuxueVMGYs3",
  "resource": {
    "resourceType": "Condition",
    "id": "eOehPddWAhK6IiU-t5zKD4q6FWpPmZWHicXuxueVMGYs3",
    "clinicalStatus": "active",
    "verificationStatus": "confirmed",
    "category": [
      {
        "coding": [
          {
            "system": "http://hl7.org/fhir/condition-category",
            "code": "problem-list-item",
            "display": "Problem List Item"
          }
        ],
        "text": "Problem List Item"
      }
    ],
    "code": {
      "coding": [
        {
          "system": "http://hl7.org/fhir/sid/icd-10-cm",
          "code": "M25.421",
          "display": "Effusion, right elbow"
        },
        {
          "system": "http://hl7.org/fhir/sid/icd-10-cm",
          "code": "M25.422",
          "display": "Effusion, left elbow"
        },
        {
          "system": "http://snomed.info/sct",
          "code": "15725321000119101",
          "display": "Bilateral Effusion Of Elbow Joints"
        }
      ],
      "text": "Effusion of both elbows"
    },
    "subject": {
      "reference": "https://hostname/instance/api/FHIR/STU3/Patient/eviuA2ieGBzz-m-3VB6H3HA3",
      "display": "Allergies, IPS"
    },
    "onsetDateTime": "2024-11-05",
    "assertedDate": "2024-11-05",
    "note": [
      {
        "text": "Overview note\r\n"
      }
```

```
        ]
      },
      "search": {
        "mode": "match"
      }
    }
  ]
}
```

Change log

Epic Version | Description | Type

February 2024 | The USCDIv3 specification for Condition will move the Condition's start date and time from onsetPeriod to onsetDatetime and end date and time from abatementPeriod to abatementDatetime fields. This could impact the evaluation of a Condition if a Condition consumer is not prepared for the transition. | Breaking change

May 2023 (Special Updates to February 2023) | Code systems in the Condition.category element now return a URL in place of an OID. Typical code systems include: System URL OID SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 LOINC http://loinc.org 2.16.840.1.113883.6.1 | Breaking change

System | URL | OID

SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96

LOINC | http://loinc.org | 2.16.840.1.113883.6.1

November 2022 | Code systems in the Condition.code element for the following code sets now return a URL in place of an OID: ActCode, ICD-10-CM, ICD-10-CA, ICD-10-NL, SNOMED System URL OID ActCode STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode 2.16.840.1.113883.5.4 ICD-10-CM http://hl7.org/fhir/sid/icd-10-cm 2.16.840.1.113883.6.90 ICD-10-CA http://hl7.org/fhir/sid/icd-10-ca 2.16.840.1.113883.6.94 ICD-10-NL http://hl7.org/fhir/sid/icd-10-nl 2.16.840.1.113883.6.3.2 SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 | Breaking change

System | URL | OID

ActCode | STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode | 2.16.840.1.113883.5.4

ICD-10-CM | http://hl7.org/fhir/sid/icd-10-cm | 2.16.840.1.113883.6.90

ICD-10-CA | http://hl7.org/fhir/sid/icd-10-ca | 2.16.840.1.113883.6.94

ICD-10-NL | http://hl7.org/fhir/sid/icd-10-nl | 2.16.840.1.113883.6.3.2

SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96

10.2.7 Flag.Read (Health Concern) (STU3)

File: site/apis/10-Clinical-General/2-STU3/10205-Flag-Read-Health-Concern-STU3.html

Epic API ID | 10205

FHIR version | STU3

Categories | Clinical General

API type | Industry-Standard

USCDI | No

Available in inpatient and other contexts | Yes

HTTP method | GET

URL template | /api/FHIR/STU3/Flag/{ID}

Supports OAuth 2.0 | True

Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients

Performs scope validation | False

Source | https://fhir.epic.com/Specifications?api=10205

Description

Request parameters

Name | Type | Optional | Array | Cardinality | Description

ID | String | False | | | The Flag FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description

Flag | Flag | False | | | A single instance of a Flag resource.

Flag.category | CodeableConcept | True | | | The Epic category ID for the condition, labeled “health-concern,” and the LOINC code for the condition, labeled “condition.”

Flag.code | CodeableConcept | False | | | The diagnosis code for the condition. Starting in the February 2024 version of Epic or with special updates to November 2023, August 2023, or May 2023 version of Epic, each code also sends its display name and the provider-entered problem name is sent if available.

Flag.extension (also: flag-detail) | Extension | True | Yes | | A reference to the Condition (Health Concern) resource.

Flag.id | String | False | | | The Flag FHIR ID.

Flag.meta | Meta | True | | | The document metadata.

Flag.meta.profile | String | True | | | Available in Netherlands environments only. This environment shows that this API conforms to the profile connected to the following canonical url: http://nictiz.nl/fhir/StructureDefinition/zib-Alert

Flag.period | Period | True | | | The period during which the condition was marked as active.

Flag.period.end | String | True | | | Date when the condition was marked as resolved.

Flag.period.start | String | True | | | Date when the condition was added.

Flag.status | String | False | | | The status of the condition: active, inactive, or entered in error.

Flag.subject | Reference | False | | | A reference to the Patient resource.

Sample request

```
https://hostname/instance/api/FHIR/STU3/Flag/eyy2ff4sfMS2r1RFpnKxJaP.VLBEBhq5JiMivDW.0klM3
```

Sample response


```
{
  "resourceType": "Flag",
  "id": "enpfRSY57pmn6IxrZD.2eDUnlKr9n60ip6xWB55Ihlws3",
  "extension": [
    {
      "valueReference": {
        "reference": " https://hostname/instance/api/FHIR/STU3/Condition/emYq8NzSEaKrCZtPWXa-GIioNncX-zeoLZG4utC3iD5w3",
        "display": "Eczema"
      },
      "url": "http://hl7.org/fhir/StructureDefinition/flag-detail"
    }
  ],
  "status": "active",
  "category": {
    "coding": [
      {
        "system": "http://open.epic.com/FHIR/StructureDefinition/flag-category",
        "code": "health-concern",
        "display": "Health Concern Flag"
      },
      {
        "system": "http://loinc.org",
        "code": "75323-6",
        "display": "condition"
      }
    ],
    "text": "Health Concern Flag"
  },
  "code": {
    "coding": [
      {
        "system": "http://hl7.org/fhir/sid/icd-10-cm",
        "code": "L30.9",
        "display": "Dermatitis, unspecified"
      },
      {
        "system": "http://snomed.info/sct",
        "code": "43116000",
        "display": "Eczema"
      }
    ],
    "text": "Eczema"
  },
  "subject": {
    "reference": "Patient/euI3cZMuLFNu6kO9mg4ScEg3",
    "display": "Test, Patient"
  },
  "period": {
    "start": "2023-06-28"
  }
}
```


Change log

Epic Version	Description	Type
November 2022	Code systems for the following code sets now return a URL in place of an OID: ActCode, ICD-10-CM, ICD-10-CA, ICD-10-NL, SNOMED System URL OID ActCode STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode 2.16.840.1.113883.5.4 ICD-10-CM http://hl7.org/fhir/sid/icd-10-cm 2.16.840.1.113883.6.90 ICD-10-CA http://hl7.org/fhir/sid/icd-10-ca 2.16.840.1.113883.6.94 ICD-10-NL http://hl7.org/fhir/sid/icd-10-nl 2.16.840.1.113883.6.3.2 SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96	Breaking change

System	URL	OID
ActCode	STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode	2.16.840.1.113883.5.4
ICD-10-CM	http://hl7.org/fhir/sid/icd-10-cm	2.16.840.1.113883.6.90
ICD-10-CA	http://hl7.org/fhir/sid/icd-10-ca	2.16.840.1.113883.6.94
ICD-10-NL	http://hl7.org/fhir/sid/icd-10-nl	2.16.840.1.113883.6.3.2
SNOMED	http://snomed.info/sct	2.16.840.1.113883.6.96

10.2.8 Flag.Read (Infection) (STU3)

File: site/apis/10-Clinical-General/2-STU3/10203-Flag-Read-Infection-STU3.html	
Epic API ID	10203
FHIR version	STU3
Categories	Clinical General
API type	Industry-Standard
USCDI	No
Available in inpatient and other contexts	Yes
HTTP method	GET

URL template | /api/FHIR/STU3/Flag/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10203

Description

This web service retrieves high-level infection information from a patient's chart. An infection might be a laboratory-confirmed infection or colonization, such as influenza or methicillin-resistant Staphylococcus aureus, or a placeholder for a suspected infection, such as respiratory rule-out or suspected COVID-19.

Like the Condition (Infection) web service, this web service evaluates infection (INF) records, but it returns only a brief overview of key information about those records, such as whether they are active and when they were resolved. If you need to return more detailed information, use the Condition (Infection) web service instead.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | String | False | | | The Flag FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Flag | Flag | False | | | A single instance of a Flag resource.
Flag.category | CodeableConcept | True | | | The Epic category ID for the infection, labeled “infection,” and the LOINC code for the infection, labeled “condition.”
Flag.code | CodeableConcept | False | | | The diagnosis code for the infection.
Flag.encounter | Reference | True | | | A reference to the Encounter resource. Used only for encounter-level infections.
Flag.extension (also: flag-detail) | Extension | True | Yes | | A reference to Condition (Infection) resource.
Flag.id | String | False | | | The Flag FHIR ID.
Flag.meta | Meta | True | | | The document metadata.
Flag.meta.profile | String | True | | | Available in Netherlands environments only. This environment shows that this API conforms to the profile connected to the following canonical url: http://nictiz.nl/fhir/StructureDefinition/zib-Alert
Flag.period | Period | True | | | The period during which the infection was marked as active.
Flag.period.end | String | True | | | Date when the infection was marked as resolved.
Flag.period.start | String | True | | | Date when the infection was added.
Flag.status | String | False | | | The status of the infection: active or inactive.
Flag.subject | Reference | False | | | A reference to the Patient resource.

Sample request

```
https://hostname/instance/api/FHIR/STU3/Flag/eFbv9LL28ikxKaUddrVLr-IVq.wf.k7BR.QGEqq8hfEM3
```

Sample response


```
{
  "resourceType": "Flag",
  "id": "eFbv9LL28ikxKaUddrVLr-IVq.wf.k7BR.QGEqq8hfEM3",
  "status": "active",
  "category": {
    "coding": [
      {
        "system": "http://open.epic.com/FHIR/StructureDefinition/flag-category",
        "code": "infection",
        "display": "Infection Flag"
      },
      {
        "system": "http://loinc.org",
        "code": "75323-6",
        "display": "condition"
      }
    ],
    "text": "Infection Flag"
  },
  "code": {
    "coding": [
      {
        "system": "urn:oid:1.2.840.114350.1.13.861.1.7.4.698084.10375",
        "code": "3",
        "display": "Influenza"
      }
    ],
    "text": "Influenza"
  },
  "subject": {
    "reference": "https://hostname/instance/api/FHIR/STU3/Patient/exnWmPch0AyoUp7ZBplVf.Q3",
    "display": "DXL, Contacts"
  },
  "period": {
    "start": "2021-07-27"
```

```
  }
}
```

10.2.9 Flag.Search (Health Concern) (STU3)

File: site/apis/10-Clinical-General/2-STU3/10206-Flag-Search-Health-Concern-STU3.html
Epic API ID | 10206
FHIR version | STU3
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/STU3/Flag?category={category}&patient={patient}&status={status}&subject={subject}&_id={_id}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10206

Description

Request parameters

Name | Type | Optional | Array | Cardinality | Description
category | String | True | | | Always "health-concern" for this web service.
patient | String | Either patient or subject must be provided. If both are provided, they must match. | | | Reference to a Patient resource the Flag is for.
status | String | True | | | The status of the condition: active, inactive, or entered in error.
subject | String | Either patient or subject must be provided. If both are provided, they must match. | | | Reference to a Patient resource the Flag is for.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | Entry | False | Yes | | A single entry in the resource bundle.
Bundle.entry.resource | ResourceContainer | False | | | A single resource returned as part of this bundle.
Bundle.entry.resource.Flag | Flag | False | | | A single instance of a Flag resource.
Bundle.entry.resource.Flag.category | CodeableConcept | False | | | The Epic category ID for the condition, labeled “health-concern,” and the LOINC code for the condition, labeled “condition.”
Bundle.entry.resource.Flag.code | CodeableConcept | False | | | The diagnosis code for the condition. Starting in the February 2024 version of Epic or with special updates to November 2023, August 2023, or May 2023 version of Epic, each code also sends its display name and the provider-entered problem name is sent if available.
Bundle.entry.resource.Flag.extension (also: flag-detail) | Extension | True | Yes | | A reference to the Condition (Health Concern) resource.
Bundle.entry.resource.Flag.id | String | False | | | The Flag FHIR ID.
Bundle.entry.resource.Flag.meta | Meta | True | | | The document metadata.
Bundle.entry.resource.Flag.meta.profile | String | True | | | Available in Netherlands environments only. This environment shows that this API conforms to the profile connected to the following canonical url: http://nictiz.nl/fhir/StructureDefinition/zib-Alert
Bundle.entry.resource.Flag.period | Period | True | | | The period during which the condition was marked as active.
Bundle.entry.resource.Flag.period.end | String | True | | | Date when the condition was marked as resolved.
Bundle.entry.resource.Flag.period.start | String | True | | | Date when the condition was added.
Bundle.entry.resource.Flag.status | String | False | | | The status of the condition: active, inactive, or entered in error.
Bundle.entry.resource.Flag.subject | Reference | False | | | A reference to the Patient resource.

Sample request

```
https://hostname/instance/api/FHIR/STU3/Flag?patient=exnWmPch0AyoUp7ZBplVf.Q3&category=health-concern
```

Sample response

```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 1,
  "link": [
    {
      "relation": "self",
      "url": " https://hostname/instance/api/FHIR/STU3/Flag?patient=euI3cZMuLFNu6kO9mg4ScEg3&category=health-concern"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url": " https://hostname/instance/api/FHIR/STU3/Flag/enpfRSY57pmn6IxrZD.2eDUnlKr9n60ip6xWB55Ihlws3"
        }
      ],
      "fullUrl": " https://hostname/instance/api/FHIR/STU3/Flag/enpfRSY57pmn6IxrZD.2eDUnlKr9n60ip6xWB55Ihlws3",
      "resource": {
```

```
        "resourceType": "Flag",
        "id": "enpfRSY57pmn6IxrZD.2eDUnlKr9n60ip6xWB55Ihlws3",
        "extension": [
          {
            "valueReference": {
              "reference": " https://hostname/instance/api/FHIR/STU3/Condition/emYq8NzSEaKrCZtPWXa-GIioNncX-
    zeoLZG4utC3iD5w3",
              "display": "Eczema"
            },
            "url": "http://hl7.org/fhir/StructureDefinition/flag-detail"
          }
        ],
        "status": "active",
        "category": {
          "coding": [
            {
              "system": "http://open.epic.com/FHIR/StructureDefinition/flag-category",
              "code": "health-concern",
              "display": "Health Concern Flag"
            },
            {
              "system": "http://loinc.org",
              "code": "75323-6",
              "display": "condition"
            }
          ],
          "text": "Health Concern Flag"
        },
        "code": {
          "coding": [
            {
              "system": "http://hl7.org/fhir/sid/icd-10-cm",
              "code": "L30.9",
              "display": "Dermatitis, unspecified"
            },
            {
              "system": "http://snomed.info/sct",
              "code": "43116000",
              "display": "Eczema"
            }
          ],
          "text": "Eczema"
        },
        "subject": {
          "reference": " https://hostname/instance/api/FHIR/STU3/Patient/euI3cZMuLFNu6kO9mg4ScEg3",
          "display": " Test, Patient"
        },
        "period": {
          "start": "2023-06-28"
        }
      },
      "search": {
        "mode": "match"
      }
    }
  ]
}
```


Change log

Epic Version | Description | Type
November 2022 | Code systems for the following code sets now return a URL in place of an OID: ActCode, ICD-10-CM, ICD-10-CA, ICD-10-NL, SNOMED System URL OID ActCode STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode 2.16.840.1.113883.5.4 ICD-10-CM http://hl7.org/fhir/sid/icd-10-cm 2.16.840.1.113883.6.90 ICD-10-CA http://hl7.org/fhir/sid/icd-10-ca 2.16.840.1.113883.6.94 ICD-10-NL http://hl7.org/fhir/sid/icd-10-nl 2.16.840.1.113883.6.3.2 SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 | Breaking change
System | URL | OID
ActCode | STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode | 2.16.840.1.113883.5.4
ICD-10-CM | http://hl7.org/fhir/sid/icd-10-cm | 2.16.840.1.113883.6.90
ICD-10-CA | http://hl7.org/fhir/sid/icd-10-ca | 2.16.840.1.113883.6.94
ICD-10-NL | http://hl7.org/fhir/sid/icd-10-nl | 2.16.840.1.113883.6.3.2
SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96

10.2.10 Flag.Search (Infection) (STU3)

File: site/apis/10-Clinical-General/2-STU3/10204-Flag-Search-Infection-STU3.html
Epic API ID | 10204
FHIR version | STU3
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET

URL template | /api/FHIR/STU3/Flag?category={category}&patient={patient}&status={status}&subject={subject}&_id={_id}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10204

Description

This web service retrieves high-level infection information from a patient's chart. An infection might be a laboratory-confirmed infection or colonization, such as influenza or methicillin-resistant Staphylococcus aureus, or a placeholder for a suspected infection, such as respiratory rule-out or suspected COVID-19.

Like the Condition (Infection) web service, this web service evaluates infection (INF) records, but it returns only a brief overview of key information about those records, such as whether they are active and when they were resolved. If you need to return more detailed information, use the Condition (Infection) web service instead.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
category | String | True | | | Always "infection" for this web service.
patient | String | Either patient or subject must be provided. If both are provided, they must match. | | | Reference to a Patient resource the Flag is for.
status | String | True | | | The status of the infection: active or inactive.
subject | String | Either patient or subject must be provided. If both are provided, they must match. | | | Reference to a Patient resource the Flag is for.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | Entry | False | Yes | | A single entry in the resource bundle.
Bundle.entry.resource | ResourceContainer | False | | | A single resource returned as part of this bundle.
Bundle.entry.resource.Flag | Flag | False | | | A single instance of a Flag resource.
Bundle.entry.resource.Flag.category | CodeableConcept | True | | | The Epic category ID for the infection, labeled “infection,” and the LOINC code for the infection, labeled “condition.”
Bundle.entry.resource.Flag.code | CodeableConcept | False | | | The diagnosis code for the infection.
Bundle.entry.resource.Flag.encounter | Reference | True | | | A reference to the Encounter resource. Used only for encounter-level infections.
Bundle.entry.resource.Flag.extension (also: flag-detail) | Extension | True | Yes | | A reference to the Condition (Infection) resource.
Bundle.entry.resource.Flag.id | String | False | | | The Flag FHIR ID.
Bundle.entry.resource.Flag.meta | Meta | True | | | The document metadata.
Bundle.entry.resource.Flag.meta.profile | String | True | | | Available in Netherlands environments only. This environment shows that this API conforms to the profile connected to the following canonical url: http://nictiz.nl/fhir/StructureDefinition/zib-Alert
Bundle.entry.resource.Flag.period | Period | True | | | The period during which the infection was marked as active.
Bundle.entry.resource.Flag.period.end | String | True | | | Date when the infection was marked as resolved.
Bundle.entry.resource.Flag.period.start | String | True | | | Date when the infection was added.
Bundle.entry.resource.Flag.status | String | False | | | The status of the infection: active or inactive.
Bundle.entry.resource.Flag.subject | Reference | False | | | A reference to the Patient resource.

Sample request

```
https://hostname/instance/api/FHIR/STU3/Flag?patient=exnWmPch0AyoUp7ZBplVf.Q3&category=infection
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 3,
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/STU3/Flag?patient=exnWmPch0AyoUp7ZBplVf.Q3&category=infection"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url": "https://hostname/instance/api/FHIR/STU3/Flag/e0k81lAJpUaaBWo6PSvirZJ8SPlUWfSwHCDA53Ab9-4A3"
        }
      ],
      "fullUrl": "https://hostname/instance/api/FHIR/STU3/Flag/e0k81lAJpUaaBWo6PSvirZJ8SPlUWfSwHCDA53Ab9-4A3",
      "resource": {
        "resourceType": "Flag",
        "id": "e0k81lAJpUaaBWo6PSvirZJ8SPlUWfSwHCDA53Ab9-4A3",
        "status": "entered-in-error",
        "category": {
          "coding": [
            {
              "system": "http://open.epic.com/FHIR/StructureDefinition/flag-category",
              "code": "infection",
```

```
          "display": "Infection Flag"
        },
        {
          "system": "http://loinc.org",
          "code": "75323-6",
          "display": "condition"
        }
      ],
      "text": "Infection Flag"
    },
    "code": {
      "coding": [
        {
          "system": "urn:oid:1.2.840.114350.1.13.861.1.7.4.698084.10375",
          "code": "66002",
          "display": "ICON EL 2"
        }
      ],
      "text": "ICON EL 2"
    },
    "subject": {
      "reference": "https://hostname/instance/api/FHIR/STU3/Patient/exnWmPch0AyoUp7ZBplVf.Q3",
      "display": "DXL, Contacts"
    },
    "period": {
      "start": "2021-09-08",
      "end": "2021-09-08T20:12:16Z"
    },
    "encounter": {
      "reference":
"https://hostname/instance/api/FHIR/STU3/Encounter/eVCZHFbcaQME1z8q58dVMPrvvP82VAgtKbe4oVzwUB4A3",
      "identifier": {
        "use": "usual",
        "system": "urn:oid:1.2.840.114350.1.13.861.1.7.3.698084.8",
        "value": "10001585941"
      },
      "display": "History"
    }
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/STU3/Flag/eG0-zfyMqT1h29FBrsnrhdaNBCP-ydWPSUr44UbL4qfY3"
    }
  ],
  "fullUrl": "https://hostname/instance/api/FHIR/STU3/Flag/eG0-zfyMqT1h29FBrsnrhdaNBCP-ydWPSUr44UbL4qfY3",
  "resource": {
    "resourceType": "Flag",
    "id": "eG0-zfyMqT1h29FBrsnrhdaNBCP-ydWPSUr44UbL4qfY3",
    "status": "active",
    "category": {
      "coding": [
        {
          "system": "http://open.epic.com/FHIR/StructureDefinition/flag-category",
          "code": "infection",
          "display": "Infection Flag"
        },
        {
          "system": "http://loinc.org",
          "code": "75323-6",
          "display": "condition"
        }
      ],
      "text": "Infection Flag"
    },
    "code": {
      "coding": [
        {
          "system": "urn:oid:1.2.840.114350.1.13.861.1.7.4.698084.10375",
          "code": "66001",
          "display": "ICON EL 1"
        },
        {
          "system": "http://snomed.info/sct",
          "code": "73430006",
          "display": "Sleep Apnea"
        }
      ],
      "text": "ICON EL 1"
    },
    "subject": {
      "reference": "https://hostname/instance/api/FHIR/STU3/Patient/exnWmPch0AyoUp7ZBplVf.Q3",
      "display": "DXL, Contacts"
    },
```

```
        "period": {
          "start": "2021-09-08"
        },
        "encounter": {
          "reference":
    "https://hostname/instance/api/FHIR/STU3/Encounter/eyAizba4J4yI-R43baHTOg7nY3a3mztFu57ZsgPsfmGI3",
          "identifier": {
            "use": "usual",
            "system": "urn:oid:1.2.840.114350.1.13.861.1.7.3.698084.8",
            "value": "10001459125"
          },
          "display": "Office Visit"
        }
      },
      "search": {
        "mode": "match"
      }
    },
    {
      "link": [
        {
          "relation": "self",
          "url": "https://hostname/instance/api/FHIR/STU3/Flag/eFbv9LL28ikxKaUddrVLr-IVq.wf.k7BR.QGEqq8hfEM3"
        }
      ],
      "fullUrl": "https://hostname/instance/api/FHIR/STU3/Flag/eFbv9LL28ikxKaUddrVLr-IVq.wf.k7BR.QGEqq8hfEM3",
      "resource": {
        "resourceType": "Flag",
        "id": "eFbv9LL28ikxKaUddrVLr-IVq.wf.k7BR.QGEqq8hfEM3",
        "status": "active",
        "category": {
          "coding": [
            {
              "system": "http://open.epic.com/FHIR/StructureDefinition/flag-category",
              "code": "infection",
              "display": "Infection Flag"
            },
            {
              "system": "http://loinc.org",
              "code": "75323-6",
              "display": "condition"
            }
          ],
          "text": "Infection Flag"
        },
        "code": {
          "coding": [
            {
              "system": "urn:oid:1.2.840.114350.1.13.861.1.7.4.698084.10375",
              "code": "3",
              "display": "Influenza"
            }
          ],
          "text": "Influenza"
        },
        "subject": {
          "reference": "https://hostname/instance/api/FHIR/STU3/Patient/exnWmPch0AyoUp7ZBplVf.Q3",
          "display": "DXL, Contacts"
        },
        "period": {
          "start": "2021-07-27"
        }
      },
      "search": {
        "mode": "match"
      }
    }
  ]
}
```


10.2.11 Observation.$lastn (Social History) (STU3)

File: site/apis/10-Clinical-General/2-STU3/10223-Observation-lastn-Social-History-STU3.html
Epic API ID | 10223
FHIR version | STU3
Categories | Clinical General, Clinical Diagnostics
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/STU3/Observation/$lastn?category={category}&code={code}&max={max}&patient={patient}&subject={subject}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10223

Description

The FHIR Observation Social History $lastn interaction returns a patient's living situation.

Living situation observations correspond to living arrangement categories in Epic, which are defined separately by each Epic organization. Work with the organization to determine supported values if you plan to exchange living situation data through FHIR.

This web service returns at most only a single living situation observation for a patient. Historical values are not available in Epic.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
category | String | True | | | Use "social-history" to search for living situation observations.
code | String | False | | | Use the following SNOMED code to search for living situation: 365508006: Living situation
max | String | True | | | The maximum amount of responses for each subtype of the category.
patient | String | Either patient or subject must be provided. If both are provided, they must reference the same patient. | | | Reference to a patient resource.
subject | String | Either patient or subject must be provided. If both are provided, they must reference the same patient. | | | The reference to a patient resource.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | Entry | False | Yes | | A single entry in the resource bundle.
Bundle.entry.resource | ResourceContainer | False | | | A single resource returned as part of this bundle.
Bundle.entry.resource.Observation | Observation | False | | | A single Observation FHIR resource.
Bundle.entry.resource.Observation.category | CodeableConcept | False | Yes | | This is always "social-history".
Bundle.entry.resource.Observation.code | CodeableConcept | False | | | Use the following SNOMED code to search for living situation: 365508006: Living situation
Bundle.entry.resource.Observation.id | id | False | | | The Observation FHIR ID.
Bundle.entry.resource.Observation.status | code | False
Bundle.entry.resource.Observation.subject | Reference | False | | | The patient FHIR ID.
Bundle.entry.resource.Observation.valueCodeableConcept | CodeableConcept | False | | | Living situation has no default SNOMED code mapping.

Sample request

```
https://hostname/instance/api/FHIR/STU3/Observation/$lastn?patient=eBYtol8OBUpN.6HoCtL4yTA3&code=365508006
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 1,
  "link": [
    {
      "relation": "self",
      "url": "https://eccp-internal.epic.com/Interconnect-
    CDE/api/FHIR/STU3/Observation/$lastn?patient=eBYtol8OBUpN.6HoCtL4yTA3&code=365508006"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url": "https://eccp-internal.epic.com/Interconnect-
    CDE/api/FHIR/STU3/Observation/e6hV1E53oN6cVWy6bWBXLSql.OgS6VmeOeObAeNZgzrKkz-2E.-CnZhBDqseSoiHt3"
        }
      ],
      "fullUrl": "https://eccp-internal.epic.com/Interconnect-
    CDE/api/FHIR/STU3/Observation/e6hV1E53oN6cVWy6bWBXLSql.OgS6VmeOeObAeNZgzrKkz-2E.-CnZhBDqseSoiHt3",
      "resource": {
        "resourceType": "Observation",
        "id": "e6hV1E53oN6cVWy6bWBXLSql.OgS6VmeOeObAeNZgzrKkz-2E.-CnZhBDqseSoiHt3",
        "status": "final",
        "category": [
          {
            "coding": [
              {
                "system": "http://hl7.org/fhir/observation-category",
                "code": "social-history",
                "display": "Social History"
              }
            ],
            "text": "Social History"
          }
        ],
        "code": {
```

```
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "365508006",
              "display": "Finding of residence and accommodation circumstances (finding)"
            }
          ],
          "text": "Finding of residence and accommodation circumstances (finding)"
        },
        "subject": {
          "reference": "https://eccp-internal.epic.com/Interconnect-CDE/api/FHIR/STU3/Patient/eBYtol8OBUpN.6HoCtL4yTA3",
          "display": "Mychart, Aaron"
        },
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "urn:oid:2.16.840.1.113883.6.96",
              "code": "257564005",
              "display": "Apartment (environment)"
            },
            {
              "system": "urn:oid:2.16.840.1.113883.2.4.3.11.60.40.4.13.1",
              "code": "FLATW",
              "display": "Appartement of flatwoning"
            }
          ],
          "text": "Apartment"
        }
      },
      "search": {
        "mode": "match"
      }
    }
  ]
}
```


Change log

Change Log

Epic Version | Description | Type
February 2026 | Observation.$lastn Sub-Resourcing Scope Enforced | Breaking change

10.2.12 Observation.Read (Social History) (STU3)

File: site/apis/10-Clinical-General/2-STU3/855-Observation-Read-Social-History-STU3.html
Epic API ID | 855
FHIR version | STU3
Categories | Clinical General
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Keywords | Smoking; Social; Social history; tobacco
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | August 2018
HTTP method | GET
URL template | api/FHIR/STU3/Observation/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=855

Description

The Observation (Social History) sub-resource searches for patient observations which include any of the following types of data:

Type of data | Description
Smoking history | The status of a patient's tobacco use. This could be used to determine if the patient is a current or former tobacco user, what type of tobacco product is/was used, and the dates the patient started and, if a former user, stopped using tobacco. This includes passive exposure information in the code response element.
Pregnancy Status | The patient’s current pregnancy status.
Lactation Status | The patient’s current lactation status.
Physical Findings of Uterus | Whether the patient has a uterus.
Estimated Delivery Date | The patient’s estimated date of delivery.
Last Menstrual Period | The patient’s last menstrual period.
Alcohol use history | The patient's alcohol use history and, if applicable, the type and amount of alcohol the patient consumes.
Drug use history | The patient's drug use history and, if applicable, the type and amount of drugs the patient consumes.
Living situation | The patient's current living situation.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | Required | | | The Observation FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Observation (STU3) | Observation | | | | A single Observation FHIR resource.
Observation (STU3).category | CodeableConcept | | Yes | 1..* | This is always "social-history".
Observation (STU3).code | CodeableConcept | | | 1..1 | The relevant LOINC or SNOMED code value(s), depending on the type of data returned: Smoking history: LOINC code 72166-2 and SNOMED code 365980008 Alcohol use history: LOINC code 11331-6 and SNOMED code 228273003 Drug use history: LOINC code 11343-1 and SNOMED code 228366006 Living situation: SNOMED code 365508006 Pregnancy status: LOINC code 82810-3 Uterus findings: LOINC code 32486-3 Lactation Status: string "Lactation Status"
Observation (STU3).comment | string | | | 0..1 | Applies only to smoking history, alcohol use history, and drug use history. The comments. For organizations in the Netherlands, these are sent in both provider and MyChart contexts. For all other organizations, these are sent only in a provider context.
Observation (STU3).component | BackboneElement | | Yes | 0..* | Can return "Packs/Day" and "Years Used" when reported data is available.
Observation (STU3).effectivePeriod | Period | | | 0..1 | Date when the observation occurred.
Observation (STU3).id | id | | | 1..1 | The Observation FHIR ID.
Observation (STU3).issued | instant | | | 0..1 | The date when this status is documented. This is set to the date that the status was assessed, if available. Otherwise, this is set to one of the following values to indicate that the response is considered up-to-date as of the specified date: * The date of the patient's most recent history encounter, if available. * The current date. The issued date might not be populated for every observation type.
Observation (STU3).performer | Reference | | Yes | 0..* | The person who documented the status. This is not populated if the status is never assessed.
Observation (STU3).status | code | | | 1..1 | This is always "final." More detailed status information, such as whether the patient's smoking status is unknown, can be found by reviewing the values specified in the code element or the valueCodeableConcept element.
Observation (STU3).subject | Reference | | | 0..1 | The patient.
Observation (STU3).valueBoolean | boolean | | | 0..1 | Yes/No answer to whether the patient has a uterus.
Observation (STU3).valueCodeableConcept | CodeableConcept | | | 0..1 | The SNOMED code or other value pertaining to smoking status, alcohol use status, drug use status, living situation, pregnancy status, or lactation status. Default Epic-to-SNOMED mappings are listed below. This value can also be modified by organization-specific mapping. Smoking history supported values: 449868002-Smokes tobacco daily (finding) 428041000124106-Current some day smoker 230059006-Occasional cigarette smoker (available in Netherlands environments only) 77176002-Smoker, current status unknown 8517006-Former smoker 266919005-Never smoker 43381005-Passive smoker (available in Netherlands environments only) 266927001-Unknown if ever smoker 266919005-Never smoker 266927001-Unknown if ever smoker 428071000124103-Heavy tobacco smoker 428061000124105-Light tobacco smoker Alcohol use history supported values: 219006-Current drinker of alcohol (finding) 105542008-Current non-drinker of alcohol (finding) OTH-Other 82581004-Ex-drinker (finding) 228274009-Lifetime non-drinker (finding) Drug use history supported values: 361055000-Misuses drugs (finding) 228367002-Does not misuse drugs (finding) OTH-Other 44870007-Misused drugs in past (finding) 228368007-Has never misused drugs Living situation has no default mapping, but can be customized by the Epic organization. Pregnancy status supported values: 77386006-Pregnancy (finding) UNK-Unknown 60001007-Not pregnant (finding) Lactation status supported values: 373066001-Yes 373067005-No 261665006-Unknown

Sample request

```
GET https://hostname/instance/api/FHIR/STU3/Observation/euvO8iq4aYWriTr-
    MA7S-IkCxqCXJDmc.rRo7MsHFYNG0HdfUGgXpHY.GcKccA1fi3
```

Sample response


```
{
  "resourceType": "Observation",
  "id": "euvO8iq4aYWriTr-MA7S-IkCxqCXJDmc.rRo7MsHFYNG0HdfUGgXpHY.GcKccA1fi3",
  "status": "final",
  "category": [
    {
      "coding": [
        {
          "system": "http://hl7.org/fhir/observation-category",
          "code": "social-history",
          "display": "Social History"
        }
      ],
      "text": "Social History"
    }
  ],
  "code": {
    "coding": [
      {
        "system": "http://loinc.org",
        "code": "72166-2",
        "display": "Tobacco smoking status"
      },
      {
        "system": "http://snomed.info/sct",
        "code": "365980008",
        "display": "Finding of tobacco use and exposure (finding)"
      }
    ],
    "text": "Smoking History"
  },
  "subject": {
```

```
  "reference": "https://hostname/instance/api/FHIR/STU3/Patient/eeiBNKuRR84uD0nIFn73BGQ3",
  "display": "Test, Patient FHIR"
},
"effectivePeriod": {
  "start": "2009-01-01"
},
"issued": "2021-05-11T05:00:00Z",
"performer": [
  {
    "reference": "https://hostname/instance/api/FHIR/STU3/Practitioner/ekk17eu0BAPwHQ6s2NDWYEQ3",
    "display": "FHIR Doc, MD"
  }
],
"valueCodeableConcept": {
  "coding": [
    {
      "system": "http://snomed.info/sct",
      "code": "449868002",
      "display": "Smokes tobacco daily"
    }
  ],
  "text": "Every Day"
},
"component": [
  {
    "code": {
      "coding": [
        {
          "system": "http://loinc.org",
          "code": "81228-9",
          "display": "Tobacco product"
        },
        {
          "system": "http://snomed.info/sct",
          "code": "53661000146106",
          "display": "Type of tobacco used (observable entity)"
        }
      ],
      "text": "Type"
    },
    "valueCodeableConcept": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "65568007",
          "display": "Cigarette Smoker"
        }
      ],
      "text": "Cigarettes"
    }
  },
  {
    "code": {
      "coding": [
        {
          "system": "http://loinc.org",
          "code": "81228-9",
          "display": "Tobacco product"
        },
        {
          "system": "http://snomed.info/sct",
          "code": "53661000146106",
          "display": "Type of tobacco used (observable entity)"
        }
      ],
      "text": "Type"
    },
    "valueCodeableConcept": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "81703003",
          "display": "Chews Tobacco"
        }
      ],
      "text": "Chew"
    }
  },
  {
    "code": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "401201003",
          "display": "Cigarette pack-years (observable entity)"
        }
      ],
      "text": "Cigarette pack-years (observable entity)"
    },
```

```
      "valueQuantity": {
        "value": 3.72,
        "unit": "{PackYears}",
        "system": "http://unitsofmeasure.org",
        "code": "{PackYears}"
      }
    },
    {
      "code": {
        "coding": [
          {
            "system": "http://loinc.org",
            "code": "88029-4",
            "display": "Tobacco use duration"
          }
        ],
        "text": "Years Used"
      },
      "valueQuantity": {
        "value": 12.4,
        "unit": "Years Used",
        "system": "http://unitsofmeasure.org",
        "code": "a"
      }
    },
    {
      "code": {
        "coding": [
          {
            "system": "http://loinc.org",
            "code": "8663-7",
            "display": "Cigarettes smoked current (pack per day) - Reported"
          },
          {
            "system": "http://snomed.info/sct",
            "code": "266918002",
            "display": "Tobacco smoking consumption (observable entity)"
          }
        ],
        "text": "Packs/Day"
      },
      "valueQuantity": {
        "value": 0.3,
        "unit": "Packs/Day",
        "system": "http://snomed.info/sct",
        "code": "228963008"
      }
    }
  ]
}
```


Change log

Epic Version	Description	Type
February 2024	Updated the mappings for the “82810-3 Pregnancy Status” to return with SNOMED values more specific to Pregnancy Status. Be sure to update any functionality that relies on these codes.	Breaking Change
November 2022	Fixed a bug that caused the "32486-3 Physical Findings of Uterus" LOINC code return with a SNOMED system value (http://snomed.info/sct) instead of a LOINC system (http://loinc.org). If you use this LOINC code, make sure you aren't relying on seeing a SNOMED system.	Breaking Change

10.2.13 Observation.Search (Social History) (STU3)

File: site/apis/10-Clinical-General/2-STU3/856-Observation-Search-Social-History-STU3.html	
Epic API ID	856
FHIR version	STU3
Categories	Clinical General
API type	USCDI
A la carte	Industry-Standard Level 1 API
USCDI	Yes
Keywords	Smoking; Social; Social history;Flowsheets
Available in inpatient and other contexts	Yes
ONC certification products	EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions	August 2018
HTTP method	GET
URL template	api/FHIR/STU3/Observation?patient={patient}&category={category}&subject={subject}&issued={issued}&code={code}
Supports OAuth 2.0	True
Supported user types	Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation	False
Source	https://fhir.epic.com/Specifications?api=856

Description

The Observation (Social History) sub-resource searches for patient observations which include any of the following types of data:

Type of data | Description

Smoking history | The status of a patient's tobacco use. This could be used to determine if the patient is a current or former tobacco user, what type of tobacco product is/was used, and the dates the patient started and, if a former user, stopped using tobacco. This includes passive exposure information in the code response element.

Pregnancy Status | The patient’s current pregnancy status.

Lactation Status | The patient’s current lactation status.

Physical Findings of Uterus | Whether the patient has a uterus.

Estimated Delivery Date | The patient’s estimated date of delivery.

Last Menstrual Period | The patient’s last menstrual period.

Alcohol use history | The patient's alcohol use history and, if applicable, the type and amount of alcohol the patient consumes.

Drug use history | The patient's drug use history and, if applicable, the type and amount of drugs the patient consumes.

Living situation | The patient's current living situation.

Request parameters

Name | Type | Optional | Array | Cardinality | Description

_count | number | Optional | Yes | | Number of results per page.

_id | token | Optional | Yes | | FHIR resource IDs for the desired resources. Can be used in combination with _count and _include. All others will be ignored.

_include | string | Optional | Yes | | Used to indicate which referenced resources to include in the search bundle.

category | token | Conditional | Yes | | Use "social-history" to search for social history and OB/Gyn details. Either a category or a code search parameter must be provided.

code | token | Conditional | Yes | | Either the LOINC or SNOMED code for a particular type of observation can be used. This can include any of the following values: Smoking history: LOINC code 72166-2 or SNOMED code 365980008 Alcohol use history: LOINC code 11331-6 or SNOMED code 228273003 Drug use history: LOINC code 11343-1 or SNOMED code 228366006 Living situation: SNOMED code 365508006 Pregnancy status: LOINC code 82810-3 Uterus findings: LOINC code 32486-3 Note: Lactation Status cannot be searched for by code with the STU3 version of this API, but it can be returned in the response elements. Either a category or a code search parameter must be provided.

issued | date | Optional | Yes | | Date when the observation occurred.

patient | reference | Conditional | Yes | | Reference to a patient resource. Either patient or a subject is required.

subject | reference | Conditional | Yes | | Reference to a patient resource. Either patient or a subject is required.

Response elements

Name | Type | Optional | Array | Cardinality | Description

Observation (STU3) | Observation | | | | A single Observation FHIR resource.

Observation (STU3).category | CodeableConcept | | Yes | 1..* | This is always "social-history".

Observation (STU3).code | CodeableConcept | | | 1..1 | The relevant LOINC or SNOMED code value(s), depending on the type of data returned: Smoking history: LOINC code 72166-2 and SNOMED code 365980008 Alcohol use history: LOINC code 11331-6 and SNOMED code 228273003 Drug use history: LOINC code 11343-1 and SNOMED code 228366006 Living situation: SNOMED code 365508006 Pregnancy status: LOINC code 82810-3 Uterus findings: LOINC code 32486-3 Lactation Status: string "Lactation Status"

Observation (STU3).comment | string | | | 0..1 | Applies only to smoking history, alcohol use history, and drug use history. The comments. For organizations in the Netherlands, these are sent in both provider and MyChart contexts. For all other organizations, these are sent only in a provider context.

Observation (STU3).component | BackboneElement | | Yes | 0..* | Can return "Packs/Day" and "Years Used" when reported data is available.

Observation (STU3).effectivePeriod | Period | | | 0..1 | Date when the observation occurred.

Observation (STU3).id | id | | | 1..1 | The Observation FHIR ID.

Observation (STU3).issued | instant | | | 0..1 | The date when this status is documented. This is set to the date that the status was assessed, if available. Otherwise, this is set to one of the following values to indicate that the response is considered up-to-date as of the specified date: * The date of the patient's most recent history encounter, if available. * The current date. The issued date might not be populated for every observation type.

Observation (STU3).performer | Reference | | Yes | 0..* | The person who documented the status. This is not populated if the status is never assessed.

Observation (STU3).status | code | | | 1..1 | This is always "final." More detailed status information, such as whether the patient’s smoking status is unknown, can be found by reviewing the values specified in the code element or the valueCodeableConcept element.

Observation (STU3).subject | Reference | | | 0..1 | The patient.

Observation (STU3).valueBoolean | boolean | | | 0..1 | Yes/No answer to whether the patient has a uterus.

Observation (STU3).valueCodeableConcept | CodeableConcept | | | 0..1 | The SNOMED code or other value pertaining to smoking status, alcohol use status, drug use status, living situation, pregnancy status, or lactation status. Default Epic-to-SNOMED mappings are listed below. This value can also be modified by organization-specific mapping. Smoking history supported values: 449868002-Smokes tobacco daily (finding) 428041000124106-Current some day smoker 230059006-Occasional cigarette smoker (available in Netherlands environments only) 77176002-Smoker, current status unknown 8517006-Former smoker 266919005-Never smoker 43381005-Passive smoker (available in Netherlands environments only) 266927001-Unknown if ever smoker 266919005-Never smoker 266927001-Unknown if ever smoker 428071000124103-Heavy tobacco smoker 428061000124105-Light tobacco smoker Alcohol use history supported values: 219006-Current drinker of alcohol (finding) 105542008-Current non-drinker of alcohol (finding) OTH-Other 82581004-Ex-drinker (finding) 228274009-Lifetime non-drinker (finding) Drug use history supported values: 361055000-Misuses drugs (finding) 228367002-Does not misuse drugs (finding) OTH-Other 44870007-Misused drugs in past (finding) 228368007-Has never misused drugs Living situation has no default mapping, but can be customized by the Epic organization. Pregnancy status supported values: 77386006-Pregnancy (finding) UNK-Unknown 60001007-Not pregnant (finding) Lactation status supported values: 373066001-Yes 373067005-No 261665006-Unknown

Sample request

```
GET https://hostname/instance/api/FHIR/STU3/Observation?patient=eeiBNKuRR84uD0nIFn73BGQ3&category=social-history
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 4,
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/STU3/Observation?patient=eeiBNKuRR84uD0nIFn73BGQ3&category=social-
    history"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url": "https://hostname/instance/api/FHIR/STU3/Observation/euvO8iq4aYWriTr-
    MA7S-IkCxqCXJDmc.rRo7MsHFYNG0HdfUGgXpHY.GcKccA1fi3"
        }
      ],
      "fullUrl": "https://hostname/instance/api/FHIR/STU3/Observation/euvO8iq4aYWriTr-
    MA7S-IkCxqCXJDmc.rRo7MsHFYNG0HdfUGgXpHY.GcKccA1fi3",
      "resource": {
        "resourceType": "Observation",
        "id": "euvO8iq4aYWriTr-MA7S-IkCxqCXJDmc.rRo7MsHFYNG0HdfUGgXpHY.GcKccA1fi3",
        "status": "final",
        "category": [
          {
            "coding": [
              {
                "system": "http://hl7.org/fhir/observation-category",
                "code": "social-history",
                "display": "Social History"
              }
            ],
            "text": "Social History"
          }
        ],
        "code": {
          "coding": [
            {
              "system": "http://loinc.org",
              "code": "72166-2",
              "display": "Tobacco smoking status"
            },
            {
              "system": "http://snomed.info/sct",
              "code": "365980008",
              "display": "Finding of tobacco use and exposure (finding)"
            }
          ],
          "text": "Smoking History"
        },
        "subject": {
          "reference": "https://hostname/instance/api/FHIR/STU3/Patient/eeiBNKuRR84uD0nIFn73BGQ3",
          "display": "Test, Patient FHIR"
        },
        "effectivePeriod": {
          "start": "2009-01-01"
        },
        "issued": "2021-05-11T05:00:00Z",
        "performer": [
          {
            "reference": "https://hostname/instance/api/FHIR/STU3/Practitioner/ekk17eu0BAPwHQ6s2NDWYEQ3",
            "display": "FHIR Doc, MD"
          }
        ],
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "449868002",
              "display": "Smokes tobacco daily"
            }
          ],
          "text": "Every Day"
        },
        "component": [
          {
            "code": {
              "coding": [
                {
                  "system": "http://loinc.org",
                  "code": "81228-9",
                  "display": "Tobacco product"
                },
                {
```

```
        "system": "http://snomed.info/sct",
        "code": "53661000146106",
        "display": "Type of tobacco used (observable entity)"
      }
    ],
    "text": "Type"
  },
  "valueCodeableConcept": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "65568007",
        "display": "Cigarette Smoker"
      }
    ],
    "text": "Cigarettes"
  }
},
{
  "code": {
    "coding": [
      {
        "system": "http://loinc.org",
        "code": "81228-9",
        "display": "Tobacco product"
      },
      {
        "system": "http://snomed.info/sct",
        "code": "53661000146106",
        "display": "Type of tobacco used (observable entity)"
      }
    ],
    "text": "Type"
  },
  "valueCodeableConcept": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "81703003",
        "display": "Chews Tobacco"
      }
    ],
    "text": "Chew"
  }
},
{
  "code": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "401201003",
        "display": "Cigarette pack-years (observable entity)"
      }
    ],
    "text": "Cigarette pack-years (observable entity)"
  },
  "valueQuantity": {
    "value": 3.72,
    "unit": "{PackYears}",
    "system": "http://unitsofmeasure.org",
    "code": "{PackYears}"
  }
},
{
  "code": {
    "coding": [
      {
        "system": "http://loinc.org",
        "code": "88029-4",
        "display": "Tobacco use duration"
      }
    ],
    "text": "Years Used"
  },
  "valueQuantity": {
    "value": 12.4,
    "unit": "Years Used",
    "system": "http://unitsofmeasure.org",
    "code": "a"
  }
},
{
  "code": {
    "coding": [
      {
        "system": "http://loinc.org",
        "code": "8663-7",
        "display": "Cigarettes smoked current (pack per day) - Reported"
      },
```

```
            {
              "system": "http://snomed.info/sct",
              "code": "266918002",
              "display": "Tobacco smoking consumption (observable entity)"
            }
          ],
          "text": "Packs/Day"
        },
        "valueQuantity": {
          "value": 0.3,
          "unit": "Packs/Day",
          "system": "http://snomed.info/sct",
          "code": "228963008"
        }
      }
    ]
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/STU3/Observation/euvO8iq4aYWriTr-
MA7S-IkCxqCXJDmc.rRo7MsHFYNH7Qg8BxrXz6rsT-bDm9pqY3"
    }
  ],
  "fullUrl": "https://hostname/instance/api/FHIR/STU3/Observation/euvO8iq4aYWriTr-
MA7S-IkCxqCXJDmc.rRo7MsHFYNH7Qg8BxrXz6rsT-bDm9pqY3",
  "resource": {
    "resourceType": "Observation",
    "id": "euvO8iq4aYWriTr-MA7S-IkCxqCXJDmc.rRo7MsHFYNH7Qg8BxrXz6rsT-bDm9pqY3",
    "status": "final",
    "category": [
      {
        "coding": [
          {
            "system": "http://hl7.org/fhir/observation-category",
            "code": "social-history",
            "display": "Social History"
          }
        ],
        "text": "Social History"
      }
    ],
    "code": {
      "coding": [
        {
          "system": "http://loinc.org",
          "code": "11331-6",
          "display": "Alcohol Use History"
        },
        {
          "system": "http://snomed.info/sct",
          "code": "228273003",
          "display": "Finding relating to alcohol drinking behavior (finding)"
        }
      ],
      "text": "Alcohol Use History"
    },
    "subject": {
      "reference": "https://hostname/instance/api/FHIR/STU3/Patient/eeiBNKuRR84uD0nIFn73BGQ3",
      "display": "Test, Patient FHIR"
    },
    "issued": "2021-05-11T05:00:00Z",
    "valueCodeableConcept": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "219006",
          "display": "Current drinker of alcohol (finding)"
        }
      ],
      "text": "Yes"
    },
    "comment": "Patient noted that they battle with addiction dependency",
    "component": [
      {
        "code": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "160573003",
              "display": "Alcohol intake (observable entity)"
            }
          ],
          "text": "Alcohol intake (observable entity)"
```

```
        },
        "valueQuantity": {
          "value": 4.79,
          "unit": "alcoholic_units/day",
          "system": "http://unitsofmeasure.org",
          "code": "{alcoholic_units}/d"
        }
      }
    ]
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/STU3/Observation/euvO8iq4aYWriTr-
MA7S-IkCxqCXJDmc.rRo7MsHFYNGu1kiHo5QU6KA81JFpeO6U3"
    }
  ],
  "fullUrl": "https://hostname/instance/api/FHIR/STU3/Observation/euvO8iq4aYWriTr-
MA7S-IkCxqCXJDmc.rRo7MsHFYNGu1kiHo5QU6KA81JFpeO6U3",
  "resource": {
    "resourceType": "Observation",
    "id": "euvO8iq4aYWriTr-MA7S-IkCxqCXJDmc.rRo7MsHFYNGu1kiHo5QU6KA81JFpeO6U3",
    "status": "final",
    "category": [
      {
        "coding": [
          {
            "system": "http://hl7.org/fhir/observation-category",
            "code": "social-history",
            "display": "Social History"
          }
        ],
        "text": "Social History"
      }
    ],
    "code": {
      "coding": [
        {
          "system": "http://loinc.org",
          "code": "11343-1",
          "display": "Drug Use History"
        },
        {
          "system": "http://snomed.info/sct",
          "code": "228366006",
          "display": "Finding relating to drug misuse behavior (finding)"
        }
      ],
      "text": "Drug Use History"
    },
    "subject": {
      "reference": "https://hostname/instance/api/FHIR/STU3/Patient/eeiBNKuRR84uD0nIFn73BGQ3",
      "display": "Test, Patient FHIR"
    },
    "issued": "2021-05-11T05:00:00Z",
    "valueCodeableConcept": {
      "coding": [
        {
          "system": "http://hl7.org/fhir/v3/NullFlavor",
          "code": "OTH",
          "display": "Other"
        }
      ],
      "text": "Defer"
    },
    "comment": "test drug use comment",
    "component": [
      {
        "code": {
          "coding": [
            {
              "system": "http://loinc.org",
              "code": "63643-1",
              "display": "Drug type used in your lifetime [AUDADIS-IV]"
            },
            {
              "system": "http://snomed.info/sct",
              "code": "410942007",
              "display": "Drug or medicament (substance)"
            }
          ],
          "text": "Type"
        },
        "valueCodeableConcept": {
```

```
            "text": "Marijuana"
          }
        },
        {
          "code": {
            "coding": [
              {
                "system": "http://snomed.info/sct",
                "code": "228390007",
                "display": "Frequency of drug misuse (observable entity)"
              }
            ],
            "text": "Frequency of drug misuse (observable entity)"
          },
          "valueString": "10"
        }
      ]
    },
    "search": {
      "mode": "match"
    }
  },
  {
    "link": [
      {
        "relation": "self",
        "url": "https://hostname/instance/api/FHIR/STU3/Observation/ebiMCwUNh-
SNrB0dsvGvnNMwKf1o3iBVQ7ylWjKIXQiezrl73wAiRGQRFdVu6q2kq3"
      }
    ],
    "fullUrl": "https://hostname/instance/api/FHIR/STU3/Observation/ebiMCwUNh-
SNrB0dsvGvnNMwKf1o3iBVQ7ylWjKIXQiezrl73wAiRGQRFdVu6q2kq3",
    "resource": {
      "resourceType": "Observation",
      "id": "ebiMCwUNh-SNrB0dsvGvnNMwKf1o3iBVQ7ylWjKIXQiezrl73wAiRGQRFdVu6q2kq3",
      "status": "final",
      "category": [
        {
          "coding": [
            {
              "system": "http://hl7.org/fhir/observation-category",
              "code": "social-history",
              "display": "Social History"
            }
          ],
          "text": "Social History"
        }
      ],
      "code": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "365508006",
            "display": "Finding of residence and accommodation circumstances (finding)"
          }
        ],
        "text": "Finding of residence and accommodation circumstances (finding)"
      },
      "subject": {
        "reference": "https://hostname/instance/api/FHIR/STU3/Patient/eeiBNKuRR84uD0nIFn73BGQ3",
        "display": "Test, Patient FHIR"
      },
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "urn:oid:2.16.840.1.113883.6.96",
            "code": "105536009",
            "display": "Living in housing without electricity (finding)"
          }
        ],
        "text": "aje arrangement"
      }
    },
    "search": {
      "mode": "match"
    }
  }
 ]
}
```


Change log

Epic Version	Description	Type
February 2024	Updated the mappings for the “82810-3 Pregnancy Status” to return with SNOMED values more specific to Pregnancy Status. Be sure to update any functionality that relies on these codes.	Breaking Change
November 2022	Fixed a bug that caused the "32486-3 Physical Findings of Uterus" LOINC code return with a SNOMED system value (http://snomed.info/sct) instead of a LOINC system (http://loinc.org). If you use this LOINC code, make sure you aren't relying on seeing a SNOMED system.	Breaking Change

10.2.14 Procedure.Read (Nursing Intervention) (STU3)

File: site/apis/10-Clinical-General/2-STU3/10477-Procedure-Read-Nursing-Intervention-STU3.html
Epic API ID | 10477
FHIR version | STU3
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/STU3/Procedure/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10477

Description

This resource can be used by only healthcare organizations in the Netherlands. This resource cannot be used in any other locales at this time.

The FHIR Procedure resource defines an activity performed on or with a patient as part of the provision of care.

The Procedure (Intervention) API is designed to summarize any care and/or treatment carried out by a nurse in response to problems that arise in relation to a patient's care plan goals.

This resource returns information about inpatient care plan interventions only.

The read interaction enables the lookup of a Procedure resource by a constant server ID. The read interaction allows clients to store only the server ID, and with a single request, retrieve the most up-to-date information about a procedure. Read interactions begin with a client having previously established a relationship, typically when following a reference to this resource from another resource or through querying for procedures through the search interaction.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | String | False | | | The Procedure FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Procedure | Procedure | False
Procedure.category | CodeableConcept | False | | | A code that classifies the procedure. This will always be SNOMED code 9632001 (Nursing procedure) for the Procedure (Intervention) API.
Procedure.code | CodeableConcept | False | | | External codes associated with the procedure.
Procedure.extension (also: Frequency) | Extension | False | Yes | | How often an intervention was performed.
Procedure.meta | Meta | True | | | The document metadata.
Procedure.note | Annotation | True | Yes | | Comment on the nursing intervention.
Procedure.performedPeriod | Period | True | | | The period of time over which the procedure was performed.
Procedure.performer | Performer | True | Yes | | Who performed the Procedure.
Procedure.performer.actor | Reference | True | | | Reference to the Practitioner that performed the procedure. The reading physician or, if unavailable, the resulting physician.
Procedure.extension (also: Requester) | Extension | False | Yes | | The user who requested the intervention.
Procedure.status | String | False | | | The status of the Procedure. Intervention Status Procedure Status in-progress in-progress completed completed cancelled entered-in-error
Intervention Status | Procedure Status
in-progress | in-progress
completed | completed
cancelled | entered-in-error
Procedure.subject | Reference | False | | | Reference to the Patient resource for whom this Procedure resource is relevant.
Procedure.extension (also: TreatmentObjective) | Extension | False | Yes | | The description of the treatment goal that the intervention decision is based on. Uses the "pertains to goal" code system.

Sample request

```
https://hostname/instance/api/FHIR/STU3/Procedure/enUqBOgIgnH0uYkGqnsBQsdFw8XZSd-axvzJVJ9ybs6E3
```

Sample response

```
{
  "resourceType": "Procedure",
  "id": "enUqBOgIgnH0uYkGqnsBQsdFw8XZSd-axvzJVJ9ybs6E3",
  "meta": {
    "profile": [
      "http://nictiz.nl/fhir/StructureDefinition/zib-NursingIntervention"
    ]
```

```
},
"extension": [
  {
    "valueReference": {
      "reference": "https://vs-icx.epic.com/Interconnect-Current-
  Stage1-Primary/api/FHIR/STU3/Goal/eFj3HGT8DXPwINZezNND31g3",
      "display": "IPQA CS Goal"
    },
    "url": "http://hl7.org/fhir/StructureDefinition/goal-pertainsToGoal"
  },
  {
    "valueTiming": {
      "repeat": {
        "frequency": 2,
        "period": 1,
        "periodUnit": "d"
      },
      "code": {
        "coding": [
          {
            "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.2.697081",
            "code": "200006",
            "display": "BID"
          }
        ],
        "text": "BID"
      }
    },
    "url": "http://hl7.org/fhir/StructureDefinition/procedure-schedule"
  },
  {
    "valueReference": {
      "reference": "https://vs-icx.epic.com/Interconnect-Current-
  Stage1-Primary/api/FHIR/STU3/Practitioner/e24exSNO1wPT5FJn2sKyTaQ3",
      "display": "John Doe, MD"
    },
    "url": "http://nictiz.nl/fhir/StructureDefinition/zib-NursingIntervention-Requester"
  }
],
"status": "entered-in-error",
"category": {
  "coding": [
    {
      "system": "http://snomed.info/sct",
      "code": "9632001",
      "display": "Nursing procedure (procedure)"
    }
  ],
  "text": "Nursing Intervention"
},
"code": {
  "coding": [
    {
      "system": "http://snomed.info/sct",
      "code": "416393001",
      "display": "Procedure time out (procedure)"
    },
    {
      "system": "http://snomed.info/sct",
      "code": "707000009",
      "display": "SNOMED CT parsable string (foundation metadata concept)"
    }
  ],
  "text": "IPQA CS Goal Int"
},
"subject": {
  "reference": "https://vs-icx.epic.com/Interconnect-Current-
  Stage1-Primary/api/FHIR/STU3/Patient/eGmXCJpemueo3cQT8TXd13A3",
  "display": "Smith, Josiah"
},
"performedPeriod": {
  "end": "2022-05-03T15:39:15Z"
},
"performer": [
  {
    "actor": {
      "reference": "https://vs-icx.epic.com/Interconnect-Current-
  Stage1-Primary/api/FHIR/STU3/Practitioner/e24exSNO1wPT5FJn2sKyTaQ3",
      "display": "John Doe, MD"
    }
  }
],
"note": [
  {
    "authorReference": {
      "reference": "https://vs-icx.epic.com/Interconnect-Current-
  Stage1-Primary/api/FHIR/STU3/Practitioner/e24exSNO1wPT5FJn2sKyTaQ3",
      "display": "John Doe, MD"
    },
```

```
      "time": "2022-05-03T15:35:59Z",
      "text": "Patient was asleep."
    }
  ]
}
```


10.2.15 Procedure.Read (Orders, Surgeries) (STU3)

File: site/apis/10-Clinical-General/2-STU3/939-Procedure-Read-Orders-Surgeries-STU3.html
Epic API ID | 939
FHIR version | STU3
Categories | Clinical General
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | August 2019
HTTP method | GET
URL template | api/FHIR/STU3/Procedure/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=939

Description

The FHIR Procedure resource defines an activity performed on or with a patient as part of the provision of care. It corresponds with surgeries and procedures performed, including endoscopies and biopsies, as well as less invasive actions like counseling and physiotherapy.

* Note: The STU3 version of this web service currently supports only order procedure records. Dental and surgica procedures are not yet supported. You can use the Procedure.Read (DSTU2) web service if you need to support dental or surgical procedures.

This resource is designed for a high-level summarization around the occurrence of a procedure, and not for specific procedure log documentation - a concept that does not yet have a defined FHIR Resource. When searching, only completed procedures are returned.

The read interaction enables the lookup of a Procedure resource by a constant server ID. The read interaction allows clients to store only the server ID, and with a single request, retrieve the most up-to-date information about a procedure. Read interactions typically begin with a client having previously established a relationship, often through querying for Procedures through the search interaction.

A user or staff member accessing this FHIR resource must have security to view orders, results, and procedure reports.

A MyChart user accessing this FHIR resource must be authorized to view procedures in MyChart.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | String | False | | | The Procedure FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Procedure | Procedure | False
Procedure.basedOn | Reference | True | Yes | | Reference to the ProcedureRequest resource that the procedure log was created from. Only supported for resources with a category of Surgical Procedure.
Procedure.bodySite | CodeableConcept | True | Yes | | External code for the Operative Region for the procedure. Laterality for the procedure may be included as a BodySite-Qualifier extension value to bodySite. Only supported for resources with a category of Surgical Procedure.
Procedure.bodySite.extension (also: bodySite-qualifier) | Extension | True | Yes | | Mapped codes and text description for the laterality of the procedure.
Procedure.category | CodeableConcept | True | | | A code that classifies the procedure. This will always be one of the following SNOMED codes: 387713003 (Surgical Procedure), 103693007 (Diagnostic Procedure)
Procedure.code | CodeableConcept | False | | | External codes associated with the procedure.
Procedure.context | Reference | True | | | Reference to an Encounter resource associated with the Procedure.
Procedure.focalDevice | FocalDevice | True | Yes | | Reference to a device marked as implanted, explanted, or adjusted during the procedure. Only supported for resources with a category of Surgical Procedure.
Procedure.identifier | Identifier | False | Yes | | System-specific ID for the procedure order record.
Procedure.location | Reference | True | | | Reference to the location resource where the procedure was performed. Only supported for resources with a category of Surgical Procedure.
Procedure.meta | Meta | True | | | The document metadata.
Procedure.meta.profile | String | True | | | Available in Netherlands environments only. This environment shows that this API conforms to the profile connected to the following canonical url: http://nictiz.nl/fhir/StructureDefinition/zib-Procedure
Procedure.notDone | Boolean | True | | | If the procedure was not performed as scheduled. True if status is aborted, otherwise false.

Procedure.outcome | CodeableConcept | True | | | The result of a clinical action. Starting in the November 2025 version of Epic, imaging and lab orders with unsuccessful attempts return the SNOMED code 385671000 with text “Unsuccessful attempt.”

Procedure.performedDateTime | String | True | | | When the procedure was performed.

Procedure.performedPeriod | Period | True | | | Period of time over which the procedure was performed.

Procedure.performer | Performer | True | Yes | | Who performed the Procedure.

Procedure.performer.actor | Reference | True | | | Reference to the Practitioner that performed the procedure.

Procedure.reasonCode | CodeableConcept | False | Yes | | External codes for diagnosis records associated with the procedure.

Procedure.reasonReference | Reference | True | Yes | | Applicable only to organizations in the Netherlands. Reference to a condition associated with the DBC episode for the procedure. Only supported for resources with a category of Surgical Procedure.

Procedure.status | String | False | | | The status of the Procedure. API Value FHIR Event Status active in-progress cancelled aborted completed completed aborted aborted *Note: Omitted FHIR event status equivalent values are not returned by this search.

API Value | FHIR Event Status

active | in-progress

cancelled | aborted

completed | completed

aborted | aborted

Procedure.subject | Reference | False | | | Reference to the Patient resource for whom this Procedure resource is relevant.

Procedure.usedReference | Reference | False | Yes | | Reference to a device marked as a wasted implant that was used during the procedure. Only supported for resources with a category of Surgical Procedure.

Sample request

```
https://hostname/instance/api/FHIR/STU3/Procedure/eMm0.ONP9yY6gU1plJr8Cloh9xVrrivStBjA6GTod-Ag3
```

Sample response

```
<Procedure xmlns="http://hl7.org/fhir">
  <id value="eMm0.ONP9yY6gU1plJr8Cloh9xVrrivStBjA6GTod-Ag3"/>
  <identifier>
    <use value="usual"/>
    <type>
      <text value="ORD"/>
    </type>
    <system value="urn:oid:1.2.840.114350.1.13.861.1.7.2.798268"/>
    <value value="3287774"/>
  </identifier>
  <identifier>
    <use value="usual"/>
    <type>
      <text value="EAP"/>
    </type>
    <system value="urn:oid:1.2.840.114350.1.13.861.1.7.2.696580"/>
    <value value="307"/>
  </identifier>
  <status value="completed"/>
  <notDone value="false"/>
  <code>
    <coding>
      <system value="http://www.ama-assn.org/go/cpt"/>
      <code value="11100"/>
      <display value="BIOPSY OF SKIN LESION"/>
    </coding>
    <coding>
      <system value="http://snomed.info/sct"/>
      <code value="129902007"/>
      <display value="BIOPSY OF SKIN LESION"/>
    </coding>
    <coding>
      <system value="http://loinc.org"/>
      <code value="10002-4"/>
      <display value="BIOPSY OF SKIN LESION"/>
    </coding>
    <text value="BIOPSY OF SKIN LESION"/>
  </code>
  <subject>
    <reference value="Patient/ej4nFxzySJAysjMuv4rTBNA3"/>
    <display value="Dan Daniels Jr., MD"/>
  </subject>
  <context>
    <reference value="Encounter/ej5R7X.4RdaFNR5vWdsTpVTtccmkRkWJcKixc5XVY4cQ3"/>
  </context>
  <performedDateTime value="2019-02-15"/>
  <reasonCode>
    <coding>
      <system value="http://snomed.info/sct"/>
      <code value="400097005"/>
      <display value="Ingrowing nail"/>
    </coding>
    <text value="Ingrowing nail"/>
  </reasonCode>
</Procedure>
```

Change log

Epic Version | Description | Type
May 2023 (Special Updates to February 2023) | Code systems for the following code sets now return a URL in place of an OID in the following elements: Procedure.bodySite, Procedure.bodySite.extension(BodySite-Qualifier).valueCodeableConceptProcedure System URL OID SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 LOINC http://loinc.org 2.16.840.1.113883.6.1 | Breaking change
System | URL | OID
SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96
LOINC | http://loinc.org | 2.16.840.1.113883.6.1

10.2.16 Procedure.Read (Restricting Intervention) (STU3)

File: site/apis/10-Clinical-General/2-STU3/10475-Procedure-Read-Restricting-Intervention-STU3.html
Epic API ID | 10475
FHIR version | STU3
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/STU3/Procedure/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10475

Description

This resource is available only for healthcare organizations in the Netherlands.

The FHIR Procedure resource defines an activity performed on or with a patient as part of the provision of care.

The Procedure (Restriction Intervention) API is designed to record interventions that are used to implement freedom restricting measures on the patient.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | String | False | | | The Procedure FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Procedure | Procedure | False
Procedure.extension (also: Assent) | Extension | True | Yes | | Records whether a patient or their legal representative provided consent.
Procedure.category | CodeableConcept | False | | | A code that classifies the procedure. Always includes SNOMED code 225317005 for this resource
Procedure.code | CodeableConcept | False | | | External codes associated with the procedure. Bound to TypeOfInterventionCodelist.
Procedure.context | Reference | True | | | Reference to the encounter in which the procedure took place.
Procedure.id | String | False | | | The Procedure FHIR ID.
Procedure.extension (also: legallyCapable) | Extension | False | Yes | | Whether a patient is legally capable. Contained within a parent zib-FreedomRestrictingIntervention-LegallyCapable extension
Procedure.extension (also: legallyCapableComment) | Extension | True | Yes | | Includes free-text instruction information on a patient’s legal capability. Contained within a parent zib-FreedomRestrictingIntervention-LegallyCapable extension
Procedure.performedPeriod | Period | False | | | The period of time over which the procedure was performed.
Procedure.reasonCode | CodeableConcept | True | Yes | | The reason for use that provides the justification for applying involuntary care. Mapped to ReasonForUseCodelist.
Procedure.status | String | False | | | The status of the procedure order. The table below describes how the status of the intervention procedure order maps to the status returned by this resource. Order Status FHIR Status Pending, Sent, Holding for Referral Preparation Verified, Pending Verify In-Progress Canceled or Discontinued (without documentation) Stopped Denied Approval Not-Done Suspend Suspended Resulted, Completed, Dispensed, Discontinued or Canceled (with documentation) Completed
Order Status | FHIR Status
Pending, Sent, Holding for Referral | Preparation
Verified, Pending Verify | In-Progress
Canceled or Discontinued (without documentation) | Stopped
Denied Approval | Not-Done
Suspend | Suspended
Resulted, Completed, Dispensed, Discontinued or Canceled (with documentation) | Completed
Procedure.subject | Reference | False | | | Reference to the Patient resource for whom this Procedure resource is relevant.

Sample request

```
https://hostname/instance/api/FHIR/STU3/Procedure/fFK-tRE1AmmsrTnHfWRlXZofEtsqrIGnyBvpRohoGPuM4
```

Sample response


```
{
  "resourceType": "Procedure",
```

```
  "id": "fFK-tRE1AmmsrTnHfWRlXZofEtsqrIGnyBvpRohoGPuM4",
  "meta": {
    "profile": [
      "http://nictiz.nl/fhir/StructureDefinition/zib-FreedomRestrictingIntervention"
    ]
  },
  "extension": [
    {
      "extension": [
        {
          "valueBoolean": false,
          "url": "legallyCapable"
        },
        {
          "valueString": "Patient's legal representative provided consent for this action",
          "url": "legallyCapableComment"
        }
      ],
      "url": "http://nictiz.nl/fhir/StructureDefinition/zib-FreedomRestrictingIntervention-LegallyCapable"
    },
    {
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "urn:oid:2.16.840.1.113883.2.4.3.11.60.40.4.6.1",
            "code": "4",
            "display": "Instemming wettelijke vertegenwoordiger"
          }
        ],
        "text": "Instemming wettelijke vertegenwoordiger"
      },
      "url": "http://nictiz.nl/fhir/StructureDefinition/zib-FreedomRestrictingIntervention-Assent"
    }
  ],
  "status": "in-progress",
  "category": {
    "coding": [
      {
        "system": "http://open.epic.com/FHIR/StructureDefinition/Procedure-category",
        "code": "freedom-restricting-intervention",
        "display": "Freedom Restricting Intervention"
      },
      {
        "system": "http://snomed.info/sct",
        "code": "225317005",
        "display": "Restriction of movement"
      }
    ],
    "text": "Restriction of movement"
  },
  "code": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "225210009",
        "display": "LOCKED DOOR PROCEDURE (PROCEDURE)"
      }
    ],
    "text": "LOCKED DOOR PROCEDURE (PROCEDURE)"
  },
  "subject": {
    "reference": "https://hostname/instance/api/FHIR/STU3/Patient/eojz6maS66J54Bwze3hFhDg3",
    "display": "Henderson, T."
  },
  "context": {
    "reference": "https://hostname/instance/api/FHIR/STU3/Encounter/eW3YHxwDTobKP5n-EptoUCA3",
    "identifier": {
      "use": "usual",
      "system": "urn:oid:1.2.840.114350.1.13.260.1.7.3.698084.8",
      "value": "239148"
    },
    "display": "Hospital Encounter"
  },
  "performedPeriod": {
    "start": "2022-07-14T20:21:00Z"
  },
  "reasonCode": [
    {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "121481000146105",
          "display": "Diversion of situation that could result in self-harm or harm to others"
        }
      ],
      "text": "Diversion of situation that could result in self-harm or harm to others"
    }
  ]
}
```

10.2.17 Procedure.Read (Surgical History) (STU3)

File: site/apis/10-Clinical-General/2-STU3/10398-Procedure-Read-Surgical-History-STU3.html
Epic API ID | 10398
FHIR version | STU3
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/STU3/Procedure/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10398

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | String | False | | | The Procedure FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Procedure | Procedure | False | | | An instance of the STU3 Procedure resource.
Procedure.bodySite | CodeableConcept | True | Yes | | Where on the body the procedure was performed.
Procedure.category | CodeableConcept | True | | | A code that classifies the procedure. This will always be SNOMED code 387713003 (Surgical procedure) for the Procedure (Surgical History) API.
Procedure.code | CodeableConcept | True | | | External codes associated with the procedure. Can include CPT codes in all environments. For organizations in the Netherlands: Can additionally show CBV, VT, or ZA codes.
Procedure.notDone | Boolean | False | | | Populated whether a procedure is positive (False) or negative (True).
Procedure.notDoneReason | CodeableConcept | True | | | 416128008 (No past history of procedure) for pertinent negatives. Element is not included for positive procedures.
Procedure.note | Annotation | True | Yes | | Comments about the surgical history or pertinent negative.
Procedure.performedDateTime | String | True | | | When the procedure was performed.
Procedure.performedPeriod | Period | True | | | Returned only for organizations in the Netherlands. When the procedure was performed.
Procedure.status | String | False | | | The status of the Procedure. Supported values include: FHIR Event Status Returned when... completed positive procedure unknown pertinent negative
FHIR Event Status | Returned when...
completed | positive procedure
unknown | pertinent negative
Procedure.subject | Reference | False | | | Reference to the Patient resource for whom this Procedure resource is relevant.
Procedure.extension (also: surgical-history-laterality) | Extension | True | Yes | | Laterality options using SNOMED codes and names: 7771000 (Left) 24028007 (Right) 51440002 (Right and left) 385432009 (Not applicable) 276727009 (Null)
Procedure.extension (also: surgical-history-source) | Extension | True | Yes | | Free-text description of the source of the information. For example “Provider” or “Patient.”

Sample request

```
https://hostname/instance/api/FHIR/STU3/Procedure/e6TLqT3DRRMilDsH7iLanDTHsQqw3qUG5Z2k2HgPmjTauY3-NGlc0EQJsWJ9ihFEB3
```

Sample response

```
<Procedure xmlns="http://hl7.org/fhir">
  <id value="e6TLqT3DRRMilDsH7iLanDTHsQqw3qUG5Z2k2HgPmjTauY3-NGlc0EQJsWJ9ihFEB3"/>
  <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/surgical-history-source">
    <valueString value="Provider"/>
  </extension>
  <status value="completed"/>
  <notDone value="false"/>
  <category>
    <coding>
      <system value="http://snomed.info/sct"/>
      <code value="387713003"/>
      <display value="Surgical procedure"/>
    </coding>
    <text value="Surgical History"/>
  </category>
  <code>
    <coding>
      <system value="http://www.ama-assn.org/go/cpt"/>
      <code value="78605"/>
      <display value="BRAIN IMAGING, COMPLETE"/>
    </coding>
    <text value="BRAIN IMAGING, COMPLETE"/>
  </code>
  <subject>
    <reference value="https://hostname/instance/api/FHIR/STU3/Patient/eWS81qlsLhrRhdd-uUESLchfNaHbZ-U5oNAK5Vth0qo83"/>
    <display value="Sparta, Demo"/>
```

```
  </subject>
  <bodySite>
    <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/surgical-history-laterality">
      <valueCodeableConcept>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="246594002"/>
          <display value="Bilateral absent plantar response"/>
        </coding>
        <text value="Bilateral"/>
      </valueCodeableConcept>
    </extension>
  </bodySite>
</Procedure>
```

Change log

Epic Version | Description | Type
May 2023 (Special Updates to February 2023) | Code systems for the following code sets now return a URL in place of an OID in the following elements: Procedure.bodySite, Procedure.bodySite.extension(BodySite-Qualifier).valueCodeableConceptProcedure System URL OID SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 LOINC http://loinc.org 2.16.840.1.113883.6.1 | Breaking change
System | URL | OID
SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96
LOINC | http://loinc.org | 2.16.840.1.113883.6.1

10.2.18 Procedure.Search (Nursing Intervention) (STU3)

File: site/apis/10-Clinical-General/2-STU3/10480-Procedure-Search-Nursing-Intervention-STU3.html
Epic API ID | 10480
FHIR version | STU3
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/STU3/Procedure?category={category}&date={date}&patient={patient}&subject={subject}&_id={_id}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10480

Description

This resource can be used only by healthcare organizations in the Netherlands. This resource cannot be used in any other locales at this time.

The FHIR Procedure resource defines an activity performed on or with a patient as part of the provision of care.

The Procedure (Intervention) API is designed to summarize any care and/or treatment carried out by a nurse in response to problems that arise in relation to a patient's care plan goals.

This resource returns information about inpatient care plan interventions only.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
category | String | True | | | A code that classifies the procedure. This will always be SNOMED code 9632001 (Nursing procedure) for the Procedure (Intervention) API.
patient | String | False | | | The Patient FHIR resource ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | Entry | False | Yes | | A single entry in the resource bundle.
Bundle.entry.resource | ResourceContainer | False | | | A single resource returned as part of this bundle.
Bundle.entry.resource.Procedure | Procedure | False
Bundle.entry.resource.Procedure.category | CodeableConcept | False | | | A code that classifies the procedure. This will always be SNOMED code 9632001 (Nursing procedure) for the Procedure (Intervention) API.
Bundle.entry.resource.Procedure.code | CodeableConcept | False | | | External codes associated with the procedure.
Bundle.entry.resource.Procedure.extension (also: Frequency) | Extension | False | Yes | | How often an intervention was performed.
Bundle.entry.resource.Procedure.meta | Meta | True | | | The document metadata.
Bundle.entry.resource.Procedure.note | Annotation | True | Yes | | Comment on the nursing intervention.
Bundle.entry.resource.Procedure.performedPeriod | Period | True | | | The period of time over which the procedure was performed.
Bundle.entry.resource.Procedure.performer | Performer | True | Yes | | Who performed the procedure.
Bundle.entry.resource.Procedure.performer.actor | Reference | True | | | Reference to the Practitioner that performed the procedure. The reading physician or, if unavailable, the resulting physician.
Bundle.entry.resource.Procedure.extension (also: Requester) | Extension | False | Yes | | The user who requested the intervention.

Bundle.entry.resource.Procedure.status | String | False | | | The status of the Procedure. Intervention Status Procedure Status in-progress in-progress completed completed cancelled entered-in-error
Intervention Status | Procedure Status
in-progress | in-progress
completed | completed
cancelled | entered-in-error
Bundle.entry.resource.Procedure.subject | Reference | False | | | Reference to the Patient resource for whom this Procedure resource is relevant.
Bundle.entry.resource.Procedure.extension (also: TreatmentObjective) | Extension | False | Yes | | The description of the treatment goal that the intervention decision is based on. Uses the "pertains to goal" code system.
Bundle.total | Int64 | True | | | Total resources that match the search criteria.

Sample request

```
https://hostname/instance/api/FHIR/STU3/Procedure?category=9632001&patient=eGmXCJpemueo3cQT8TXd13A3
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 3,
  "link": [
    {
      "relation": "self",
      "url": "https://vs-icx.epic.com/Interconnect-Current-
    Stage1-Primary/api/FHIR/STU3/Procedure?category=9632001&patient=eGmXCJpemueo3cQT8TXd13A3"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url": "https://vs-icx.epic.com/Interconnect-Current-
    Stage1-Primary/api/FHIR/STU3/Procedure/eagzGKlJ8ngd5AM8WfQhG8XhF3SR2DI.ku.0JXnhuI.M3"
        }
      ],
      "fullUrl": "https://vs-icx.epic.com/Interconnect-Current-
    Stage1-Primary/api/FHIR/STU3/Procedure/eagzGKlJ8ngd5AM8WfQhG8XhF3SR2DI.ku.0JXnhuI.M3",
      "resource": {
        "resourceType": "Procedure",
        "id": "eagzGKlJ8ngd5AM8WfQhG8XhF3SR2DI.ku.0JXnhuI.M3",
        "extension": [
          {
            "valueReference": {
              "reference": "https://vs-icx.epic.com/Interconnect-Current-
    Stage1-Primary/api/FHIR/STU3/Goal/eIm9Mc1OQCtIsgtM7jEFk.A3",
              "display": "IPQA CS Goal"
            },
            "url": "http://hl7.org/fhir/StructureDefinition/goal-pertainsToGoal"
          },
          {
            "valueTiming": {
              "repeat": {
                "frequency": 2,
                "period": 1,
                "periodUnit": "d"
              },
              "code": {
                "coding": [
                  {
                    "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.2.697081",
                    "code": "200006",
                    "display": "BID"
                  }
                ],
                "text": "BID"
              }
            },
            "url": "http://hl7.org/fhir/StructureDefinition/procedure-schedule"
          },
          {
            "valueReference": {
              "reference": "https://vs-icx.epic.com/Interconnect-Current-
    Stage1-Primary/api/FHIR/STU3/Practitioner/e24exSNO1wPT5FJn2sKyTaQ3",
              "display": "John Doe, MD"
            },
            "url": "http://nictiz.nl/fhir/StructureDefinition/zib-NursingIntervention-Requester"
          }
        ],
        "status": "in-progress",
        "category": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "9632001",
              "display": "Nursing procedure (procedure)"
```

```
        }
      ],
      "text": "Nursing Intervention"
    },
    "code": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "416393001",
          "display": "Procedure time out (procedure)"
        },
        {
          "system": "http://snomed.info/sct",
          "code": "707000009",
          "display": "SNOMED CT parsable string (foundation metadata concept)"
        }
      ],
      "text": "IPQA CS Goal Int"
    },
    "subject": {
      "reference": "https://vs-icx.epic.com/Interconnect-Current-
Stage1-Primary/api/FHIR/STU3/Patient/eGmXCJpemueo3cQT8TXd13A3",
      "display": "Smith, Josiah"
    },
    "performedPeriod": {
      "start": "2022-05-11"
    },
    "performer": [
      {
        "actor": {
          "reference": "https://vs-icx.epic.com/Interconnect-Current-
Stage1-Primary/api/FHIR/STU3/Practitioner/e24exSNO1wPT5FJn2sKyTaQ3",
          "display": "John Doe, MD"
        }
      }
    ],
    "note": [
      {
        "authorReference": {
          "reference": "https://vs-icx.epic.com/Interconnect-Current-
Stage1-Primary/api/FHIR/STU3/Practitioner/e24exSNO1wPT5FJn2sKyTaQ3",
          "display": "John Doe, MD"
        },
        "time": "2022-05-05T20:16:24Z",
        "text": "Patient was asleep."
      }
    ]
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url": "https://vs-icx.epic.com/Interconnect-Current-
Stage1-Primary/api/FHIR/STU3/Procedure/enUqBOgIgnH0uYkGqnsBQsdFw8XZSd-axvzJVJ9ybs6E3"
    }
  ],
  "fullUrl": "https://vs-icx.epic.com/Interconnect-Current-
Stage1-Primary/api/FHIR/STU3/Procedure/enUqBOgIgnH0uYkGqnsBQsdFw8XZSd-axvzJVJ9ybs6E3",
  "resource": {
    "resourceType": "Procedure",
    "id": "enUqBOgIgnH0uYkGqnsBQsdFw8XZSd-axvzJVJ9ybs6E3",
    "meta": {
      "profile": [
        "http://nictiz.nl/fhir/StructureDefinition/zib-NursingIntervention"
      ]
    },
    "extension": [
      {
        "valueReference": {
          "reference": "https://vs-icx.epic.com/Interconnect-Current-
Stage1-Primary/api/FHIR/STU3/Goal/eFj3HGT8DXPwINZezNND31g3",
          "display": "IPQA CS Goal"
        },
        "url": "http://hl7.org/fhir/StructureDefinition/goal-pertainsToGoal"
      },
      {
        "valueTiming": {
          "repeat": {
            "frequency": 2,
            "period": 1,
            "periodUnit": "d"
          },
          "code": {
            "coding": [
              {
```

```
              "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.2.697081",
              "code": "200006",
              "display": "BID"
            }
          ],
          "text": "BID"
        }
      },
      "url": "http://hl7.org/fhir/StructureDefinition/procedure-schedule"
    },
    {
      "valueReference": {
        "reference": "https://vs-icx.epic.com/Interconnect-Current-
Stage1-Primary/api/FHIR/STU3/Practitioner/e24exSNO1wPT5FJn2sKyTaQ3",
        "display": "John Doe, MD"
      },
      "url": "http://nictiz.nl/fhir/StructureDefinition/zib-NursingIntervention-Requester"
    }
  ],
  "status": "entered-in-error",
  "category": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "9632001",
        "display": "Nursing procedure (procedure)"
      }
    ],
    "text": "Nursing Intervention"
  },
  "code": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "416393001",
        "display": "Procedure time out (procedure)"
      },
      {
        "system": "http://snomed.info/sct",
        "code": "707000009",
        "display": "SNOMED CT parsable string (foundation metadata concept)"
      }
    ],
    "text": "IPQA CS Goal Int"
  },
  "subject": {
    "reference": "https://vs-icx.epic.com/Interconnect-Current-
Stage1-Primary/api/FHIR/STU3/Patient/eGmXCJpemueo3cQT8TXd13A3",
    "display": "Smith, Josiah"
  },
  "performedPeriod": {
    "end": "2022-05-03T15:39:15Z"
  },
  "performer": [
    {
      "actor": {
        "reference": "https://vs-icx.epic.com/Interconnect-Current-
Stage1-Primary/api/FHIR/STU3/Practitioner/e24exSNO1wPT5FJn2sKyTaQ3",
        "display": "John Doe, MD"
      }
    }
  ],
  "note": [
    {
      "authorReference": {
        "reference": "https://vs-icx.epic.com/Interconnect-Current-
Stage1-Primary/api/FHIR/STU3/Practitioner/e24exSNO1wPT5FJn2sKyTaQ3",
        "display": "John Doe, MD"
      },
      "time": "2022-05-03T15:35:59Z",
      "text": "Patient was asleep"
    }
  ]
 },
 "search": {
  "mode": "match"
 }
},
{
 "link": [
  {
    "relation": "self",
    "url": "https://vs-icx.epic.com/Interconnect-Current-
Stage1-Primary/api/FHIR/STU3/Procedure/e.mGoO8iwbH0l95uEZOtDQE0MPcfjbStXA.SEkRxFfcw3"
  }
 ],
 "fullUrl": "https://vs-icx.epic.com/Interconnect-Current-
Stage1-Primary/api/FHIR/STU3/Procedure/e.mGoO8iwbH0l95uEZOtDQE0MPcfjbStXA.SEkRxFfcw3",
 "resource": {
```

```
    "resourceType": "Procedure",
    "id": "e.mGoO8iwbH0l95uEZOtDQE0MPcfjbStXA.SEkRxFfcw3",
    "extension": [
      {
        "valueReference": {
          "reference": "https://vs-icx.epic.com/Interconnect-Current-
Stage1-Primary/api/FHIR/STU3/Goal/eusgRhTxMPqNHaAVKxDOWjw3",
          "display": "IPQA CS Goal"
        },
        "url": "http://hl7.org/fhir/StructureDefinition/goal-pertainsToGoal"
      },
      {
        "valueTiming": {
          "repeat": {
            "frequency": 2,
            "period": 1,
            "periodUnit": "d"
          },
          "code": {
            "coding": [
              {
                "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.2.697081",
                "code": "200006",
                "display": "BID"
              }
            ],
            "text": "BID"
          }
        },
        "url": "http://hl7.org/fhir/StructureDefinition/procedure-schedule"
      },
      {
        "valueReference": {
          "reference": "https://vs-icx.epic.com/Interconnect-Current-
Stage1-Primary/api/FHIR/STU3/Practitioner/e24exSNO1wPT5FJn2sKyTaQ3",
          "display": "John Doe, MD"
        },
        "url": "http://nictiz.nl/fhir/StructureDefinition/zib-NursingIntervention-Requester"
      }
    ],
    "status": "completed",
    "category": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "9632001",
          "display": "Nursing procedure (procedure)"
        }
      ],
      "text": "Nursing Intervention"
    },
    "code": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "416393001",
          "display": "Procedure time out (procedure)"
        },
        {
          "system": "http://snomed.info/sct",
          "code": "707000009",
          "display": "SNOMED CT parsable string (foundation metadata concept)"
        }
      ],
      "text": "IPQA CS Goal Int"
    },
    "subject": {
      "reference": "https://vs-icx.epic.com/Interconnect-Current-
Stage1-Primary/api/FHIR/STU3/Patient/eGmXCJpemueo3cQT8TXd13A3",
      "display": "Smith, Josiah"
    },
    "performedPeriod": {
      "start": "2022-05-06",
      "end": "2022-05-06"
    },
    "performer": [
      {
        "actor": {
          "reference": "https://vs-icx.epic.com/Interconnect-Current-
Stage1-Primary/api/FHIR/STU3/Practitioner/e24exSNO1wPT5FJn2sKyTaQ3",
          "display": "John Doe, MD"
        }
      }
    ]
  },
  "search": {
    "mode": "match"
  }
}
```

```
  ]
}
```

10.2.19 Procedure.Search (Orders, Surgeries) (STU3)

File: site/apis/10-Clinical-General/2-STU3/940-Procedure-Search-Orders-Surgeries-STU3.html
Epic API ID | 940
FHIR version | STU3
Categories | Clinical General
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | August 2019
HTTP method | GET
URL template | api/FHIR/STU3/Procedure?patient={patient}&subject={subject}&date={date}&category={category}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=940

Description

The FHIR Procedure resource defines an activity performed on or with a patient as part of the provision of care. It corresponds with surgeries and procedures performed, including endoscopies and biopsies, as well as less invasive actions like counseling and physiotherapy.

This resource is designed for a high-level summarization around the occurrence of a procedure, and not for specific procedure log documentation - a concept that does not yet have a defined FHIR Resource. When searching, only completed procedures are returned.

A user or staff member accessing this FHIR resource must have at security to view orders, results, and procedure reports.

A MyChart user accessing this FHIR resource must be authorized to view procedures in MyChart.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
category | String | True | | | A code that classifies the procedure. The following SNOMED codes are acceptable values: 387713003 (Surgical procedure), 103693007 (Diagnostic procedure)
date | String | True | | | Date or period that the procedure was performed, using the FHIR date parameter format.
patient | String | True | | | The Patient FHIR resource ID.
subject | String | True | | | Subject of the procedure.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | Entry | False | Yes | | A single entry in the resource bundle.
Bundle.entry.resource | ResourceContainer | False | | | A single resource returned as part of this bundle.
Bundle.entry.resource.Procedure | Procedure | False
Bundle.entry.resource.Procedure.basedOn | Reference | True | Yes | | Reference to the ProcedureRequest resource that the procedure was created from. Only supported for resources with a category of Surgical Procedure.
Bundle.entry.resource.Procedure.bodySite | CodeableConcept | True | Yes | | External code for the Operative Region for the procedure. Laterality for the procedure may be included as a BodySite-Qualifier extension value to bodySite. Only supported for resources with a category of Surgical Procedure.
Bundle.entry.resource.Procedure.bodySite.extension (also: bodySite-qualifier) | Extension | False | Yes | | Mapped codes and text description for the laterality of the procedure.
Bundle.entry.resource.Procedure.category | CodeableConcept | True | | | A code that classifies the procedure. The following SNOMED codes are acceptable values: 387713003 (Surgical procedure), 103693007 (Diagnostic procedure)
Bundle.entry.resource.Procedure.code | CodeableConcept | False | | | External codes associated with the procedure.
Bundle.entry.resource.Procedure.context | Reference | True | | | Reference to an Encounter resource associated with the Procedure
Bundle.entry.resource.Procedure.focalDevice | FocalDevice | True | Yes | | Reference to a device marked as implanted, explanted, or adjusted during the procedure. Only supported for resources with a category of Surgical Procedure.
Bundle.entry.resource.Procedure.identifier | Identifier | False | Yes | | System-specific ID for the procedure order record.
Bundle.entry.resource.Procedure.location | Reference | True | | | Reference to the location resource where the procedure was performed. Only supported for resources with a category of Surgical Procedure.
Bundle.entry.resource.Procedure.meta | Meta | True | | | The document metadata.
Bundle.entry.resource.Procedure.meta.profile | String | True | | | Available in Netherlands environments only. This environment shows that this API conforms to the profile connected to the following canonical url: http://nictiz.nl/fhir/StructureDefinition/zib-Procedure
Bundle.entry.resource.Procedure.notDone | Boolean | True | | | If the procedure was not performed as scheduled. True if status is aborted, otherwise false.
Bundle.entry.resource.Procedure.outcome | CodeableConcept | True | | | The result of a clinical action. Starting in the November 2025 version of Epic, imaging and lab orders with unsuccessful attempts return the SNOMED code 385671000 with text “Unsuccessful attempt.”
Bundle.entry.resource.Procedure.performedDateTime | String | True | | | When the procedure was performed.

Bundle.entry.resource.Procedure.performedPeriod | Period | True | | | Period of time over which the procedure was performed.
Bundle.entry.resource.Procedure.performer | Performer | True | Yes | | Who performed the Procedure.
Bundle.entry.resource.Procedure.performer.actor | Reference | True | | | Reference to the Practitioner that performed the procedure.
Bundle.entry.resource.Procedure.reasonCode | CodeableConcept | False | Yes | | External codes for diagnosis records associated with the procedure.
Bundle.entry.resource.Procedure.reasonReference | Reference | True | Yes | | Applicable only to organizations in the Netherlands. Reference to a condition associated with the DBC episode for the procedure. Only supported for resources with a category of Surgical Procedure.
Bundle.entry.resource.Procedure.status | String | False | | | The status of the Procedure. API Value FHIR Event Status active in-progress cancelled aborted completed completed aborted aborted *Note: Omitted FHIR event status equivalent values are not returned by this search.
API Value | FHIR Event Status
active | in-progress
cancelled | aborted
completed | completed
aborted | aborted
Bundle.entry.resource.Procedure.subject | Reference | False | | | The Patient FHIR ID for whom this Procedure resource is relevant.
Bundle.entry.resource.Procedure.usedReference | Reference | True | Yes | | Reference to a device marked as a wasted implant that was used during the procedure. Only supported for resources with a category of Surgical Procedure.
Bundle.total | Int64 | True | | | Total resources that match the search criteria.

Sample request

```
https://hostname/instance/api/FHIR/STU3/Procedure?patient=ezHbeFCm6Ih8kgxuHUAdvJQ3&date=gt2019-03-01
```

Sample response

```
<Bundle xmlns="http://hl7.org/fhir">
  <type value="searchset"/>
  <total value="1"/>
  <link>
    <relation value="self"/>
    <url
    value="https://hostname/instance/api/FHIR/STU3/Procedure?patient=ezHbeFCm6Ih8kgxuHUAdvJQ3&amp;date=gt2019-03-01"/>
  </link>
  <entry>
    <link>
      <relation value="self"/>
      <url value="https://hostname/instance/api/FHIR/STU3/Procedure/eU5gdYpvdZq2L35a.KCwkMmnEHbTo5PRumcpj19md9Qc3"/>
    </link>
    <fullUrl value="https://hostname/instance/api/FHIR/STU3/Procedure/eU5gdYpvdZq2L35a.KCwkMmnEHbTo5PRumcpj19md9Qc3"/>
    <resource>
      <Procedure>
        <id value="eU5gdYpvdZq2L35a.KCwkMmnEHbTo5PRumcpj19md9Qc3"/>
        <identifier>
          <use value="usual"/>
          <type>
            <text value="ORD"/>
          </type>
          <system value="urn:oid:1.2.840.114350.1.13.861.1.7.2.798268"/>
          <value value="3328252"/>
        </identifier>
        <identifier>
          <use value="usual"/>
          <type>
            <text value="EAP"/>
          </type>
          <system value="urn:oid:1.2.840.114350.1.13.861.1.7.2.696580"/>
          <value value="5929"/>
        </identifier>
        <status value="aborted"/>
        <notDone value="true"/>
        <code>
          <coding>
            <system value="http://www.ama-assn.org/go/cpt"/>
            <code value="78300"/>
            <display value="BONE IMAGING, LIMITED AREA"/>
          </coding>
          <text value="BONE IMAGING, LIMITED AREA - Unsuccessful Attempt"/>
        </code>
        <subject>
          <reference value="Patient/ezHbeFCm6Ih8kgxuHUAdvJQ3"/>
          <display value="Dan Daniels"/>
        </subject>
        <context>
          <reference value="Encounter/eBehjCsFzgC8JsbmSD8afaHfmaTDucWCzx.oSH5Sp1uw3"/>
        </context>
        <performedDateTime value="2019-03-05"/>
        <reasonCode>
          <coding>
            <system value="http://snomed.info/sct"/>
            <code value="102614006"/>
            <display value="Generalized abdominal pain"/>
          </coding>
          <text value="Generalized abdominal pain"/>
        </reasonCode>
      </Procedure>
```

```
    </resource>
    <search>
      <mode value="match"/>
    </search>
  </entry>
</Bundle>
```

Change log

Epic Version | Description | Type
May 2023 (Special Updates to February 2023) | Code systems for the following code sets now return a URL in place of an OID in the following elements: Procedure.bodySite, Procedure.bodySite.extension(BodySite-Qualifier).valueCodeableConceptProcedure System URL OID SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 LOINC http://loinc.org 2.16.840.1.113883.6.1 | Breaking change
System | URL | OID
SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96
LOINC | http://loinc.org | 2.16.840.1.113883.6.1

10.2.20 Procedure.Search (Restricting Intervention) (STU3)

File: site/apis/10-Clinical-General/2-STU3/10473-Procedure-Search-Restricting-Intervention-STU3.html
Epic API ID | 10473
FHIR version | STU3
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/STU3/Procedure?category={category}&patient={patient}&subject={subject}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10473

Description

This resource is available only for healthcare organizations in the Netherlands.

The FHIR Procedure resource defines an activity performed on or with a patient as part of the provision of care.

The Procedure (Restriction Intervention) API is designed to record interventions that are used to implement freedom restricting measures on the patient.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
category | String | True | | | A code that classifies the procedure. Use SNOMED code “225317005” or custom Epic category value “freedom-restricting-intervention” to search for restricting interventions.
patient | String | Either "patient" or "subject" is required. If both are specified, they must match. | | | A Patient FHIR resource ID representing the subject of the intervention.
subject | String | Either "patient" or "subject" is required. If both are specified, they must match. | | | A Patient FHIR resource ID representing the subject of the intervention.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | Entry | False | Yes | | A single entry in the resource bundle.
Bundle.entry.resource | ResourceContainer | False | | | A single resource returned as part of this bundle.
Bundle.entry.resource.Procedure | Procedure | False
Bundle.entry.resource.Procedure.extension (also: Assent) | Extension | True | Yes | | Records whether a patient or their legal representative provided consent.
Bundle.entry.resource.Procedure.category | CodeableConcept | False | | | A code that classifies the procedure. Always includes SNOMED code “225317005” and the custom Epic category “freedom-restricting-intervention” for this resource.
Bundle.entry.resource.Procedure.code | CodeableConcept | False | | | External codes associated with the procedure. Bound to TypeOfInterventionCodelist.
Bundle.entry.resource.Procedure.context | Reference | True | | | Reference to the encounter in which the procedure took place.
Bundle.entry.resource.Procedure.extension (also: legallyCapable) | Extension | False | Yes | | Whether a patient is legally capable. Contained within a parent zib-FreedomRestrictingIntervention-LegallyCapable extension.
Bundle.entry.resource.Procedure.extension (also: legallyCapableComment) | Extension | True | Yes | | Includes free-text instruction information on a patient’s legal capability. Contained within a parent zib-FreedomRestrictingIntervention-LegallyCapable extension.
Bundle.entry.resource.Procedure.performedPeriod | Period | False | | | The period of time over which the procedure was performed.
Bundle.entry.resource.Procedure.reasonCode | CodeableConcept | True | Yes | | The reason for use that provides the justification for applying involuntary care. Mapped to ReasonForUseCodelist.
Bundle.entry.resource.Procedure.status | String | False | | | The status of the procedure order. The table below describes how the status of the intervention procedure order maps to the status returned by this resource. Order Status FHIR Status Pending, Sent, Holding for Referral Preparation Verified, Pending Verify In-Progress Canceled or Discontinued (without documentation) Stopped Denied Approval Not-Done Suspend Suspended Resulted, Completed, Dispensed, Discontinued or Canceled (with documentation) Completed

Order Status | FHIR Status
Pending, Sent, Holding for Referral | Preparation
Verified, Pending Verify | In-Progress
Canceled or Discontinued (without documentation) | Stopped
Denied Approval | Not-Done
Suspend | Suspended
Resulted, Completed, Dispensed, Discontinued or Canceled (with documentation) | Completed
Bundle.entry.resource.Procedure.subject | Reference | False | | | Reference to the Patient resource for whom this Procedure resource is relevant.

Sample request

```
https://hostname/instance/api/FHIR/STU3/Procedure?category=freedom-restricting-
    intervention&subject=eojz6maS66J54Bwze3hFhDg3
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 1,
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/STU3/Procedure?category=freedom-restricting-
    intervention&subject=eojz6maS66J54Bwze3hFhDg3"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url": "https://hostname/instance/api/FHIR/STU3/Procedure/fFK-tRE1AmmsrTnHfWRlXZofEtsqrIGnyBvpRohoGPuM4"
        }
      ],
      "fullUrl": "https://hostname/instance/api/FHIR/STU3/Procedure/fFK-tRE1AmmsrTnHfWRlXZofEtsqrIGnyBvpRohoGPuM4",
      "resource": {
        "resourceType": "Procedure",
        "id": "fFK-tRE1AmmsrTnHfWRlXZofEtsqrIGnyBvpRohoGPuM4",
        "meta": {
          "profile": [
            "http://nictiz.nl/fhir/StructureDefinition/zib-FreedomRestrictingIntervention"
          ]
        },
        "extension": [
          {
            "extension": [
              {
                "valueBoolean": false,
                "url": "legallyCapable"
              },
              {
                "valueString": "Patient's legal representative provided consent for this action",
                "url": "legallyCapableComment"
              }
            ],
            "url": "http://nictiz.nl/fhir/StructureDefinition/zib-FreedomRestrictingIntervention-LegallyCapable"
          },
          {
            "valueCodeableConcept": {
              "coding": [
                {
                  "system": "urn:oid:2.16.840.1.113883.2.4.3.11.60.40.4.6.1",
                  "code": "4",
                  "display": "Instemming wettelijke vertegenwoordiger"
                }
              ],
              "text": "Instemming wettelijke vertegenwoordiger"
            },
            "url": "http://nictiz.nl/fhir/StructureDefinition/zib-FreedomRestrictingIntervention-Assent"
          }
        ],
        "status": "in-progress",
        "category": {
          "coding": [
            {
              "system": "http://open.epic.com/FHIR/StructureDefinition/Procedure-category",
              "code": "freedom-restricting-intervention",
              "display": "Freedom Restricting Intervention"
            },
            {
              "system": "http://snomed.info/sct",
              "code": "225317005",
              "display": "Restriction of movement"
            }
          ],
          "text": "Restriction of movement"
```

```
        },
        "code": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "225210009",
              "display": "LOCKED DOOR PROCEDURE (PROCEDURE)"
            }
          ],
          "text": "LOCKED DOOR PROCEDURE (PROCEDURE)"
        },
        "subject": {
          "reference": "https://hostname/instance/api/FHIR/STU3/Patient/eojz6maS66J54Bwze3hFhDg3",
          "display": "Henderson, T."
        },
        "context": {
          "reference": "https://hostname/instance/api/FHIR/STU3/Encounter/eW3YHxwDTobKP5n-EptoUCA3",
          "identifier": {
            "use": "usual",
            "system": "urn:oid:1.2.840.114350.1.13.260.1.7.3.698084.8",
            "value": "239148"
          },
          "display": "Hospital Encounter"
        },
        "performedPeriod": {
          "start": "2022-07-14T20:21:00Z"
        },
        "reasonCode": [
          {
            "coding": [
              {
                "system": "http://snomed.info/sct",
                "code": "121481000146105",
                "display": "Diversion of situation that could result in self-harm or harm to others"
              }
            ],
            "text": "Diversion of situation that could result in self-harm or harm to others"
          }
        ]
      },
      "search": {
        "mode": "match"
      }
    }
  ]
}
```


10.2.21 Procedure.Search (Surgical History) (STU3)

File: site/apis/10-Clinical-General/2-STU3/10399-Procedure-Search-Surgical-History-STU3.html
Epic API ID | 10399
FHIR version | STU3
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/STU3/Procedure?category=387713003&patient={patient}&subject={subject}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10399

Description

The FHIR Procedure resource defines an activity performed on or with a patient as part of the provision of care. It corresponds with surgeries and procedures performed, including endoscopies and biopsies, as well as less invasive actions like counseling and physiotherapy.

The Procedure (Surgical History) API is designed to summarize historical surgical procedures a patient has had, as well as pertinent negatives (indicating a patient has not had a particular surgery). Other types of procedures can be returned in other APIs, such as the Procedure (Orders) API for procedures created from orders.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
category | String | True | | | A code that classifies the procedure. This will always be SNOMED code 387713003 (Surgical procedure) for the Procedure (Surgical history) API.
patient | String | Either this or subject must be provided. | | | The Patient FHIR resource ID.
subject | String | Either this or patient must be provided. | | | Subject of the Procedure.

Response elements

Name | Type | Optional | Array | Cardinality | Description

Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.

Bundle.entry | Entry | False | Yes | | A single entry in the resource bundle.

Bundle.entry.resource | ResourceContainer | False | | | A single resource returned as part of this bundle.

Bundle.entry.resource.Procedure | Procedure | False | | | An instance of the STU3 Procedure resource.

Bundle.entry.resource.Procedure.bodySite | CodeableConcept | True | Yes | | Where on the body the procedure was performed.

Bundle.entry.resource.Procedure.category | CodeableConcept | True | | | A code that classifies the procedure. Always SNOMED code 387713003 (Surgical procedure) for the Procedure (Surgical History) resource.

Bundle.entry.resource.Procedure.code | CodeableConcept | True | | | External codes associated with the procedure. Can include CPT codes in all environments. For organizations in the Netherlands: Can additionally show CBV, VT, or ZA codes.

Bundle.entry.resource.Procedure.notDone | Boolean | False | | | Populated whether a procedure is positive (False) or negative (True).

Bundle.entry.resource.Procedure.notDoneReason | CodeableConcept | True | | | 416128008 (No past history of procedure) for pertinent negatives. Element is not included for positive procedures.

Bundle.entry.resource.Procedure.note | Annotation | True | Yes | | Comments about the surgical history or pertinent negative.

Bundle.entry.resource.Procedure.performedDateTime | String | False | | | When the procedure was performed.

Bundle.entry.resource.Procedure.status | String | False | | | The status of the Procedure. Supported values include: FHIR Event Status Returned when... completed positive procedure Unknown pertinent negative

FHIR Event Status | Returned when...

completed | positive procedure

Unknown | pertinent negative

Bundle.entry.resource.Procedure.subject | Reference | False | | | Reference to the Patient resource for whom this Procedure resource is relevant.

Bundle.entry.resource.Procedure.extension (also: surgical-history-laterality) | Extension | True | Yes | | Laterality options using SNOMED codes and names: 7771000 (Left) 24028007 (Right) 51440002 (Right and left) 385432009 (Not applicable) 276727009 (Null)

Bundle.entry.resource.Procedure.extension (also: surgical-history-source) | Extension | True | Yes | | Free-text description of the source of the information. For example “Provider” or “Patient.”

Bundle.total | Int64 | True | | | Total resources that match the search criteria.

Sample request

```
https://hostname/instance/api/FHIR/STU3/Procedure?category=387713003&Patient=eWS81qlsLhrRhdd-
    uUESLchfNaHbZ-U5oNAK5Vth0qo83
```

Sample response

```
<Bundle xmlns="http://hl7.org/fhir">
  <type value="searchset"/>
  <total value="2"/>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/STU3/Procedure?Patient=eWS81qlsLhrRhdd-
    uUESLchfNaHbZ-U5oNAK5Vth0qo83"/>
  </link>
  <entry>
    <link>
      <relation value="self"/>
      <url value="https://hostname/instance/api/FHIR/STU3/Procedure/e6TLqT3DRRMilDsH7iLanDTHsQqw3qUG5Z2k2HgPmjTauY3-
    NGlc0EQJsWJ9ihFEB3"/>
    </link>
    <fullUrl value="https://hostname/instance/api/FHIR/STU3/Procedure/e6TLqT3DRRMilDsH7iLanDTHsQqw3qUG5Z2k2HgPmjTauY3-
    NGlc0EQJsWJ9ihFEB3"/>
    <resource>
      <Procedure>
        <id value="e6TLqT3DRRMilDsH7iLanDTHsQqw3qUG5Z2k2HgPmjTauY3-NGlc0EQJsWJ9ihFEB3"/>
        <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/surgical-history-source">
          <valueString value="Provider"/>
        </extension>
        <status value="completed"/>
        <notDone value="false"/>
        <category>
          <coding>
            <system value="http://snomed.info/sct"/>
            <code value="387713003"/>
            <display value="Surgical procedure"/>
          </coding>
          <text value="Surgical History"/>
        </category>
        <code>
          <coding>
            <system value="http://www.ama-assn.org/go/cpt"/>
            <code value="78605"/>
            <display value="BRAIN IMAGING, COMPLETE"/>
          </coding>
          <text value="BRAIN IMAGING, COMPLETE"/>
        </code>
        <subject>
          <reference value="https://hostname/instance/api/FHIR/STU3/Patient/eWS81qlsLhrRhdd-
    uUESLchfNaHbZ-U5oNAK5Vth0qo83"/>
          <display value="Sparta, Demo"/>
        </subject>
        <bodySite>
          <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/surgical-history-laterality">
```

```
          <valueCodeableConcept>
            <coding>
              <system value="http://snomed.info/sct"/>
              <code value="246594002"/>
              <display value="Bilateral absent plantar response"/>
            </coding>
            <text value="Bilateral"/>
          </valueCodeableConcept>
        </extension>
      </bodySite>
    </Procedure>
  </resource>
  <search>
    <mode value="match"/>
  </search>
 </entry>
 <entry>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/STU3/Procedure/e6TLqT3DRRMilDsH7iLanDTHsQqw3qUG5Z2k2HgPmjTbpW5cYe-
  pjCtRLeEtFVd3D3"/>
  </link>
  <fullUrl
  value="https://hostname/instance/api/FHIR/STU3/Procedure/e6TLqT3DRRMilDsH7iLanDTHsQqw3qUG5Z2k2HgPmjTbpW5cYe-
  pjCtRLeEtFVd3D3"/>
  <resource>
    <Procedure>
      <id value="e6TLqT3DRRMilDsH7iLanDTHsQqw3qUG5Z2k2HgPmjTbpW5cYe-pjCtRLeEtFVd3D3"/>
      <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/surgical-history-source">
        <valueString value="Provider"/>
      </extension>
      <status value="unknown"/>
      <notDone value="true"/>
      <notDoneReason>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="416128008"/>
          <display value="No past history of procedure"/>
        </coding>
        <text value="No past history of procedure"/>
      </notDoneReason>
      <category>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="387713003"/>
          <display value="Surgical procedure"/>
        </coding>
        <text value="Surgical History"/>
      </category>
      <code>
        <coding>
          <system value="http://www.ama-assn.org/go/cpt"/>
          <code value="29860"/>
          <display value="HIP ARTHROSCOPY, DX"/>
        </coding>
        <text value="HIP ARTHROSCOPY, DX"/>
      </code>
      <subject>
        <reference value="https://hostname/instance/api/FHIR/STU3/Patient/eWS81qlsLhrRhdd-
  uUESLchfNaHbZ-U5oNAK5Vth0qo83"/>
        <display value="Sparta, Demo"/>
      </subject>
      <performedDateTime value="2022-02-02"/>
    </Procedure>
  </resource>
  <search>
    <mode value="match"/>
  </search>
 </entry>
</Bundle>
```


Change log

Epic Version | Description | Type

May 2023 (Special Updates to February 2023) | Code systems for the following code sets now return a URL in place of an OID in the following elements: Procedure.bodySite, Procedure.bodySite.extension(BodySite-Qualifier).valueCodeableConceptProcedure System URL OID SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 LOINC http://loinc.org 2.16.840.1.113883.6.1 | Breaking change

System | URL | OID

SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96

LOINC | http://loinc.org | 2.16.840.1.113883.6.1

10.3 Clinical General: DSTU2

File: site/apis/10-Clinical-General/3-DSTU2/index.html

* AllergyIntolerance.Read (DSTU2)

* AllergyIntolerance.Search (DSTU2)
* Condition.Read (Problems) (DSTU2)
* Condition.Search (Problems) (DSTU2)
* FamilyMemberHistory.Search (DSTU2)
* Observation.Read (Social History) (DSTU2)
* Observation.Search (Social History) (DSTU2)
* Procedure.Read (Orders) (DSTU2)
* Procedure.Search (Orders) (DSTU2)

10.3.1 AllergyIntolerance.Read (DSTU2)

File: site/apis/10-Clinical-General/3-DSTU2/1-AllergyIntolerance-Read-DSTU2.html
Epic API ID | 1
FHIR version | DSTU2
Categories | Clinical General
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Keywords | Allergies
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | Epic 2015
HTTP method | GET
URL template | api/FHIR/DSTU2/AllergyIntolerance/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=1

Description

The FHIR AllergyIntolerance resource defines clinical information about a patient's allergic response to a substance. The AllergyIntolerance resource defines the substance that elicited the response, as well as when the reaction occured, the severity, and the type of reaction noted. The AllergyIntolerance resource can also accommodate search by ID and by patient, allowing it to return a list of allergies. If a patient has no active allergies, an AllergyIntolerance resource will be returned indicating whether the patient's allergies have never been reviewed (are not on file), or if they have been reviewed and it has been determined that they have no known allergies.

The AllergyIntolerance resource corresponds to the following Common Clinical Data Set (CCDS) Elements: Mediation Allergies.

The Read interaction enables the look up of an AllergyIntolerance resource by a constant server ID. Clients can store only the server ID, and with a single request, retrieve the most up-to-date information about the AllergyIntolerance. Read interactions typically begin with a client having previously established a relationship, often through querying for a patient's allergies through the Search interaction.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | String | False | | | The AllergyIntolerance FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
AllergyIntolerance | AllergyIntolerance | False | | | A single AllergyIntolerance resource.
AllergyIntolerance.category | String | True | | | Refers to food, medication, or environmental allergies.
AllergyIntolerance.criticality | String | True | | | Refers to the risk of clinical harm of a reaction to the substance.
AllergyIntolerance.id | String | False | | | The AllergyIntolerance FHIR ID.
AllergyIntolerance.note | Annotation | True | | | Any comments entered by the recorder.
AllergyIntolerance.onset | String | False | | | Date allergy was first noticed.
AllergyIntolerance.patient | ReferencedResource | False | | | Reference to the patient associated with the allergy.
AllergyIntolerance.reaction | Event | False | Yes | | List of known reactions the patient had to the substance.
AllergyIntolerance.reaction.certainty | String | True | | | Will always be "confirmed". All allergies documented within Epic are known allergies and are confirmed.
AllergyIntolerance.reaction.manifestation | CodeableConcept | False | Yes | | List of known reactions the patient had to the substance.
AllergyIntolerance.reaction.onset | String | True | | | Date allergy was first noticed.
AllergyIntolerance.reaction.substance | CodeableConcept | True | | | The substance that the patient has the specified reactions to.
AllergyIntolerance.recordedDate | DateTime | True | | | The date the allergy was added to the patient.
AllergyIntolerance.recorder | ReferencedResource | True | | | Reference to the person to last modify the allergy.
AllergyIntolerance.status | String | True | | | Confirmation status of the resource. The only options we will send for this field will be "confirmed" for active allergies or "resolved" for expired allergies. Otherwise, we will send nothing.

AllergyIntolerance.substance | CodeableConcept | False | | | The substance that the patient has the specified reactions to.

Sample request

```
https://hostname/instance/api/FHIR/DSTU2/AllergyIntolerance/T8akyavrsha11onGpAcKERQB
```

Sample response

```
<AllergyIntolerance xmlns="http://hl7.org/fhir">
  <recordedDate value="2016-02-16T12:11:18"/>
  <status value="confirmed"/>
  <criticality value="CRITL"/>
  <onset value="2015-10-16T00:00:00"/>
  <id value="T8akyavrsha11onGpAcKERQB"/>
  <recorder>
    <display value="Helen Claire Wilson, MD"/>
    <reference value="https://hostname/instance/api/FHIR/DSTU2/Practitioner/TrTk7Zaw96VJM5sIX95IYHAB"/>
  </recorder>
  <patient>
    <display value="Fiona Hailey Ingrid Robertson"/>
    <reference value="https://hostname/instance/api/FHIR/DSTU2/Patient/Tb19ikuzxOh6YglUN28cuQ-d.3AjOduTYFVxlRM82bqEB"/>
  </patient>
  <substance>
    <text value="PENICILLINS"/>
    <coding>
      <system value="http://www.nlm.nih.gov/research/umls/rxnorm/"/>
      <code value="7980"/>
      <display value="PENICILLINS"/>
    </coding>
    <coding>
      <system value="http://hl7.org/fhir/ndfrt"/>
      <code value="N0000005840"/>
      <display value="PENICILLINS"/>
    </coding>
  </substance>
  <reaction>
    <certainty value="confirmed"/>
    <onset value="2015-10-16T00:00:00"/>
    <manifestation>
      <text value="Weal"/>
      <coding>
        <system value="http://snomed.info/sct"/>
        <code value="247472004"/>
        <display value="Weal"/>
      </coding>
    </manifestation>
    <manifestation>
      <text value="Swelling"/>
      <coding>
        <system value="http://snomed.info/sct"/>
        <code value="65124004"/>
        <display value="Swelling"/>
      </coding>
    </manifestation>
    <note>
      <text value="Patient reported hives and swelling in their throat when administered penicillin in a previous ED
    visit"/>
    </note>
  </reaction>
  <note>
    <text value="Patient reported hives and swelling in their throat when administered penicillin in a previous ED
    visit"/>
  </note>
</AllergyIntolerance>
```

10.3.2 AllergyIntolerance.Search (DSTU2)

File: site/apis/10-Clinical-General/3-DSTU2/2-AllergyIntolerance-Search-DSTU2.html
Epic API ID | 2
FHIR version | DSTU2
Categories | Clinical General
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Keywords | Allergies
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | Epic 2015
HTTP method | GET
URL template | /api/FHIR/DSTU2/AllergyIntolerance?patient={patient}&subject={subject}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=2

Description

The FHIR AllergyIntolerance resource defines clinical information about a patient's allergic response to a substance. The AllergyIntolerance resource defines the substance that elicited the response, as well as when the reaction occured, the severity, and the type of reaction noted. The AllergyIntolerance resource can also accommodate search by ID and by patient, allowing it to return a list of allergies. If a patient has no active allergies, an AllergyIntolerance resource will be returned indicating whether the patient's allergies have never been reviewed (are not on file), or if they have been reviewed and it has been determined that they have no known allergies.

The AllergyIntolerance resource corresponds to the following Common Clinical Data Set (CCDS) Elements: Mediation Allergies.

The search interaction enables the client to query for all allergies associated with a given patient's medical history. The client, having established the patient in question, now wishes to retrieve the list of that patient's allergies.

To ensure clinical safety for a patient, searches filtered with a search parameter will return all active allergies in addition to other data returned in the search.

Note:

onset

Remarks

FHIR AllergyIntolerance Resources can be returned in either XML or JSON format through either Read or Search interactions. Read interactions refer to the lookup of a single AllergyIntolerance resource by its external FHIR ID. Search interactions can return multiple resources according to the parameters used, which for now will be limited to AllergyIntolerance IDs and/or a Patient ID.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
patient | String | Either patient or subject must be provided. If both are provided, they must reference the same patient. | | | Reference to a patient resource.
subject | String | Either patient or subject must be provided. If both are provided, they must reference the same patient. | | | Reference to a patient resource.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | Entry | False | Yes | | A single entry in the resource bundle.
Bundle.entry.resource | Resource | False | | | A single resource returned as part of this bundle.
Bundle.entry.resource.AllergyIntolerance | AllergyIntolerance | False | | | A single AllergyIntolerance resource.
Bundle.entry.resource.AllergyIntolerance.category | String | False | | | Refers to food, medication, or environmental allergies.
Bundle.entry.resource.AllergyIntolerance.criticality | String | False | | | Refers to the risk of clinical harm of a reaction to the substance.
Bundle.entry.resource.AllergyIntolerance.note | Annotation | False | | | Any comments entered by the recorder.
Bundle.entry.resource.AllergyIntolerance.patient | ReferencedResource | False | | | Reference to a patient resource for whom the allergy is relevant.
Bundle.entry.resource.AllergyIntolerance.reaction | Event | False | Yes | | List of known reactions the patient had to the substance.
Bundle.entry.resource.AllergyIntolerance.reaction.certainty | String | False | | | Will always be "confirmed". All allergies documented within Epic are known allergies and are confirmed.
Bundle.entry.resource.AllergyIntolerance.reaction.manifestation | CodeableConcept | False | Yes | | A single reaction the patient had to the substance.
Bundle.entry.resource.AllergyIntolerance.reaction.onset | String | False | | | Date allergy was first noticed.
Bundle.entry.resource.AllergyIntolerance.reaction.substance | CodeableConcept | False | | | The substance that the patient has the specified reactions to.
Bundle.entry.resource.AllergyIntolerance.recordedDate | DateTime | True | | | The date the allergy was added to the patient.
Bundle.entry.resource.AllergyIntolerance.recorder | ReferencedResource | False | | | Reference to the person to last modify the allergy.
Bundle.entry.resource.AllergyIntolerance.status | String | False | | | Confirmation status of the resource. The only options we will send for this field will be "confirmed" for active allergies or "resolved" for expired allergies. Otherwise, we will send nothing.
Bundle.entry.resource.AllergyIntolerance.substance | CodeableConcept | False | | | The substance the patient had the specified reactions to.
Bundle.total | Int32 | True | | | Total number of resources that matched the search criteria.

Sample request

```
https://hostname/instance/api/FHIR/DSTU2/AllergyIntolerance?patient=Tb19ikuzxOh6YglUN28cuQ-d.3AjOduTYFVxlRM82bqEB
```

Sample response

```
<Bundle xmlns="http://hl7.org/fhir">
  <type value="searchset"/>
  <total value="3"/>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/DSTU2/AllergyIntolerance?_id=&amp;patient=Tb19ikuzxOh6YglUN28cuQ-
    d.3AjOduTYFVxlRM82bqEB"/>
  </link>
  <entry>
```

```
  <fullUrl value="https://hostname/instance/api/FHIR/DSTU2/AllergyIntolerance/T8akyavrsha11onGpAcKERQB"/>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/DSTU2/AllergyIntolerance/T8akyavrsha11onGpAcKERQB"/>
  </link>
  <resource>
    <AllergyIntolerance>
      <recordedDate value="2016-02-16T12:11:18"/>
      <status value="confirmed"/>
      <criticality value="CRITL"/>
      <onset value="2015-10-16T00:00:00"/>
      <id value="T8akyavrsha11onGpAcKERQB"/>
      <recorder>
        <display value="Helen Claire Wilson, MD"/>
        <reference value="https://hostname/instance/api/FHIR/DSTU2/Practitioner/TrTk7Zaw96VJM5sIX95IYHAB"/>
      </recorder>
      <patient>
        <display value="Fiona Hailey Ingrid Robertson"/>
        <reference
  value="https://hostname/instance/api/FHIR/DSTU2/Patient/Tb19ikuzxOh6YglUN28cuQ-d.3AjOduTYFVxlRM82bqEB"/>
      </patient>
      <substance>
        <text value="PENICILLINS"/>
        <coding>
          <system value="http://www.nlm.nih.gov/research/umls/rxnorm/"/>
          <code value="7980"/>
          <display value="PENICILLINS"/>
        </coding>
        <coding>
          <system value="http://hl7.org/fhir/ndfrt"/>
          <code value="N0000005840"/>
          <display value="PENICILLINS"/>
        </coding>
      </substance>
      <reaction>
        <certainty value="confirmed"/>
        <onset value="2015-10-16T00:00:00"/>
        <manifestation>
          <text value="Weal"/>
          <coding>
            <system value="http://snomed.info/sct"/>
            <code value="247472004"/>
            <display value="Weal"/>
          </coding>
        </manifestation>
        <manifestation>
          <text value="Swelling"/>
          <coding>
            <system value="http://snomed.info/sct"/>
            <code value="65124004"/>
            <display value="Swelling"/>
          </coding>
        </manifestation>
        <note>
          <text value="Patient reported hives and swelling in their throat when administered penicillin in a previous
  ED visit"/>
        </note>
      </reaction>
      <note>
        <text value="Patient reported hives and swelling in their throat when administered penicillin in a previous ED
   visit"/>
      </note>
    </AllergyIntolerance>
  </resource>
  <search>
    <mode value="match"/>
  </search>
</entry>
<entry>
  <fullUrl value="https://hostname/instance/api/FHIR/DSTU2/AllergyIntolerance/TJ5dAeG4I9i-JrSbaGgURXwB"/>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/DSTU2/AllergyIntolerance/TJ5dAeG4I9i-JrSbaGgURXwB"/>
  </link>
  <resource>
    <AllergyIntolerance>
      <recordedDate value="2016-02-16T12:12:46"/>
      <status value="confirmed"/>
      <criticality value="CRITL"/>
      <onset value="2016-02-16T00:00:00"/>
      <id value="TJ5dAeG4I9i-JrSbaGgURXwB"/>
      <recorder>
        <display value="Helen Claire Wilson, MD"/>
        <reference value="https://hostname/instance/api/FHIR/DSTU2/Practitioner/TrTk7Zaw96VJM5sIX95IYHAB"/>
      </recorder>
      <patient>
        <display value="Fiona Hailey Ingrid Robertson"/>
        <reference
  value="https://hostname/instance/api/FHIR/DSTU2/Patient/Tb19ikuzxOh6YglUN28cuQ-d.3AjOduTYFVxlRM82bqEB"/>
```

```
      </patient>
      <substance>
        <text value="SALICYLATES"/>
        <coding>
          <system value="http://www.nlm.nih.gov/research/umls/rxnorm/"/>
          <code value="9522"/>
          <display value="SALICYLATES"/>
        </coding>
        <coding>
          <system value="http://hl7.org/fhir/ndfrt"/>
          <code value="N0000005042"/>
          <display value="SALICYLATES"/>
        </coding>
      </substance>
      <reaction>
        <certainty value="confirmed"/>
        <onset value="2016-02-16T00:00:00"/>
        <manifestation>
          <text value="Itching (SNOMED-CT: 279333002)"/>
        </manifestation>
        <manifestation>
          <text value="Rash (SNOMED-CT: 271807003)"/>
        </manifestation>
      </reaction>
    </AllergyIntolerance>
  </resource>
  <search>
    <mode value="match"/>
  </search>
</entry>
<entry>
  <fullUrl value="https://hostname/instance/api/FHIR/DSTU2/AllergyIntolerance/T5WGI1HfetaUJHW2o5zDKsAB"/>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/DSTU2/AllergyIntolerance/T5WGI1HfetaUJHW2o5zDKsAB"/>
  </link>
  <resource>
    <AllergyIntolerance>
      <recordedDate value="2016-02-16T12:17:23"/>
      <status value="confirmed"/>
      <criticality value="CRITH"/>
      <onset value="2016-02-16T00:00:00"/>
      <id value="T5WGI1HfetaUJHW2o5zDKsAB"/>
      <recorder>
        <display value="Helen Claire Wilson, MD"/>
        <reference value="https://hostname/instance/api/FHIR/DSTU2/Practitioner/TrTk7Zaw96VJM5sIX95IYHAB"/>
      </recorder>
      <patient>
        <display value="Fiona Hailey Ingrid Robertson"/>
        <reference
  value="https://hostname/instance/api/FHIR/DSTU2/Patient/Tb19ikuzxOh6YglUN28cuQ-d.3AjOduTYFVxlRM82bqEB"/>
      </patient>
      <substance>
        <text value="PEANUT-CONTAINING DRUG PRODUCTS"/>
        <coding>
          <system value="http://www.nlm.nih.gov/research/umls/rxnorm/"/>
          <code value="1540357"/>
          <display value="PEANUT-CONTAINING DRUG PRODUCTS"/>
        </coding>
      </substance>
      <reaction>
        <certainty value="confirmed"/>
        <onset value="2016-02-16T00:00:00"/>
        <manifestation>
          <text value="Anaphylaxis (SNOMED-CT: 39579001)"/>
        </manifestation>
      </reaction>
    </AllergyIntolerance>
  </resource>
  <search>
    <mode value="match"/>
  </search>
</entry>
<entry>
  <resource>
    <OperationOutcome>
      <issue>
        <severity value="warning"/>
        <details value="This response includes information available to the authorized user at the time of the
  request. It may not contain the entire record available in the system."/>
        <code>
          <text value="informational"/>
          <coding>
            <system value="http://hl7.org/fhir/issue-type"/>
            <code value="informational"/>
            <display value="informational"/>
          </coding>
        </code>
      </issue>
```

```
      </OperationOutcome>
    </resource>
    <search>
      <mode value="outcome"/>
    </search>
  </entry>
</Bundle>
```

10.3.3 Condition.Read (Problems) (DSTU2)

File: site/apis/10-Clinical-General/3-DSTU2/6-Condition-Read-Problems-DSTU2.html
Epic API ID | 6
FHIR version | DSTU2
Categories | Clinical General
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Keywords | Health Concerns;Problems;Problem List
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | Epic 2015
HTTP method | GET
URL template | api/FHIR/DSTU2/Condition/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=6

Description

The FHIR Condition resource defines detailed information about a Condition, problem, or diagnosis relevant to the health of the patient. Conditions are typically associated with other resources, such as the reason for a prescription, procedure, diagnostic test, etc. A Condition might be acute or chronic, but it is not used to represent an adverse reaction or an allergy intolerance.

The Condition resource corresponds to the following Common Clinical Data Set (CCDS) Elements: Problems and Health Concerns.

The Read interaction enables the look up of a Condition resource by a constant server ID. Clients can store only the server ID, and with a single request, retrieve the most up-to-date information about the Condition. Read interactions typically begin with a client having previously established a relationship, often through querying for patients through the search interaction.

Note that this resource retrieves only data stored in problem list records. As a result, medical history data documented outside of a patient's problem list isn't available to applications using this service unless that data is retrieved using another method.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | String | False | | | The Condition FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Condition | Condition | False | | | A single Condition resource.
Condition.abatementDateTime | DateTime | True | | | Date the problem was resolved.
Condition.asserter | ReferencedResource | False | | | User who first added the problem.
Condition.category | CodeableConcept | False | | | This is always Diagnosis.
Condition.clinicalStatus | String | False | | | The status of the problem. Can be active, inactive, or resolved.
Condition.code | CodeableConcept | False | | | The external codes associated with the problem. Uses SNOMED codes, or returns a name only if no SNOMED code is mapped.
Condition.dateRecorded | DateTime | True | | | Date the problem was added to the patient’s chart.
Condition.id | String | False | | | The Condition FHIR ID.
Condition.notes | String | False | | | Free text comment.
Condition.onsetDateTime | String | False | | | Date the problem began.
Condition.patient | ReferencedResource | False | | | Reference to the Patient resource associated with the Condition.
Condition.severity | CodeableConcept | False | | | The priority of the problem.

Sample request

```
http://hostname/instance/api/FHIR/DSTU2/Condition/TKucVI9PtvImxUTclJmmLDwB
```

Sample response

```
<Condition xmlns="http://hl7.org/fhir">
  <dateRecorded value="2016-01-19T00:00:00"/>
  <clinicalStatus value="active"/>
```

```
  <onsetDateTime value="2016-01-19T00:00:00"/>
  <verificationStatus value="confirmed"/>
  <id value="TKucVI9PtvImxUTclJmmLDwB"/>
  <patient>
    <display value="John Condition Doe"/>
    <reference value="https://hostname/instance/api/FHIR/DSTU2/Patient/T0hY1cLFnO.TSydQJmt.mneK9ATxa9uysPCZFPmn2KygB"/>
  </patient>
  <asserter>
    <display value="John Doe, MD"/>
    <reference value="https://hostname/instance/api/FHIR/DSTU2/Practitioner/TiOmkdUgh7pJl0-mpSeeFtQB"/>
  </asserter>
  <code>
    <text value="Broken leg"/>
    <coding>
      <system value="2.16.840.1.113883.6.90"/>
      <code value="S82.90XA"/>
      <display value="Broken leg"/>
    </coding>
    <coding>
      <system value="http://snomed.info/sct"/>
      <code value="46866001"/>
      <display value="Fracture of lower limb (disorder)"/>
    </coding>
  </code>
  <category>
    <text value="Diagnosis"/>
    <coding>
      <system value="http://www.loinc.org"/>
      <code value="29308-4"/>
      <display value="Diagnosis"/>
    </coding>
    <coding>
      <system value="http://snomed.info/sct"/>
      <code value="439401001"/>
      <display value="Diagnosis"/>
    </coding>
    <coding>
      <system value="http://hl7.org/fhir/condition-category"/>
      <code value="diagnosis"/>
      <display value="Diagnosis"/>
    </coding>
  </category>
  <severity>
    <text value="Med"/>
    <coding>
      <system value="http://snomed.info/sct"/>
      <code value="6736007"/>
      <display value="Moderate"/>
    </coding>
  </severity>
</Condition>
```

10.3.4 Condition.Search (Problems) (DSTU2)

File: site/apis/10-Clinical-General/3-DSTU2/7-Condition-Search-Problems-DSTU2.html
Epic API ID | 7
FHIR version | DSTU2
Categories | Clinical General
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Keywords | Health Concerns;Problems;Problem List
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | Epic 2015
HTTP method | GET
URL template | api/FHIR/DSTU2/Condition?patient={patient}&_id={_id}&clinicalStatus={clinicalStatus}&category={category}&subject={subject}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=7

Description

The FHIR Condition resource defines detailed information about a Condition, problem, or diagnosis relevant to the health of the patient. Conditions are typically associated with other resources, such as the reason for a prescription, procedure, diagnostic test, etc. A Condition might be acute or chronic, but it is not used to represent an adverse reaction or an allergy intolerance.

The Condition resource corresponds to the following Common Clinical Data Set (CCDS) Elements: Problems and Health Concerns.

The Search interaction enables the client to query for all problems associated with a given patient's Conditions. The client, having established the patient in question, now wishes to retrieve the list of that patient's problems or documented health concerns.

Note that this resource retrieves only data stored in problem list records. As a result, medical history data documented outside of a patient's problem list isn't available to applications using this service unless that data is retrieved using another method.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
category | String | True | | | Type of condition – must be “Diagnosis”.
clinicalStatus | String | True | | | Status of the condition FHIR Status Epic Status active active conditions resolved resolved conditions inactive deleted conditions If no status is specified all active (confirmed) conditions will be returned. *Note: Any FHIR Status not listed in the above table is not returned.
FHIR Status | Epic Status
active | active conditions
resolved | resolved conditions
inactive | deleted conditions
patient | String | Either patient or subject must be specified. If both are specified, they must reference the same patient. | | | The Patient FHIR resource ID.
subject | String | Either subject or patient must be specified. If both are specified, they must reference the same patient. | | | Subject of the condition.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False
Bundle.entry | Entry | False | Yes
Bundle.entry.resource | Resource | False
Bundle.entry.resource.Condition | Condition | False | | | A single Condition resource.
Bundle.entry.resource.Condition.abatementDateTime | DateTime | True | | | Date the problem was resolved.
Bundle.entry.resource.Condition.asserter | ReferencedResource | True | | | User who first added the problem.
Bundle.entry.resource.Condition.category | CodeableConcept | False | | | This is always Diagnosis.
Bundle.entry.resource.Condition.clinicalStatus | String | False | | | The status of the problem. Can be active, inactive, or resolved.
Bundle.entry.resource.Condition.code | CodeableConcept | False | | | The external codes associated with the problem. Uses SNOMED codes, or returns a name only if no SNOMED code is mapped.
Bundle.entry.resource.Condition.dateRecorded | DateTime | True | | | Date the problem was added to the patient’s chart.
Bundle.entry.resource.Condition.id | String | False | | | The Condition FHIR ID.
Bundle.entry.resource.Condition.notes | String | True | | | Free text comment.
Bundle.entry.resource.Condition.onsetDateTime | String | True | | | Date the problem began.
Bundle.entry.resource.Condition.patient | ReferencedResource | False | | | Reference to the Patient associated with the Condition.
Bundle.entry.resource.Condition.severity | CodeableConcept | True | | | The priority of the problem.
Bundle.total | Int32 | True

Sample request

```
http://hostname/instance/api/FHIR/DSTU2/Condition?patient=T0hY1cLFnO.TSydQJmt.mnZlFY0oi78v5x9iaxuyWOmUB&category=diagnos
    is
```

Sample response

```
<Bundle xmlns="http://hl7.org/fhir">
  <type value="searchset"/>
  <total value="2"/>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/DSTU2/Condition?patient=T0hY1cLFnO.TSydQJmt.mnZlFY0oi78v5x9iaxuyWOmUB
    &amp;category=diagnosis"/>
  </link>
  <entry>
    <fullUrl value="https://hostname/instance/api/FHIR/DSTU2/Condition/TFS10M6fD5mCLGAkbXhbQ7AB"/>
    <link>
      <relation value="self"/>
      <url value="https://hostname/instance/api/FHIR/DSTU2/Condition/TFS10M6fD5mCLGAkbXhbQ7AB"/>
    </link>
    <resource>
      <Condition>
        <dateRecorded value="2016-01-13T00:00:00"/>
        <clinicalStatus value="active"/>
        <onsetDateTime value="2016-01-13T00:00:00"/>
        <verificationStatus value="confirmed"/>
        <id value="TFS10M6fD5mCLGAkbXhbQ7AB"/>
        <patient>
          <display value="John Condition Doe"/>
          <reference
    value="https://hostname/instance/api/FHIR/DSTU2/Patient/T0hY1cLFnO.TSydQJmt.mneK9ATxa9uysPCZFPmn2KygB"/>
        </patient>
        <asserter>
          <display value="John Doe, MD"/>
          <reference value="https://hostname/instance/api/FHIR/DSTU2/Practitioner/TiOmkdUgh7pJl0-mpSeeFtQB"/>
```

```
      </asserter>
      <code>
        <text value="Pneumonia"/>
        <coding>
          <system value="2.16.840.1.113883.6.90"/>
          <code value="J18.9"/>
          <display value="Pneumonia"/>
        </coding>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="233604007"/>
          <display value="Pneumonia (disorder)"/>
        </coding>
      </code>
      <category>
        <text value="Diagnosis"/>
        <coding>
          <system value="http://www.loinc.org"/>
          <code value="29308-4"/>
          <display value="Diagnosis"/>
        </coding>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="439401001"/>
          <display value="Diagnosis"/>
        </coding>
        <coding>
          <system value="http://hl7.org/fhir/condition-category"/>
          <code value="diagnosis"/>
          <display value="Diagnosis"/>
        </coding>
      </category>
      <severity>
        <text value="High"/>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="24484000"/>
          <display value="Severe"/>
        </coding>
      </severity>
    </Condition>
  </resource>
  <search>
    <mode value="match"/>
  </search>
</entry>
<entry>
  <fullUrl value="https://hostname/instance/api/FHIR/DSTU2/Condition/TKucVI9PtvImxUTclJmmLDwB"/>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/DSTU2/Condition/TKucVI9PtvImxUTclJmmLDwB"/>
  </link>
  <resource>
    <Condition>
      <dateRecorded value="2016-01-19T00:00:00"/>
      <clinicalStatus value="active"/>
      <onsetDateTime value="2016-01-19T00:00:00"/>
      <verificationStatus value="confirmed"/>
      <id value="TKucVI9PtvImxUTclJmmLDwB"/>
      <patient>
        <display value="John Condition Doe"/>
        <reference
  value="https://hostname/instance/api/FHIR/DSTU2/Patient/T0hY1cLFnO.TSydQJmt.mneK9ATxa9uysPCZFPmn2KygB"/>
      </patient>
      <asserter>
        <display value="John Doe, MD"/>
        <reference value="https://hostname/instance/api/FHIR/DSTU2/Practitioner/TiOmkdUgh7pJl0-mpSeeFtQB"/>
      </asserter>
      <code>
        <text value="Broken leg"/>
        <coding>
          <system value="2.16.840.1.113883.6.90"/>
          <code value="S82.90XA"/>
          <display value="Broken leg"/>
        </coding>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="46866001"/>
          <display value="Fracture of lower limb (disorder)"/>
        </coding>
      </code>
      <category>
        <text value="Diagnosis"/>
        <coding>
          <system value="http://www.loinc.org"/>
          <code value="29308-4"/>
          <display value="Diagnosis"/>
        </coding>
        <coding>
```

```
            <system value="http://snomed.info/sct"/>
            <code value="439401001"/>
            <display value="Diagnosis"/>
          </coding>
          <coding>
            <system value="http://hl7.org/fhir/condition-category"/>
            <code value="diagnosis"/>
            <display value="Diagnosis"/>
          </coding>
        </category>
        <severity>
          <text value="Med"/>
          <coding>
            <system value="http://snomed.info/sct"/>
            <code value="6736007"/>
            <display value="Moderate"/>
          </coding>
        </severity>
      </Condition>
    </resource>
    <search>
      <mode value="match"/>
    </search>
  </entry>
</Bundle>
```

10.3.5 FamilyMemberHistory.Search (DSTU2)

File: site/apis/10-Clinical-General/3-DSTU2/337-FamilyMemberHistory-Search-DSTU2.html
Epic API ID | 337
FHIR version | DSTU2
Categories | Clinical General
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | Epic 2015
HTTP method | GET
URL template | api/FHIR/DSTU2/FamilyMemberHistory?patient={patient}&subject={subject}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=337

Description

Returns the medical and surgical history of a patient's relatives using RESTful URLs. For now, we only support Search by FHIR Patient ID, which will return all FamilyMemberHistory resources associated with a single patient.

A user or staff member accessing this FHIR resource must have security to access the patient's history.

A MyChart user accessing this FHIR resource must be authorized to view family medical history in MyChart.

Remarks

FHIR FamilyMemberHistory resources can be retrieved through a Search interaction with a Patient ID, and can be consumed as either XML or JSON.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
patient | String | Either patient or subject must be provided. If both are provided, they must reference the same patient. | | | The Patient resource with the associated FamilyMemberHistory.
subject | String | Either patient or subject must be provided. If both are provided, they must reference the same patient. | | | The Patient resource with the associated FamilyMemberHistory.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | Entry | False | Yes | | A single entry in the resource bundle.
Bundle.entry.resource | Resource | False | | | A single resource returned as part of this bundle.
Bundle.entry.resource.FamilyMemberHistory | FamilyMemberHistory | False | | | A single FamilyMemberHistory resource.
Bundle.entry.resource.FamilyMemberHistory.ageString | String | False | | | Either the relative’s death age or their approximate current age.
Bundle.entry.resource.FamilyMemberHistory.condition | Condition | False | Yes | | Information about the family member's condition.
Bundle.entry.resource.FamilyMemberHistory.condition.code | CodeableConcept | False | | | A codeable concept.
Bundle.entry.resource.FamilyMemberHistory.condition.onsetString | String | False | | | Age of onset.
Bundle.entry.resource.FamilyMemberHistory.date | DateTime | True | | | Date the FamilyMemberHistory resource was recorded or updated.
Bundle.entry.resource.FamilyMemberHistory.deceasedBoolean | Boolean | False | | | Whether or not relative is deceased.

Bundle.entry.resource.FamilyMemberHistory.name | String | True | | | Name of the relative.
Bundle.entry.resource.FamilyMemberHistory.patient | ReferencedResource | False | | | The Patient resource with the associated FamilyMemberHistory
Bundle.entry.resource.FamilyMemberHistory.relationship | CodeableConcept | False | | | The type of relationship between the patient and their relative.
Bundle.total | Int32 | False | | | Total number of resources that matched the search criteria.

Sample request

```
https://hostname/instance/api/FHIR/DSTU2/FamilyMemberHistory?patient=Tb19ikuzxOh6YglUN28cuQ-d.3AjOduTYFVxlRM82bqEB
```

Sample response

```
<Bundle xmlns="http://hl7.org/fhir">
  <type value="searchset"/>
  <total value="3"/>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/DSTU2/FamilyMemberHistory?patient=Tb19ikuzxOh6YglUN28cuQ-
    d.3AjOduTYFVxlRM82bqEB"/>
  </link>
  <entry>
    <resource>
      <FamilyMemberHistory>
        <date value="2016-03-04T00:00:00"/>
        <name value="Robertson,Deborah"/>
        <deceasedBoolean value="false"/>
        <patient>
          <display value="Fiona Hailey Ingrid Robertson"/>
          <reference
    value="https://hostname/instance/api/FHIR/DSTU2/Patient/Tb19ikuzxOh6YglUN28cuQ-d.3AjOduTYFVxlRM82bqEB"/>
        </patient>
        <relationship>
          <text value="Mother"/>
          <coding>
            <system value="http://hl7.org/fhir/ValueSet/v3-FamilyMember"/>
            <code value="MTH"/>
            <display value="Mother"/>
          </coding>
        </relationship>
        <condition>
          <onsetString value="53"/>
          <code>
            <text value="Breast cancer"/>
          </code>
        </condition>
      </FamilyMemberHistory>
    </resource>
    <search>
      <mode value="match"/>
    </search>
  </entry>
  <entry>
    <resource>
      <FamilyMemberHistory>
        <date value="2016-03-04T00:00:00"/>
        <name value="Robertson,Jack"/>
        <deceasedBoolean value="false"/>
        <patient>
          <display value="Fiona Hailey Ingrid Robertson"/>
          <reference
    value="https://hostname/instance/api/FHIR/DSTU2/Patient/Tb19ikuzxOh6YglUN28cuQ-d.3AjOduTYFVxlRM82bqEB"/>
        </patient>
        <relationship>
          <text value="Father"/>
          <coding>
            <system value="http://hl7.org/fhir/ValueSet/v3-FamilyMember"/>
            <code value="FTH"/>
            <display value="Father"/>
          </coding>
        </relationship>
        <condition>
          <onsetString value="55"/>
          <code>
            <text value="Heart attack"/>
          </code>
        </condition>
      </FamilyMemberHistory>
    </resource>
    <search>
      <mode value="match"/>
    </search>
  </entry>
  <entry>
    <resource>
      <FamilyMemberHistory>
        <date value="2016-03-04T00:00:00"/>
```

```
        <name value="Casilla,Hailey"/>
        <deceasedBoolean value="true"/>
        <patient>
          <display value="Fiona Hailey Ingrid Robertson"/>
          <reference
    value="https://hostname/instance/api/FHIR/DSTU2/Patient/Tb19ikuzxOh6YglUN28cuQ-d.3AjOduTYFVxlRM82bqEB"/>
        </patient>
        <relationship>
          <text value="Maternal Grandmother"/>
          <coding>
            <system value="http://hl7.org/fhir/ValueSet/v3-FamilyMember"/>
            <code value="MGRMTH"/>
            <display value="Maternal Grandmother"/>
          </coding>
        </relationship>
        <condition>
          <onsetString value="62"/>
          <code>
            <text value="Breast cancer"/>
          </code>
        </condition>
        <condition>
          <code>
            <text value="Pulmonary hypertension"/>
            <coding>
              <system value="http://snomed.info/sct"/>
              <code value="70995007"/>
              <display value="Pulmonary hypertension"/>
            </coding>
          </code>
        </condition>
        <condition>
          <onsetString value="82"/>
          <code>
            <text value="Stroke"/>
          </code>
        </condition>
      </FamilyMemberHistory>
    </resource>
    <search>
      <mode value="match"/>
    </search>
  </entry>
  <entry>
    <resource>
      <OperationOutcome>
        <issue>
          <severity value="warning"/>
          <details value="This response includes information available to the authorized user at the time of the
    request. It may not contain the entire record available in the system."/>
          <code>
            <text value="informational"/>
            <coding>
              <system value="http://hl7.org/fhir/issue-type"/>
              <code value="informational"/>
              <display value="informational"/>
            </coding>
          </code>
        </issue>
      </OperationOutcome>
    </resource>
    <search>
      <mode value="outcome"/>
    </search>
  </entry>
</Bundle>
```

10.3.6 Observation.Read (Social History) (DSTU2)

File: site/apis/10-Clinical-General/3-DSTU2/447-Observation-Read-Social-History-DSTU2.html
Epic API ID | 447
FHIR version | DSTU2
Categories | Clinical General
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Keywords | Smoking; Social; Social history;Flowsheets
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | Epic 2015
HTTP method | GET
URL template | api/FHIR/DSTU2/Observation/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients

Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=447

Description

The FHIR Observation resource defines information about a measurement or simple assertion made about a patient. The FHIR Observation Smoking History resource retrieves the status of a patient's tobacco use. This could be used to determine if the patient is a current or former tobacco user, what type of tobacco product is/was used, and the dates the patient started and, if a former user, stopped using tobacco.

The Observation resource corresponds to the following Common Clinical Data Set (CCDS) Elements: Smoking Status

The Read interaction enables the look up of an Observation resource by a constant server ID. Clients can store only the server ID, and with a single request, retrieve the most up-to-date information about the observation. Read interactions typically begin with a client having previously established a relationship, often through querying for observations through the search interaction.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | String | False

Response elements

Name | Type | Optional | Array | Cardinality | Description
Observation | Observation | False | | | A single Observation FHIR resource.
Observation.category | CodeableConcept | False | | | Always "social-history".
Observation.code | CodeableConcept | False | | | The LOINC code for smoking history (72166-2).
Observation.comments | String | True | | | The comments. These are not sent in a MyChart context.
Observation.effectivePeriod | Period | True | | | Dates when a patient starts or ends smoking.
Observation.effectivePeriod.end | DateTime | True | | | The date the patient quit smoking.
Observation.effectivePeriod.start | DateTime | True | | | The date the patient started smoking.
Observation.id | String | False | | | The Observation FHIR ID.
Observation.issued | DateTime | True | | | The date this observation was documented.
Observation.status | String | False | | | This value is always "final". If Smoking Tobacco Use is 6 - Never Assessed, this value is "unknown".
Observation.subject | ReferencedResource | False | | | The patient FHIR ID.
Observation.valueCodeableConcept | CodeableConcept | True | | | SNOMED coded value of smoking status.

Sample request

```
http://hostname/instance/api/FHIR/DSTU2/Observation/TE8W3rK2.OY-eAQIFbovGO-5EtqnRnHScvBDg3VhNWFkB
```

Sample response

```
<Observation xmlns="http://hl7.org/fhir">
  <comments value="This is a comment on Smoker."/>
  <status value="final"/>
  <id value="TE8W3rK2.OY-eAQIFbovGO-5EtqnRnHScvBDg3VhNWFkB"/>
  <code>
    <text value="Smoking History"/>
    <coding>
      <system value="http://loinc.org"/>
      <code value="72166-2"/>
      <display value="Smoking History"/>
    </coding>
  </code>
  <subject>
    <display value="John Smoker Doe"/>
    <reference value="https://hostname/instance/api/FHIR/DSTU2/Patient/TE8W3rK2.OY-eAQIFbovGOw3YxmqXp-GmS6WnZ0Q.ZwEB"/>
  </subject>
  <valueCodeableConcept>
    <text value="Current Some Day Smoker"/>
    <coding>
      <system value="http://snomed.info/sct"/>
      <code value="428041000124106"/>
      <display value="Current some day smoker"/>
    </coding>
  </valueCodeableConcept>
</Observation>
```

10.3.7 Observation.Search (Social History) (DSTU2)

File: site/apis/10-Clinical-General/3-DSTU2/448-Observation-Search-Social-History-DSTU2.html
Epic API ID | 448
FHIR version | DSTU2
Categories | Clinical General
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes

Keywords | Smoking; Social; Social history;tobacco
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | Epic 2015
HTTP method | GET
URL template | api/FHIR/DSTU2/Observation?code={code}&patient={patient}&subject={subject}&category={category}&issued={issued}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=448

Description

The FHIR Observation resource defines information about a measurement or simple assertion made about a patient. The FHIR Observation Smoking History resource retrieves the status of a patient's tobacco use. This could be used to determine if the patient is a current or former tobacco user, what type of tobacco product is/was used, and the dates the patient started and, if a former user, stopped using tobacco.

The Observation resource corresponds to the following Common Clinical Data Set (CCDS) Elements: Smoking Status.

The Search interaction enables the client to query for all observations of a specific name or list of names that have been filed for a given patient. The client, having established the patient in question, now wishes to retrieve observations describing documented smoking history of a patient. Clients can further refine the search to specify a date for the Observation resources in question.

Remarks

When searching, a patient resource ID and LOINC code must be supplied.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
category | String | Either this or code must be provided. | | | Use "social-history" to search for smoking history observations.
code | String | Either this or category must be provided. | | | LOINC code of smoking history observation (72166-2).
issued | String | True | | | The date the observation was recorded.
patient | String | Either this or subject must be provided | | | Reference to a patient resource.
subject | String | Either this or subject must be provided | | | Reference to a patient resource.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | Entry | False | Yes | | A single entry in the resource bundle.
Bundle.entry.resource | Resource | False | | | A single resource returned as part of this bundle.
Bundle.entry.resource.Observation | Observation | False | | | A single Observation FHIR resource.
Bundle.entry.resource.Observation.category | CodeableConcept | False | | | Always "social-history".
Bundle.entry.resource.Observation.code | CodeableConcept | False | | | The LOINC code for smoking history (72166-2).
Bundle.entry.resource.Observation.comments | String | True | | | The comments. These are not sent in a MyChart context.
Bundle.entry.resource.Observation.effectivePeriod | Period | True | | | Dates when a patient starts or ends smoking.
Bundle.entry.resource.Observation.effectivePeriod.end | DateTime | True | | | The date the patient quit smoking.
Bundle.entry.resource.Observation.effectivePeriod.start | DateTime | True | | | The date the patient started smoking.
Bundle.entry.resource.Observation.id | String | False | | | The Observation FHIR ID.
Bundle.entry.resource.Observation.issued | DateTime | True | | | The date this observation was documented.
Bundle.entry.resource.Observation.status | String | False | | | If smoking tobacco use is documented as "never assessed", this value is "unknown". Otherwise the value will always be "final".
Bundle.entry.resource.Observation.subject | ReferencedResource | False | | | The patient FHIR ID.
Bundle.entry.resource.Observation.valueCodeableConcept | CodeableConcept | True | | | SNOMED coded value of smoking status.

Sample request

```
http://hostname/instance/api/FHIR/DSTU2/Observation?patient=TE8W3rK2.OY-eAQIFbovGOw3YxmqXp-GmS6WnZ0Q.ZwEB&code=72166-2
```

Sample response

```
<Bundle xmlns="http://hl7.org/fhir">
  <type value="searchset"/>
  <total value="1"/>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/DSTU2/Observation?patient=TE8W3rK2.OY-eAQIFbovGOw3YxmqXp-
    GmS6WnZ0Q.ZwEB&amp;code=72166-2"/>
  </link>
  <entry>
    <fullUrl value="https://hostname/instance/api/FHIR/DSTU2/Observation/TE8W3rK2.OY-
    eAQIFbovGO-5EtqnRnHScvBDg3VhNWFkB"/>
    <link>
      <relation value="self"/>
      <url value="https://hostname/instance/api/FHIR/DSTU2/Observation/TE8W3rK2.OY-eAQIFbovGO-5EtqnRnHScvBDg3VhNWFkB"/>
```

```
    </link>
    <resource>
      <Observation>
        <comments value="This is a comment on Smoker."/>
        <status value="final"/>
        <id value="TE8W3rK2.OY-eAQIFbovGO-5EtqnRnHScvBDg3VhNWFkB"/>
        <code>
          <text value="Smoking History"/>
          <coding>
            <system value="http://loinc.org"/>
            <code value="72166-2"/>
            <display value="Smoking History"/>
          </coding>
        </code>
        <subject>
          <display value="John Smoker Doe"/>
          <reference value="https://hostname/instance/api/FHIR/DSTU2/Patient/TE8W3rK2.OY-eAQIFbovGOw3YxmqXp-
    GmS6WnZ0Q.ZwEB"/>
        </subject>
        <valueCodeableConcept>
          <text value="Current Some Day Smoker"/>
          <coding>
            <system value="http://snomed.info/sct"/>
            <code value="428041000124106"/>
            <display value="Current some day smoker"/>
          </coding>
        </valueCodeableConcept>
      </Observation>
    </resource>
    <search>
      <mode value="match"/>
    </search>
  </entry>
</Bundle>
```


10.3.8 Procedure.Read (Orders) (DSTU2)

File: site/apis/10-Clinical-General/3-DSTU2/33-Procedure-Read-Orders-DSTU2.html
Epic API ID | 33
FHIR version | DSTU2
Categories | Clinical General
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Keywords | Procedures
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | Epic 2017
HTTP method | GET
URL template | api/FHIR/DSTU2/Procedure/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=33

Description

The FHIR Procedure resource defines an activity performed on or with a patient as part of the provision of care. It corresponds with surgeries and procedures performed, including endoscopies and biopsies, as well as less invasive actions like counseling and physiotherapy. This resource is designed for a high-level summary around the occurrence of a procedure, and not for specific procedure log documentation. The below documentation describes surgical, dental, and diagnostic procedures performed at a particular organization, but does not include historical documentation of procedures on a patient.

The Procedure resource corresponds to the following Common Clinical Data Set (CCDS) Elements: Procedures.

The read interaction enables the lookup of a Procedure resource by a constant server ID. The read interaction allows clients to store only the server ID, and with a single request, retrieve the most up-to-date information about a procedure. Read interactions typically begin with a client having previously established a relationship, often through querying for Procedures through the search interaction.

Remarks

The various procedure types have different unique identifiers that will be used for generating a FHIR Resource REST ID.

* Procedure Orders - Based on "ORD" and ORD .1 and EAP .1

* Dental Procedures - Based on "RES" and RES .1

* Surgeries - Based on "ORL" and ORL .1 and Panel # and ORP .1

The data is otherwise stored in a similar format between all procedure types. Exceptions are noted within the read response grid below.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | String | False | | | The Procedure FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Procedure | Procedure | False
Procedure.code | CodeableConcept | False | | | External codes associated with the procedure.
Procedure.identifier | Identifier | False | Yes | | System-specific ID for the procedure order record.
Procedure.performedDateTime | DateTime | This or performedPeriod will be populated. | | | When the procedure was performed.
Procedure.performedPeriod | Period | This or performedDateTime will be populated. | | | Period of time over which the procedure was performed.
Procedure.performer | performer | True | Yes | | Who performed the Procedure.
Procedure.performer.actor | ReferencedResource | True | | | Reference to the Practitioner that performed the procedure.
Procedure.reasonCodeableConcept | CodeableConcept | False | | | External codes for diagnosis records associated with the procedure.
Procedure.reasonReference | ReferencedResource | True | | | Reference to Condition resources associated with the Procedure.
Procedure.status | String | False | | | The status of the Procedure. API Value FHIR Event Status cancelled aborted completed completed *Note: Omitted FHIR event status equivalent values are not returned by this search.
API Value | FHIR Event Status
cancelled | aborted
completed | completed
Procedure.subject | ReferencedResource | False | | | The Patient FHIR ID for whom this Procedure resource is relevant.

Sample request

```
https://hostname/instance/api/FHIR/DSTU2/Procedure/Tam-Bgf-I1ezzHyQSPI9Xuy0S3u4A64a3ntxGIXjTfOgB
```

Sample response

```
<Procedure xmlns="http://hl7.org/fhir">
  <status value="completed"/>
  <notPerformed value="false"/>
  <id value="Tam-Bgf-I1ezzHyQSPI9Xuy0S3u4A64a3ntxGIXjTfOgB"/>
  <subject>
    <display value="Fhir A Doe"/>
    <reference value="https://hostname/instance/api/FHIR/DSTU2/Patient/TGwyi7uWQngTh8wlsxLyWPi6.8wgRuUnMqMfRuwJhsFkB"/>
  </subject>
  <code>
    <text value="LAB CBC"/>
    <coding>
      <system value="2.16.840.1.113883.6.14"/>
      <code value="LAB1003"/>
      <display value="LAB CBC"/>
    </coding>
  </code>
  <performer>
    <actor>
      <display value="John Doe, MD"/>
      <reference value="https://hostname/instance/api/FHIR/DSTU2/Practitioner/T261TVJilfMxfUGfuXuDQpQB"/>
    </actor>
  </performer>
  <performedPeriod>
    <start value="2016-01-22T21:54:00"/>
    <end value="2016-01-22T21:54:00"/>
  </performedPeriod>
</Procedure>
```

10.3.9 Procedure.Search (Orders) (DSTU2)

File: site/apis/10-Clinical-General/3-DSTU2/34-Procedure-Search-Orders-DSTU2.html
Epic API ID | 34
FHIR version | DSTU2
Categories | Clinical General
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Keywords | Procedures
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | Epic 2017
HTTP method | GET
URL template | api/FHIR/DSTU2/Procedure?patient={patient}&subject={subject}&date={date}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=34

Description

The FHIR Procedure resource defines an activity performed on or with a patient as part of the provision of care. It corresponds with surgeries and procedures performed, including endoscopies and biopsies, as well as less invasive actions like counseling and physiotherapy. This resource is designed for a high-level summary around the occurrence of a procedure, and not for specific procedure log documentation. The below documentation describes surgical, dental, and diagnostic procedures performed at a particular organization, but does not include historical documentation of procedures on a patient. When searching, only completed procedures are returned.

The Procedure resource corresponds to the following Common Clinical Data Set (CCDS) Elements: Procedures.

The Search interaction enables the client to query for all of a patient's completed procedures by providing the patient's FHIR ID. The client can also look up a list of known Procedure resources with the "_id" parameter. The client, having established the patient in question, now wishes to retrieve all procedure data for the patient.

Some healthcare activities could be classified as either a diagnostic report or a procedure (e.g. endoscopic procedures, biopsies, etc.). Individual health systems can define which activities they classify as a procedure vs. a diagnostic report. As a result, you may see some procedures returned as a DiagnosticReport resource at some health systems, but as a Procedure resource at others.

Note:

Request parameters

Name | Type | Optional | Array | Cardinality | Description
date | String | False | | | The date or date range the Procedure was performed.
patient | String | Either patient or subject must be specified. If both are specified, they must reference the same patient. | | | The Patient FHIR resource ID.
subject | String | Either subject or patient must be specified. If both are specified, they must reference the same patient. | | | Subject of the procedure.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Bundle of resources that met the search criteria.
Bundle.entry | Entry | False | Yes | | One Procedure resource in the bundle.
Bundle.entry.resource | Resource | False
Bundle.entry.resource.Procedure | Procedure | False
Bundle.entry.resource.Procedure.code | CodeableConcept | False | | | External codes associated with the procedure.
Bundle.entry.resource.Procedure.identifier | Identifier | False | Yes | | System-specific ID for the procedure order record.
Bundle.entry.resource.Procedure.performedDateTime | DateTime | This or performedPeriod will be populated. | | | When the procedure was performed.
Bundle.entry.resource.Procedure.performedPeriod | Period | This or performedDateTime will be populated. | | | Period of time over which the procedure was performed.
Bundle.entry.resource.Procedure.performer | performer | True | Yes | | Who performed the Procedure.
Bundle.entry.resource.Procedure.performer.actor | ReferencedResource | True | | | Reference to the Practitioner that performed the Procedure.
Bundle.entry.resource.Procedure.reasonCodeableConcept | CodeableConcept | False | | | External codes for diagnosis records associated with the procedure.
Bundle.entry.resource.Procedure.reasonReference | ReferencedResource | True | | | Reference to Condition resources associated with the Procedure.
Bundle.entry.resource.Procedure.status | String | False | | | The status of the Procedure. API Value FHIR Event Status cancelled aborted completed completed *Note: Omitted FHIR event status equivalent values are not returned by this search.
API Value | FHIR Event Status
cancelled | aborted
completed | completed
Bundle.total | Int32 | True | | | Total number of Procedure resources that met the search criteria.

Sample request

```
https://hostname/instance/api/FHIR/DSTU2/Procedure?patient=Tam-Bgf-I1ezzHyQSPI9Xuy0S3u4A64a3ntxGIXjTfOgB
```

Sample response

```
<Bundle xmlns="http://hl7.org/fhir">
  <type value="searchset"/>
  <total value="5"/>
  <link>
    <relation value="self"/>
    <url
    value="https://hostname/instance/api/FHIR/DSTU2/Procedure?patient=TGwyi7uWQngTh8wlsxLyWPi6.8wgRuUnMqMfRuwJhsFkB"/>
  </link>
  <entry>
    <fullUrl value="https://hostname/instance/api/FHIR/DSTU2/Procedure/Tam-Bgf-I1ezzHyQSPI9Xuy0S3u4A64a3ntxGIXjTfOgB"/>
    <link>
      <relation value="self"/>
      <url value="https://hostname/instance/api/FHIR/DSTU2/Procedure/Tam-Bgf-I1ezzHyQSPI9Xuy0S3u4A64a3ntxGIXjTfOgB"/>
    </link>
    <resource>
      <Procedure>
        <status value="completed"/>
```

```
      <notPerformed value="false"/>
      <id value="Tam-Bgf-I1ezzHyQSPI9Xuy0S3u4A64a3ntxGIXjTfOgB"/>
      <subject>
        <display value="Fhir A Doe"/>
        <reference
  value="https://hostname/instance/api/FHIR/DSTU2/Patient/TGwyi7uWQngTh8wlsxLyWPi6.8wgRuUnMqMfRuwJhsFkB"/>
      </subject>
      <code>
        <text value="LAB CBC"/>
        <coding>
          <system value="2.16.840.1.113883.6.14"/>
          <code value="LAB1003"/>
          <display value="LAB CBC"/>
        </coding>
      </code>
      <performer>
        <actor>
          <display value="John Doe, MD"/>
          <reference value="https://hostname/instance/api/FHIR/DSTU2/Practitioner/T261TVJilfMxfUGfuXuDQpQB"/>
        </actor>
      </performer>
      <performedPeriod>
        <start value="2016-01-22T21:54:00"/>
        <end value="2016-01-22T21:54:00"/>
      </performedPeriod>
    </Procedure>
  </resource>
  <search>
    <mode value="match"/>
  </search>
</entry>
<entry>
  <fullUrl value="https://hostname/instance/api/FHIR/DSTU2/Procedure/THHQnW48YiA.fTWoSsw6jwwB"/>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/DSTU2/Procedure/THHQnW48YiA.fTWoSsw6jwwB"/>
  </link>
  <resource>
    <Procedure>
      <status value="completed"/>
      <notPerformed value="false"/>
      <id value="THHQnW48YiA.fTWoSsw6jwwB"/>
      <subject>
        <display value="Fhir A Doe"/>
        <reference
  value="https://hostname/instance/api/FHIR/DSTU2/Patient/TGwyi7uWQngTh8wlsxLyWPi6.8wgRuUnMqMfRuwJhsFkB"/>
      </subject>
      <code>
        <text value="DENTAL GOLD FOIL ONE SURFACE"/>
        <coding>
          <system value="2.16.840.1.113883.6.13"/>
          <code value="D2410"/>
          <display value="DENTAL GOLD FOIL ONE SURFACE"/>
        </coding>
      </code>
      <reasonCodeableConcept>
        <text value="Activities involving golf"/>
        <coding>
          <system value="2.16.840.1.113883.6.96"/>
          <code value="282291009"/>
          <display value="Activities involving golf"/>
        </coding>
        <coding>
          <system value="2.16.840.1.113883.6.103"/>
          <code value="E006.2"/>
          <display value="Activities involving golf"/>
        </coding>
      </reasonCodeableConcept>
      <performer>
        <actor>
          <display value="Gordon Lofgren, D.D.S."/>
          <reference value="https://hostname/instance/api/FHIR/DSTU2/Practitioner/TDi0c-wTdSkamGZ03pmjCQQB"/>
        </actor>
      </performer>
      <performedPeriod>
        <start value="2016-01-25T21:44:52"/>
        <end value="2016-01-25T21:44:52"/>
      </performedPeriod>
    </Procedure>
  </resource>
  <search>
    <mode value="match"/>
  </search>
</entry>
<entry>
  <fullUrl value="https://hostname/instance/api/FHIR/DSTU2/Procedure/T5VFSb4MEF7yP.XtlhXr89gB"/>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/DSTU2/Procedure/T5VFSb4MEF7yP.XtlhXr89gB"/>
```

```
  </link>
  <resource>
    <Procedure>
      <status value="completed"/>
      <notPerformed value="false"/>
      <id value="T5VFSb4MEF7yP.XtlhXr89gB"/>
      <subject>
        <display value="Fhir A Doe"/>
        <reference
  value="https://hostname/instance/api/FHIR/DSTU2/Patient/TGwyi7uWQngTh8wlsxLyWPi6.8wgRuUnMqMfRuwJhsFkB"/>
      </subject>
      <code>
        <text value="DENTAL PROPHYLAXIS ADULT"/>
        <coding>
          <system value="2.16.840.1.113883.6.13"/>
          <code value="D1110"/>
          <display value="DENTAL PROPHYLAXIS ADULT"/>
        </coding>
      </code>
      <performer>
        <actor>
          <display value="Gordon Lofgren, D.D.S."/>
          <reference value="https://hostname/instance/api/FHIR/DSTU2/Practitioner/TDi0c-wTdSkamGZ03pmjCQQB"/>
        </actor>
      </performer>
      <performedPeriod>
        <start value="2016-01-25T21:44:15"/>
        <end value="2016-01-25T21:44:15"/>
      </performedPeriod>
    </Procedure>
  </resource>
  <search>
    <mode value="match"/>
  </search>
</entry>
<entry>
  <fullUrl value="https://hostname/instance/api/FHIR/DSTU2/Procedure/Tul-uG7hfat0IJz4JLX1ltlV8-UC59.qQBkHe7gp.KC4B"/>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/DSTU2/Procedure/Tul-uG7hfat0IJz4JLX1ltlV8-UC59.qQBkHe7gp.KC4B"/>
  </link>
  <resource>
    <Procedure>
      <status value="completed"/>
      <notPerformed value="false"/>
      <id value="Tul-uG7hfat0IJz4JLX1ltlV8-UC59.qQBkHe7gp.KC4B"/>
      <subject>
        <display value="Fhir A Doe"/>
        <reference
  value="https://hostname/instance/api/FHIR/DSTU2/Patient/TGwyi7uWQngTh8wlsxLyWPi6.8wgRuUnMqMfRuwJhsFkB"/>
      </subject>
      <code>
        <text value="APPENDECTOMY OPEN"/>
        <coding>
          <system value="2.16.840.1.113883.6.12"/>
          <code value="44955"/>
          <display value="APPENDECTOMY OPEN"/>
        </coding>
      </code>
      <reasonCodeableConcept>
        <text value="Acute appendicitis with generalized peritonitis"/>
        <coding>
          <system value="2.16.840.1.113883.6.103"/>
          <code value="540.0"/>
          <display value="Acute appendicitis with generalized peritonitis"/>
        </coding>
      </reasonCodeableConcept>
      <performedPeriod>
        <start value="2016-01-26T16:50:00"/>
      </performedPeriod>
    </Procedure>
  </resource>
  <search>
    <mode value="match"/>
  </search>
</entry>
<entry>
  <fullUrl value="https://hostname/instance/api/FHIR/DSTU2/Procedure/TzS7jvRDZocjM4.9VAaiLZvV87NSC4dxveMFO.dFHtKIB"/>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/DSTU2/Procedure/TzS7jvRDZocjM4.9VAaiLZvV87NSC4dxveMFO.dFHtKIB"/>
  </link>
  <resource>
    <Procedure>
      <status value="completed"/>
      <notPerformed value="false"/>
      <id value="TzS7jvRDZocjM4.9VAaiLZvV87NSC4dxveMFO.dFHtKIB"/>
      <subject>
        <display value="Fhir A Doe"/>
```

```
          <reference
      value="https://hostname/instance/api/FHIR/DSTU2/Patient/TGwyi7uWQngTh8wlsxLyWPi6.8wgRuUnMqMfRuwJhsFkB"/>
          </subject>
          <code>
            <text value="APPENDECTOMY OPEN"/>
            <coding>
              <system value="2.16.840.1.113883.6.12"/>
              <code value="44950"/>
              <display value="APPENDECTOMY OPEN"/>
            </coding>
          </code>
          <performedPeriod>
            <start value="2016-01-26T14:15:00"/>
          </performedPeriod>
        </Procedure>
      </resource>
      <search>
        <mode value="match"/>
      </search>
    </entry>
  </Bundle>
```